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Maximum-Entropy Sampling

Algorithms and Application

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Marcia Fampa • Jon Lee

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Dedicated to Kurt, on the occasion of his $65+\delta$ *birthday. A continuous or discrete celebration? All one!*

Preface

> "Don't worry about the overall importance of the problem; work on it if it looks interesting. I think there's a sufficient correlation between interest and importance."
>
> - David Blackwell, 1984 [DeG86]

This book is aimed at optimization students and researchers looking for something a little different. The TSP (traveling-salesperson problem) is a great problem for learning about 0/1 *linear* optimization (see [LLKS85], for example). The MESP (maximum-entropy sampling problem) is a great problem for learning about 0/1 *nonlinear* optimization. Even if you are not directly interested in our problem, study this book — we guarantee that it will open your mind!

Briefly, this book is mainly about algorithms for MESP: find a maximum (log-)determinant principal submatrix $C[S, S]$ of a given order s, from an input positive-semidefinite matrix C of order $n > s$. The problem is nice to study for a variety of reasons:

- MESP has a simple statement (like the TSP).
- If C is the covariance matrix for a multivariate Gaussian vector $\mathcal{Y} \in \mathbb{R}^n$, then MESP is equivalent to finding a maximum differential-entropy subvector $\mathcal{Y}_S$ of size s from $\mathcal{Y}$.
- MESP has application to the problem of designing environmental-monitoring networks and to spatial statistics in general.
- MESP is a "submodular maximization" problem, so it has links to a well-studied (NP-Hard) problem amenable to approximation algorithms.
- MESP is NP-Hard (like the TSP) and already very challenging with $n = 100$ (*not* like the TSP!).
- MESP is a very challenging and unusual problem from the perspective of 0/1 optimization; the constraints are simple, and all of the complexity is in the (non-separable) objective function (the reverse of the usual formulation of the TSP).
- Like the TSP, useful mathematical-optimization techniques for MESP include: convex optimization, semidefinite optimization, Lagrangian relaxation, Lagrangian

duality, integer-linear optimization, matching, dynamic programming, branch-and-bound, and local search.
- Methods for MESP use lots of matrix algebra and matrix calculus ☺.

We have tried to make the book as self-contained as is practical. Some previous exposure to linear, integer-linear, and nonlinear optimization will make the going easier. We have an appendix on tools that we use from matrix algebra and calculus; if you are unfamiliar with or rusty on those subjects, then at least breeze through the appendix to start. We do not include results of computational experiments; but for many of the results that we discuss, results of computational experiments can be found in the source papers.

> "No, this is something that I can't understand, positively can't understand. But the strangest, the most incomprehensible thing of all, is how authors can choose such subjects. I confess that this is quite inconceivable; it is indeed . . . no, no, I just can't understand it at all!"
>
> - Nikolai Gogol (The Nose)

Jon first got *infected* with MESP, by email (but not a virus), via Maurice Queyranne (University of British Columbia), who had heard about it from his *infected* UBC colleague Jim Zidek, an environmental statistician. This was during Jon's 1991-2 stay at the Center for Operations Research and Econometrics (CORE) at Université Catholique de Louvain (Belgium). Jon's work on MESP that year, with Chun Ko (then a post-doctoral fellow at Centrum Wiskunde & Informatica (CWI), Amsterdam; now *recovered*, a Trader at Citigroup) and Maurice, was seminal in Jon's now long interest in mixed-integer nonlinear optimization. Soon after, Jon had a doctoral student at the University of Kentucky, Joy Williams. Kurt Anstreicher (University of Iowa), a long-time friend and collaborator of Jon's, had a visiting Doutorado Sanduíche student from Universidade Federal do Rio de Janeiro (UFRJ), Marcia Fampa. The four collaborated on MESP, leading to Ph.D.s for Joy ([Wil98]) and Marcia ([Fam96]), and so now three more were *infected* with MESP (Joy *recovered*, now Vice President for Academic Affairs at the University of Sioux Falls).

Another key moment was when Jon gave a talk on MESP at the Iowa Center for Global Environmental Research, University of Iowa, April 1994. There he met Dale Zimmerman (Iowa) and Werner Müller (Johannes Kepler University, Linz), which lead to his participation at MODA 1995 (Spetses), where he met Henry Wynn, and at MODA 2001 (Puchberg/Schneeberg). Via all of this, some of the mathematical-optimization work of Jon and his MESP collaborators was exposed to the optimal-design community.

> DeGroot: One component of the interest is probably that others are interested in it, anyway.
> Blackwell: That's a big component. You want to tell somebody about it after you've done it.
>
> - see [DeG86]

We conceived the idea of this monograph in the pre-pandemic autumn of 2019, and started it in earnest during the pandemic summer of 2020. We continued working hard on it during the pandemic summer of 2021, completed a first draft in the pandemic autumn of 2021, and refined and finalized it during the pandemic first half of 2022. Hopefully the reader is at a post-pandemic time.



Entropy (and cannabis) "to go" at Ann Arbor in pandemic times

Marcia's work on MESP was supported in part by CNPq grants 303898/2016-0 and 434683/2018-3. Jon was supported over the years to study MESP by NSF grants DMI-9401424 and CMMI-1160915, ONR grant N00014-17-1- 2296, AFOSR grants FA9550-19-1-0175 and FA9550-22-1-0172, and additionally by Conservatoire National des Arts et Métiers. We were further supported by funding from the Simons Foundation and the Centre de Recherches Mathématiques (at Université de Montréal), through the Simons-CRM *scholar-in-residence program* (in connection with the thematic activity *Mixed-Integer Nonlinear Programming: Theory and Computation,* organized by Andrea Lodi and Bruce Shepherd, October 2019).

We also gratefully acknowledge Leo Liberti, Sebastian Sager, and Angelika Wiegele, for organizing the 2019 Oberwolfach workshop on Mixed-Integer Nonlinear Optimization, which catalyzed our renewed interest in maximum-entropy sampling, via the talk of Kurt Anstreicher on the topic.

We thank our collaborators on MESP: Hessa Al-Thani, Kurt Anstreicher, Sam Burer, Zhongzhu Chen, Alan Hoffman, Chun Ko, Amélie Lambert, Yongchun Li, Maurice Queyranne, Joy Williams, and Weijun Xie. Additionally, on the related topic of submodular maximization, Jon wishes to acknowledge his collaborators: Vahab Mirrokni, Viswanath Nagarajan, Andrew Orso, Siqian Shen, Maxim Sviridenko, and Jan Vondrák. And on the closely-related topic of convex MINLO, Marcia wishes to acknowledge her close collaborators Wendel Melo and Fernanda Pereira.

Ann Arbor, Rio de Janeiro, and Paris *Marcia Fampa*
June 2022 *Jon Lee*

Overview

- **Notation.** Tables of much of the notation that we use.
- **Chapter 1: The problem and basic properties.** We define and give basic properties of *differential entropy*, and then we define the *maximum-entropy sampling problem* (MESP) and its constrained version (CMESP). The reminder of the chapter is devoted to basic properties of MESP/CMESP. We demonstrate that MESP is NP-Hard, via its relationship with the stable-set (graph) problem, and we establish the efficiently solvable case in which the input matrix C or its inverse is tridiagonal. We introduce the equivalent complementary problem of MESP/CMESP. We introduce the equivalent scaled version of MESP/CMESP. We introduce the important idea of "masking" a covariance matrix (i.e., taking its Hadamard product with a correlation matrix), an operation that never decreases entropy. Finally, we demonstrate that entropy is a submodular set function.
- **Chapter 2: Branch-and-bound.** Our main algorithmic framework for attacking MESP/CMESP is of *branch-and-bound* type. We describe the framework and some methods for MESP for getting good lower bounds, i.e., good feasible solutions. The methods are in the categories of greedy solutions, local-search solutions, and a subtle combination of those ideas to get an efficient approximation algorithm with a performance guarantee.
- **Chapter 3: Upper bounds.** A large part of the book is on upper bounds for MESP/CMESP, a key ingredient for making branch-and-bound practical. Upper bounds are based on many different ideas including convex relaxations, semidefinite optimization, eigenvalues, Lagrangian relaxation, Lagrangian duality, 0/1 linear optimization, matching, and dynamic programming.
- **Chapter 4: Environmental monitoring.** An important application for all of the algorithmic work on MESP/CMESP is in the area of environmental monitoring. We go into some depth for this application, including the details of how to pass from raw data to a covariance matrix that serves as input for MESP/CMESP.
- **Chapter 5: Opportunities.** There are still many challenges for MESP/CMESP. In this chapter, we describe some promising directions for further research.
- **Appendix A: Basic formulae and inequalities.** A collection of formulae and inequalities that we make use of. This is more than a list — we put a lot of

effort into (i) making accessible proofs of most of the results, and (ii) fitting the results together in a coherent manner. So, while the impatient reader may choose to just refer to the appendix "on demand", there is a benefit to studying the entire appendix at the outset.

- **References.** Cited references.
- **Index.** An index of terms, definitions, etc.

Notation

Basic sets and set-related notation:

$\mathbb{R}$	reals
$\mathbb{R}_+$	nonnegative reals
$\mathbb{R}_{++}$	positive reals
$\mathbb{Z}$	integers
$\mathbb{Z}_+$	nonnegative integers
$\mathbb{Z}_{++}$	positive integers
N	$\{1, 2, \ldots, n\}$
$S + j$	shorthand for $S \cup \{j\}$ (for $j \notin S$)
$T \setminus S$	set subtraction of S from T ($\{j \in T \ : \ j \notin S\}$)
$S - j$	shorthand for $S \setminus \{j\}$ (for $j \in S$)
$\emptyset$	empty set
2^N	power set of N (i.e., the set of all subsets of N)
$x(S)$	0/1 characteristic vector of $S \subset N$ ($x_j(S) = 1$ if $j \in S$; $x_j(S) = 0$ if $j \in N \setminus S$)
$S(x)$	support of $x \in \mathbb{R}^n$ ($j \in S(x)$ iff $x_j \neq 0$)
$\text{conv}(X)$	convex hull of finite $X \subset \mathbb{R}^n$

Matrix- and vector-related notation:

$\text{diag}(X)$	extracts the diagonal of an $m \times n$ matrix X as a $\min\{m, n\}$-vector
$\text{Diag}(x)$	makes an $n \times n$ diagonal matrix from an n-vector x
$\text{Diag}(X)$	$:= \text{Diag}(\text{diag}(X))$ (i.e., zeros out the off-diagonal entries of X)
det	determinant
ldet	log det (of a positive-definite matrix)
$\circ$	Hadamard (i.e., element-wise) product of identically-shaped matrices
$A^{(2)}$	$A \circ A$
A^T	transpose of matrix A
n.s.(A)	null space of matrix A
$\bullet$	matrix dot product ($A \bullet B := \text{tr}(A^\mathsf{T} B)$)
A^{-1}	inverse of a square nonsingular matrix A

$A^{-\mathsf{T}}$	inverse transpose of a square nonsingular matrix A
$\lambda_i(A)$	i-th greatest eigenvalue of a symmetric $n \times n$ matrix A
$\lambda_{\max}(A)$	$:= \lambda_1(A)$ for a symmetric $n \times n$ matrix A
$\lambda_{\min}(A)$	$:= \lambda_n(A)$ for a symmetric $n \times n$ matrix A
$\sigma_i(A)$	i-th greatest singular value of a real matrix A
$A[S, T]$	submatrix of A with rows/cols indexed by S/T
A_{ij}	shorthand for $A[\{i\}, \{j\}]$
$A[\cdot, T]$	columns T of matrix A
$A[S, \cdot]$	rows S of matrix A
$A[\cdot, j]$	shorthand for column j of matrix A
$A[i, \cdot]$	shorthand row i of matrix A
$A \succeq 0$	symmetric A is positive semidefinite (i.e., nonnegative eigenvalues)
$A \succ 0$	symmetric A is positive definite (i.e., positive eigenvalues)
$\mathbb{S}^n$	real order-n symmetric matrices
$\mathbb{S}^n_+$	real order-n symmetric positive-semidefinite matrices
$\mathbb{S}^{n,s}_+$	real order-n symmetric positive-semidefinite matrices of rank at least s
$\mathbb{S}^n_{++}$	real order-n symmetric positive-definite matrices ($\mathbb{S}^n_{++} := \mathbb{S}^{n,n}_+$)
I_n	identity matrix (n omitted if clear from context)
J_n	matrix of all ones (n omitted if clear from context)
$\mathbf{e}_i$	i-th standard unit vector
E_{ij}	$:= (\mathbf{e}_i \mathbf{e}_j^\mathsf{T} + \mathbf{e}_j \mathbf{e}_i^\mathsf{T})/2$ (in particular, $E_{ii} := \mathbf{e}_i \mathbf{e}_i^\mathsf{T}$)
$\mathbf{e}$	vector of all ones
u^v	for $u, v \in \mathbb{R}^n$, $(u^v)_i := u_i^{v_i}$, for $i = 1, 2, \ldots, n$
$u_{[\ell]}$	ℓ-th greatest entry of the vector $u \in \mathbb{R}^n$
$u_{\max}$	$:= u_{[1]}$ for $u \in \mathbb{R}^n$
$u_{\min}$	$:= u_{[n]}$ for $u \in \mathbb{R}^n$
$[u]^+$	the positive part of $u \in \mathbb{R}^n$ (i.e., $[u]^+_i := \max\{0, u_i\}$)
$[u]^-$	the negative part of $u \in \mathbb{R}^n$ (i.e., $[u]^-_i := -\min\{0, u_i\}$)

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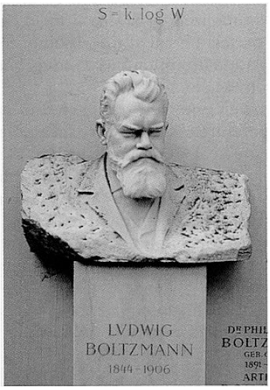
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Chapter 1
The problem and basic properties

> ”Chance and chance alone has a message for us. Everything that occurs out of necessity, everything expected, repeated day in and day out, is mute. Only chance can speak to us. We read its message much as gypsies read the images made by coffee grounds at the bottom of a cup.”
>
> - Milan Kundera (The Unbearable Lightness of Being, 1984)

The term “entropy” was coined by the physicist Rudolf Clausius in 1865, in his study of thermodynamics, and in particular his version of the Second Law: the ratio of the heat content of a closed system to its absolute temperature cannot decrease — he called this ratio the *entropy*. Maxwell, Boltzmann, and Gibbs developed the idea mathematically, as they founded and developed statistical mechanics. Notably, Boltzmann, starting from a combinatorial description of the number of “microstates” corresponding to a “macrostate”, arrived at a mathematical definition of entropy, eventually carved on his tombstone.[1]



Boltzmann was able to mathematically demonstrate that his entropy was the same one as that of Clausius.

[1] It is at the Zentralfriedhof, in Vienna, where we also find the graves of many famous musicians.

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Notably, Shannon developed entropy in information theory. The idea to borrow the term entropy from statistical mechanics was due to von Neumann (see [Ave12]):

> "Why don't you call it entropy? In the first place, a mathematical development very much like yours already exists in Boltzmann's statistical mechanics, and in the second place, no one understands entropy very well, so in any discussion you will be in a position of advantage!"

In the next subsection, we attempt to decrease the "advantage" of Shannon over the patient reader, as we describe *differential entropy* and its basic properties. Although Shannon is broadly known for popularizing discrete entropy, he did in his seminal paper also advance the idea of differential (or continuous) entropy; see [Sha48].

1.1 Differential entropy

For integer $n \geq 1$, let $N := \{1, 2, \ldots, n\}$, and let $\mathcal{Y}_N := (\mathcal{Y}_1, \mathcal{Y}_2, \ldots, \mathcal{Y}_n)^\mathsf{T}$ be a random vector supported on $\mathbb{R}^n$ with joint density function f. For our purposes, we will be interested in nonempty subsets $S \subset N$, the associated subvectors $\mathcal{Y}_S$, and the marginal density functions f_S (obtained by integrating f over the variables indexed from $N \setminus S$).

The (differential or continuous) **entropy** of $\mathcal{Y}_S$ is defined as

$$h(\mathcal{Y}_S) := -E[\log f_S(\mathcal{Y}_S)] = -\int f_S(y) \log f_S(y) dy,$$

where $E[\cdot]$ denotes the expected-value function. Roughly speaking, the entropy $h(\mathcal{Y}_S)$ measures the "information" or "disorder" associated with $\mathcal{Y}_S$.

It is a simple exercise in multivariable calculus to see the following result.

Proposition 1.1.1 If $\mathcal{Z}_S := A\mathcal{Y}_S + b$, where $A \in \mathbb{R}^{|S|\times|S|}$ is invertible, and $b \in \mathbb{R}^{|S|}$, then $h(\mathcal{Z}_S) = h(\mathcal{Y}_S) + \log|\det A|$.

Remark 1.1.2 From Proposition 1.1.1, it is clear that the entropy can be *negative*, and what is further exposed is that the entropy itself is not a meaningful number, but rather *differences in entropy* are meaningful.

The **mean** (vector) of $\mathcal{Y}_S$ is

$$\mu_S := E[\mathcal{Y}_S] \in \mathbb{R}^{|S|},$$

and the **covariance matrix** of $\mathcal{Y}_S$ is

$$C[S,S] := E[(\mathcal{Y}_S - \mu_S)(\mathcal{Y}_S - \mu_S)^\mathsf{T}] \in \mathbb{S}_+^{|S|} .$$

Remark 1.1.3 The mean and covariance behave nicely under affine transformations. If $\mathcal{Z}_S := A\mathcal{Y}_S + b$, where $A \in \mathbb{R}^{|S|\times|S|}$ is invertible, and $b \in \mathbb{R}^{|S|}$, then $E[\mathcal{Z}_S] = A\mu_S + b$, and $E[(\mathcal{Z}_S - E[\mathcal{Z}_S])(\mathcal{Z}_S - E[\mathcal{Z}_S])^\mathsf{T}] = A\, C[S,S]\, A^\mathsf{T}$.

The random vector $\mathcal{Y}_S$ has a **(non-degenerate) joint Gaussian distribution** with mean vector μ_S and covariance matrix $C[S,S] \in \mathbb{S}_{++}^{|S|}$ if it has density function

$$f_S(\mathcal{Y}_S) = \phi_S(\mathcal{Y}_S) := \frac{\exp\left\{-\frac{1}{2}(y-\mu_S)^\mathsf{T}(C[S,S])^{-1}(y-\mu_S)\right\}}{\sqrt{(2\pi)^{|S|}\det(C[S,S])}}, \text{ for } y \in \mathbb{R}^{|S|}.$$

Remark 1.1.4 Considering, again, affine transformations $\mathcal{Z}_S := A\mathcal{Y}_S + b$ (see Remark 1.1.3), we obtain a random vectors having a **degenerate joint Gaussian distribution**, by making non-invertible a linear transformations of a random vector having a non-degenerate joint Gaussian distributions; in such a use, the linear transformation matrix A need not be square. In this way, we can obtain joint Gaussian distributions with singular covariance matrices. Strictly speaking, such distributions do not have densities, and we define $h(\mathcal{Y}_S) := -\infty$ in such cases. Note that we can have $h(\mathcal{Y}_N) = -\infty$ and $h(\mathcal{Y}_S)$ finite for some $S \subset N$.

If Y has a joint Gaussian distribution with covariance matrix $C[N,N]$, then for all nonempty $S \subset N$, the random vector $\mathcal{Y}_S$ has a joint Gaussian distribution with covariance matrix $C[S,S]$. Furthermore, when Y is Gaussian, we get a very nice and easily computable form for $h(\mathcal{Y}_S)$, depending only (linearly) on $\operatorname{ldet} C[S,S]$.

Theorem 1.1.5 If $\mathcal{Y}_S$ has a non-degenerate joint Gaussian distribution with covariance matrix $C[S,S]$, then

$$h(\mathcal{Y}_S) = \frac{1}{2}\operatorname{ldet}(2\pi\mathrm{e}\, C[S,S]) = \frac{1}{2}\Big((1+\log(2\pi))|S| + \operatorname{ldet} C[S,S]\Big).$$

Remark 1.1.6 From Theorem 1.1.5, we can see consistency between Proposition 1.1.1 and the expression for the covariance expression in Remark 1.1.3.

For the proof of Theorem 1.1.5, we first establish a simple lemma, which does *not* depend on $\mathcal{Y}_S$ having a joint Gaussian distribution. The lemma concerns the expected

value of the square of the so-called **Mahalanobis distance** between random $\mathcal{Y}_S$ and its mean μ_S.

Lemma 1.1.7

$$E[(\mathcal{Y}_S - \mu_S)^\top (C[S,S])^{-1}(\mathcal{Y}_S - \mu_S)] = |S|.$$

Proof (Lemma 1.1.7)

$$\begin{aligned} E[(\mathcal{Y}_S - \mu_S)^\top (C[S,S])^{-1}(\mathcal{Y}_S - \mu_S)] &= E[(C[S,S])^{-1} \bullet (\mathcal{Y}_S - \mu_S)(\mathcal{Y}_S - \mu_S)^\top] \\ &= C[S,S])^{-1} \bullet E[(\mathcal{Y}_S - \mu_S)(\mathcal{Y}_S - \mu_S)^\top] \\ &= (C[S,S])^{-1} \bullet C[S,S] \\ &= \mathrm{tr}((C[S,S])^\top (C[S,S])^{-1}) \\ &= \mathrm{tr}(I_{|S|}) = |S|. \end{aligned}$$

□

Proof (Theorem 1.1.5)

$$\begin{aligned} h(\mathcal{Y}_S) &:= -E[\log \phi_S(\mathcal{Y}_S)] = -\int \phi_S(y) \log \phi_S(y)\, dy \\ &= -\int \phi_S(y) \log \left(\frac{\exp\left\{-\frac{1}{2}(y-\mu_S)^\top (C[S,S])^{-1}(y-\mu_S)\right\}}{\sqrt{(2\pi)^{|S|}\det(C[S,S])}} \right) dy \\ &= -\frac{1}{2}\int \phi_S(y) \left(-\log\left((2\pi)^{|S|}\det(C[S,S])\right) - (y-\mu_S)^\top (C[S,S])^{-1}(y-\mu_S) \right) dy \\ &= \frac{1}{2}\Bigg(\int \phi_S(y) \log\left((2\pi)^{|S|}\det(C[S,S])\right) dy \\ &\qquad + E[(\mathcal{Y}_S - \mu_S)^\top (C[S,S])^{-1}(\mathcal{Y}_S - \mu_S)] \Bigg) \\ &= \frac{1}{2}\left(\log\left((2\pi)^{|S|}\det(C[S,S])\right) + |S| \right) \\ &= \frac{1}{2}\Big((1+\log(2\pi))|S| + \mathrm{ldet}\, C[S,S] \Big). \end{aligned}$$

□

Looking closely at the proof of Theorem 1.1.5 and considering that Lemma 1.1.7 does not depend on the distribution of $\mathcal{Y}_S$, we can easily see the following technical lemma which we will soon make use of.

Lemma 1.1.8 If $\mathcal{Y}_S$ has a non-degenerate joint Gaussian density ϕ_S, and $\mathcal{Z}_S$ having density f_S satisfies $E[\mathcal{Z}_S] = E[\mathcal{Y}_S] =: \mu_S$ and $E[(\mathcal{Z}_S - \mu_S)(\mathcal{Z}_S - \mu_S)^\mathsf{T}] = E[(\mathcal{Y}_S - \mu_S)(\mathcal{Y}_S - \mu_S)^\mathsf{T}]$, then

$$-\int f_S(y) \log \phi_S(y)\, dy = h(\mathcal{Y}_S).$$

A very important consequence of Theorem 1.1.5 is that $h(\mathcal{Y}_S)$ is increasing in $\operatorname{ldet} C[S,S]$ when $|S|$ is fixed. Further, we have the following very useful and intriguing property.

Theorem 1.1.9 Among all continuous random vectors $\mathcal{Y}_S$ with covariance matrix $C[S,S] \in \mathbb{S}_{++}^{|S|}$, the ones with maximum entropy are Gaussian.

To prove Theorem 1.1.9, we need some preliminaries. A key useful quantity is a well-known (asymmetric) "distance" from one density to another:

The **relative entropy** (or **Kullback-–Leibler distance**) between density f and density g is defined as

$$D(f||g) := \int f(y) \log \frac{f(y)}{g(y)} dy,$$

where we integrate over the support of f, which we assume is contained in the support of g.

Lemma 1.1.10 (Gibb's inequality)

$$D(f||g) \geq 0,$$

with equality if and only if $f = g$ almost everywhere.

Proof (Lemma 1.1.10)

$$\begin{aligned} -D(f||g) &= -\int f(y) \log \left(\frac{f(y)}{g(y)} \right) dy \\ &= \int f(y) \log \left(\frac{g(y)}{f(y)} \right) dy \\ &\leq \log \left(\int f(y) \frac{g(y)}{f(y)} dy \right) \quad \text{(by Jensen's inequality)} \end{aligned}$$

$$= \log\left(\int g(y)dy\right) = \log(1) = 0.$$

Note that because $-\log(\cdot)$ is strictly convex, the only case for equality in Jensen's inequality is when $f = g$ almost everywhere. □

Recall that ϕ_S denotes the density function of a non-degenerate joint Gaussian vector $\mathcal{Y}_S$ with mean vector μ_S and covariance matrix $C[S,S]$.

Proof (Theorem 1.1.9) Let f_S be the density function of any random $\mathcal{Z}_S$ with support on $\mathbb{R}^{|S|}$, having mean vector μ_S and covariance matrix $C[S,S]$.

$$\begin{aligned}
0 &\le D(f_S||\phi_S) \quad \text{(by Lemma 1.1.10)} \\
&= \int f_S(y)\log\left(\frac{f_S(y)}{\phi_S(y)}\right)dy \\
&= \int f_S(y)\left(\log f_S(y) - \log\phi_S(y)\right)dy \\
&= \int f_S(y)\log f_S(y)dy - \int f_S(y)\log\phi_S(y)dy \\
&= -h(\mathcal{Z}_S) + h(\mathcal{Y}_S) \quad \text{(by Lemma 1.1.8).}
\end{aligned}$$

□

Finally, so as to better understand differential entropy, we wish to relate it, in the Gaussian case, to a probabilistic statement concerning locating a random $\mathcal{Y}_S$. We have already seen that

$$E[(\mathcal{Y}_S - \mu_S)^{\mathsf{T}}(C[S,S])^{-1}(\mathcal{Y}_S - \mu_S)] = |S|$$

(Lemma 1.1.7), in general. But is the Gaussian case, we can say more; we have that

$$(\mathcal{Y}_S - \mu_S)^{\mathsf{T}}(C[S,S])^{-1}(\mathcal{Y}_S - \mu_S) \sim \chi^2_{|S|}$$

(a χ^2 random variable with $|S|$ degrees of freedom), and so we have

$$\Pr\left\{(\mathcal{Y}_S - \mu_S)^{\mathsf{T}}(C[S,S])^{-1}(\mathcal{Y}_S - \mu_S) \le \chi^2_{|S|}(\alpha)\right\} = 1-\alpha,$$

where the value $\chi^2_{|S|}(\alpha) \in \mathbb{R}$ is defined so that this statement is true. The set of $\mathcal{Y}_S$ satisfying the inequality in the probability statement (i.e., with squared Mahalanobis distance from μ_S being no more than $\chi^2_{|S|}(\alpha)$) is an ellipsoid, with volume

$$\frac{\pi^{n/2}}{\Gamma(n/2+1)}\left(\chi^2_{|S|}(\alpha)\right)^{n/2}\sqrt{\det C[S,S]}\,.$$

So we can see the role that $\det C[S,S]$ (and hence indirectly $h(\mathcal{Y}_S)$) plays in measuring the size of the (ellipsoidal) set that locates a random Gaussian $\mathcal{Y}_S$ with high

probability. In the non-Gaussian case, differential entropy continues to play a role, but the corresponding subject of "typical sets" (see [CT06], for example) is beyond what we wish to cover.

1.2 The MESP and the CMESP

Let C be a symmetric positive semidefinite matrix, with nonzero diagonal entries, and with rows/columns indexed from $N := \{1, 2, \ldots, n\}$, with $n \geq 2$. For $0 < s \leq \min\{\text{rank}(C),\ n-1\}$, we define the **maximum-entropy sampling problem**

$$\begin{aligned} \text{(MESP)} \qquad z(C,s) &:= \max\{\ldet C[S,S] \ :\ |S| = s\}\,. \\ &= \max\left\{\ldet C[S(x),S(x)] \ :\ \mathbf{e}^{\mathsf{T}}x = s,\ x \in \{0,1\}^n\right\}, \end{aligned}$$

where $S(x) := \{j \in N \ :\ x_j \neq 0\}$. We write MESP two different ways because in some contexts it is convenient to think of a feasible solution as an $S \subset N$, and in other contexts, we prefer to think of a feasible solution as an $x \in \{0,1\}^n$. Note that in MESP, as is common, we define $\log(0) := -\infty$.

Because of Theorem 1.1.5, MESP is equivalent to maximizing the entropy $h(\mathcal{Y}_S)$, over $S \subset N$ having $|S| = s$, when $\mathcal{Y}$ has a joint Gaussian distribution with covariance matrix C. Further, because of Remark 1.1.2, maximizing the entropy is meaningful.

The term "maximum-entropy sampling" comes from the experimental-design literature (see [SW87]). It has subsequently been developed in that literature, and more broadly in statistics (see [SW00]).

As a finite discrete problem, statisticians employed simple interchange heuristics, which actually do quite well on instances of modest size. Seeking a more principled approach than afforded by heuristics, the problem was first approached by the mathematical-optimization community in [KLQ95].

We also consider a generalization of MESP. For $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$, we define the **constrained maximum-entropy sampling problem**

$$\begin{aligned} \text{(CMESP)} \qquad z(C,s,A,b) &:= \max\Big\{\ldet C[S,S] \ :\ |S| = s; \\ &\qquad\qquad \textstyle\sum_{j\in S} a_{ij} \leq b_i,\ i = 1,2,\ldots,m\Big\} \\ &= \max\left\{\ldet C[S(x),S(x)] \ :\ \mathbf{e}^{\mathsf{T}}x = s,\right. \\ &\qquad\qquad \left. Ax \leq b,\ x \in \{0,1\}^n\right\}, \end{aligned}$$

first approached by [Lee98a] and [AFLW96, AFLW99, Fam96, Wil98].

A key application for us is in the area of environmental monitoring. In this application area, the set N corresponds to spatially distributed sites that collect precipitation, over time, with the aim of measuring concentrations of some pollutant. Then the matrix C corresponds to the sample covariance matrix for the observations of concentrations of this pollutant. Collecting observations is assumed to be costly, and so our goal is to contract the set N is a subset of s sites (for collecting future observations). MESP aims at choosing the set of s sites having maximum information, as measured by differential entropy. For CMESP, side constraints can be, for example, budget constraints, or perhaps constraints limiting the number of sites selected with sub-geographies. There are a lot of details to consider in preparing a meaningful covariance matrix C, and we discuss this application much more in Chapter 4. Also see Section 5.3 for references to other applications.

Besides the interest in environmental monitoring and, more generally, experimental design, CMESP can be seen as a generalization of *0/1 (integer) linear programming* (with a cardinality constraint), that case being when C is a diagonal matrix. As such, the general case provided an early and challenging example of a *non-factorable* integer *nonlinear* optimization problem, for the so-called MINLO/MINLP (mixed-integer nonlinear optimization/programming) community. A relatively straightforward re-formulation of CMESP is as

$$\max \left\{ \operatorname{ldet}\left(\operatorname{Diag}(x) C \operatorname{Diag}(x) + \operatorname{Diag}(\mathbf{e} - x)\right) : \mathbf{e}^{\mathsf{T}} x = s,\ Ax \le b,\ x \in \{0,1\}^n \right\}.$$

"Factorable" refers to functions that can be reasonably decomposed via a library of nonlinear functions of very few variables (e.g., a^x, $\log_a(x)$, x^a, $\sin(x)$, xy, x^y, xyz) and affine functions in an arbitrary number of variables. "Reasonably" means a low-depth decomposition. Factorable programming, introduced by [McC72, McC76, McC85], and the associated "spatial" branch-and-bound algorithmic framework, is the foundation of all successful "general purpose" software aimed at global optimization of MINLO problems (for examples, see BARON: [TS02, Sah96]; SCIP: [Ach09, GAB+20]; Couenne: [BLL+09]; ANTIGONE: [MF14]; SHOT: [KLW16]). A basic requirement is that for each library function, we have a tractable convex relaxation for the graph of the function on box subsets of the function domain. It is an important challenge to develop general-purpose techniques for broad classes of non-factorable MINLO problems. Clearly, the objective function $\operatorname{ldet}\left(\operatorname{Diag}(x) C \operatorname{Diag}(x) + \operatorname{Diag}(\mathbf{e} - x)\right)$ is generally non-factorable, due to the det function. Of course it becomes linear when C is a diagonal matrix, but even then CMESP is still "NP-Hard" (see Proposition 1.3.4).

1.3 Hardness

We assume some familiarity with complexity theory; see [GJ79]. For the uninitiated, we prove hardness for a problem by demonstrating that (even a special case of the problem) is a *known hard problem*. There is *a lot* more to this (e.g., what precisely is a "known hard problem"?), but it gives enough of the idea to understand what follows. A problem is *not* hard if we have an algorithm to solve it that runs in polynomial time in the the number of bits needed to encode the problem. At a high level, we map each instance of a "known hard decision" problem (i.e., a so-called "NP-Complete" problem) on an n-vertex graph to an instance of MESP on an order-n covariance matrix C. Thus, if we could efficiently solve MESP for order-n covariance matrices, we could then efficiently solve n-vertex instances of our "known hard decision". This does not truly mean that we will have proved that MESP cannot be solved in polynomial time. We will have simply established that if we could solve MESP in polynomial time, then we could solve all "NP-Complete" problems in polynomial time — which is generally considered to be highly unlikely.

To prove hardness for MESP, we follow the ideas in [KLQ95], though we slightly modify the construction and simply the proof.

Theorem 1.3.1 (see [KLQ95, Theorem 1.iv]) MESP is NP-Hard, even if $\mathrm{diag}(C) = n\mathbf{e}$ with off-diagonal entries in $\{0, 1\}$ (hence even if C is diagonally dominant).

We employ a couple of lemmas to establish Theorem 1.3.1.

Lemma 1.3.2 Let $A(G)$ be the (vertex-vertex) adjacency matrix of an undirected graph G having vertex set N. Let $B_G := A(G) + nI_n$. Let G' be obtained from G by deleting edges (at least one), and let $B_{G'} := A(G') + nI_n$. Then $\det B_{G'} > \det B_G$.

Lemma 1.3.2 is a variation on [KLQ95, Lemma 1], now with a simpler proof.

Proof (Lemma 1.3.2) Without loss of generality, we may assume that G' is G with precisely one edge $\{i, j\}$ deleted. Notice that B_G, and $B_{G'}$, and their principal submatrices are symmetric and diagonally dominant, hence positive definite. The matrices B_G and $B_{G'}$ have the form

$$C := \left(\begin{array}{c|c} C[\{i,j\},\{i,j\}] & C[\{i,j\},N-i-j] \\ \hline C[N-i-j,\{i,j\}] & C[N-i-j,N-i-j] \end{array} \right),$$

with the only difference being that

$$C[\{i,j\},\{i,j\}] = \begin{cases} \begin{pmatrix} n & 1 \\ 1 & n \end{pmatrix}, & \text{for } G; \\ \begin{pmatrix} n & 0 \\ 0 & n \end{pmatrix}, & \text{for } G'. \end{cases}$$

Employing the Schur complement determinant formula (Theorem A.2.2), we have

$$\det C = \det C[N-i-j, N-i-j] \times \det\Big(\underbrace{C[\{i,j\},\{i,j\}]}_{(*)} - C[\{i,j\}, N-i-j](C[N-i-j, N-i-j])^{-1} C[N-i-j, \{i,j\}]\Big).$$

The only dependence (in the right-hand side of this equation) on $C[i,j] = C[j,i]$ (which is the only part that is different for G vs. G') is in $(*)$. The entire second multiplicand is of the form

$$\det \begin{pmatrix} \alpha_1 & \beta \\ \beta & \alpha_2 \end{pmatrix} = \alpha_1\alpha_2 - \beta^2$$

for G, and

$$\det \begin{pmatrix} \alpha_1 & \beta-1 \\ \beta-1 & \alpha_2 \end{pmatrix} = \alpha_1\alpha_2 - (\beta-1)^2 > \alpha_1\alpha_2 - \beta^2$$

for G'. The result follows. □

Relative to an undirected graph, we use the idea of a **stable set**: a set of vertices with no edges between them.

Lemma 1.3.3 (see [KLQ95, Lemma 2 and 3]) For an undirected graph G, and $S \subset N$ with $|S| = s \geq 1$, we have

$$\det(B_G[S,S]) \quad \begin{cases} = n^s, & \text{if } S \text{ is a stable set of } G; \\ \leq n^s - n^{s-2}, & \text{if } S \text{ is not a stable set of } G. \end{cases}$$

Proof (Lemma 1.3.3) It is trivial that if S is a stable set, then $\det(B_G[S,S]) = \det(nI_n) = n^s$. For the second part, because of Lemma 1.3.2, we can assume that the subgraph of G induced by S has a single edge, say $\{i,j\}$. Then

$$\begin{aligned} \det(B_G[S,S]) &= \det B_G[\{i,j\},\{i,j\}] \times \det B_G[S-i-j, S-i-j] \\ &= \det \begin{pmatrix} n & 1 \\ 1 & n \end{pmatrix} \det(nI_{s-2}) = (n^2-1)n^{s-2} = n^s - n^{s-2}. \end{aligned}$$

□

Proof (Theorem 1.3.1) We simply use the fact the the problem of determining if a graph G has a stable set of size s (part of the input) is NP-Complete (see [GJ79]). For a

graph G, we produce a rational instance of MESP by letting $C := B_G := A(G)+nI_n$. Using Lemma 1.3.3, if $z(C,s) = \log(n^s)$, then G has a stable set of cardinality s. Alternatively, if $z(C,s) \leq \log(n^s - n^{s-2})$, then G does not have a stable set of cardinality s. Notice that: (i) we can compute any needed determinants in polynomial time in the input size, and (ii) the separation between n^s and $n^s - n^{s-2}$ is convenient (for the "Turing-machine model" of computation) because it means that we do not need to compute logarithms exactly. □

From NP-Hardness of the corresponding version of the knapsack problem (see [GJ79]), we easily establish hardness for CMESP.

Proposition 1.3.4 CMESP is NP-Hard, even if C is diagonal and $m = 1$.

Though NP-Hard, we should keep in mind that a general MESP or CMESP problem is already practically hard at $n = 100$, while we think of hard instances of practical knapsack problems as having n much larger.

1.4 A solvable case

If C is a diagonal matrix, MESP is trivially solvable by choosing S to comprise the elements i having the s greatest values of $C[i,i]$. But we can go further. [ATL21a] demonstrated the following.

Theorem 1.4.1 MESP is solvable in polynomial time when C is tridiagonal.

Proof Suppose that C is tridiagonal. Let S be an ordered subset of N. Then we can write $C[S,S]$ uniquely as $C[S,S] = \text{Diag}(C[S_1,S_1], C[S_2,S_2], \ldots, C[S_p,S_p])$, with $p \geq 1$, where each S_i is a *maximal ordered contiguous subset* of S, and for all $1 \leq i < j \leq p$, all elements of S_i are less than all elements of S_j. We call the S_i the *blocks* of S, and in particular S_p is the *last block*.

It is easy to see that

$$\det C[S,S] = \prod_{i=1}^{p} \det C[S_i,S_i] = \det C[S_p,S_p] \times \det C[S \setminus S_p, S \setminus S_p,].$$

With this way of thinking, we define

$$f(k,\ell,t) := \max\{\det C[S,S] \ : \ |S| = t,\ S \subset N, \text{ and the last piece of } S \text{ is } [k,\ell]\},$$

for $1 \le \ell - k + 1 \le t \le s$. Note that the last block of S (in the definition of f) has $\ell - k + 1$ elements, which is why t must be at least that large. We have that

$$z(C, s) = \log \max_{k,\ell} \{ f(k, \ell, s) \ : \ 1 \le k \le \ell \le n, \ \ell - k + 1 \le s \},$$

where we are simply maximizing over the possible (quadratic number of) last pieces.

Our dynamic-programming recursion is

$$f(k, \ell, t) = \det C[[k, \ell], [k, \ell]] \times \max_{i,j} \{ f(i, j, t - (\ell - k + 1)) \ : \\ 1 \le i \ \le j \le k - 2, \ j - i + 1 \le t - (\ell - k + 1) \}.$$

The idea is that if $[k, \ell]$ is the last block of an optimal selection of t elements, then we have to pick $t - (\ell - k + 1)$ more elements, and *element* $k - 1$ *is ruled out* (because $[k, \ell]$ is maximal).

To get the recursion started, we calculate

$$f(k, \ell, \ell - k + 1) = \det C[[k, \ell], [k, \ell]],$$

for $1 \le k \le \ell \le n$, $\ell - k + 1 \le s$, observing that in such a boundary case, there is only one feasible solution.

Already, it appears that the initialization requires $O(n^5)$ basic arithmetic operations, but in fact we can do this part in $O(n^2)$ operations, using the tridiagonal-determinant formula: If

$$T_r := \begin{pmatrix} a_1 & b_1 & & & \\ b_1 & a_2 & b_2 & & \\ & b_2 & \ddots & \ddots & \\ & & \ddots & \ddots & b_{r-1} \\ & & & b_{r-1} & a_r \end{pmatrix},$$

then $\det T_r = a_r \det T_{r-1} - b_{r-1}^2 \det T_{r-2}$, for $r \ge 2$, where $\det T_0 := 1$ and $\det T_1 = a_1$.

Next, we compute, using the recursion, for $t = 1, 2, \ldots, s$, $f(k, \ell, t)$ for all $1 \le k \le \ell \le n$ such that $\ell - k + 1 < t$. It is not hard to see that this gives an $O(n^5)$ algorithm for MESP, when C is tridiagonal. □

1.5 The complementary problem

There is a nice and simple formula relating a principal minor of an invertible matrix with the complementary minor of the inverse (Corollary A.2.4; see [HJ85, Section 0.8.4]):

$$\det C[S, S] = \det C \times \det(C^{-1})[N \setminus S, N \setminus S].$$

With this identity, we have the following result.

Proposition 1.5.1 For $C \in \mathbb{S}^n_{++}$,

$$z(C, s) = z(C^{-1}, n - s) + \operatorname{ldet} C,$$

and so upper bounds for $z(C^{-1}, n - s)$ yield upper bounds for $z(C, s)$, shifting by $\operatorname{ldet} C$ (first exploited in [AFLW96, AFLW99]). We refer to the MESP having optimal value $z(C^{-1}, n - s)$ as the **complementary problem** defined by the data C and s.

Combining Theorem 1.4.1 and Proposition 1.5.1, we immediately obtain the following result.

Theorem 1.5.2 MESP is solvable in polynomial time when C^{-1} is tridiagonal.

For the constrained case, we have the following result.

Proposition 1.5.3 For $C \in \mathbb{S}^n_{++}$,

$$z(C, s, A, b) = z(C^{-1}, n - s, \bar{A}, \bar{b}) + \operatorname{ldet} C,$$

where $\bar{A} := -A$, and $\bar{b} := b - A\mathbf{e}$. Then, we define the **complementary problem** of CMESP as the CMESP having optimal value $z(C^{-1}, n - s, \bar{A}, \bar{b})$ (i.e., CMESP where C is replaced C^{-1}, s by $n - s$, A by $\bar{A}$, and b by $\bar{b}$).

1.6 Scaling

For a positive constant γ, and S with $|S| = s$, we have that

$$\det(\gamma C)[S, S] = \gamma^s \det C[S, S].$$

With this identity, we very easily have the following result.

Proposition 1.6.1

$$z(C, s, A, b) = z(\gamma C, s, A, b) - s \log \gamma,$$

and so upper bounds for $z(\gamma C, s, A, b)$ yield upper bounds for $z(C, s, A, b)$, shifting by $-s \log \gamma$ (first exploited in [AFLW96, AFLW99]).

1.7 Masks

Given a positive integer n, we define a **mask** as any $M \in \mathbb{S}^n_+$ with $\operatorname{diag}(M) = \mathbf{e}$. Masks are better known as "correlation matrices", but the term "mask" better captures our use. Masks for MESP were introduced in [AL04] (also see [BL07]), where it was observed, by direct application of Oppenheim's inequality (Theorem A.4.11), that for any $S \subset N$ and mask M,

$$\det C[S, S] \leq \det\big((C \circ M)[S, S]\big).$$

That is, masking cannot decrease entropy. Therefore,

Theorem 1.7.1 For any mask M,

$$z(C, s) \leq z(M \circ C, s),$$

and so upper bounds for $z(M \circ C, s)$ are also upper bounds for $z(C, s)$.

Remark 1.7.2 We note that when M is a matrix that is all zero except for square diagonal blocks of all ones, a so-called "combinatorial mask" (see [HLW01] and §3.1.1), then Oppenheim's inequality reduces to what is known as Fischer's inequality (see [Fis08] and [HJ85, Theorem 7.8.3]).

1.8 Submodularity

A function $f : 2^N \to \mathbb{R}$ is a **submodular function** if *any* of these four properties hold:

$$f(S \cup T) + f(S \cap T) \leq f(S) + f(T), \text{ for all } S, T \subset N. \tag{1.1}$$

$$f(T + i) - f(T) \leq f(S + i) - f(S), \text{ for all } S \subset T \subset N, i \in N \setminus T. \tag{1.2}$$

$$f(S \cup T) - f(S) \leq \sum_{i \in T \setminus S} (f(S + i) - f(S)), \text{ for all } S, T \subset N. \tag{1.3}$$

$$f(S \cup T) - f(T) \geq \sum_{i \in S \setminus T} (f(S) - f(S - i)), \text{ for all } S, T \subset N. \tag{1.4}$$

(1.1) is the most common definition and its equivalence with (1.2) is very well known. (1.3) and (1.4) are well known for experts, and can be found in [NWF78, Proposition 2.1].

Submodular functions are well-studied objects in discrete optimization (see [Fuj05]), and part of that interest stems from the following connection with entropy.

Theorem 1.8.1 (The Hadamard-Fischer inequalities; see [HJ85, Section 7.8, Problem 14]; also see [KK83])

For a matrix $C[N, N] \in \mathbb{S}_+^n$, and $S, T \subset N$, we have

$$\det C[S \cup T, S \cup T] \times \det C[S \cap T, S \cap T] \leq \det C[S, S] \times \det C[T, T].$$

Consequently, when $C \in \mathbb{S}_{++}^n$, the function $f(S) := \operatorname{ldet} C[S, S]$ is submodular.

Remark 1.8.2 Note that $\det C[\emptyset, \emptyset] := 1$. The special case of $S \cap T = \emptyset$ for Theorem 1.8.1 is just Fischer's inequality. But the proof of the general case uses Fischer's inequality, after taking a Schur complement.

Proof (Theorem 1.8.1)

The result is trivial when $C[S \cap T, S \cap T]$ is singular, so we assume otherwise. Now consider $C[S \cup T, S \cup T]$ in partitioned form:

$$\left(\begin{array}{c|c|c} C[S \cap T, S \cap T] & C[S \cap T, S \setminus T] & C[S \cap T, T \setminus S] \\ \hline C[S \setminus T, S \cap T] & C[S \setminus T, S \setminus T] & C[S \setminus T, T \setminus S] \\ \hline C[T \setminus S, S \cap T] & C[T \setminus S, S \setminus T] & C[T \setminus S, T \setminus S] \end{array}\right),$$

and after taking a Schur complement on $C[S \cap T, S \cap T]$:

$$\left(\begin{array}{c|c|c} C[S\cap T, S\cap T] & 0 & 0 \\ \hline 0 & C_{S\cap T}[S\setminus T, S\setminus T] & C_{S\cap T}[S\setminus T, T\setminus S] \\ \hline 0 & C_{S\cap T}[T\setminus S, S\setminus T] & C_{S\cap T}[T\setminus S, T\setminus S] \end{array}\right).$$

We have

$$\begin{aligned}
&\det C[S\cap T, S\cap T]\times \det C[S\cup T, S\cup T]\\
&= \left(\det C[S\cap T, S\cap T]\right)^2 \times \det\left(\begin{array}{c|c} C_{S\cap T}[S\setminus T, S\setminus T] & C_{S\cap T}[S\setminus T, T\setminus S] \\ \hline C_{S\cap T}[T\setminus S, S\setminus T] & C_{S\cap T}[T\setminus S, T\setminus S] \end{array}\right)\\
&\qquad\qquad \text{(by the Schur complement determinant formula (Theorem A.2.2))}\\
&\le \left(\det C[S\cap T, S\cap T]\right)^2 \times \det C_{S\cap T}[S\setminus T, S\setminus T]\times \det C_{S\cap T}[T\setminus S, T\setminus S]\\
&\qquad\qquad \text{(by Fischer's inequality)}\\
&= \left(\det C[S\cap T, S\cap T]\times \det(C_{S\cap T}[S\setminus T, S\setminus T])\right)\\
&\qquad \times \left(\det C[S\cap T, S\cap T]\times \det(C_{S\cap T}[T\setminus S, T\setminus S])\right)\\
&= \det C[S,S]\times \det C[T,T].
\end{aligned}$$

□

We will return to submodular functions in §2.3.3, when we study an efficient approximation algorithm for MESP.

Chapter 2
Branch-and-bound

Number Six: Where am I?
Number Two: In the Village.
Number Six: What do you want?
Number Two: Information.
Number Six: Whose side are you on?
Number Two: That would be telling. We want information. . . information. . . information.
Number Six: You won't get it.
Number Two: By hook or by crook, we will.

- The Prisoner, 1967–8

Branch-and-bound for mixed-integer linear optimization (see [Lee22], for example) is a very well-known and successful algorithmic framework. The branch-and-bound approach to MESP was initiated with [KLQ95] (also see[Lee04, pp. 191–2]). Here, in the main flow, we focus on MESP, but along the way we describe how to handle CMESP (first introduced with [Lee98a]; also see [AFLW96] and [AFLW99]).

2.1 The branch-and-bound algorithmic framework for MESP

We consider "subproblems" of MESP of the form:

$$(L(F_1, F_0)) \qquad z(C, s; F_1, F_0) := \max \left\{ \operatorname{ldet} C[S, S] \ : \ |S| = s, \ F_1 \subset S \subset N \setminus F_0 \right\},$$

where $F_1, F_0 \subset N$, $F_1 \cap F_0 = \emptyset$, $|F_1| \leq s$, $|F_0| \leq n - s$.

Generally, we think of F_1 as indices fixed into a potential solution S, and F_0 as indices fixed out of a potential solution S. By definition, we have $z(C, s; \emptyset, \emptyset) = z(C, s)$, or equivalently $L(\emptyset, \emptyset) =$ MESP.

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The **branch-and-bound** algorithm maintains an unordered list $\mathcal{L}$ of subproblems. Additionally, the algorithm maintains LB, a real number or $-\infty$, which is a lower bound on $z(C, s)$. As the algorithm progresses, it maintains two key invariant properties:

Key invariant properties for MESP branch-and-bound

- Every subproblem on $\mathcal{L}$ is feasible but does not have a unique feasible solution.
- If there is a feasible solution S of MESP with $\operatorname{ldet} C[S, S] > \text{LB}$ then S is a feasible solution for some subproblem on $\mathcal{L}$.

At the start of the algorithm, we simply set $\mathcal{L} := \{\text{MESP}\}$ and $\text{LB} := \operatorname{ldet}(C[S, S])$ for any feasible solution S. Clearly this satisfies the key invariant property. Note that because we assume that $0 < s < n$, MESP does not itself have a unique feasible solution. If the algorithm ever reaches the state of $\mathcal{L} = \emptyset$, then the key invariant property implies that $z(C, s) = \text{LB}$, and we have solved MESP. So what remains is to explain how $\mathcal{L}$ and LB evolve during the algorithm, and to check that eventually we will reach the end state of $\mathcal{L} = \emptyset$.

In general, an iteration of the algorithm chooses and removes some subproblem $L(F_1, F_0) \in \mathcal{L}$. Next, we apply some upper bounding method (see §3) to the chosen subproblem $L(F_1, F_0)$. If the calculated upper bound for the subproblem $L(F_1, F_0)$ is less than or equal to LB, we simply discard $L(F_1, F_0)$, and the key invariant properties are maintained — this is the good case, where the list $\mathcal{L}$ shrinks. If this is not the case, we choose a **branching index** $j \in N \setminus F_1 \setminus F_0$ (see [BLLW13], for example). From this, we define the **in child** as $L(F_1 + j, F_0)$ and the **out child** as $L(F_1, F_0 + j)$ — this is the bad case, where the list $\mathcal{L}$ grows. It is easy to see that every feasible solution of $L(F_1, F_0)$ is feasible for either its in child or its out child.

How we treat our children for MESP branch-and-bound

- If $|F_1 + j| = s$, then the in child has the unique feasible solution $F_1 + j$, and we discard the in child and update $\text{LB} := \max\{\text{LB}, \operatorname{ldet} C[F_1 + j, F_1 + j]\}$.
- If $|F_0 + j| = n - s$, then the out child has the unique feasible solution $N \setminus (F_0 + j)$, and we discard the out child and update $\text{LB} := \max\{\text{LB}, \operatorname{ldet} C[N \setminus (F_0 + j), N \setminus (F_0 + j)]\}$.

It is easy to see that with this processing of children, we maintain the key invariant properties. We note that even though we hope to discard children, no child is a bad child.

Noticing that a child that is put on the list $\mathcal{L}$ has one more index fixed than its parent, eventually the list must become empty, and the algorithm concludes.

This branch-and-bound algorithm is already a complete algorithmic framework for MESP. An important part that we have not yet described is how to calculate an upper bound for a subproblem from the list $\mathcal{L}$ — this will come later, in §3. But first, in §2.2 and §2.3, we cover some practical ways to enhance the framework.

A subproblem $L(F_1, F_0)$ is itself equivalent to a MESP via Schur complementation: S is an optimal solution of $L(F_1, F_0)$ if and only if $S \setminus F_1$ is an optimal solution of the MESP for which we want to select a principal submatrix of order $s - |F_1|$ from the covariance matrix $C_{F_1}[N \setminus F_0 \setminus F_1, N \setminus F_0 \setminus F_1]$. Note that this Schur complement is the covariance matrix for the Y_j, for $j \in N \setminus F_0 \setminus F_1$, *conditioned* on the Y_j, for $j \in F_1$. That is, $C_{F_1}[N \setminus F_0 \setminus F_1, N \setminus F_0 \setminus F_1]$ is the Schur complement of $C[F_1, F_1]$ in $C[N \setminus F_0, N \setminus F_0]$. Moreover, via the Schur complement determinant formula (Theorem A.2.2), we have that

$$z(C, s; F_1, F_0) \;=\; \operatorname{ldet} C[F_1, F_1] \;+\; z\left(C_{F_1}[N \setminus F_0 \setminus F_1, N \setminus F_0 \setminus F_1], s - |F_1|\right) \;.$$

So it is easy to see how we can apply any upper bounding method for MESP by first calculating an upper bound for $z(C_{F_1}[N \setminus F_0 \setminus F_1, N \setminus F_0 \setminus F_1], s - |F_1|)$ and then adding $\operatorname{ldet} C[F_1, F_1]$ to the resulting bound, thus obtaining an upper bound for $z(C, s; F_1, F_0)$

Next, we demonstrate how our in and out children for our branching scheme reverse under complementation (see §1.5 for the definition of the complementary problem). The optimal objective value of MESP is $z(C, s)$. Suppose that we branch on $i \in N$. The out-child has optimal value

$$\zeta(\text{orig}, \text{out}) := z\left(C[N - i, N - i], s\right),$$

and the in-child has optimal value

$$\zeta(\text{orig}, \text{in}) := \log C[i, i] + z\left(C_i[N - i, N - i], s - 1\right).$$

Via complementation, we have $z(C, s) = \operatorname{ldet}(C) + z\left(C^{-1}, n - s\right)$. The MESP problem associated with $z(C^{-1}, n - s)$ also has an out-child and an in-child. To translate the objective value of those children back to the original MESP, we add $\operatorname{ldet} C$. The leads to the translated objective value of the out-child of the complementary problem:

$$\zeta(\text{comp}, \text{out}) := \operatorname{ldet} C + z\left(\left(C^{-1}\right)[N - i, N - i], n - s\right).$$

Similarly, we have the translated objective value of the in-child of the complementary problem:

$$\zeta(\text{comp}, \text{in}) := \operatorname{ldet} C + \log\left(\left(C^{-1}\right)[i, i]\right) + z\left(\left(C^{-1}\right)_i[N - i, N - i], n - s - 1\right).$$

Applying complementation, we obtain:

$$\begin{aligned}\zeta(\text{orig},\text{out}) &:= z\left(C[N-i,N-i],s\right)\\ &= \operatorname{ldet} C[N-i,N-i] + z\left(\left(C[N-i,N-i]\right)^{-1}, n-1-s\right)\\ &=: \bar{\zeta}(\text{orig},\text{out});\end{aligned}$$

$$\begin{aligned}\zeta(\text{orig},\text{in}) &:= \log C[i,i] + z\left(C_i[N-i,N-i],s-1\right)\\ &= \log C[i,i] + \operatorname{ldet} C_i[N-i,N-i]\\ &\quad + z\left(\left(C_i[N-i,N-i]\right)^{-1}, (n-1)-(s-1) = n-s\right)\\ &=: \bar{\zeta}(\text{orig},\text{in});\end{aligned}$$

$$\begin{aligned}\zeta(\text{comp},\text{out}) &= \operatorname{ldet} C + z\left(\left(C^{-1}\right)[N-i,N-i], n-s\right)\\ &= \operatorname{ldet} C + \operatorname{ldet}\left(C^{-1}\right)[N-i,N-i]\\ &\quad + z\left(\left(\left(C^{-1}\right)[N-i,N-i]\right)^{-1}, (n-1)-(n-s) = s-1\right)\\ &=: \bar{\zeta}(\text{comp},\text{out});\end{aligned}$$

$$\begin{aligned}\zeta(\text{comp},\text{in}) &= \operatorname{ldet} C + \log\left(\left(C^{-1}\right)[i,i]\right) + z\left(\left(C^{-1}\right)_i[N-i,N-i], n-s-1\right)\\ &= \operatorname{ldet} C + (C^{-1})[i,i] + \operatorname{ldet}\left((C^{-1})_i[N-i,N-i]\right)\\ &\quad + z\left(\left(\left(C^{-1}\right)_i[N-i,N-i]\right)^{-1}, (n-1)-(n-s-1) = s\right)\\ &=: \bar{\zeta}(\text{comp},\text{in}).\end{aligned}$$

Proposition 2.1.1

1. $\zeta(\text{orig},\text{out}) = \bar{\zeta}(\text{comp},\text{in})$;
2. $\zeta(\text{orig},\text{in}) = \bar{\zeta}(\text{comp},\text{out})$;
3. $\zeta(\text{comp},\text{out}) = \bar{\zeta}(\text{orig},\text{in})$;
4. $\zeta(\text{comp},\text{in}) = \bar{\zeta}(\text{orig},\text{out})$.

Proof We only prove the first part; the others are very similar.

$$\left((C^{-1})_i[N-i,N-i]\right)^{-1} = C[N-i,N-i].$$

This can be checked by redefining C as C^{-1}, and then instead we have to check that

$$\left(C_i[N-i,N-i]\right)^{-1} = \left(C^{-1}\right)[N-i,N-i],$$

which follows directly from Theorem A.2.3. Therefore we have that

$$z\left(\left(\left(C^{-1}\right)_i[N-i,N-i]\right)^{-1},s\right)=z\big(C[N-i,N-i],s\big).$$

So it remains to check that

$$\det C\times(C^{-1})[i,i]\times\det\left(\left(C^{-1}\right)_i[N-i,N-i]\right)=1.$$

Equivalently,

$$\det C^{-1}=(C^{-1})[i,i]\times\det\left(\left(C^{-1}\right)_i[N-i,N-i]\right).$$

Again, redefining C as C^{-1}, we need to show that

$$\det C=C[i,i]\times\det C_i[N-i,N-i].$$

But this comes immediately from Theorem A.2.2. □

2.2 Global upper bound for early termination

It is useful to have a global upper bound, UB, on $z(C,s)$. With this, we could decide to stop the algorithm early, with a solution of value LB, and we might be satisfied with such a solution if $UB-LB$ is within some given tolerance.

Let ub(L) be the value of *any upper-bounding method* applied to $L\in\mathcal{L}$. The we let

$$\text{UB}:=\max\left\{\text{LB},\ \max_{L\in\mathcal{L}}\{\text{ub(L)}\}\right\}.$$

It is important to notice that the value of UB is not used in any way besides evaluating the possibility of early termination. But how can we really calculate this UB (if we indeed want to have it), when we do not apply an upper-bounding method to subproblems until they are *removed* from $\mathcal{L}$? The answer is that if we want to calculate and maintain this UB, we simply need to apply an upper-bounding method (and save its value with a subproblem) at the moment a subproblem is *added* to $\mathcal{L}$, rather than at the moment it is *removed* from $\mathcal{L}$. If we do this, then when we remove a subproblem L from $\mathcal{L}$, we should immediately discard it if ub(L) (now already calculated) is less than or equal to LB (which might have increased after L was added to $\mathcal{L}$).

2.3 Good lower bounds

An important issue in the performance of branch-and-bound is the calculation of good lower bounds. The basic algorithm described in §2.1 only has the possibility to improve the lower bound LB when a child has a unique feasible solution. But there are many simple heuristics for generating very good lower bounds, and they can be employed on MESP or even on any subproblem on $\mathcal{L}$. Furthermore, some heuristics are not invariant under complementation (see §1.5), so in fact we can apply such heuristics to the complementary problem and sometimes gain better lower bounds.

Good lower bounds are important for controlling the size of the list $\mathcal{L}$. For a subproblem $L(F_1, F_0)$ selected and removed from $\mathcal{L}$, we do not consider children of the subproblem if the upper bound for $L(F_1, F_0)$ is less than or equal to LB. So a greater value for LB enables us to discard more children.

In the following subsections, two lower-bounding heuristics are described for MESP, utilized already in [KLQ95]. These heuristics can also be applied to the complementary problem (see §1.5 and [KLQ95]), and this tends to be fruitful when the fraction s/n is large. For a subproblem $L(F_1, F_0)$ of MESP, as would be on the list $\mathcal{L}$, we proceed as follows. We apply the heuristics with $C \leftarrow C_{F_1}[N \setminus F_0 \setminus F_1, N \setminus F_0 \setminus F_1]$, $n \leftarrow n - |F_0| - |F_1|$, and $s \leftarrow s - |F_1|$. For any heuristically-obtained solution S, we use $S \cup F_1$ as the associated heuristic solution of MESP.

2.3.1 Greedy

Algorithm 1 is a basic "greedy" algorithm. It can be quite good at producing optimal or near-optimal solutions on many practical instances of MESP. For cases in which s is a large fraction of n and C is nonsingular, it can be better to apply the algorithm to the complementary problem, and then complement the output.

Algorithm 1: Greedy

Input: $C \in \mathbb{S}^n_+$; $1 < s < n$;
Output: S;
$S := \emptyset$;
while $|S| < s$ **do**
  let $j^* := \operatorname{argmax}\left\{ \operatorname{ldet} C[S + j, S + j] \ : \ j \in N \setminus S \right\}$;
  let $S := S + j$;
end while

Because we assume that $\operatorname{rank}(C) \geq s$, then at each iteration of the Greedy algorithm, we have that $C[S, S]$ is nonsingular.

Implemented in the most naïve manner, the number of elementary operations of the greedy algorithm is $O(n^5)$; there are $O(n)$ iterations of the while loop, each argmax can require $O(n)$ comparisons, and finally each determinant is calculated in

$O(n^3)$ elementary operations. But actually, we can do far better. We will maintain the Schur complement defined as

$$C_S := C[N \setminus S, N \setminus S] - C[N \setminus S, S] \, (C[S, S])^{-1} \, C[S, N \setminus S],$$

but we will not calculate it using the defining formula.

It is easy to check that

$$\operatorname{ldet} C[S + j, S + j] - \operatorname{ldet} C[S, S] = \log C_S[j, j],$$

so in fact we can get the needed information to calculate the argmax in $O(n)$ operations. Next, we update the Schur complement; that is, we will describe how to efficiently calculate C_{S+j} from C_S, using only $O(n^2)$ operations. We simply use the formula:

$$C_{S+j} = C_S[N \setminus S - j, N \setminus S - j] - \frac{1}{C_S[j, j]} C_S[N \setminus S - j, j] C_S[j, N \setminus S - j].$$

We can see that C_{S+j} is just a rank-1 change to $C_S[N \setminus S - j, N \setminus S - j]$ (which is just a submatrix of C_S).

Overall, we can see that carefully implemented, the number of elementary operations of the greedy algorithm can be $O(n^3)$.

Remark 2.3.1 It can be that round-off errors accumulate, and the iterations could become numerically unstable. So it can make sense to occasionally compute the Schur complement of the current solution from scratch. A good way to do this, without matrix inversion, is to first calculate the Cholesky decomposition $LL^\mathsf{T} = C[S, S]$. Then solve the triangular system $LW = C[S, N \setminus S]$ for W. Then $C_S := C[N \setminus S, N \setminus S] - W^\mathsf{T} W$. All of this work is $O(n^3)$. If we only do this a constant number of times while executing the greedy algorithm, then we only worsen its complexity by a constant factor.

2.3.2 Swapping

A swapping/interchange heuristic can often provide a significant improvement over a feasible solution, e.g., one produced using the greedy algorithm. The work per iteration is polynomially bounded, but we cannot give any useful bound on the number of iterations. This does not bother us, because we can always stop when we are satisfied with the current solution.

Algorithm 2: Swapping

Input: $C \in \mathbb{S}^n_+$; $S \subset N$ with $1 < |S| < n$ and $C[S,S]$ nonsingular;
Output: S;
improving := TRUE;
while improving **do**
  let $(i^*, j^*) := \operatorname{argmax} \left\{ \operatorname{ldet} C[S - i + j, S - i + j] \ : \ i \in S, \ j \in N \setminus S \right\}$;
  if $\operatorname{ldet} C[S - i + j, S - i + j] > \operatorname{ldet} C[S,S]$ **then**
    $S := S - i + j$;
  else
    improving := FALSE;
  end if
end while

Implemented in the most naïve manner, the number of elementary operations of the swapping algorithm *per iteration* is $O(n^5)$; there are $O(n^2)$ pairs of (i, j) to check, and for each one we have an $O(n^3)$ determinant calculation. But working carefully, we can reduce the work per iteration significantly. We can implement the search over pairs (i, j) as an outer loop over i and an inner loop over j. Within the outer loop, but before the inner loop, we calculate C_{S-i}, in $O(n^3)$ elementary operations. Then, within the inner loop, iterating on j, we calculate C_{S-i+j} from C_{S-i} using $O(n^2)$ operations; we do this in exactly the same manner for which we calculated C_{S+j} from C_S in the greedy algorithm (see §2.3.1). Overall, we have $O(n)$ choices of i, and for each choice, we can evaluate all possible j in $O(n^3)$ elementary operations. So, we can locate the best improving swap using $O(n^4)$ elementary operations.

Remark 2.3.2 From a practical viewpoint, it can make sense to not compute the best swap, but rather to accept the best swap over j, once an i is found for which there is *some* j that is an improving swap.

2.3.3 Approximation algorithm

In this subsection, we will establish that a simple and efficient local-search scheme gives us a solution for MESP with a constant-factor performance guarantee, as compared to z^*. In fact, the algorithm applies more generally to maximizing a nonnegative submodular function subject to a cardinality constraint.

Theorem 2.3.3 ([LMNS10]) Given a submodular function $f : 2^N \to \mathbb{R}_+$, an integer $0 < s < n$, and a fixed $0 < \epsilon < 1$, there is an efficient local-search algorithm for calculating a feasible solution for

$$z^* := \max\{f(S) \ : \ S \subset N, \ |S| = s\}$$

that is at least $\frac{1}{6}(1 - \epsilon)$ of z^*.

Before proving Theorem 2.3.3, we wish to explain a bit about the connection with MESP and the assumption that f is nonnegative.

Recall (Theorem 1.1.5) that when $\mathcal{Y}$ has a joint Gaussian distribution with covariance matrix $C[N, N]$, the differential entropy of $\mathcal{Y}_S$ is

$$h(\mathcal{Y}_S) = \frac{1}{2} \operatorname{ldet}(2\pi \mathrm{e} C[S, S]) = \frac{1}{2}\Big((1 + \log(2\pi))|S| + \operatorname{ldet} C[S, S]\Big).$$

Further, Theorem 1.8.1 established that $\operatorname{ldet} C[S, S]$ is submodular, and from that it is an easy exercise to check that $f(S) := h(\mathcal{Y}_S)$ is submodular — it is the sum of a modular function and a submodular function.

Therefore, Theorem 2.3.3 applies to MESP whenever $f(S) := h(\mathcal{Y}_S)$ is nonnegative. Unfortunately, the differential entropy function is not nonnegative. But we can easily find a simple and natural checkable condition (that C is "sufficiently positive definite") to guarantee this.

Proposition 2.3.4 If $\lambda_n(C) \geq \frac{1}{2\pi \mathrm{e}}$, then $h(Y_S) \geq 0$ for all $S \subset N$.

Proof This follows easily from noting that

$$\det C[S, S] = \prod_{\ell=1}^{|S|} \lambda_\ell(C[S, S]) \geq (\lambda_n(C))^s$$

using Cauchy's eigenvalue interlacing inequalities (Theorem A.3.3), and then using $h(Y_S) = \frac{1}{2} \operatorname{ldet}(2\pi \mathrm{e}\, C[S, S])$. □

Before getting to the ingredients for establishing Theorem 2.3.3, we will justify a bit why we pursue it. Let S^* be an optimal solution of MESP. With $f(S) := h(Y_S)$, we have $z(C, s) = f(S^*)$. We have

$$f(S^*) - f(S) \geq 0, \text{ for all feasible solutions } S \text{ of MESP}, \tag{2.1}$$

and this quantity is meaningful because it is the *difference* of (differential) entropies (see Remark 1.1.2). What we want to establish is that we can *efficiently* construct a feasible solution $\hat{S}$ satisfying

$$0 \geq \alpha f(S^*) - f(\hat{S}), \tag{2.2}$$

for some α satisfying $0 < \alpha \leq 1$. If we could somehow do this with $\alpha = 1$, then combining inequalities (2.1) and (2.2), we would have that $\hat{S}$ is an optimal solution. Well, we do not know how to do this (and doing so would prove that P=NP!); so we do the best that we can, which means for the greatest α that we can muster; which turns out for us to be $\alpha = \frac{1}{6}(1 - \epsilon)$. We do note that there are much better constant-factor approximation guarantees for this problem, using more complicated algorithms and analysis (see [BFNS14], where $\alpha = 0.356$ is achieved. Also see [NS16] and [LX20] for related work).

Our proof of Theorem 2.3.3 (which is adapted from [LMNS09, LMNS10]) needs a couple of lemmas concerning simple local searches.

Lemma 2.3.5 Let S be *any* locally-optimal solution under swapping. For any $T \subset N$ satisfying $|T| = s$, we have

$$2f(S) \geq f(S \cup T) + f(S \cap T).$$

Proof Let $\pi : T \setminus S \to S \setminus T$ be any bijection. For all $j \in T \setminus S$, we have

$$\begin{aligned} f(S + j) - f(S) &\leq f(S - \pi(j) + j) - f(S - \pi(j)) \quad \text{(submodularity)} \\ &\leq f(S) - f(S - \pi(j)) \quad \text{(local optimality of } S\text{)}. \end{aligned}$$

Adding over $j \in T \setminus S$, we deduce

$$\sum_{j \in T \setminus S} (f(S + j) - f(S)) \leq \sum_{j \in T \setminus S} (f(S) - f(S - \pi(j))),$$

and then using submodularity on the left-hand side (applying (1.3) appropriately), we obtain

$$f(S \cup T) - f(S) \leq \sum_{j \in T \setminus S} (f(S) - f(S - \pi(j))).$$

Finally, we have

$$\begin{aligned} \sum_{j \in T \setminus S} (f(S) - f(S - \pi(j))) &= \sum_{i \in S \setminus T} (f(S) - f(S - i)) \\ &\leq f(S) - f(S \cap T) \quad \text{(applying (1.4) appropriately)}, \end{aligned}$$

and the result follows. □

In a similar manner, we can establish the following result.

Lemma 2.3.6 Let S be *any* locally-optimal solution under swapping and deletions. For any $T \subset N$ satisfying $|T| \leq s$, we have

$$2f(S) \geq f(S \cup T) + f(S \cap T).$$

We are now ready to prove our main theorem.

Proof (Theorem 2.3.3) At a high level, we will either work with $f(S)$, if $s \leq n/2$, or the complementary problem if $s \geq n/2$. For the complementary problem, we mean

$$\max\{f(N \setminus T) \ : \ T \subset N, \ |T| = n - s\} = z^* \, ,$$

which is clearly equivalent to our original problem, and it is easy to check that $g(T) := f(N \setminus T)$ is submodular and nonnegative when f is.

So, without loss of generality, we assume that $s \leq n/2$.

Next, starting with any feasible solution, we apply the swapping algorithm, to obtain a feasible solution S_1 satisfying

$$2f(S^1) \geq f(S^1 \cup S^*) + f(S^1 \cap S^*), \tag{2.3}$$

where S^* is an optimal solution.

Next, we run a local search allowing swaps and deletions, just within $N \setminus S^1$, to obtain a set $B \subset N \setminus S^1$ satisfying

$$2f(B) \geq f(B \cup (S^* \setminus S^1)) + f(B \cap (S^* \setminus S^1)). \tag{2.4}$$

Next, if $|B| < s$, we extend B to feasible solutions $F^1 := B \cup B^1$ and $F^2 := B \cup B^2$ disjointly; that is, with $B^1 \cap B^2 = \emptyset$. That we can do this relies on $s \leq n/2$. Finally, we output S as the best of S^1, F^1, and F^2. We have

$$\begin{aligned}
6f(S) &\geq 2\left(f(S^1) + f(F^1) + f(F^2)\right) && (S \text{ is the best of the three}) \\
&\geq 2\left(f(S^1) + f(F^1 \cap F^2) + f(F^1 \cup F^2)\right) && (\text{submodularity of } f) \\
&\geq 2\left(f(S^1) + f(B) + f(F^1 \cup F^2)\right) && (\text{definition of } F^1 \text{ and } F^2) \\
&\geq 2\left(f(S^1) + f(B)\right) && (\text{nonnegativity of } f) \\
(*) \quad &\geq f(S^1 \cup S^*) + f(S^1 \cap S^*) + f(B \cup (S^* \setminus S^1)) + f(B \cap (S^* \setminus S^1)) \\
&\qquad && (\text{by inequalities (2.3) and (2.4)}) \\
&\geq f(S^1 \cup S^*) + f(B \cup S^*) + f(\emptyset) + f(B \cap (S^* \setminus S^1)) && (\text{submodularity of } f) \\
&\geq f(S^1 \cup S^*) + f(B \cup S^*) && (\text{nonnegativity of } f) \\
&\geq f(S^1 \cup S^* \cup B) + f(S^*) && (\text{submodularity of } f) \\
&\geq f(S^*) && (\text{nonnegativity of } f).
\end{aligned}$$

If instead we have $|B| = s$, then we simply return as S the best of S^1 and B. As for the performance guarantee, we start with $4f(S) \geq 2\left(f(S^1) + f(B)\right)$, and then continue as above, starting with $(*)$.

The final issue is the polynomial running time. In fact, the algorithm will not run in polynomial time as presented. What we need to do is to do the local search until there is no improvement available to within a fixed approximation factor, and then we easily get approximate versions of Lemmas 2.3.5 and 2.3.6 (with a different fixed approximation factor). We use those revised lemmas to get the final result. Details are in [LMNS10]. □

Remark 2.3.7 It is interesting to observe that both complementation (see §1.5) and swapping (see §2.3.2), two techniques with very practical motivation, play key roles in the approximation algorithm.

2.4 The branch-and-bound algorithmic framework for CMESP

Very little needs to be changed in our branch-and-bound algorithmic framework for MESP if we are instead interested in CMESP. For a subproblem determined by $F_1, F_0 \subset N$, $F_1 \cap F_0 = \emptyset$, $|F_1| \leq s$, $|F_0| \leq n - s$, we consider the **continuous relaxation of its feasible region**:

$$\begin{aligned}\big\{x \in \mathbb{R}^n \;:\; & \mathbf{e}^\mathsf{T} x = s,\; Ax \leq b,\; 0 \leq x \leq \mathbf{e},\\ & x_j = 0 \text{ for } j \in F_0,\; x_j = 1 \text{ for } j \in F_1\big\}.\end{aligned}$$

Next, we modify the first of the "key invariant properties":

Key invariant properties for CMESP branch-and-bound

- Every subproblem on $\mathcal{L}$ is feasible, but *the continuous relaxation of the corresponding CMESP* does not have a unique feasible solution.
- If there is a feasible solution S of *CMESP* with $\operatorname{ldet} C[S, S] > \mathrm{LB}$ then S is a feasible solution for some subproblem on $\mathcal{L}$.

To enforce the modified key invariant properties, we have to treat our children with more care.

How we treat our children for CMESP branch-and-bound

- If $|F_1 + j| = s$, then $F_1 + j$ is the unique set S satisfying $F_1 + j \subset S \subset N \setminus F_0$, $|S| = s$; so in this case, we discard the in-child, and if $\sum_{k \in F_1 + j} A[\cdot, k] \leq b$, then we update LB $:= \max\{\text{LB}, \operatorname{ldet} C[F_1 + j, F_1 + j]\}$.
- If $|F_0 + j| = n - s$, then $N \setminus (F_0 + j)$ is the unique set S satisfying $F_1 \subset S \subset N \setminus (F_0 + j)$, $|S| = s$; so in this case, we discard the out-child, and if $\sum_{k \in N \setminus (F_0 + j)} A[\cdot, k] \leq b$, then we update LB $:= \max\{\text{LB}, \operatorname{ldet} C[N \setminus (F_0 + j), N \setminus (F_0 + j)]\}$.
- If a child cannot be discarded based on one of the these rules above, we go further. We consider the feasible region of the continuous relaxation of the CMESP associated with a child. With a single linear-optimization problem, we can determine a maximal set of linear equations satisfied by all points of the continuous relaxation (see [FRT85]). If this is a single point $\hat{x}$, then we discard the child, and if $\hat{x}$ is binary, then we update LB $:= \max\{\text{LB}, \operatorname{ldet} C[S(\hat{x}), S(\hat{x})]\}$.

This is enough to have a complete branch-and-bound algorithmic framework for CMESP. The methodology for global upper bounds (see §2.2) does not change at all for CMESP. The ideas for lower bounds (§2.3) completely changes, as heuristics for locally searching over a discrete feasible region depend quite a lot on the nature of the constraints $Ax \leq b$. A simple idea is to somehow linearize our objective function, apply some of the arsenal of techniques for integer *linear* optimization (keeping in mind that these ILPs would be quite small compared to what is usually handled), and then take resulting solutions $\hat{x}$ and evaluate them according to the actual CMESP objective $\operatorname{ldet} C[S(\hat{x}), S(\hat{x})]$. In the simplest version of this, we could use the linearized objective function $\max \sum_{j \in N} C[j, j] x_j$.

Chapter 3
Upper bounds

"Too much information running through my brain
Too much information driving me insane
Too much information running through my brain
Too much information driving me insane"

- The Police (Too much information, from Ghost in the Machine, 1981)

In this chapter, we focus immediately on CMESP (which subsumes MESP), whenever the analysis does not simplify much for MESP. This is the case, often, for results involving upper bounds based on convex relaxations. But in other places, the methodology for treating CMESP really is different than for MESP (e.g., §3.1), and we then start by analyzing MESP.

Theorems 1.1.5 and 1.1.9 imply that from any upper bound on $z(C, s)$, we can immediately get an upper bound on the corresponding problem where $\mathcal{Y}$ has *any* distribution. So even though all of the upper bounds that we will give are designed for the multivariate Gaussian, they are valid in general. That is,

$$h(\mathcal{Y}_S) \leq \frac{1}{2} \operatorname{ldet}(2\pi \mathrm{e}\, C[S,S]) = \frac{1}{2}\Big((1+\log(2\pi))|S| + \operatorname{ub}(C,s)\Big),$$

for any distribution of $\mathcal{Y}_S$, where $\operatorname{ub}(C, s)$ is *any* upper bound on $z(C, s)$.

3.1 Spectral bounds

The "spectral bound" for MESP was introduced in [KLQ95]. Combinatorial masking for the spectral bound was introduced in [HLW01]. Non-combinatorial masking for it was introduced in [AL04] and studied further in [BL07]. The spectral bound for MESP was extended to CMESP in [Lee98a].

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3.1.1 Unconstrained

For MESP, we have the **spectral bound**:

Theorem 3.1.1 ([KLQ95])

$$z(C,s) \le \sum_{\ell=1}^{s} \log \lambda_\ell(C),$$

Proof Let S^* be an optimal solution of MESP. Then

$$\det C[S^*,S^*] = \prod_{\ell=1}^{s} \lambda_\ell\left(C[S^*,S^*]\right) \le \prod_{\ell=1}^{s} \lambda_\ell(C),$$

by Cauchy's eigenvalue interlacing inequalities (Theorem A.3.3). The result follows by taking logarithms. □

Via the next result, we see that there is no possible gain in applying the spectral bound to the complementary problem (see §1.5).

Proposition 3.1.2 The spectral bound is invariant under complementation.

Proof

$$\det C \times \prod_{\ell=1}^{n-s} \lambda_\ell(C^{-1}) \;=\; \prod_{\ell=1}^{n} \lambda_\ell(C) \times \prod_{\ell=s+1}^{n} 1/\lambda_\ell(C) \;=\; \prod_{\ell=1}^{s} \lambda_\ell(C).$$

The result follows by taking logarithms. □

Using a mask M (see §1.7), we get the "masked spectral bound" $\sum_{\ell=1}^{s} \log \lambda_\ell(M \circ C)$. For example, with $M = I_n$, we get the **diagonal bound**:

Proposition 3.1.3 ([HLW01])

$$z(C,s) \le \sum_{\ell=1}^{s} \log \operatorname{diag}(C)_{[\ell]} \,,$$

which can be directly seen from Hadamard's inequality.

It is interesting to observe that the spectral bound and diagonal bound are not generally comparable.

Example 3.1.4 Let n be even, and let $s = n/2$. Let C comprise $n/2$ 2×2 blocks, all equal to J_2 . Because $\mathrm{Diag}(C) = \mathbf{e}$, the diagonal bound is 0. Also, it is easy to see that the optimal value of this instance of MESP is 0 — simply take one element from each pair. But the eigenvalues of J_2 are 0 and 2, and therefore the eigenvalues of C are 0 with multiplicity $n/2$, and 0 with multiplicity $n/2$. Therefore the spectral bound is $\frac{n}{2}\log(2)$, much worse than the diagonal bound. ♠

Example 3.1.5 Let $C := J_n + \epsilon I_n$, with $\epsilon > 0$. The diagonal bound is $s\log(1+\epsilon)$, which goes to 0 as $\epsilon \to 0^+$. The eigenvalues of C are $n+\epsilon$ with multiplicity 1, and ϵ with multiplicity $n-1$. Hence the spectral bound is $\log(n+\epsilon)+(s-1)\log(\epsilon)$, which goes to $-\infty$ as $\epsilon \to 0^+$, as long as $s > 1$. It is easy to see that every feasible solution to this instance of MESP has objective value $\log(s+\epsilon)+(s-1)\log(\epsilon)$. Therefore the objective gap for the spectral bound is tending to $\log(n)-\log(s)$, while the objective gap for the diagonal bound is going to ∞. ♠

Generalizing the construction of the diagonal bound from the spectral bound using the mask $M := I_n$, we define a **combinatorial mask** M as one in which M is all zero except for $1 \le p \le n$ square diagonal blocks of all ones. With $p = 1$, we have $M = J_n$, and with $p = n$, we have $M = I_n$. The spectral bound applied to $M \circ C$, where M is a combinatorial mask, is known as a **spectral partition bound** (see [HLW01, Section 3]).

Example 3.1.6 Let n be even, $s = n/2$, and

$$C[N,N] = \left(\begin{array}{c|c} C[S,S] & C[S,N\setminus S] \\ \hline C[N\setminus S,S] & C[N\setminus S,N\setminus S] \end{array}\right) := \left(\begin{array}{c|c} J_s + nI_s & 0 \\ \hline 0 & J_s + \frac{3n}{4}I_s \end{array}\right).$$

The eigenvalues of $C[S,S] = J_s + nI_s$ are $3n/2$ with multiplicity 1, and n with multiplicity $s-1$. The diagonal entries of $C[N\setminus S, N\setminus S] = J_s + \frac{3n}{4}I_s$ are all $1 + 3n/4$. The key observation is that all eigenvalues of $C[S,S]$ exceed all diagonal entries of $C[N\setminus S, N\setminus S]$. This is enough to imply that the spectral partition bound associated with the combinatorial mask

$$M[N,N] = \left(\begin{array}{c|c} M[S,S] & M[S,N\setminus S] \\ \hline M[N\setminus S,S] & M[N\setminus S,N\setminus S] \end{array}\right) := \left(\begin{array}{c|c} J_s & 0 \\ \hline 0 & I_s \end{array}\right)$$

proves the optimality of S. It is an exercise to check that this does not hold for either $M := J_n$ or $M := I_n$. ♠

Distilling out the idea of this example, we have the following sufficient optimality criterion.

Proposition 3.1.7 If $|S| = s$ and

$$\lambda_{\min}(C[S,S]) \geq \max_{\ell \in N \setminus S} C[\ell,\ell],$$

then S is optimal for MESP.

In general, it is a challenge to find a good combinatorial mask for a given instance of MESP. A simple idea is to do a local-search on the space of combinatorial masks (i.e., partitions) , seeking a good one (see [HLW01, Figure 2]). Formally, we write partitions of N as $\mathcal{N} := (N_1, \ldots, N_p)$, where the N_i are nonempty, disjoint sets whose union is N. Associated with the partition $\mathcal{N}$ is a combinatorial mask M, with $M[N_i, N_i] = J_{|N_i|}$ and all other entries equal to zero. For a local-search aimed at decreasing the spectral partition bound, we can suggest three reasonable starting partitions:

1. $\mathcal{N} := (N)$ (associated with the spectral bound),
2. $\mathcal{N} := (\{1\}, \{2\}, \ldots, \{n\})$ (associated with the spectral bound),
3. $\mathcal{N} := (S, \{j_1\}, \{j_2\}, \ldots, \{j_{n-s}\})$, where S is a heuristically generated solution of MESP and $N \setminus S = \{j_1, j_2, \ldots, j_{n-s}\}$ (associated with Proposition 3.1.7).

From any starting partition, we can suggest reasonable local moves:

1. single-element move: move an element from one block to another block, or create a new singleton block from it.
2. two-element swap: swap two elements from different blocks.
3. (one new 2-block): take two elements from different blocks or the same block, and create a new 2-element block.
4. merge two blocks into a single block.

[AL04] introduced non-combinatorial masks, in the context of the spectral bound for MESP. They considered the **masked spectral bound**

$$\xi(M) := \sum_{\ell=1}^{s} \log \lambda_\ell(M \circ C)$$

and locally minimizing the bound over masks M. Generally, this is a challenging problem because the function to be minimized is not generally convex nor differentiable, and the set of masks naturally uses a semidefiniteness constraint in its description. In any case, we need to be able to compute the gradient of $\xi(M)$, at a mask $\hat{M}$, to cook up a descent algorithm for locally minimizing $\xi(M)$.

Theorem 3.1.8 ([AL04]) If $\lambda_s(\hat{M} \circ C) > \lambda_{s+1}(\hat{M} \circ C)$, then

$$\frac{\partial \xi(\hat{M})}{\partial M} = C \circ \sum_{\ell=1}^{s} \frac{1}{\lambda_\ell(\hat{M} \circ C)} \hat{u}_\ell \hat{u}_\ell^\mathsf{T} ,$$

where $\sum_{\ell=1}^{n} \lambda_\ell(\hat{M} \circ C)\hat{u}_\ell \hat{u}_\ell^\mathsf{T}$ is a spectral decomposition of $\hat{M} \circ C$.

Proof Our strategy is to apply Theorem A.4.16. For $\lambda \in \mathbb{R}^n_{++}$, let

$$\phi(\lambda) := \sum_{\ell=1}^{s} \log \lambda_\ell .$$

We have

$$\frac{\partial \xi(\hat{M})}{\partial M_{ij}} = \sum_{\ell=1}^{n} \frac{\partial \phi(\lambda(\hat{M} \circ C))}{\partial \lambda_\ell} h_{ij}^\ell(\hat{M}),$$

where

$$h_{ij}^\ell(\hat{M}) := \hat{u}_\ell^\mathsf{T} \frac{\partial(\hat{M} \circ C)}{\partial M_{ij}} \hat{u}_\ell = C_{ij} \hat{u}_{\ell i} \hat{u}_{\ell j} ,$$

and

$$\frac{\partial \phi(\lambda(\hat{M} \circ C))}{\partial \lambda_\ell} = \begin{cases} 1/\lambda_\ell(\hat{M} \circ C), & \text{for } \ell \le s; \\ 0, & \text{for } \ell > s. \end{cases}$$

So

$$\frac{\partial \xi(\hat{M})}{\partial M_{ij}} = C_{ij} \sum_{\ell=1}^{s} \frac{1}{\lambda_\ell(\hat{M} \circ C)} \hat{u}_{\ell i} \hat{u}_{\ell j} .$$

□

Remark 3.1.9 We could devise various descent algorithms based on Theorem 3.1.8, aimed at locally minimizing $\xi(M)$. Unfortunately, ξ is not generally differentiable at $\hat{M}$ when $\lambda_s(\hat{M} \circ C) = \lambda_{s+1}(\hat{M} \circ C)$. In such a case, by Theorem A.4.25, $C \circ \sum_{\ell=1}^{s} \frac{1}{\lambda_\ell(\hat{M} \circ C)} \hat{u}_\ell \hat{u}_\ell^\mathsf{T}$ is a subgradient of ξ at $\hat{M}$. In practice, we do not expect to locally minimize ξ "exactly", because our context is in applying this to subproblems in branch-and-bound (see §2.1). In such a context, if after a few descent steps we cannot get the objective value of ξ below LB (see §2.1), then we may as well branch. In case, as we will see next, there is a more efficient approach that we recommend.

[BL07] considered similar methods but avoided the semidefiniteness constraint. Let

$$H(V) := \mathrm{Diag}\left((\mathrm{diag}(VV^\mathsf{T}))^{-\frac{1}{2}} \right) V,$$

with $V \in \mathbb{R}^{n\times n}$. The function H normalizes the rows of a matrix. We have $H_{i\cdot}(V) = \|V_{i\cdot}\|^{-1}V_{i\cdot}$, for $i = 1, \ldots, n$. Notice that $M := H(V)H^{\mathsf{T}}(V)$ is always a mask, and all masks arise this way. Next, we let

$$g(W) := \xi(WW^{\mathsf{T}}),$$

where W is row-normalized. That is, $g(W)$ is the masked spectral bound corresponding to the mask $M = WW^{\mathsf{T}}$. Finally, it is convenient to define

$$f(V) := g(H(V)),$$

that is the masked spectral bound using the mask $H(V)H^{\mathsf{T}}(V)$.

Theorem 3.1.10 ([BL07]) If

$$\lambda_s\left(\left(H(\hat{V})H^{\mathsf{T}}(\hat{V})\right)\circ C\right) > \lambda_{s+1}\left(\left(H(\hat{V})H^{\mathsf{T}}(\hat{V})\right)\circ C\right),$$

then, for $1 \le i, j \le n$,

$$\frac{\partial f(\hat{V})}{\partial V_{ij}} = \|V_{i\cdot}\|^{-1}\frac{\partial g(H(\hat{V}))}{\partial H_{ij}(V)} - V_{ij}\sum_{k=1}^{n}\frac{\partial g(H(\hat{V}))}{\partial H_{ik}(V)}V_{ik}\|V_{i\cdot}\|^{-3},$$

where

$$\frac{\partial g(H(\hat{V}))}{\partial H(V)} = 2\left(C\circ\left(\sum_{\ell=1}^{s}\frac{1}{\lambda_\ell\left(\left(H(\hat{V})H^{\mathsf{T}}(\hat{V})\right)\circ C\right)}\hat{u}_\ell\hat{u}_\ell^{\mathsf{T}}\right)\right)H(\hat{V}),$$

and $\sum_{\ell=1}^{n}\lambda_\ell\left(\left(H(\hat{V})H^{\mathsf{T}}(\hat{V})\right)\circ C\right)\hat{u}_\ell\hat{u}_\ell^{\mathsf{T}}$ is a spectral decomposition of $\left(H(\hat{V})H^{\mathsf{T}}(\hat{V})\right)\circ C$.

Proof First, we observe that because $H_{ik}(V) = \|V_{i\cdot}\|^{-1}V_{ik}$, we can calculate

$$\frac{\partial H_{ik}(V)}{\partial V_{ij}} = \begin{cases} -V_{ij}V_{ik}\|V_{i\cdot}\|^{-3}, & \text{for } k \neq j; \\ -V_{ik}^2\|V_{i\cdot}\|^{-3} + \|V_{i\cdot}\|^{-1}, & \text{for } k = j. \end{cases}$$

Assuming (for now) that $\frac{\partial g(H(\hat{V}))}{\partial H_{ik}(V)}$ exists, for all i and k, then by the chain rule, we have

$$\begin{aligned}\frac{\partial f(\hat{V})}{\partial V_{ij}} &= \sum_{k=1}^{n}\frac{\partial g(H(\hat{V}))}{\partial H_{ik}(V)}\frac{\partial H_{ik}(\hat{V})}{\partial V_{ij}} \\ &= \|V_{i\cdot}\|^{-1}\frac{\partial g(H(\hat{V}))}{\partial H_{ij}(V)} - V_{ij}\sum_{k=1}^{n}\frac{\partial g(H(\hat{V}))}{\partial H_{ik}(V)}V_{ik}\|V_{i\cdot}\|^{-3}.\end{aligned}$$

Next, let $W := H(V)$, and $M := WW^{\mathsf{T}}$. Because $M_{\ell k} = W_{\ell\cdot}(W_{k\cdot})^{\mathsf{T}}$, we have

$$\frac{\partial \hat{M}_{\ell k}}{\partial W_{ij}} = \begin{cases} 0, & \text{for } \ell \neq i,\ k \neq i; \\ \hat{W}_{kj}, & \text{for } \ell = i; \\ \hat{W}_{\ell j}, & \text{for } k = i, \end{cases}$$

and hence

$$\begin{aligned} \frac{\partial g(\hat{W})}{\partial W_{ij}} &= \sum_{\ell=1}^{n}\sum_{k=1}^{n} \frac{\partial g(\hat{W})}{\partial M_{\ell k}} \frac{\partial \hat{M}_{\ell k}}{\partial W_{ij}} = \sum_{k=1}^{n} \frac{\partial g(\hat{W})}{\partial M_{ik}} \hat{W}_{kj} + \sum_{\ell=1}^{n} \frac{\partial g(\hat{W})}{\partial M_{\ell i}} \hat{W}_{\ell j} \\ &= 2\sum_{k=1}^{n} \frac{\partial g(\hat{W})}{\partial M_{ik}} \hat{W}_{kj} = 2\sum_{k=1}^{n} \frac{\partial \xi(\hat{M})}{\partial M_{ik}} \hat{W}_{kj} \ . \end{aligned}$$

As long as $\lambda_s\left(H(\hat{V})H^{\mathsf{T}}(\hat{V}) \circ C\right) > \lambda_{s+1}\left(H(\hat{V})H^{\mathsf{T}}(\hat{V}) \circ C\right)$, we can employ Theorem 3.1.10 to calculate. So

$$\begin{aligned} \frac{\partial g(\hat{W})}{\partial W_{ij}} &= 2\sum_{k=1}^{n} \left(C_{ik} \sum_{\ell=1}^{s} \frac{1}{\lambda_\ell(\hat{W}\hat{W}^{\mathsf{T}} \circ C)} \hat{u}_{\ell i}\hat{u}_{\ell k} \right) \hat{W}_{kj} \\ &= 2\left(C_{i\cdot} \circ \left(\sum_{\ell=1}^{s} \frac{1}{\lambda_\ell(\hat{W}\hat{W}^{\mathsf{T}} \circ C)} \hat{u}_{\ell i}\hat{u}_{\ell}^{\mathsf{T}} \right) \right) \hat{W}_{\cdot j} \ . \end{aligned}$$

It is easy now to check, that this last expression is the entry in row i and column j of

$$2\left(C \circ \left(\sum_{\ell=1}^{s} \frac{1}{\lambda_\ell\left(\hat{W}\hat{W}^{\mathsf{T}} \circ C\right)} \hat{u}_\ell \hat{u}_\ell^{\mathsf{T}} \right) \right) \hat{W},$$

The result follows. □

[BL07] found that a basic steepest-descent algorithm with back tracking is effective; see Algorithm 3. Parameter values of $k_{\max} := 2$ and $\beta := 1/3$ are practical.

Algorithm 3: Steepest descent for the spectral bound with factored mask

Input: V, $t_{\max}$, $k_{\max}$, $0 < \beta < 1$;
Output: V;
let $t := 1$;
while $t \leq t_{\max}$ **do**
  let $k := 0$;
  while $f(WW^{\mathsf{T}}) > f(VV^{\mathsf{T}})$ *and* $k < k_{\max}$ **do**
    let $k := k + 1$ and $W := V - \beta^k \nabla f(V)$;
  end while
  let $V := W$ and $t := t + 1$;
end while

3.1.2 Constrained

Now we turn to the CMESP. For $\pi \in \mathbb{R}^m_+$, we define a diagonal matrix $D_\pi \in \mathbb{S}^n_{++}$ by

$$D_\pi[k,j] := \begin{cases} \exp\{-\frac{1}{2}\sum_{i=1}^m \pi_i a_{ij}\}, & \text{for } k = j; \\ 0, & \text{for } k \neq j. \end{cases}$$

Next, we define

$$v(\pi) := \pi^\mathsf{T} b + \sum_{\ell=1}^{s} \log \lambda_\ell \left(D_\pi C D_\pi\right),$$

the **Lagrangian spectral bound**.

Theorem 3.1.11 ([Lee98a])

$$z(C,s,A,b) \leq v(\pi), \text{ for all } \pi \in \mathbb{R}^m_+ .$$

Proof Let $S \subset N$ be an arbitrary feasible solution of CMESP, and consider arbitrary $\pi \in \mathbb{R}^m_+$. Then

$$\begin{aligned} v(\pi) &\geq \pi^\mathsf{T} b + \operatorname{ldet}(D_\pi[S,S]C[S,S]D_\pi[S,S]) \\ &\qquad \text{(from the definition of } v \text{ and Theorem 3.1.1).} \\ &= \pi^\mathsf{T} b + \log\left(\det\left((D_\pi[S,S])^2\right) \times \det C[S,S]\right) \\ &= \pi^\mathsf{T} b + \log\left(\prod_{j\in S} \exp\left\{-\sum_{i=1}^m \pi_i a_{ij}\right\}\right) + \operatorname{ldet} C[S,S] \\ &= \pi^\mathsf{T}\left(b - \sum_{j\in S} A[\cdot,j]\right) + \operatorname{ldet} C[S,S] \\ &\geq \operatorname{ldet} C[S,S], \end{aligned}$$

with the last inequality using the nonnegativity of π and the feasibility of S. □

Theorem 3.1.12 ([Lee98a]) v is a convex function on $\mathbb{R}^m_+$.

Our proof of Theorem 3.1.12 uses the following **multiplicative inequality for leading singular values**.

Theorem 3.1.13 (See [HJ94, Theorem 3.3.4, p.171–2]) For compatible matrices A and B,

$$\prod_{\ell=1}^{s} \sigma_\ell(AB) \leq \prod_{\ell=1}^{s} \sigma_\ell(A)\sigma_\ell(B) \,.$$

Proof (Theorem 3.1.12) It suffices to prove the convexity of $\tilde{v}(\pi) := v(\pi) - \pi^{\mathsf{T}} b$.

$$\begin{aligned}
&2\tilde{v}(\tfrac{1}{2}\pi^1 + \tfrac{1}{2}\pi^2) \\
&\quad = \log \prod_{\ell=1}^{s} \lambda_\ell^2 \left(D_{\frac{1}{2}\pi^1+\frac{1}{2}\pi^2} C D_{\frac{1}{2}\pi^1+\frac{1}{2}\pi^2} \right) \\
&\quad = \log \prod_{\ell=1}^{s} \lambda_\ell \left(D_{\frac{1}{2}\pi^1+\frac{1}{2}\pi^2} C D_{\frac{1}{2}\pi^1+\frac{1}{2}\pi^2} D_{\frac{1}{2}\pi^1+\frac{1}{2}\pi^2} C D_{\frac{1}{2}\pi^1+\frac{1}{2}\pi^2} \right) \\
&\quad = \log \prod_{\ell=1}^{s} \lambda_\ell \left(D_{-\frac{1}{2}\pi^1+\frac{1}{2}\pi^2} D_{\pi^1} C D_{\pi^1} D_{\pi^2} C D_{\pi^2} D_{\frac{1}{2}\pi^1-\frac{1}{2}\pi^2} \right) \\
&\quad = \log \prod_{\ell=1}^{s} \sigma_\ell \left(D_{\pi^1} C D_{\pi^1} D_{\pi^2} C D_{\pi^2} \right) \\
&\quad \leq \log \prod_{\ell=1}^{s} \sigma_\ell \left(D_{\pi^1} C D_{\pi^1} \right) \sigma_\ell \left(D_{\pi^2} C D_{\pi^2} \right) \qquad \text{(by Theorem 3.1.13)} \\
&\quad = \log \prod_{\ell=1}^{s} \lambda_\ell \left(D_{\pi^1} C D_{\pi^1} \right) \lambda_\ell \left(D_{\pi^2} C D_{\pi^2} \right) \\
&\quad = \log \prod_{\ell=1}^{s} \lambda_\ell \left(D_{\pi^1} C D_{\pi^1} \right) + \log \prod_{\ell=1}^{s} \lambda_\ell \left(D_{\pi^2} C D_{\pi^2} \right) \\
&\quad = \tilde{v}(\pi^1) + \tilde{v}(\pi^2).
\end{aligned}$$

□

Because of Proposition 3.1.11 and Theorem 3.1.12, it is an attractive idea to seek to solve the related Lagrangian dual:

$$\min \left\{ v(\pi) \;:\; \pi \in \mathbb{R}^m_+ \right\}.$$

For $\hat{\pi} \in \mathbb{R}^m$, let $\sum_{\ell=1}^{n} \lambda_\ell(D_{\hat{\pi}} C D_{\hat{\pi}}) u_\ell u_\ell^{\mathsf{T}}$ be a spectral decomposition of $D_{\hat{\pi}} C D_{\hat{\pi}}$. We write the j-th component of u_ℓ as $u_{\ell j}$, and we we define $x(\hat{\pi})$ by $x_j(\hat{\pi}) := \sum_{\ell=1}^{s} (\hat{u}_{\ell j})^2$, for $j = 1, 2, \ldots, n$, Equivalently, let $\hat{U}$ be the matrix comprising the $\hat{u}^\ell$, $\ell = 1, \ldots, s$, as its ordered set of columns. Then $x(\hat{\pi}) := \hat{U}^{(2)}\mathbf{e}$.

Proposition 3.1.14 ([Lee98a]) For all $\hat{\pi} \in \mathbb{R}^m$, the vector $x(\hat{\pi})$ satisfies

(i) $\mathbf{e}^\mathsf{T} x(\hat{\pi}) = s$;
(ii) $0 \leq x(\hat{\pi}) \leq \mathbf{e}$.

Proof

$$\mathbf{e}^\mathsf{T} x(\hat{\pi}) = \mathbf{e}^\mathsf{T} \hat{U}^{(2)} \mathbf{e} = \mathrm{tr}(\hat{U}^\mathsf{T} \hat{U}) = \mathrm{tr}(I_s) = s.$$

Clearly we have that $x(\hat{\pi}) \geq 0$ as its entries are sums of squares. Also, we have that $\hat{U}\hat{U}^\mathsf{T} = I$ has as its j-th diagonal entry $\sum_{\ell=1}^n (\hat{u}_{\ell j})^2 = 1$. Therefore, $x(\hat{\pi}) \leq 1$. □

Proposition 3.1.14 indicates how $x(\hat{\pi})$ is a "continuous solution" of MESP. This helps to motivate the following result.

Theorem 3.1.15 ([Lee98a]) If the $\lambda_s(D_{\hat{\pi}} C D_{\hat{\pi}}) > \lambda_{s+1}(D_{\hat{\pi}} C D_{\hat{\pi}})$, then $\frac{\partial}{\partial \pi} v(\hat{\pi}) = b - A x(\hat{\pi})$.

Proof Our strategy is to apply Theorem A.4.16. For $\lambda \in \mathbb{R}^n_{++}$, let

$$\phi(\lambda) := \sum_{\ell=1}^{s} \log \lambda_\ell \, .$$

We have

$$\frac{\partial v(\hat{\pi})}{\partial \pi_i} = b_i + \sum_{\ell=1}^{n} \frac{\partial \phi(\lambda(D_{\hat{\pi}} C D_{\hat{\pi}}))}{\partial \lambda_\ell} h_i^\ell(\hat{\pi}),$$

where

$$\begin{aligned} h_i^\ell(\hat{\pi}) &= \hat{u}_\ell^\mathsf{T} \frac{\partial D_{\hat{\pi}} C D_{\hat{\pi}}}{\partial \pi_i} \hat{u}_\ell \\ &= -\frac{1}{2} \hat{u}_\ell^\mathsf{T} \Big(\mathrm{Diag}(A[i,\cdot]) D_{\hat{\pi}} C D_{\hat{\pi}} + D_{\hat{\pi}} C D_{\hat{\pi}} \, \mathrm{Diag}(A[i,\cdot]) \Big) \hat{u}_\ell \, , \end{aligned}$$

and

$$\frac{\partial \phi(\lambda(D_{\hat{\pi}} C D_{\hat{\pi}}))}{\partial \lambda_\ell} = \begin{cases} 1/\lambda_\ell(D_{\hat{\pi}} C D_{\hat{\pi}}), & \text{for } \ell \leq s; \\ 0, & \text{for } \ell > s. \end{cases}$$

So,

$$\frac{\partial v(\hat{\pi})}{\partial \pi_i} = b_i - \frac{1}{2}\sum_{\ell=1}^{s} \frac{\hat{u}_\ell^{\mathsf{T}}\Big(\mathrm{Diag}(A[i,\cdot])D_{\hat{\pi}}CD_{\hat{\pi}} + D_{\hat{\pi}}CD_{\hat{\pi}}\,\mathrm{Diag}(A[i,\cdot])\Big)\hat{u}_\ell}{\lambda_\ell(D_{\hat{\pi}}CD_{\hat{\pi}})},$$

Using the eigenvector equation

$$(D_{\hat{\pi}}CD_{\hat{\pi}})\hat{u}_\ell = \lambda_\ell(D_{\hat{\pi}}CD_{\hat{\pi}})\hat{u}_\ell\ ,$$

we obtain

$$\hat{u}_\ell^{\mathsf{T}}(D_{\hat{\pi}}CD_{\hat{\pi}})\,\mathrm{Diag}(A[i,\cdot])\hat{u}_\ell = \hat{u}_\ell^{\mathsf{T}}\lambda_\ell(D_{\hat{\pi}}CD_{\hat{\pi}})\,\mathrm{Diag}(A[i,\cdot])\hat{u}_\ell\ .$$

Consequently,

$$\begin{aligned}
\frac{\partial v(\hat{\pi})}{\partial \pi_i} &= b_i - \sum_{\ell=1}^{s} \frac{\lambda_\ell(D_{\hat{\pi}}CD_{\hat{\pi}})\hat{u}_\ell^{\mathsf{T}}\,\mathrm{Diag}(A[i,\cdot])\hat{u}_\ell}{\lambda_\ell(D_{\hat{\pi}}CD_{\hat{\pi}})} \\
&= b_i - \sum_{\ell=1}^{s} \hat{u}_\ell^{\mathsf{T}}\,\mathrm{Diag}(A[i,\cdot])\hat{u}_\ell \\
&= b_i - \sum_{\ell=1}^{s}\sum_{j=1}^{n} A[i,j]\hat{u}_{\ell j}^2 \\
&= b_i - A[i,\cdot])x(\hat{\pi}).
\end{aligned}$$

□

Remark 3.1.16 We could concoct various descent algorithms based on Theorem 3.1.15, aimed at solving the Lagrangian dual (for example, see [Lee98a]). The unfortunate issue is that v is not generally differentiable at $\hat{\pi}$ when $\lambda_s(D_{\hat{\pi}}CD_{\hat{\pi}}) = \lambda_{s+1}(D_{\hat{\pi}}CD_{\hat{\pi}})$. In such a case, by Theorem A.4.25, $b - Ax(\hat{\pi})$ is a subgradient of v at $\hat{\pi}$. But in practice, we do not expect to solve the Lagrangian dual "exactly". Rather, our context is in applying this to subproblems in branch-and-bound (see §2.1). In such a context, if after a few descent steps we cannot get the objective value of the Lagrangian dual below LB (see §2.1), then we may as well stop working on the instance of the Lagrangian dual (for the subproblem at hand) and branch.

3.2 Integer linear optimization

Integer linear optimization/programming (ILP) is a potentially attractive tool for MESP and CMESP. Typically, an n of interest for MESP would be rather small in the ILP world, and so it makes sense to try to and make use of formulations

and computational tools from ILP. This direction was pursued in [HLW01] and in [LW03].

3.2.1 An ILP-based diagonal bound for CMESP

In §3.1.1, we presented the diagonal bound for MESP as a manifestation of a masked spectral bound. But we can extend the diagonal bound in another way, to handle CMESP, via 0/1 linear optimization (see [HLW01]):

$$\max \left\{ \sum_{j=1}^{n} \log(C[j,j]) x_j \ : \ \mathbf{e}^{\mathsf{T}} x = s, \ Ax \leq b, \ x \in \{0,1\}^n \right\}.$$

When there are no constraints $Ax \leq b$, this coincides with the diagonal bound. In cases where the ILP is hard to solve, we could perform a partial ILP branch-and-bound search and use the resulting global upper bound from that.

When $Ax \leq b$ is a knapsack constraint, we could apply standard dynamic programming to calculate the upper bound. Suppose that $Ax \leq b$ is the single constraint $\alpha^{\mathsf{T}} x \leq \beta$, with $\alpha \in \mathbb{Z}_+^n$ and $\beta \in \mathbb{Z}_+$. For $\eta \in \mathbb{Z}_+$ with $\eta \leq \beta$ and $t \in \mathbb{Z}_+$ with $t \leq s$, let

$$f_k(\eta, t) := \max \left\{ \sum_{j=1}^{k} \log(C[j,j]) x_j \ : \ \mathbf{e}^{\mathsf{T}} x = t, \ \alpha^{\mathsf{T}} x \leq \eta, \ x \in \{0,1\}^k \right\}.$$

We have the recursion

$$f_k(\eta, t) = \max \{ f_{k-1}(\eta, t), \ f_{k-1}(\eta - a_k, t-1) + \log C[k,k] \},$$

with the boundary conditions

$$f_k(\eta, t) := \begin{cases} 0, & \text{for } \eta \geq 0, t = 0, \text{ or } \eta = 0, t \geq 0, \text{ or } k = 0, \eta \geq 0, t \geq 0; \\ -\infty, & \text{for } \eta < 0 \text{ or } t < 0. \end{cases}$$

With this recursion, starting from $k = 0$ and incrementing k up to n, we can compute the bound $f_n(\beta, s)$ in $O(n^2\beta)$ arithmetic operations.

3.2.2 An ILP-based partition bound for MESP

Returning to MESP, we can organize another 0/1 linear optimization (see [LW03]). Let $\mathcal{N} := (N_1, \ldots, N_p)$ be a partition of N into p nonempty parts. We define

$$f_k(N_i) := z(C[N_i, N_i], k), \ \textit{or an upper bound on that,}$$

for $i = 1, \ldots, p$ and $k = 1, 2, \ldots, |N_i|$; and we define $f_0(N_i) = 0$ for for $i = 1, \ldots, p$. From a practical point of view, we assume for each i that we can easily calculate $z(C[N_i, N_i], k)$ for values of k such that $\min\{k, |N_i| - k\}$ is small. When $z(C[N_i, N_i], k)$ is not easily calculated, we simply use an upper bound on it for $f_k(N_i)$ (for example, any of the upper bounds in this section). From this, we formulate the bound

$$
\begin{aligned}
g(C, s; \mathcal{N}) := \max & \sum_{i=1}^{p} \sum_{k=1}^{|N_i|} f_k(N_i) x_k(i) \\
& \text{subject to:} \\
& \sum_{k=1}^{|N_i|} x_k(i) \leq 1, \text{ for } i = 1, 2, \ldots, p, \\
& \sum_{i=1}^{p} \sum_{k=1}^{|N_i|} k x_k(i) = s, \\
& x_k(i) \in \{0, 1\}, \text{ for } i = 1, 2, \ldots, p, \ k = 1, 2, \ldots, |N_i|.
\end{aligned}
$$

The interpretation of $x_k(i)$ is that it is an indicator variable for picking the optimal k-element subset of N_i . The fact that this is a valid upper bound again follows from Fischer's inequality. When all N_i are singletons, this bound becomes the diagonal bound.

When $|N_i| \leq 2$ for all $i = 1, 2, \ldots, p$, we can calculate the bound via a maximum-weight capacitated b-matching problem, thus it can be calculated in polynomial time (see [GK04], for example). We set up a graph $G(C, s; \mathcal{N})$ (refer to Figure 3.1), as follows.

Vertices: We have a vertex v_i for each i. There are two additional vertices: Z ("ZERO") and T ("TWO").

Edges: For each i, we have an edge between v_i and Z. For i such that $|N_i| = 2$, we have an edge between v_i and T. There is one more edge, between Z and T.

A capacitated b-matching is an assignment of integers to the edges of the graph, satisfying given bounds, further satisfying given requirements (i.e., bounds) on the edge sums at each vertex.

Edges bounds: All edges have lower bounds of 0 and upper bounds of 1, except for the edge between Z and T, which has no bounds.

Vertex requirements: The v_i all have lower bounds of 0 and upper bounds of 1. Vertex Z has an exact requirement of p, and vertex T has an exact requirement of s.

Finally, we define edge weights, finalizing the input for the maximum-weight capacitated b-matching problem.

Edge weights: The weight of the edge between v_i and Z is $-f_1(N_i)$. The weight of the edge between v_i and T is $f_2(N_i) - f_1(N_i)$. The weight of the edge between Z and T is 0.

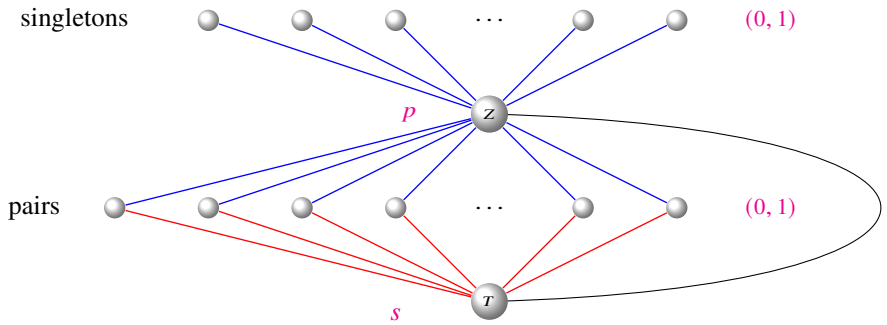


Fig. 3.1 $G(C, s; \mathcal{N})$

Theorem 3.2.1 ([LW03]) $\sum_{i=1}^{p} f_1(N_i)$ plus the optimal value of the maximum-weight capacitated b-matching problem on $G(C, s; \mathcal{N})$ is precisely $g(C, s; \mathcal{N})$. Hence we can calculate $g(C, s; \mathcal{N})$ in polynomial time, when $|N_i| \leq 2$ for $i = 1, 2, \ldots, p$.

Proof We refer to Figure 3.1. Vertices v_i corresponding to sets N_i with $|N_i| = 1$ (respectively, 2) are called "singletons" ("pairs"). Edges between vertex Z (respectively, T and the v_i are colored blue (red). We refer to feasible solutions of the graph problem. Because of the edge bounds, such solutions are 0/1 valued on the blue and red edges. Let F be the support of such a solution, restricted to the blue and red edges. Let k_Z (respectively, k_T) be the number of red edges in F. Because of the exact vertex requirements at Z and T, we have $p - k_Z = s - k_T$ (the equation enforced by the edge between Z and T), which we rewrite as $s = p + k_T - k_Z$.

The "baseline solution" is $F = \emptyset$. In this case, $k_Z = k_T = 0$, and so $s = p$. This corresponds to choosing one element from each pair. For each blue edge (v_i, Z) in F, we are choosing zero elements from N_i, one fewer than the baseline choice, and this is why we subtract k_Z . For each red edge (v_i, T) in F, we are choosing two elements from N_i, one more than the baseline choice, and this is why we add k_T . Therefore the vertex required force the correct accounting for F, so that it really corresponds to choosing s elements from N. The baseline solution corresponds to choosing one element from each N_i, and this is why we have the constant $\sum_{i=1}^{p} f_1(N_i)$ (in the statement of the theorem). A blue edge (v_i, Z) corresponds to choosing zero elements from N_i, and this is why its weight is $-f_1(N_i)$, eliminating the contribution of $f_1(N_i)$ in the baseline solution. A red edge (v_i, T) corresponds to choosing two elements from N_i, and this is why its weight is $f_2(N_i) - f_1(N_i)$, eliminating the contribution

of $f_1(N_i)$ in the baseline solution, and replacing the contribution by $f_2(N_i)$. Overall, we get that the weight of F plus $\sum_{i=1}^{p} f_1(N_i)$ is $\sum_{i=1}^{p} f_{|N_i \cap F|}(N_i) = g(C, s; \mathcal{N})$. □

We can seek to find a good partition with $|N_i| \leq 2$ for all i. Just as we advocated doing local search on the space of partitions for locally minimizing the spectral partition bound (see §3.1.1), we can proceed similarly here, but restricting to partitions with $|N_i| \leq 2$ for all i. In this way, we can execute each local-search move here in polynomial time.

3.3 linx bound

The "linx bound" for CMESP was introduced in [Ans20]. It has many nice features that we will expose. It is a convex-programming based bound, in the natural variable $x \in \mathbb{R}^n$. We will see that it is "self complementary", like the spectral bound. It is not invariant under scaling, but there is a good methodology, introduced in [CFLL21], for finding an optimal scaling. The derivation of the linx bound is simple, yet subtle. And, practically speaking, it turns out to be one of the best upper bounds available.

For $x \in [0, 1]^n$, we define

$$f(x) := \tfrac{1}{2} \operatorname{ldet} L(x),$$
$$\text{where } L(x) := C \operatorname{Diag}(x) C + \operatorname{Diag}(\mathbf{e} - x).$$

We define the **linx bound** for CMESP:

$$\text{(linx)} \qquad z_{\text{linx}} := \max \left\{ f(x) \ : \ \mathbf{e}^{\mathsf{T}} x = s; \ Ax \leq b; \ 0 \leq x \leq \mathbf{e} \right\}.$$

We are justified in using maximum rather than supremum, using the fact that $L(x) \in \mathbb{S}^n_+$ for all $x \in [0, 1]^n$ and Proposition A.1.4 (using $\mathcal{K} := \{x \in \mathbb{R}^n \ : \ L(x) \in \mathbb{S}^n_{++}\}$ and $\mathcal{Q} := \left\{x \in \mathbb{R}^n \ : \ \mathbf{e}^{\mathsf{T}} x = s; \ Ax \leq b; \ 0 \leq x \leq \mathbf{e}\right\}$).

Anstreicher chose the name "linx" because $L(x)$ is linear in x. The name "linx bound" is justified by the following result.

Theorem 3.3.1 ([Ans20])

$$z(C, s, A, b) \leq z_{\text{linx}} .$$

Proof It suffices to demonstrate that for any $S \subset N$ with $|S| = s$, we have $f(x(S)) = \operatorname{ldet} C[S, S]$. Let $T := N \setminus S$. After a symmetric permutation of indices, we can write

$$\begin{aligned} C \operatorname{Diag}(x(S))C &= \left(\begin{array}{c|c} C[S,S] & C[S,T] \\ \hline C[S,T]^{\mathsf{T}} & C[T,T] \end{array}\right)\left(\begin{array}{c|c} I & 0 \\ \hline 0 & 0 \end{array}\right)\left(\begin{array}{c|c} C[S,S] & C[S,T] \\ \hline C[S,T]^{\mathsf{T}} & C[T,T] \end{array}\right) \\ &= \left(\begin{array}{c|c} C[S,S]^2 & C[S,S][S,T] \\ \hline C[S,T]^{\mathsf{T}}C[S,S] & C[S,T]^{\mathsf{T}}C[S,T] \end{array}\right). \end{aligned}$$

Therefore,

$$C \operatorname{Diag}(x(S))C + \operatorname{Diag}(\mathbf{e} - x(S)) = \left(\begin{array}{c|c} C[S,S]^2 & C[S,S][S,T] \\ \hline C[S,T]^{\mathsf{T}}C[S,S] & C[S,T]^{\mathsf{T}}C[S,T] + I \end{array}\right).$$

If $C[S, S]$ is singular, then we can see that $L(x(S))$ is singular. So we have that $f(x(S)) = \operatorname{ldet} C[S, S] = -\infty$ in this case.

So, we can now assume that $C[S, S]$ is nonsingular. Employing the Schur complement determinant formula (Theorem A.2.2) and taking logarithms, we have

$$\begin{aligned} &\operatorname{ldet}(C \operatorname{Diag}(x(S))C + \operatorname{Diag}(\mathbf{e} - x(S))) \\ &= 2 \operatorname{ldet} C[S, S] + \\ &\quad \operatorname{ldet}(C[S,T]^{\mathsf{T}}C[S,T] + I - C[S,T]^{\mathsf{T}}C[S,S]C[S,S]^{-2}C[S,S]C[S,T]) \\ &= 2 \operatorname{ldet} C[S, S]. \end{aligned}$$

Therefore, the objective value of $x(S)$ in linx is precisely $\operatorname{ldet} C[S, S]$. □

3.3.1 Convexity of linx

Lemma 3.3.2

(i) For $C \succeq 0$, we have $L(x) \succ 0$, for all $0 \leq x < \mathbf{e}$.
(ii) For $C \succ 0$, we have $L(x) \succ 0$, for all $0 \leq x \leq \mathbf{e}$.

Proof In both cases, if $x < \mathbf{e}$, we have $C \operatorname{Diag}(x)C \succeq 0$ and $\operatorname{Diag}(\mathbf{e} - x) \succ 0$. Therefore $L(x) \succ 0$. In particular, we have established case (i).

For case (ii), if $x = \mathbf{e}$, we have $C \operatorname{Diag}(x)C \succ 0$ and $\operatorname{Diag}(\mathbf{e} - x) = 0$. Therefore, $L(x) \succ 0$. Now suppose that, $x_i = 1$, for $i \in U \subset N$, $U \neq N$, and $x_i < 1$, for

$i \in V := N \setminus U$. After a symmetric permutation of indices we can write

$$
\begin{aligned}
&C\,\mathrm{Diag}(x)C + \mathrm{Diag}(\mathbf{e}-x) \\
&\quad = \left(\begin{array}{c|c} C[U,U] & C[U,V] \\ \hline C[U,V]^{\mathsf{T}} & C[V,V] \end{array}\right)\left(\begin{array}{c|c} I & 0 \\ \hline 0 & \mathrm{Diag}(x[V]) \end{array}\right)\left(\begin{array}{c|c} C[U,U] & C[U,V] \\ \hline C[U,V]^{\mathsf{T}} & C[V,V] \end{array}\right) \\
&\qquad + \left(\begin{array}{c|c} 0 & 0 \\ \hline 0 & \mathrm{Diag}(\mathbf{e}-x[V]) \end{array}\right) \\
&\quad = \left(\begin{array}{c} C[U,U] \\ \hline C[U,V]^{\mathsf{T}} \end{array}\right)\left(\begin{array}{c|c} C[U,U] & C[U,V] \end{array}\right) \qquad (3.1) \\
&\qquad + \left(\begin{array}{c} C[U,V] \\ \hline C[V,V] \end{array}\right)\mathrm{Diag}(x[V])\left(\begin{array}{c|c} C[U,V]^{\mathsf{T}} & C[V,V] \end{array}\right) \qquad (3.2) \\
&\qquad + \left(\begin{array}{c|c} 0 & 0 \\ \hline 0 & \mathrm{Diag}(\mathbf{e}-x[V]) \end{array}\right). \qquad (3.3)
\end{aligned}
$$

Clearly, the matrices (3.1), (3.2), and (3.3) are positive semidefinite. Moreover, adding (3.1) and (3.3), we have

$$
\begin{aligned}
\Theta := &\left(\begin{array}{c} C[U,U] \\ \hline C[U,V]^{\mathsf{T}} \end{array}\right)\left(\begin{array}{c|c} C[U,U] & C[U,V] \end{array}\right) + \left(\begin{array}{c|c} 0 & 0 \\ \hline 0 & \mathrm{Diag}(\mathbf{e}-x[V]) \end{array}\right) \\
= &\left(\begin{array}{c|c} C[U,U]^2 & C[U,U]C[U,V] \\ \hline C[U,V]^{\mathsf{T}}C[U,U] & C[U,V]^{\mathsf{T}}C[U,V] + \mathrm{Diag}(\mathbf{e}-x[V]) \end{array}\right).
\end{aligned}
$$

Then applying the Schur complement determinant formula (Theorem A.2.2), we obtain

$$
\begin{aligned}
\det(\Theta) &= \det(C[U,U]^2) \times \det(C[U,V]^{\mathsf{T}}C[U,V] + \mathrm{Diag}(\mathbf{e}-x[V]) \\
&\qquad - C[U,V]^{\mathsf{T}}C[U,U]C[U,U]^{-2}C[U,U]C[U,V]) \\
&= \det(C[U,U]^2) \times \det(\mathrm{Diag}(\mathbf{e}-x[V])) > 0.
\end{aligned}
$$

Therefore, case (ii) holds. □

Theorem 3.3.3 The function $\mathrm{ldet}\, L(\cdot)$ is concave on $\{x \in [0,1]^n \,:\, L(x) \succ 0\}$.

Proof For each $x \in [0,1]^n$ such that $L(x) \succ 0$, we can write

$$
\mathrm{ldet}\, L(x) = \max_{Z \succ 0} \{\mathrm{ldet}\, Z \,:\, Z \preceq L(x)\}. \qquad (3.4)
$$

Considering the monotonicity of ldet, we have that the maximum $Z \succ 0$ satisfying the equality in the constraint in (3.4) gives the value of ldet $L(x)$, for each $x \in [0,1]^n$ such that $L(x) \succ 0$. Considering that ldet is strictly concave on $\mathbb{S}^n_{++}$ (see Theorem A.4.12) and Proposition A.1.4, we have that the optimal solution of (3.4), denoted in the following by $Z^*(x)$, exists and is unique for all $x \in [0,1]^n$ such that $L(x) \succ 0$.

Now, considering the linearity of the function L, the optimality of $Z^*(\cdot)$, and the concavity of ldet, we have

$$\begin{aligned} \operatorname{ldet} L(\tau x + (1-\tau)y) &= \max_{Z \succ 0} \{\operatorname{ldet} Z \ : \ Z \preceq L(\tau x + (1-\tau)y)\} \\ &= \max_{Z \succ 0} \{\operatorname{ldet} Z \ : \ Z \preceq \tau L(x) + (1-\tau)L(y)\} \\ &\geq \operatorname{ldet}(\tau Z^*(x) + (1-\tau)Z^*(y)) \\ &\geq \tau \operatorname{ldet} Z^*(x) + (1-\tau) \operatorname{ldet} Z^*(y), \end{aligned} \tag{3.5}$$

for all $\tau \in [0,1]$ and $x, y \in [0,1]^n$ such that $L(x), L(y) \succ 0$. The result follows. □

Corollary 3.3.4 linx is a convex optimization problem.

Proof As the constraints of linx are all linear, it suffices to demonstrate that the objective function is concave on the feasible set. We proved in Theorem 3.3.3 that the objective function is concave on the convex set $\mathcal{F} := \{x \in [0,1]^n \ : \ L(x) \succ 0\}$, and we have that $f(x)$ is $-\infty$ on $\{x \in [0,1]^n \ : \ L(x) \succeq 0\} \setminus \mathcal{F}$. Therefore, the objective function is concave (in the extended sense) on all of the feasible set. □

In Corollary 3.3.4, we established that linx is a convex relaxation of CMESP. In Proposition 3.3.7, we will see that under certain assumptions, we can establish that linx has a unique optimal solution as well. In Lemma 3.3.5, we introduce these assumptions.

Lemma 3.3.5 L is a one-to-one function on $\mathbb{R}^n$ if and only if $\lambda_i(C \circ C) \neq 1$ for all $i \in N$.

Proof For $x, y \in \mathbb{R}^n$, we have

$$\begin{aligned} & C \operatorname{Diag}(x) C + \operatorname{Diag}(\mathbf{e} - x) = C \operatorname{Diag}(y) C + \operatorname{Diag}(\mathbf{e} - y) \iff \\ & C \operatorname{Diag}(x - y) C - \operatorname{Diag}(x - y) = 0. \end{aligned} \tag{3.6}$$

Note that

$$\operatorname{diag}(C \operatorname{Diag}(x - y) C - \operatorname{Diag}(x - y)) = (C \circ C - I)(x - y).$$

Therefore, (3.6) only holds for $x = y$, if and only if $C \circ C - I$ has full rank. The result follows. □

Remark 3.3.6 Note that Lemma 3.3.5 and its proof only need the symmetry of C (not $C \succeq 0$).

Proposition 3.3.7 If $\lambda_i(C \circ C) \neq 1$ for all $i \in N$, and $z_{\text{linx}} > -\infty$, then linx has a unique optimal solution.

Proof With $z_{\text{linx}} > -\infty$, any optimal solution $\hat{x}$ of linx has $L(\hat{x}) \succ 0$. Therefore, it suffices to prove that linx has a unique optimal solution on the convex set $\mathcal{F} := \{x \in \mathbb{R}^n \ : \ \mathbf{e}^T x = s;\ Ax \leq b;\ 0 \leq x \leq \mathbf{e};\ L(x) \succ 0\}$.

Considering that ldet is strictly concave on $\mathbb{S}^n_{++}$ (see Theorem A.4.12), the last inequality in (3.5) is strict for all $\tau \in (0,1)$ and x, y with $L(x) \neq L(y)$. Therefore, if $\lambda_i(C \circ C) \neq 1$ for all $i \in N$, then L is a one-to-one function on $\mathcal{F}$, and then we have that ldet $L(\cdot)$ is strictly concave on $\mathcal{F}$. The result thus follows from Proposition A.1.4. □

3.3.2 Duality for linx

We have established that linx is a convex optimization problem. Next, we formulate the dual problem of linx. We rewrite it as

$$z_{\text{linx}} := \max \left\{ \tfrac{1}{2} \operatorname{ldet} W \ : \ L(x) = W,\ \mathbf{e}^T x = s;\ Ax \leq b;\ 0 \leq x \leq \mathbf{e} \right\}.$$

We consider the Lagrangian function

$$\begin{aligned} &\mathcal{L}(W, x, \Theta, \upsilon, \nu, \pi, \tau) := \\ &\quad \tfrac{1}{2} \operatorname{ldet} W + \Theta \bullet (L(x) - W) + \upsilon^T x + \nu^T(\mathbf{e} - x) + \pi^T(b - Ax) + \tau(s - \mathbf{e}^T x), \end{aligned}$$

with dom $\mathcal{L} = \mathbb{S}^n_{++} \times \mathbb{R}^n \times \mathbb{S}^n \times \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}$. The corresponding dual function is

$$\mathcal{L}^*(\Theta, \upsilon, \nu, \pi, \tau) := \sup_{W \succ 0,\ x} \mathcal{L}(W, x, \Theta, \upsilon, \nu, \pi, \tau), \tag{3.7}$$

and the Lagrangian dual problem of linx is

$$\min\{\mathcal{L}^*(\Theta, \upsilon, \nu, \pi, \tau) \ : \ \upsilon \geq 0,\ \nu \geq 0,\ \pi \geq 0\}.$$

Assuming that $z_{\text{linx}} > -\infty$, we are justified to use minimum in the formulation of the Lagrangian dual problem, rather than infimum, because, in this case, Slater's condition holds for the primal, and so the optimal value of the Lagrangian dual problem is attained.

We have that

$$(C \operatorname{Diag}(x) C)_{ij} = \sum_{p \in N} C_{ip} C_{pj} x_p \ .$$

So,

$$L(x)_{ij} = \begin{cases} \sum_{p\in N} C_{ip}C_{pj}x_p\,, & \text{for } i \neq j, \\ \sum_{p\in N} C_{ip}^2 x_p + 1 - x_i\,, & \text{for } i = j. \end{cases}$$

Then, we consider

$$L(x) = L_0 + L_1x_1 + L_2x_2 + \cdots + L_nx_n\,.$$

where

$$L_0 = I \quad \text{and} \quad L_k[i,j] = \left\{ \begin{array}{ll} C_{ik}C_{kj}\,, & \text{for } i \neq k \text{ or} j \neq k, \\ C_{ii}^2 - 1, & \text{for } i = j = k, \end{array} \right\} \text{ for } i,j,k \in N. \tag{3.8}$$

We note that the Lagrangian function is concave in x and $W \succ 0$. Then the supremum in (3.7) occurs where the gradient of the Lagrangian function with respect to x and W is equal to zero. So, considering Lemma A.4.13, the Lagrangian dual problem may be written as

$$\begin{array}{ll} \min & \frac{1}{2}\operatorname{ldet} W + \Theta \bullet (L(x) - W) + \upsilon^{\mathsf{T}}x + \nu^{\mathsf{T}}(\mathbf{e} - x) + \pi^{\mathsf{T}}(b - Ax) + \tau(s - \mathbf{e}^{\mathsf{T}}x) \\ & \text{subject to:} \\ & \frac{1}{2}W^{-1} - \Theta = 0, \\ & \Theta \bullet L_i + \upsilon_i - \nu_i - \pi^{\mathsf{T}}A[\cdot,i] - \tau = 0, \qquad i \in N, \\ & W \succ 0,\ \upsilon \geq 0,\ \nu \geq 0,\ \pi \geq 0. \end{array} \tag{3.9}$$

Theorem 3.3.8 The following equation is valid for all feasible solutions of (3.9):

$$\begin{aligned} &\Theta \bullet (L(x) - W) + \upsilon^{\mathsf{T}}x + \nu^{\mathsf{T}}(\mathbf{e} - x) + \pi^{\mathsf{T}}(b - Ax) + \tau(s - \mathbf{e}^{\mathsf{T}}x) \\ &\quad = \Theta \bullet L_0 + \nu^{\mathsf{T}}\mathbf{e} + \pi^{\mathsf{T}}b + \tau s - n/2. \end{aligned}$$

Proof From the first constraint in (3.9) we have that $W = (2\Theta)^{-1} = \Theta^{-1}/2$. And from the group of constraints for all $i \in N$, we have

$$\begin{aligned} &\Theta \bullet \sum_{i=1}^{n} L_i x_i + \upsilon^{\mathsf{T}}x - \nu^{\mathsf{T}}x - \pi^{\mathsf{T}}Ax - \tau\mathbf{e}^{\mathsf{T}}x = 0 \quad \Rightarrow \\ &\Theta \bullet (L(x) - W) + \upsilon^{\mathsf{T}}x + \nu^{\mathsf{T}}(\mathbf{e} - x) + \pi^{\mathsf{T}}(b - Ax) + \tau(s - \mathbf{e}^{\mathsf{T}}x) = \\ &\quad \Theta \bullet L_0 + \nu^{\mathsf{T}}\mathbf{e} + \pi^{\mathsf{T}}b + \tau s - n/2. \end{aligned}$$

□

Considering Theorem 3.3.8, we may write the Lagrangian dual problem of linx, in (3.9), as

$$\begin{aligned} \min\ & -\tfrac{1}{2}\operatorname{ldet}(2\Theta) + \Theta \bullet L_0 + v^{\mathsf{T}}\mathbf{e} + \pi^{\mathsf{T}} b + \tau s - n/2 \\ & \text{subject to:} \\ & \Theta \bullet L_i + \upsilon_i - \nu_i - \pi^{\mathsf{T}} A[\cdot, i] - \tau = 0, \qquad i \in N, \\ & \Theta \succ 0,\ \upsilon \geq 0,\ \nu \geq 0,\ \pi \geq 0. \end{aligned}$$

Or, using (3.8),

(Dlinx)

$$\begin{aligned} \min\ & -\tfrac{1}{2}\operatorname{ldet}(2\Theta) + \operatorname{tr}(\Theta) + v^{\mathsf{T}}\mathbf{e} + \pi^{\mathsf{T}} b + \tau s - n/2 \\ & \text{subject to:} \\ & \operatorname{diag}(C\Theta C - \Theta) + \upsilon - \nu - A^{\mathsf{T}}\pi - \tau\mathbf{e} = 0, \\ & \Theta \succ 0,\ \upsilon \geq 0,\ \nu \geq 0,\ \pi \geq 0. \end{aligned}$$

Dlinx is always strictly feasible, because $\hat{\Theta} := I$, $\hat{\upsilon} := \mathbf{e}$, $\hat{\nu} := \operatorname{diag}(CC)$, $\hat{\pi} := 0$, $\hat{\tau} := 0$ is a strictly feasible solution.[2] Therefore, if linx has a feasible solution with finite objective value, then we have strong duality.

3.3.3 Fixing variables in linx

Generally, with respect to a convex relaxation of a discrete minimization problem: if we have a good lower bound on the objective value of a discrete optimum, and if we have a dual-feasible solution of the relaxation in which the variable for a primal constraint is large enough, then we may be able to infer that the constraint is satisfied by an equation in every discrete optimal solution. Such a principle is generally called **variable fixing**, and it is a key enhancement of branch-and-bound based on convex relaxation. In this section, we apply this principle of variable fixing to the linx relaxation of CMESP.

[2] a point is *strictly feasible* for a convex program if it satisfies the nonlinear inequalities strictly.

Theorem 3.3.9 Let

- LB be the objective-function value of a feasible solution for CMESP,
- $(\hat{\Theta}, \hat{\upsilon}, \hat{\nu}, \hat{\pi}, \hat{\tau})$ be a feasible solution for Dlinx with objective-function value $\hat{\zeta}$.

Then, for every optimal solution x^* for CMESP, we have:

$$x_k^* = 0, \ \forall\, k \in N \text{ such that } \hat{\zeta} - \text{LB} < \hat{\upsilon}_k \ ,$$
$$x_k^* = 1, \ \forall\, k \in N \text{ such that } \hat{\zeta} - \text{LB} < \hat{\nu}_k \ .$$

Proof Consider linx with the additional constraint $x_k = 1$. The dual becomes then

$$\begin{aligned} \min\ & -\tfrac{1}{2}\,\text{ldet}(2\Theta) + \text{tr}(\Theta) + \nu^{\mathsf{T}}\mathbf{e} + \pi^{\mathsf{T}} b + \tau s - n/2 - \omega \\ & \text{subject to:} \\ & \text{diag}(C\Theta C - \Theta) + \upsilon - \nu - A^{\mathsf{T}}\pi - \tau\mathbf{e} + \omega\mathbf{e}_k = 0, \\ & \Theta \succ 0,\ \upsilon \geq 0,\ \nu \geq 0,\ \pi \geq 0, \end{aligned}$$

where ω is the new dual variable. Notice that if $(\hat{\Theta}, \hat{\upsilon}, \hat{\nu}, \hat{\pi}, \hat{\tau})$ is a feasible solution for Dlinx, then $(\hat{\Theta}, \hat{\upsilon} - \omega\mathbf{e}_k, \hat{\nu}, \hat{\pi}, \hat{\tau}, \omega)$ is a feasible solution of this modified dual, *as long as* $\hat{\upsilon}_k - \omega \geq 0$.The objective value of this modified dual is $\hat{\zeta} - \omega$. So, to minimize the objective value of our feasible solution of this modified dual, we set ω equal to $\hat{\upsilon}_k$. We conclude that $\hat{\zeta} - \hat{\upsilon}_k$ is an upper bound on objective value of all solutions of CMESP that satisfy $x_k = 1$. So if $\hat{\zeta} - \hat{\upsilon}_k < LB$, then no optimal solution of CMESP can have $x_k = 1$.

Similarly, consider linx with the additional constraint $x_k = 0$. The dual becomes

$$\begin{aligned} \min\ & -\tfrac{1}{2}\,\text{ldet}(2\Theta) + \text{tr}(\Theta) + \nu^{\mathsf{T}}\mathbf{e} + \pi^{\mathsf{T}} b + \tau s - n/2 \\ & \text{subject to:} \\ & \text{diag}(C\Theta C - \Theta) + \upsilon - \nu - A^{\mathsf{T}}\pi - \tau\mathbf{e} + \omega\mathbf{e}_k = 0, \\ & \Theta \succ 0,\ \upsilon \geq 0,\ \nu \geq 0,\ \pi \geq 0, \end{aligned}$$

where ω is again the new dual variable. Notice that if $(\hat{\Theta}, \hat{\upsilon}, \hat{\nu}, \hat{\pi}, \hat{\tau})$ is a feasible solution for Dlinx, then $(\hat{\Theta}, \hat{\upsilon}, \hat{\nu} + \omega\mathbf{e}_k, \hat{\pi}, \hat{\tau}, \omega)$ is a feasible solution of this modified dual, *as long as* $\hat{\nu}_k + \omega \geq 0$.The objective value of this modified dual is $\hat{\zeta} + \omega$. So, to minimize the objective value of our feasible solution of this modified dual, we set ω equal to $-\hat{\nu}_k$. We conclude that $\hat{\zeta} - \hat{\nu}_k$ is an upper bound on objective value of all solutions of CMESP that satisfy $x_k = 0$. So if $\hat{\zeta} - \hat{\nu}_k < LB$, then no optimal solution of CMESP can have $x_k = 0$. □

Remark 3.3.10 Of course when we apply Theorem 3.3.9, it is in our interest to use the *optimal* dual solution when it is available, and we can only expect that some variables can be fixed if LB is a reasonably good lower bound on the optimal value of CMESP.

3.3.4 Computing linx and Dlinx solutions

3.3.4.1 A good feasible solution of Dlinx from linx

Variable fixing (see Theorem 3.3.9) relies on a good feasible solution of Dlinx. Moreover, certifying the quality of a feasible solution of linx also requires a good feasible solution of Dlinx. Motivated by these points, we show how to construct a feasible solution of Dlinx from a feasible solution $\hat{x}$ of linx such that $L(\hat{x}) \succ 0$, with the goal of producing a small gap.

We define $\hat{\Theta} := \frac{1}{2}(L(\hat{x}))^{-1}$, and we see that $\frac{1}{2}\operatorname{ldet} L(\hat{x}) = -\frac{1}{2}\operatorname{ldet}(2\hat{\Theta})$. The minimum duality gap between $\hat{x}$ in linx and feasible solutions of Dlinx of the form $(\hat{\Theta}, \upsilon, \nu, \pi, \tau)$ is the optimal value of the linear program

$$
\begin{array}{ll}
(G(\hat{\Theta})) & \min \ \operatorname{tr}(\hat{\Theta}) + \nu^{\mathsf{T}}\mathbf{e} + \pi^{\mathsf{T}} b + \tau s - n/2 \\
& \quad \text{subject to:} \\
& \quad \upsilon - \nu - A^{\mathsf{T}}\pi - \tau\mathbf{e} = \operatorname{diag}(\hat{\Theta} - C\hat{\Theta}C), \\
& \quad \upsilon \geq 0, \ \nu \geq 0, \ \pi \geq 0.
\end{array}
$$

Note that $G(\hat{\Theta})$ has a simple closed-form optimal solution for MESP, that is when there are no $Ax \leq b$ constraints. To construct this solution, we consider the following optimality conditions for $G(\hat{\Theta})$ in this particular case,

$$
\begin{aligned}
& \nu + \tau\mathbf{e} \geq \operatorname{diag}(C\hat{\Theta}C - \hat{\Theta}), \ \nu \geq 0, \\
& \mathbf{e}^{\mathsf{T}} x = s, \ 0 \leq x \leq \mathbf{e}, \\
& \nu^{\mathsf{T}}\mathbf{e} + \tau s = \operatorname{diag}(C\hat{\Theta}C - \hat{\Theta})^{\mathsf{T}} x,
\end{aligned}
\tag{3.10}
$$

and we define σ as the permutation of the indices $\{1, \ldots, n\}$, such that $\operatorname{diag}(C\hat{\Theta}C - \hat{\Theta})_{\sigma(1)} \geq \cdots \geq \operatorname{diag}(C\hat{\Theta}C - \hat{\Theta})_{\sigma(n)}$. Then, we can see that

$$
\begin{aligned}
\tau^* &:= \operatorname{diag}(C\hat{\Theta}C - \hat{\Theta})_{\sigma(s)}, \\
\nu^*_{\sigma(\ell)} &:= \begin{cases} \operatorname{diag}(C\hat{\Theta}C - \hat{\Theta})_{\sigma(\ell)} - \operatorname{diag}(C\hat{\Theta}C - \hat{\Theta})_{\sigma(s)}, & \text{for } 1 \leq \ell \leq s; \\ 0, & \text{for } s < \ell \leq n, \end{cases} \\
x^*_{\sigma(\ell)} &:= \begin{cases} 1, & \text{for } 1 \leq \ell \leq s; \\ 0, & \text{for } s < \ell \leq n \end{cases}
\end{aligned}
$$

satisfies (3.10). Therefore, when there are no $Ax \leq b$ constraints, $(\upsilon^* := \nu^* + \tau^* \mathbf{e} - \mathrm{diag}(C\hat{\Theta}C - \hat{\Theta}), \nu^*, \tau^*)$ is an optimal solution to $G(\hat{\Theta})$, with objective value $\mathrm{tr}(\hat{\Theta}) + \mathbf{e}^{\mathsf{T}}\nu^* + s\tau^* - n/2$, corresponding to minimum duality gap between $\hat{x}$ in linx and feasible solutions of Dlinx of the form $(\hat{\Theta}, \upsilon, \nu, \tau)$.

From the Lagrangian dual of linx, formulated as (3.9), we see that if $\hat{x}$ is an optimal solution to linx, then we have $\Theta = \frac{1}{2}(L(\hat{x}))^{-1}$ in an optimal solution of Dlinx. Therefore, an optimal solution of $G(\hat{\Theta})$ gives optimal values of the remaining variables $(\upsilon, \nu, \pi, \tau)$ of Dlinx. In this case, due to strong duality for linx and Dlinx, the optimal objective value of $G(\hat{\Theta})$ is equal to zero.

3.3.4.2 Solving linx

A convenient and efficient means for solving linx is with YALMIP/SDPT3. YALMIP (a Matlab toolbox) in conjunction with the conic-optimization solver SDPT3 (see [TTT99, TTT12]) provides a very efficient algorithmic treatment of the ldet function. YALMIP also returns to us an optimal solution of Dlinx. Of course, primal and dual solutions are only feasible and optimal to computed tolerances, so neither primal nor dual solution may give us a rigorous bound for CMESP.

Alternatively, we can solve linx using a general-purpose nonlinear-programming solver (e.g., Ipopt [WB06]). A general-purpose solver will typically return a solution of the KKT system, but the feasibility and optimality of the primal and dual solution are only to computed tolerances. Feasibility of a primal solution for linx, which has only linear constraints, can be relied upon. So we suggest using the methodology of the previous section to produce a dual-feasible solution for variable fixing, and to get a rigorous bound for CMESP.

If we wish to use such a general-purpose solver in a convenient manner, through a general-purpose mathematical-optimization modeling language (e.g., AMPL (see [FGK93]) and GAMS (see [GAM13])), we are out of luck because such languages cannot handle ldet maximization terms. But we can employ such solvers with other interfaces (e.g., via Matlab or C++), and in such a case, the user must supply function-value, function-gradient, and optionally function-Hessian subroutines to the solver. For the linx objective function, we have the following gradient and Hessian calculation.

Lemma 3.3.11 For $x \in [0,1]^n$ such that $L(x) \succ 0$, we have

$$\nabla f(x) = \frac{1}{2}\left(\mathrm{diag}\left(CL(x)^{-1}C\right) - \mathrm{diag}\left(L(x)^{-1}\right)\right).$$

Proof We have

$$(C\,\mathrm{Diag}(x)C)_{ij} = \sum_{\ell \in N} C_{i\ell}C_{\ell j}x_\ell\ .$$

So,

$$L(x)_{ij} = \begin{cases} \sum_{\ell \in N} C_{i\ell} C_{\ell j} x_\ell \,, & \text{for } i \neq j; \\ \sum_{\ell \in N} C_{i\ell}^2 x_\ell + 1 - x_i \,, & \text{for } i = j, \end{cases}$$

and

$$\frac{\partial L(x)_{ij}}{\partial x_k} = \begin{cases} C_{ik} C_{kj} \,, & \text{for } (i,j) \neq (k,k); \\ C_{kk}^2 - 1, & \text{for } (i,j) = (k,k). \end{cases}$$

Then, using the chain rule and Lemma A.4.13, we have

$$\begin{aligned} \frac{\partial \operatorname{ldet} L(x)}{\partial x_k} &= \sum_{i,j \in N} \frac{\partial \operatorname{ldet} L(x)}{\partial L(x)_{ij}} \frac{\partial L(x)_{ij}}{\partial x_k} \\ &= \sum_{i,j \in N} L(x)_{ij}^{-1} \frac{\partial L(x)_{ij}}{\partial x_k} \\ &= \sum_{i,j \in N} L(x)_{ij}^{-1} C_{ik} C_{kj} - L(x)_{kk}^{-1} \\ &= (CL(x)^{-1}C)_{kk} - L(x)_{kk}^{-1} \,. \end{aligned}$$

□

Additionally, we can work out the Hessian for the linx objective function.

Lemma 3.3.12 For $x \in [0,1]^n$ such that $L(x) \succ 0$, we have

$$\begin{aligned} \nabla^2 f(x) = \tfrac{1}{2} \Big(& - \big(CL(x)^{-1}C\big) \circ \big(CL(x)^{-1}C\big) \\ & + \big(L(x)^{-1}C\big) \circ \big(L(x)^{-1}C\big) + \big(L(x)^{-1}C\big)^{\mathsf{T}} \circ \big(L(x)^{-1}C\big)^{\mathsf{T}} \\ & - L(x)^{-1} \circ L(x)^{-1} \Big). \end{aligned}$$

Proof Using the chain rule and Lemma A.2.5, for $\ell = 1, \ldots, n$, we have

$$\begin{aligned} \frac{\partial}{\partial x_\ell} L(x)_{kk}^{-1} &= \sum_{i,j \in N} \frac{\partial L(x)_{kk}^{-1}}{\partial L(x)_{ij}} \frac{\partial L(x)_{ij}}{\partial x_\ell} \\ &= \sum_{i,j \in N} - \Big(L(x)^{-1} \mathbf{e}_i \mathbf{e}_j^{\mathsf{T}} L(x)^{-1} \Big)_{kk} \frac{\partial L(x)_{ij}}{\partial x_\ell} \,, \end{aligned}$$

and

$$\begin{aligned}\frac{\partial}{\partial x_\ell}\left(CL(x)^{-1}C\right)_{kk} &= \sum_{i,j\in N}\frac{\partial\left(CL(x)^{-1}C\right)_{kk}}{\partial L(x)_{ij}}\frac{\partial L(x)_{ij}}{\partial x_\ell}\\ &= \sum_{i,j\in N} -\left(CL(x)^{-1}\mathbf{e}_i\mathbf{e}_j^{\mathsf{T}}L(x)^{-1}C\right)_{kk}\frac{\partial L(x)_{ij}}{\partial x_\ell}.\end{aligned}$$

So, defining $g(x) := 2\nabla f(x)$, we have

$$\begin{aligned}\frac{\partial g(x)_k}{\partial x_\ell} &= \sum_{i,j\in N}\left(-\left(CL(x)^{-1}E_{ij}L(x)^{-1}C\right)_{kk}C_{i\ell}C_{\ell j} + \left(L(x)^{-1}E_{ij}L(x)^{-1}\right)_{kk}C_{i\ell}C_{\ell j}\right)\\ &\quad + \left(CL(x)^{-1}E_{\ell\ell}L(x)^{-1}C\right)_{kk} - \left(L(x)^{-1}E_{\ell\ell}L(x)^{-1}\right)_{kk}\\ &= \sum_{i,j\in N}\left(-\left(CL(x)^{-1}\right)_{ki}\left(L(x)^{-1}C\right)_{jk}C_{i\ell}C_{\ell j} + L(x)^{-1}_{ki}L(x)^{-1}_{jk}C_{i\ell}C_{\ell j}\right)\\ &\quad + \left(CL(x)^{-1}E_{\ell\ell}L(x)^{-1}C\right)_{kk} - \left(L(x)^{-1}E_{\ell\ell}L(x)^{-1}\right)_{kk}\\ &= -\left(CL(x)^{-1}\right)_{k\cdot}C_{\cdot\ell}C_{\ell\cdot}\left(L(x)^{-1}C\right)_{\cdot k} + L(x)^{-1}_{k\cdot}C_{\cdot\ell}C_{\ell\cdot}L(x)^{-1}_{\cdot k}\\ &\quad + \left(CL(x)^{-1}E_{\ell\ell}L(x)^{-1}C\right)_{kk} - \left(L(x)^{-1}E_{\ell\ell}L(x)^{-1}\right)_{kk}\\ &= -\left(\left(CL(x)^{-1}C\right)\circ\left(CL(x)^{-1}C\right)\right)_{k\ell} + \left(\left(L(x)^{-1}C\right)\circ\left(L(x)^{-1}C\right)\right)_{k\ell}\\ &\quad + \left(CL(x)^{-1}E_{\ell\ell}L(x)^{-1}C\right)_{kk} - \left(L(x)^{-1}E_{\ell\ell}L(x)^{-1}\right)_{kk}\\ &= -\left(\left(CL(x)^{-1}C\right)\circ\left(CL(x)^{-1}C\right)\right)_{k\ell} + \left(\left(L(x)^{-1}C\right)\circ\left(L(x)^{-1}C\right)\right)_{k\ell}\\ &\quad + \left(\left(CL(x)^{-1}\right)\circ\left(CL(x)^{-1}\right)\right)_{k\ell} - \left(\left(L(x)^{-1}\right)\circ\left(L(x)^{-1}\right)\right)_{k\ell}.\end{aligned}$$

Therefore,

$$\begin{aligned}\nabla^2 f(x) = \tfrac{1}{2}\Bigg(&-\left(CL(x)^{-1}C\right)\circ\left(CL(x)^{-1}C\right) + \left(L(x)^{-1}C\right)\circ\left(L(x)^{-1}C\right)\\ &+\left(CL(x)^{-1}\right)\circ\left(CL(x)^{-1}\right) - L(x)^{-1}\circ L(x)^{-1}\Bigg).\end{aligned}$$

The result follows. □

3.3.5 Scaling for linx

Considering the scaling introduced in §1.6, we define, for a positive constant γ,

$$f(\gamma;x) := \tfrac{1}{2}\Big(\operatorname{ldet} L(\gamma;x) - s\log(\gamma)\Big),$$
$$\text{where } L(\gamma;x) := \gamma C \operatorname{Diag}(x) C + \operatorname{Diag}(\mathbf{e}-x).$$

We note that $f(x) = f(\gamma;x)$ for all x such that $\mathbf{e}^T x = s$, $x \in \{0,1\}^n$. We can therefore, replace $f(x)$ by $f(\gamma;x)$ in linx, defining the **scaled linx bound**

$$(\text{linx}_\gamma) \qquad z_{\text{linx}_\gamma} := \max\left\{ f(\gamma;x) \;:\; \mathbf{e}^T x = s;\; Ax \le b;\; 0 \le x \le \mathbf{e} \right\}.$$

As mentioned in [Ans20], the linx bound is very sensitive to the choice of the scaling factor, probably due to the fact that the bound is derived by bounding the *square* of the determinant of an order-s principal submatrix of C. Therefore, a judicious selection of γ is particularly relevant to obtain a good bound.

In the following, we discuss how we select the scaling parameter γ in linx_γ , aiming at the best possible bound.

We define $\psi := \log(\gamma)$ and formulate the problem

$$\min_{\psi} \{h(\psi)\}, \tag{3.11}$$

where

$$h(\psi) := f(\exp(\psi); x^*),$$

and where $x^* := x^*(\psi)$ is a maximizer of linx_γ , with $\gamma(=\exp(\psi))$ fixed.

In Theorem 3.3.13, we show that the objective function of (3.11) is convex in the logarithm of the scaling parameter, which is particularly relevant for the computation of the optimal scaled linx bound.

Theorem 3.3.13 The function h is convex on $\mathbb{R}$.

Proof $h(\psi)$ is the point-wise maximum of $f(\exp(\psi);x)$ over feasible x for linx_γ . That is, $h(\psi) = \max\{f(\exp(\psi);x) \;:\; \mathbf{e}^T x = s;\; Ax \le b;\; 0 \le x \le \mathbf{e}\}$. So, to

show convexity of h, is suffices to show that $f(\exp(\psi); x)$ is convex in ψ, for all fixed x in the feasible set of linx_γ , such that $L(x) \succ 0$ (which holds if and only if $L(\exp(\psi); x) \succ 0$ for all $\psi \in \mathbb{R}$).

Using the chain rule and Lemma A.4.13, we have

$$\begin{aligned}\frac{\partial f(\gamma; x)}{\partial \gamma} &= L(\gamma; x)^{-1} \bullet C \operatorname{Diag}(x) C - s/\gamma \\ &= \frac{1}{\gamma}\left(L(\gamma; x)^{-1} \bullet \left(L(\gamma; x) - I + \operatorname{Diag}(x)\right) - s\right) \\ &= \frac{1}{\gamma}\left(L(\gamma; x)^{-1} \bullet \left(\operatorname{Diag}(x) - I\right) + n - s\right).\end{aligned}$$

Letting $\psi := \log(\gamma)$, by the chain rule we have

$$\frac{\partial f}{\partial \gamma} = \frac{\partial f}{\partial \psi}\frac{d\psi}{d\gamma} = \frac{\partial f}{\partial \psi}\frac{1}{\gamma}.$$

Then,

$$\begin{aligned}\frac{\partial f(\gamma; x)}{\partial \psi} &= \gamma \frac{\partial f(\gamma; x)}{\partial \gamma} \\ &= L(\exp(\psi); x)^{-1} \bullet \left(\operatorname{Diag}(x) - I\right) + n - s.\end{aligned}$$

Next, we calculate

$$\begin{aligned}\frac{\partial^2 f(\gamma; x)}{\partial \gamma^2} &= \frac{\partial}{\partial \gamma}\left(\frac{1}{\gamma}(L(\gamma; x)^{-1} \bullet \left(\operatorname{Diag}(x) - I\right) + n - s)\right) \\ &= -\frac{1}{\gamma^2}\left(L(\gamma; x)^{-1} \bullet \left(\operatorname{Diag}(x) - I\right) + n - s\right) \\ &\quad + \frac{1}{\gamma}(\mathbf{e} - x)^{\mathsf{T}} \operatorname{diag}\left(L(\gamma; x)^{-1}\left(C \operatorname{Diag}(x) C\right) L(\gamma; x)^{-1}\right).\end{aligned}$$

Then,

$$\begin{aligned}\gamma^2 \frac{\partial^2 f(\gamma; x)}{\partial \gamma^2} &= -\left(L(\gamma; x)^{-1} \bullet \left(\operatorname{Diag}(x) - I\right) + n - s\right) \\ &\quad + \gamma(\mathbf{e} - x)^{\mathsf{T}} \operatorname{diag}\left(L(\gamma; x)^{-1}\left(C \operatorname{Diag}(x) C\right) L(\gamma; x)^{-1}\right).\end{aligned}$$

Again taking $\psi := \log \gamma$ and using the chain rule we have

$$\frac{\partial^2 f}{\partial \psi^2} = \frac{\partial g}{\partial \psi} = \gamma \frac{\partial g}{\partial \gamma} = \gamma\left(\frac{\partial f}{\partial \gamma} + \gamma \frac{\partial^2 f}{\partial \gamma^2}\right) = \gamma \frac{\partial f}{\partial \gamma} + \gamma^2 \frac{\partial^2 f}{\partial \gamma^2}.$$

Finally, we have

$$\begin{aligned}\frac{\partial^2 f(\gamma;x)}{\partial\psi^2} &= L(\exp(\psi);x)^{-1} \bullet \big(\operatorname{Diag}(x) - I\big) + n - s \\ &\quad - \Big(L(\exp(\psi);x)^{-1} \bullet \big(\operatorname{Diag}(x) - I\big) + n - s\Big) \\ &\quad + \exp(\psi)(\mathbf{e}-x)^{\mathsf{T}} \operatorname{diag}\Big(L(\exp(\psi);x)^{-1}\big(C\operatorname{Diag}(x)C\big)L(\exp(\psi);x)^{-1}\Big) \\ &= \exp(\psi)(\mathbf{e}-x)^{\mathsf{T}} \operatorname{diag}\Big(L(\exp(\psi);x)^{-1}\big(C\operatorname{Diag}(x)C\big)L(\exp(\psi);x)^{-1}\Big).\end{aligned}$$

It remains to demonstrate that this last expression is nonnegative. We have $C \geq 0$ and $\operatorname{Diag}(x) \geq 0$. Then, by Lemma A.4.2, we have $C\operatorname{Diag}(x)C \geq 0$ and

$$L(\exp(\psi);x)^{-1}(C\operatorname{Diag}(x)C)L(\exp(\psi);x)^{-1} \geq 0.$$

So we have

$$\frac{\partial^2}{\partial\psi^2} f(\exp(\psi);x) \geq 0,$$

for all $\psi \in \mathbb{R}$, and we can conclude that $f(\exp(\psi);x)$ is convex in ψ. □

Considering Theorem 3.3.13, we apply a quasi-Newton method to solve (3.11). We note that the objective function h is not everywhere differentiable, but BFGS has nice convergence properties even in nonsmooth situations (see [LO13]), and Proposition 3.3.14 shows how to obtain a subgradient for h.

Proposition 3.3.14 For $\psi \in \mathbb{R}$, we have that $g_\psi(\psi) := \frac{\partial}{\partial\psi} f(\exp(\psi);x^*)$ is a subgradient of h at ψ.

Proof From the proof of Theorem 3.3.13, we have that f is convex in ψ for the optimal x^* of linx_γ . Then, for arbitrary $\tilde{\psi}$,

$$\begin{aligned}h(\psi) + (\tilde{\psi}-\psi)g_\psi(\psi) &= f(\exp(\psi);x^*) + (\tilde{\psi}-\psi)g_\psi(\psi) \\ &\leq f(\exp(\tilde{\psi});x^*) \leq h(\tilde{\psi}),\end{aligned}$$

where the second inequality holds because x^* need not be an optimal solution of linx_γ when $\gamma = \exp(\tilde{\psi})$. □

We can use the subgradient presented in Proposition 3.3.14 to replace the gradient in a quasi-Newton method that we could apply to select the scaling parameter γ. However, we should note that as f is continuously differentiable in $\psi(= \log(\gamma))$, then h is differentiable at every ψ for which linx_γ has a *unique* optimum (see [Fia83, Corollary 3.4.2] or [FI90, Theorem 4.1]). In this case, the subgradient of h is the gradient. In Proposition 3.3.7, we have presented conditions over which we can guarantee that linx has unique optimal solution. From the result, it is straightforward

to see that for each given order-n covariance matrix C, there are at most n values of ψ where h is not differentiable. This result is stated in the next corollary.

Corollary 3.3.15 If $z_{\text{linx}} > -\infty$, then h is differentiable at every $\psi \in \mathbb{R}$ such that $\psi \neq \log(1/\lambda_i(C \circ C))$, for all $i \in N$.

3.3.6 The complementary problem of linx_γ

For $C \in \mathbb{S}^n_{++}$, we can define the complementary problem for linx_γ as

$$(\overline{\text{linx}}_\gamma) \qquad z_{\overline{\text{linx}}_\gamma} := \max \left\{ \bar{f}(\bar{\gamma}; y) \ : \ \mathbf{e}^\mathsf{T} y = \bar{s},\ \bar{A} y \leq \bar{b},\ 0 \leq y \leq \mathbf{e} \right\},$$

where $\bar{s} := n - s$, $\bar{A} := -A$, $\bar{b} := b - A\mathbf{e}$, and $\bar{f}(\bar{\gamma}; y) := \operatorname{ldet} C + \frac{1}{2}\left(\operatorname{ldet} \bar{L}(y) - \bar{s} \log \bar{\gamma}\right)$, with $\bar{L}(y) := \bar{\gamma} C^{-1} \operatorname{Diag}(y) C^{-1} + \operatorname{Diag}(\mathbf{e} - y)$ and $\bar{\gamma} := 1/\gamma$. $\overline{\text{linx}}_\gamma$ is the $\text{linx}_{\bar{\gamma}}$ relaxation of the complementary problem for CMESP (see §1.5), with the constant $\operatorname{ldet} C$ added to the objective function.

We can easily verify that if $\hat{y}$ is a feasible solution of $\overline{\text{linx}}_\gamma$, then $\mathbf{e} - \hat{y}$ is a feasible solution of linx_γ. So, considering Proposition 1.5.3, we conclude that if y^* is an optimal solution to $\overline{\text{linx}}_\gamma$, then $\bar{f}(\bar{\gamma}; y^*)$ is an upper bound for CMESP.

Nevertheless, in the next theorem, we see that for linx, the bound based on the complementary problem is identical to the bound based on the original problem.

Proposition 3.3.16 (see [Ans20] for the case of $\gamma = 1$). For every $C \in \mathbb{S}^n_{++}$, s, A, b, and $\gamma > 0$, we have $z_{\overline{\text{linx}}_\gamma} = z_{\text{linx}_\gamma}$.

Proof It is easy to check that the involution $x \mapsto \mathbf{e} - x$ is a bijection between the feasible regions of linx and $\overline{\text{linx}}_\gamma$. So it suffices to demonstrate that $\bar{f}(\bar{\gamma}; y) = f(\gamma, x)$, for $y = \mathbf{e} - x$, where $f(\gamma; x) := \frac{1}{2}\Big(\operatorname{ldet} L(\gamma; x) - s \log(\gamma)\Big)$, and $L(\gamma; x) := \gamma C \operatorname{Diag}(x) C + \operatorname{Diag}(\mathbf{e} - x)$. We have

$$
\begin{aligned}
\bar{f}(\bar{\gamma}; y) &= \frac{1}{2}\left(\operatorname{ldet}\left(\bar{\gamma} C^{-1}\operatorname{Diag}(y) C^{-1} + \operatorname{Diag}(\mathbf{e} - y)\right) - \bar{s}\log\bar{\gamma}\right) + \operatorname{ldet} C \\
&= \frac{1}{2}\left(\operatorname{ldet}\left(\frac{1}{\gamma} C^{-1}\operatorname{Diag}(\mathbf{e} - x) C^{-1} + \operatorname{Diag}(x)\right) - (n - s)\log\frac{1}{\gamma}\right) + \operatorname{ldet} C \\
&= \frac{1}{2}\left(\operatorname{ldet}\left(\frac{1}{\gamma} C^{-1}\operatorname{Diag}(\mathbf{e} - x) C^{-1} + \operatorname{Diag}(x)\right) - (n - s)\log\frac{1}{\gamma}\right) \\
&\quad - \frac{n}{2}\log\gamma + \operatorname{ldet}(\sqrt{\gamma} C) \\
&= \frac{1}{2}\left(\operatorname{ldet}\left(\frac{1}{\gamma} C^{-1}\operatorname{Diag}(\mathbf{e} - x) C^{-1} + \operatorname{Diag}(x)\right) - s\log\gamma\right) + \operatorname{ldet}(\sqrt{\gamma} C) \\
&= \frac{1}{2}\left(\operatorname{ldet}\left(\operatorname{Diag}(\mathbf{e} - x) + \gamma C\operatorname{Diag}(x) C\right) - s\log\gamma\right) \\
&= f(\gamma; x).
\end{aligned}
$$

□

3.4 Factorization bound

In this section, we begin by introducing a mild generalization of a non-convex programming bound developed by [Nik15] and [LX20] (also see [Wil98, LL20]).

Suppose that the rank of C is $r \geq s$. Then we factorize $C = FF^{\mathsf{T}}$, with $F \in \mathbb{R}^{n\times k}$, for some k satisfying $r \leq k \leq n$. We note that this could be a Cholesky-type factorization (i.e., $k := r$ and F lower triangular), as in [Nik15] and [LX20]. But it could alternatively be derived from a spectral decomposition of C; that is, $C = \sum_{i=1}^{r} \lambda_i v_i v_i^{\mathsf{T}}$, where we put $\sqrt{\lambda_i} v_i$ as column i of F, $i = 1, \ldots, k := r$. Another very useful possibility is to let $k := n$, and choose F to be the matrix square root, $C^{1/2}$, which is always symmetric.

Next, for $x \in [0, 1]^n$, we define

$$
f(x) := \sum_{\ell=1}^{s} \log\left(\lambda_\ell(F(x))\right), \text{ where}
$$

$$
F(x) := \sum_{i\in N} F_{i\cdot}^{\mathsf{T}} F_{i\cdot}\, x_i \;= F^{\mathsf{T}}\operatorname{Diag}(x) F.
$$

(Fact) $\quad z_{\text{Fact}}(C, s, A, b; F) := \max\left\{f(x) \;:\; \mathbf{e}^{\mathsf{T}} x = s,\; Ax \leq b,\; 0 \leq x \leq \mathbf{e}\right\},$

We are justified in using maximum rather than supremum by employing Proposition A.1.4. For that, we take $Q := \{x \in \mathbb{R}^n \ : \ \mathbf{e}^\mathsf{T} x = s;\ Ax \leq b;\ 0 \leq x \leq \mathbf{e}\}$ and $\mathcal{K} := \{x \in \mathbb{R}^n \ : \ F(x) \in \mathbb{S}_+^{k,s}\}$. Hence $\bar{\mathcal{K}} := \{x \in \mathbb{R}^n \ : \ F(x) \in \mathbb{S}_+^k\}$.

Next, we see that Fact gives an upper bound for MESP.

Theorem 3.4.1

$$z(C, s, A, b) \leq z_{\text{Fact}}(C, s, A, b; F).$$

Proof It suffices to demonstrate that for any feasible solution S of CMESP, we have $\det C[S, S] = f(x(S))$.

For $S \subset N$ with $|S| = s$ and $\det C[S, S] > 0$, we have

$$F(x(S)) = \sum_{i \in S} F_{i\cdot}^\mathsf{T} F_{i\cdot} = F[S, \cdot]^\mathsf{T} F[S, \cdot] \in \mathbb{S}_+^k \ .$$

Also, we have

$$C[S, S] = F[S, \cdot] F[S, \cdot]^\mathsf{T} \in \mathbb{S}_{++}^s \ .$$

Now, we observe that the nonzero eigenvalues of $F[S, \cdot]^\mathsf{T} F[S, \cdot]$ and $F[S, \cdot] F[S, \cdot]^\mathsf{T}$ are identical. Finally, the rank of these matrices is exactly s, and so the s largest eigenvalues of these two matrices are positive. The result follows. □

From the proof, we can see that replacing $0 \leq x \leq \mathbf{e}$ with $x \in \{0, 1\}^n$ in Fact, we get an exact MINLO formulation for CMESP. But Fact is not generally a convex program, so we cannot make direct use of such a formulation and Theorem 3.4.1. In the next section, using Lagrangian duality, we get past this difficulty.

3.4.1 The Lagrangian dual of Fact

Considering Lagrangian duality, we will obtain an upper bound for z_{Fact} . We first re-cast Fact as

$$\max \left\{ \sum_{\ell=1}^{s} \log\left(\lambda_\ell\left(W\right)\right) \ : \ F(x) = W,\ \mathbf{e}^\mathsf{T} x = s;\ Ax \leq b;\ 0 \leq x \leq \mathbf{e} \right\}.$$

and consider the Lagrangian function

$$\mathcal{L}(W, x, \Theta, \upsilon, \nu, \pi, \tau) :=$$

$$\sum_{\ell=1}^{s} \log\left(\lambda_\ell\left(W\right)\right) + \Theta \bullet (F(x) - W) + \upsilon^\mathsf{T} x + \nu^\mathsf{T}(\mathbf{e} - x) + \pi^\mathsf{T}(b - Ax) + \tau(s - \mathbf{e}^\mathsf{T} x),$$

with $\operatorname{dom} \mathcal{L} = \mathbb{S}^{k,s}_+ \times \mathbb{R}^n \times \mathbb{S}^k \times \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}$, where $\mathbb{S}^{k,s}_+$ denotes the convex set of $k \times k$ positive semidefinite matrices with rank at least s (see Proposition A.4.1).

The corresponding dual function is

$$\mathcal{L}^*(\Theta, \upsilon, \nu, \pi, \tau) := \sup_{W \in \mathbb{S}^{k,s}_+,\ x} \mathcal{L}(W, x, \Theta, \upsilon, \nu, \pi, \tau),$$

and the corresponding Lagrangian dual problem is

$$z_{\text{DFact}}(C, s, A, b; F) := \inf\{\mathcal{L}^*(\Theta, \upsilon, \nu, \pi, \tau) \ :\ \upsilon \geq 0,\ \nu \geq 0,\ \pi \geq 0\}.$$

We call $z_{\text{DFact}} := z_{\text{DFact}}(C, s, A, b; F)$ the **factorization bound**. We note that

$$\begin{aligned} &\sup_{W \in \mathbb{S}^{k,s}_+,\ x} \left\{ \sum_{\ell=1}^{s} \log\left(\lambda_\ell\left(W\right)\right) + \Theta \bullet \left(F(x) - W\right) \right. \\ &\qquad \left. + \upsilon^{\mathsf{T}} x + \nu^{\mathsf{T}}(\mathbf{e} - x) + \pi^{\mathsf{T}}(b - Ax) + \tau(s - \mathbf{e}^{\mathsf{T}} x) \right\} \\ &= \sup_{W \in \mathbb{S}^{k,s}_+} \left\{ \sum_{\ell=1}^{s} \log\left(\lambda_\ell\left(W\right)\right) - \Theta \bullet W \right\} \qquad (3.12) \\ &\qquad + \sup_{x} \left\{ \Theta \bullet F(x) + \upsilon^{\mathsf{T}} x - \nu^{\mathsf{T}} x - \pi^{\mathsf{T}} A x - \tau \mathbf{e}^{\mathsf{T}} x + \nu^{\mathsf{T}} \mathbf{e} + \pi^{\mathsf{T}} b + \tau s \right\}. \qquad (3.13) \end{aligned}$$

In Theorems 3.4.3 and 3.4.4 we analytically characterize the suprema in (3.12) and (3.13). The next lemma will be used in the proof of Theorem 3.4.3.

Lemma 3.4.2 (see [LX20, Lemma 11]) Given $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_k$ and $0 < \hat{\beta}_1 \leq \hat{\beta}_2 \leq \cdots \leq \hat{\beta}_k$, we have that $\hat{\theta}_\ell = \hat{\beta}_\ell$, for $\ell = 1, \ldots, k$, solves

$$\min_{\substack{\theta \in \mathbb{R}^k_+ \\ \theta_1 \leq \cdots \leq \theta_k}} \left\{ \sum_{\ell=1}^{k} \hat{\lambda}_\ell \theta_\ell \ :\ \sum_{\ell=t+1}^{k} \theta_\ell \leq \sum_{\ell=t+1}^{k} \hat{\beta}_\ell\ ,\ t = 1, \ldots, k-1, \sum_{\ell=1}^{k} \theta_\ell = \sum_{\ell=1}^{k} \hat{\beta}_\ell \right\}. \qquad (3.14)$$

Proof We prove the result by induction on k. For the base case, $k = 1$, (3.14) has a unique solution $\hat{\theta}_1 = \hat{\beta}_1$, and the result holds. Suppose now that the result holds for all $k \leq \hat{k}$, where $\hat{k} \geq 1$. Let $k := \hat{k} + 1$, and let $\hat{\theta}$ be an optimal solution to the bounded linear program (3.14), that is an extreme point of its feasible set,

$$\begin{aligned} \mathcal{P} := \Big\{ \theta \in \mathbb{R}^k \ :\ &\sum_{\ell=t+1}^{k} \theta_\ell \leq \sum_{\ell=t+1}^{k} \hat{\beta}_\ell\ ,\ t = 1, \ldots, k-1, \\ &\sum_{\ell=1}^{k} \theta_\ell = \sum_{\ell=1}^{k} \hat{\beta}_\ell\ ,\ 0 \leq \theta_1 \leq \cdots \leq \theta_k \Big\}. \end{aligned}$$

First, we assume that there exists an integer $\hat{t} \in [1, k-1]$, such that $\sum_{\ell=1}^{\hat{t}} \hat{\theta}_\ell = \sum_{\ell=1}^{\hat{t}} \hat{\beta}_\ell$, and we define

$$\hat{z}_1 := \min_{\substack{\theta \in \mathbb{R}^{\hat{t}}_+ \\ \theta_1 \le \cdots \le \theta_{\hat{t}}}} \left\{ \sum_{\ell=1}^{\hat{t}} \hat{\lambda}_\ell \theta_\ell : \sum_{\ell=t+1}^{\hat{t}} \theta_\ell \le \sum_{\ell=t+1}^{\hat{t}} \hat{\beta}_\ell \ , \ 1 \le t < \hat{t}, \sum_{\ell=1}^{\hat{t}} \theta_\ell = \sum_{\ell=1}^{\hat{t}} \hat{\beta}_\ell \right\}, \tag{3.15}$$

and

$$\hat{z}_2 := \min_{\substack{\theta \in \mathbb{R}^{k-\hat{t}}_+ \\ \theta_{\hat{t}+1} \le \cdots \le \theta_k}} \left\{ \sum_{\ell=\hat{t}+1}^{k} \hat{\lambda}_\ell \theta_\ell : \sum_{\ell=t+1}^{k} \theta_\ell \le \sum_{\ell=t+1}^{k} \hat{\beta}_\ell \ , \ \hat{t} < t < k, \sum_{\ell=\hat{t}+1}^{k} \theta_\ell = \sum_{\ell=\hat{t}+1}^{k} \hat{\beta}_\ell \right\}. \tag{3.16}$$

We can verify that $\hat{\theta}_1, \ldots, \hat{\theta}_{\hat{t}}$ is a feasible solution for (3.15) and $\hat{\theta}_{\hat{t}+1}, \ldots, \hat{\theta}_k$ is a feasible solution for (3.16). Therefore, (3.14) is lower bounded by

$$\hat{z}_1 + \hat{z}_2 = \sum_{\ell=1}^{\hat{t}} \hat{\lambda}_\ell \hat{\beta}_\ell + \sum_{\ell=\hat{t}+1}^{k} \hat{\lambda}_\ell \hat{\beta}_\ell \ ,$$

where the last equality holds because of the induction hypothesis. Now, note that this lower bound is achieved at the feasible solution $\hat{\theta}_\ell = \hat{\beta}_\ell$, for $\ell = 1, \ldots, k$, of (3.14), proving that this is also an optimal solution for (3.14).

Next, we assume instead that $\sum_{\ell=1}^{\hat{t}} \hat{\theta}_\ell \neq \sum_{\ell=1}^{\hat{t}} \hat{\beta}_\ell$, for all $t = 1, \ldots, k-1$. In this case, as $\hat{\theta}$ is an extreme point of $\mathcal{P}$, at least $k-1$ of the remaining inequalities in $\mathcal{P}$, namely $\{0 \le \theta_1 \le \cdots \le \theta_k\}$, must hold as equations at $\hat{\theta}$. Clearly $\hat{\theta}_1 > 0$, otherwise, from the equality constraint in $\mathcal{P}$, we would have $\sum_{\ell=2}^{k} \hat{\theta}_\ell = \sum_{\ell=1}^{k} \hat{\beta}_\ell > \sum_{\ell=2}^{k} \hat{\beta}_\ell$, contradicting its first inequality. Therefore, we must have $\hat{\theta}_1 = \cdots = \hat{\theta}_k$, which leads to the optimal solution $\hat{\theta}_\ell = (\sum_{i=1}^{k} \hat{\beta}_i)/k$, for $\ell = 1, \ldots, k$, with objective value $\hat{z} := (\sum_{\ell,i=1}^{k} \hat{\lambda}_\ell \hat{\beta}_i)/k$. Finally, we conclude from Theorem A.1.3 (the Hardy-Littlewood-Pólya rearrangement inequalities), that $\hat{z} \ge \sum_{\ell=1}^{k} \hat{\lambda}_\ell \hat{\beta}_\ell$, and consequently, $\hat{\theta}_\ell = \hat{\beta}_\ell$, for $\ell = 1, \ldots, k$, is also an optimal solution to (3.14).

Therefore, in both cases, the result of the lemma also holds for $k = \hat{k} + 1$. □

The following lemma and proof is from [LX20, Lemma 1], but we include some additional details.

Theorem 3.4.3 (see [LX20, Lemma 1]) For $\Theta \in \mathbb{S}^k$, we have

$$\sup_{W \in \mathbb{S}^{k,s}_+} \left(\sum_{\ell=1}^{s} \log\left(\lambda_\ell\left(W\right)\right) - W \bullet \Theta \right) = \begin{cases} -s - \sum_{\ell=k-s+1}^{k} \log\left(\lambda_\ell\left(\Theta\right)\right), & \text{if } \Theta \succ 0; \\ +\infty, & \text{otherwise.} \end{cases} \tag{3.17}$$

Proof In the proof, we write $\hat{\Theta}$ for Θ to emphasize that it is fixed. We first assume that $\hat{\Theta} \succ 0$. In this case, it suffices to show that there is an optimal solution $\hat{W}$ to (3.17), such that

$$\sum_{\ell=1}^{s} \log\left(\lambda_\ell\left(\hat{W}\right)\right) - \hat{W} \bullet \hat{\Theta} = -s - \sum_{\ell=k-s+1}^{k} \log(\lambda_\ell(\hat{\Theta})). \tag{3.18}$$

We consider the spectral decomposition of $\hat{\Theta}$,

$$\hat{\Theta} = \sum_{\ell=1}^{k} \hat{\beta}_\ell \hat{v}_\ell \hat{v}_\ell^{\mathsf{T}},$$

where we assume that $0 < \hat{\beta}_1 \leq \hat{\beta}_2 \leq \cdots \leq \hat{\beta}_k$.

We define

$$\hat{W} := \sum_{\ell=1}^{k} \hat{\lambda}_\ell \hat{v}_\ell \hat{v}_\ell^{\mathsf{T}}, \tag{3.19}$$

where

$$\hat{\lambda}_\ell := \begin{cases} 1/\hat{\beta}_\ell, & \text{for } 1 \leq \ell \leq s; \\ 0, & \text{for } s+1 \leq \ell \leq k. \end{cases}$$

Note that $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_k$ and that (3.18) holds for $\hat{W}$.

It remains to show that $\hat{W}$ is an optimal solution to (3.17). Let us consider the spectral decomposition of our matrix variable W as

$$W = \sum_{\ell=1}^{k} \lambda_\ell(W) u_\ell(W) u_\ell(W)^{\mathsf{T}},$$

where we assume that $\lambda_1(W) \geq \lambda_2(W) \geq \cdots \geq \lambda_k(W)$.

In the following we consider $U(W) := (u_1(W), \ldots, u_k(W))$ and $\lambda(W) := (\lambda_1(W), \ldots, \lambda_k(W))^{\mathsf{T}}$. So, we have

$$\begin{aligned} W \bullet \hat{\Theta} &= (U(W)\operatorname{Diag}(\lambda(W))U(W)^{\mathsf{T}}) \bullet \hat{\Theta} = \operatorname{Diag}(\lambda(W)) \bullet (U(W)^{\mathsf{T}}\hat{\Theta}U(W)) \\ &= \operatorname{Diag}(\lambda(W)) \bullet \operatorname{Diag}(\theta) = \sum_{\ell=1}^{k} \lambda_\ell(W)\theta_\ell, \end{aligned}$$

where $\theta := \operatorname{diag}(U(W)^{\mathsf{T}}\hat{\Theta}U(W))$. Then, defining the vector variable $\lambda \in \mathbb{R}^k$, and the matrix variable $U \in \mathbb{R}^{k\times k}$, we replace W with $U\operatorname{Diag}(\lambda)U^{\mathsf{T}}$ in (3.17), reformulating it as

$$\sup_{\substack{\lambda\in\mathbb{R}^k_+ \\ \lambda_1\geq\cdots\geq\lambda_k}} \left\{ \sum_{\ell=1}^{s} \log(\lambda_\ell) - \min_{U,\theta\in\mathbb{R}^k_+} \left\{ \sum_{\ell=1}^{k} \lambda_\ell\theta_\ell : \theta = \operatorname{diag}(U^{\mathsf{T}}\hat{\Theta}U), U^{\mathsf{T}}U = I \right\} \right\}. \tag{3.20}$$

Note that we are justified in writing minimum rather than infimum in the inner problem in (3.20), because the infimum is over a compact set.

Finally, considering that U can be selected as any permutation matrix and $\lambda_1 \geq \cdots \geq \lambda_k$ in the inner problem in (3.20), we conclude from Theorem A.1.3 (the Hardy-Littlewood-Pólya rearrangement inequalities) that $\theta_1 \leq \cdots \leq \theta_k$ holds at its optimal solution. Also considering that the eigenvalues of $U^{\mathsf{T}}\hat{\Theta}U$ and $\hat{\Theta}$ are the same, we see by Theorem A.3.4 (Schur's Theorem: majorization by eigenvalues) that the following problem is a relaxation of the inner problem in (3.20).

$$\min_{\substack{\theta \in \mathbb{R}^k_+ \\ \theta_1 \leq \cdots \leq \theta_k}} \left\{ \sum_{\ell=1}^{k} \lambda_\ell \theta_\ell : \sum_{\ell=t+1}^{k} \theta_\ell \leq \sum_{\ell=t+1}^{k} \hat{\beta}_\ell \, , 1 \leq t < k, \ \sum_{\ell=1}^{k} \theta_\ell = \sum_{\ell=1}^{k} \hat{\beta}_\ell \right\}.$$

Now, using Lemma 3.4.2, we conclude that $\sum_{\ell=1}^{k} \lambda_\ell \hat{\beta}_\ell$ is a lower bound for the optimal value of the inner minimization problem in (3.20). In fact, this lower bound is achieved by selecting $\theta_\ell = \hat{\beta}_\ell$, for $\ell = 1, \ldots, k$, and $U = (\hat{v}_1, \ldots, \hat{v}_k)$, which is then, an optimal solution for the inner problem. Therefore, we may finally rewrite (3.20) as

$$\sup_{\substack{\lambda \in \mathbb{R}^k_+ \\ \lambda_1 \geq \cdots \geq \lambda_k}} \left\{ \sum_{\ell=1}^{s} \log(\lambda_\ell) - \sum_{\ell=1}^{k} \lambda_\ell \hat{\beta}_\ell \right\}. \tag{3.21}$$

The Lagrangian function associated to (3.21) is

$$\mathcal{L}(\lambda, \mu) := \sum_{\ell=1}^{s} \log(\lambda_\ell) - \sum_{\ell=1}^{k} \lambda_\ell \hat{\beta}_\ell + \sum_{\ell=1}^{k} \mu_\ell \lambda_\ell \, ,$$

with $\operatorname{dom} \mathcal{L} = \mathbb{R}^{k,s}_+ \times \mathbb{R}^k$, where $\mathbb{R}^{k,s}_+$ denotes the set of vectors in $\mathbb{R}^k_+$ with at least s positive components. We note that the Lagrangian function is concave in λ, so the supremum in (3.21) occurs where its gradient with respect to λ is equal to zero. Then, the optimality conditions for (3.21) are given by

$$\begin{aligned}
&\frac{1}{\lambda_\ell} - \hat{\beta}_\ell + \mu_\ell = 0, && \text{for } \ell = 1, \ldots, s, \\
&-\hat{\beta}_\ell + \mu_\ell = 0, && \text{for } \ell = s+1, \ldots, k, \\
&\mu_\ell \lambda_\ell = 0, && \text{for } \ell = 1, \ldots, k, \\
&\mu_\ell \geq 0, && \text{for } \ell = 1, \ldots, k, \\
&\lambda_1 \geq \cdots \geq \lambda_k \geq 0. &&
\end{aligned}$$

We see that the optimality conditions hold for $\lambda_\ell = \hat{\lambda}_\ell$, for $\ell = 1, \ldots, k$, $\mu_\ell = 0$, for $\ell = 1, \ldots, s$ and $\mu_\ell = \hat{\beta}_\ell$, for $\ell = s+1, \ldots, k$. Therefore, $\hat{\lambda}_\ell$, for $\ell = 1, \ldots, k$, is an optimal solution to (3.21), and equivalently, to (3.20). So, we finally see that $\hat{W}$ is an optimal solution to (3.17).

Next, we assume instead that $\hat{\Theta} \not\succ 0$. Then, we consider the spectral decomposition $\hat{\Theta} := \sum_{\ell=1}^{k} \hat{\lambda}_\ell \hat{u}_\ell \hat{u}_\ell^\top$ with $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_p > 0 \geq \hat{\lambda}_{p+1} \geq \hat{\lambda}_{p+2} \geq \cdots \geq \lambda_k$, where we have $p < k$. For a given $t \in \mathbb{R}$, we define

$$\hat{W} := \sum_{\ell=1}^{s-1} (\hat{u}_\ell \hat{u}_\ell^\top) + t \hat{u}_k \hat{u}_k^\top .$$

We can verify that, for any $t > 0$, $\hat{W}$ is a feasible solution for (3.12) with objective value

$$\log(t) - \sum_{\ell=1}^{s-1} \hat{\lambda}_\ell + t|\hat{\lambda}_k|. \tag{3.22}$$

Clearly, (3.22) goes to $+\infty$, as $t \to +\infty$, which completes the proof. □

Theorem 3.4.4 For $(\Theta, \upsilon, \nu, \pi, \tau) \in \mathbb{S}^k \times \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}$, we have

$$\sup_x \left(\Theta \bullet F(x) + \upsilon^\top x - \nu^\top x - \pi^\top A x - \tau \mathbf{e}^\top x + \nu^\top \mathbf{e} + \pi^\top b + \tau s \right)$$
$$= \begin{cases} \nu^\top \mathbf{e} + \pi^\top b + \tau s, & \text{if } F_{i\cdot} \Theta F_{i\cdot}^\top + \upsilon_i - \nu_i - (A_{\cdot i})^\top \pi - \tau = 0, \quad \text{for } i \in N; \\ +\infty, & \text{otherwise.} \end{cases}$$

Proof We have

$$\sup_x \left(\Theta \bullet F(x) + \upsilon^\top x - \nu^\top x - \pi^\top A x - \tau \mathbf{e}^\top x + \nu^\top \mathbf{e} + \pi^\top b + \tau s \right)$$
$$= \sup_x \left(\sum_{i \in N} \left(F_{i\cdot} \Theta F_{i\cdot}^\top + \upsilon_i - \nu_i - (A_{\cdot i})^\top \pi - \tau \right) x_i + \nu^\top \mathbf{e} + \pi^\top b + \tau s \right).$$

Then, the result follows from the fact that a linear function is bounded above only when it is identically zero. □

Considering Theorems 3.4.3 and 3.4.4, we see that the Lagrangian dual problem of Fact is equivalent to

(DFact)
$$
\begin{aligned}
&z_{\text{DFact}}(C,s,A,b;F) = \\
&\min \; -\sum_{\ell=k-s+1}^{k} \log\left(\lambda_\ell\left(\Theta\right)\right) + \nu^{\mathsf{T}}\mathbf{e} + \pi^{\mathsf{T}} b + \tau s - s \\
&\text{subject to:} \\
&\operatorname{diag}(F\Theta F^{\mathsf{T}}) + \upsilon - \nu - A^{\mathsf{T}}\pi - \tau\mathbf{e} = 0, \\
&\Theta \succ 0,\ \upsilon \geq 0,\ \nu \geq 0,\ \pi \geq 0.
\end{aligned}
$$

Theorem 3.4.5 (see [Nik15, Lemma 11]) DFact is a convex optimization problem.

Proof From Proposition A.4.24, we have that $-\sum_{\ell=k-s+1}^{k} \log\left(\lambda_\ell\left(\Theta\right)\right)$ is convex and continuous in $\Theta \succ 0$. □

Because Fact is not generally a convex program, we will not generally have strong duality between Fact and DFact. We are however justified in using minimum rather than infimum by employing Proposition A.1.4, noting that the objective function goes to $+\infty$ as Θ approaches the boundary of $\mathbb{S}^k_+$.

3.4.2 Duality for DFact

Consider the Lagrangian function corresponding to DFact, after eliminating the slack variable υ,

$$
\begin{aligned}
\mathcal{L}(\Theta,\nu,\pi,\tau,x,y,t) := \; & -\sum_{\ell=k-s+1}^{k} \log\left(\lambda_\ell\left(\Theta\right)\right) + \nu^{\mathsf{T}}\mathbf{e} + \pi^{\mathsf{T}} b + \tau s - s \\
& + x^{\mathsf{T}}\left(\operatorname{diag}(F\Theta F^{\mathsf{T}}) - \nu - A^{\mathsf{T}}\pi - \tau\mathbf{e}\right) - y^{\mathsf{T}}\nu - t^{\mathsf{T}}\pi,
\end{aligned}
$$

with $\operatorname{dom}\mathcal{L} = \mathbb{S}^k_{++} \times \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^m$.

The corresponding dual function is

$$
\mathcal{L}^*(x,y,t) := \inf_{\Theta \in \mathbb{S}^k_{++},\nu,\pi,\tau} \mathcal{L}(\Theta,\nu,\pi,\tau,x,y,t), \tag{3.23}
$$

and the Lagrangian dual problem of DFact is

$$
z_{\text{DDFact}}(C,s,A,b;F) := \max\{\mathcal{L}^*(x,y,t) \;:\; x \geq 0,\ y \geq 0,\ t \geq 0\}.
$$

Because DFact has a strictly feasible solution (e.g., given by $(\hat{\Theta} := I, \hat{\upsilon} := 0, \hat{\nu} := \operatorname{diag}(FF^{\mathsf{T}}) = \operatorname{diag}(C), \hat{\pi} := 0, \hat{\tau} := 0)$), we are justified to use minimum in the

formulation of the Lagrangian dual problem, rather than infimum, because then Slater's condition holds for DFact, and so the optimal value of the Lagrangian dual problem is attained.

We note that

$$\inf_{\Theta\in\,\mathbb{S}^k_{++},\nu,\pi,\tau} \mathcal{L}(\Theta,\nu,\pi,\tau,x,y,t) =$$

$$\inf_{\Theta\in\,\mathbb{S}^k_{++}} \left\{ -\sum_{\ell=k-s+1}^{k} \log\left(\lambda_\ell\left(\Theta\right)\right) + x^{\mathsf{T}}\,\mathrm{diag}(F\Theta F^{\mathsf{T}}) - s \right\} \tag{3.24}$$

$$+ \inf_{\nu,\pi,\tau} \left\{ y^{\mathsf{T}}(\mathbf{e}-x-y) + \pi^{\mathsf{T}}(b-Ax-t) + \tau(s-e^{\mathsf{T}}x) \right\}. \tag{3.25}$$

In Theorems 3.4.7 and 3.4.8 we analytically characterize the infima in (3.24) and (3.25). The next lemma is used in the proof of Theorem 3.4.7.

Lemma 3.4.6 (see [Nik15, Lemma 13]) Let $\lambda \in \mathbb{R}^k_+$ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k$ and let $0 < s \leq k$. There exists a unique integer ι, with $0 \leq \iota < s$, such that

$$\lambda_\iota > \frac{1}{s-\iota}\sum_{\ell=\iota+1}^{k} \lambda_\ell \geq \lambda_{\iota+1}\,, \tag{3.26}$$

with the convention $\lambda_0 = +\infty$.

Proof First, we prove the existence of ι. Let us denote the strict and the non-strict inequalities in (3.26), respectively by (3.26.a) and (3.26.b). Note that

1. (3.26.a) holds for $\iota = 0$, because $\lambda_0 = +\infty$,
2. If (3.26.b) does not hold for $\iota = \hat{\iota}$, then (3.26.a) holds for $\iota = \hat{\iota}+1$, for all $\hat{\iota} \in [0, s-1)$,
3. (3.26.b) holds for $\iota = s-1$, because $\sum_{\ell=s}^{k} \lambda_\ell \geq \lambda_s$.

Therefore, if (3.26) does not hold for any $\iota < s-1$, it certainly holds for $\iota = s-1$, completing the proof of existence.

Now, we prove the uniqueness of ι by contradiction. Assume that (3.26) holds for $\iota = i$ and $\iota = j$, with $0 \leq i < j < s$. By the hypothesis, we have $\lambda_{i+1} \geq \lambda_j$; therefore,

$$\begin{aligned}
(s-j)\lambda_j &\leq (s-j)\lambda_{i+1} \\
&= (s-i)\lambda_{i+1} - (j-i)\lambda_{i+1} \\
&\leq \textstyle\sum_{\ell=i+1}^{k} \lambda_\ell - (j-i)\lambda_{i+1} \\
&= \textstyle\sum_{\ell=i+1}^{j} (\lambda_\ell - \lambda_{i+1}) + \sum_{\ell=j+1}^{k} \lambda_\ell \\
&\leq \textstyle\sum_{\ell=j+1}^{k} \lambda_\ell\,,
\end{aligned}$$

which contradicts (3.26.a), for $\iota = j$. □

Suppose that $\lambda \in \mathbb{R}^k_+$, and assume that $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k$. Given integer s with $0 < s \leq k$, let ι be the unique integer defined by Lemma 3.4.6. We define

$$\phi_s(\lambda) := \sum_{\ell=1}^{\iota} \log(\lambda_\ell) + (s-\iota)\log\left(\frac{1}{s-\iota}\sum_{\ell=\iota+1}^{k}\lambda_\ell\right).$$

Next, for $X \in \mathbb{S}^k_+$, we define $\Gamma_s(X) := \phi_s(\lambda(X))$.

The following theorem and proof is from [Nik15, Lemma 16], but we include some additional details.

Theorem 3.4.7 (see [Nik15, Lemma 16]) For $x \in \mathbb{R}^n$, we have

$$\inf_{\Theta \in \mathbb{S}^k_{++}} -\sum_{\ell=k-s+1}^{k} \log(\lambda_\ell(\Theta)) + x^\mathsf{T}\operatorname{diag}(F\Theta F^\mathsf{T}) - s$$
$$= \begin{cases} \Gamma_s(F(x)), & \text{if } F(x) \succeq 0 \text{ and } \operatorname{rank}(F(x)) \geq s; \\ -\infty, & \text{otherwise.} \end{cases}$$

Proof In the proof, we write $\hat{x}$ for x to emphasize that it is fixed. We have

$$\inf_{\Theta \in \mathbb{S}^k_{++}} -\sum_{\ell=k-s+1}^{k} \log(\lambda_\ell(\Theta)) + \hat{x}^\mathsf{T}\operatorname{diag}(F\Theta F^\mathsf{T}) - s$$
$$= \inf_{\Theta \in \mathbb{S}^k_{++}} -\sum_{\ell=k-s+1}^{k} \log(\lambda_\ell(\Theta)) + F(\hat{x}) \bullet \Theta - s, \tag{3.27}$$

where $F(\hat{x}) = \sum_{i=1}^n F_i^\mathsf{T} F_i \cdot \hat{x}_i$ and $\lambda_\ell(\Theta)$ denotes the ℓ-th greatest eigenvalue of Θ.

We first assume that $F(\hat{x}) \succeq 0$ and $\hat{r} := \operatorname{rank}(F(\hat{x})) \geq s$. In this case, it suffices to show that there is an optimal solution $\hat{\Theta}$ to (3.27), such that

$$-\sum_{\ell=k-s+1}^{k} \log(\lambda_\ell(\hat{\Theta})) + F(\hat{x}) \bullet \hat{\Theta} - s = \Gamma_s(F(\hat{x})). \tag{3.28}$$

We consider the spectral decomposition of $F(\hat{x})$,

$$F(\hat{x}) = \sum_{\ell=1}^{k} \hat{\lambda}_\ell \hat{u}_\ell \hat{u}_\ell^\mathsf{T},$$

with $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_{\hat{r}} > \hat{\lambda}_{\hat{r}+1} = \cdots = \hat{\lambda}_k = 0$, and the spectral decomposition of our matrix variable Θ as

$$\Theta = \sum_{\ell=1}^{k} \beta_\ell(\Theta) v_\ell(\Theta) v_\ell(\Theta)^\mathsf{T},$$

where here, we conveniently assume that $\beta_1(\Theta) \le \beta_2(\Theta) \le \cdots \le \beta_k(\Theta)$.

Now, we define

$$\hat{\Theta} := \sum_{\ell=1}^{k} \hat{\beta}_\ell \hat{u}_\ell \hat{u}_\ell^{\mathsf{T}} ,$$

where

$$\hat{\beta}_\ell := \begin{cases} 1/\hat{\lambda}_\ell , & \text{for } 1 \le \ell \le \hat{\iota}; \\ 1/\hat{\delta}, & \text{for } \hat{\iota} < \ell \le \hat{r}; \\ (1+\epsilon)/\hat{\delta}, & \text{for } \hat{r} < \ell \le k, \end{cases}$$

where ϵ is any positive number, and $\hat{\iota}$ is the unique integer defined in Lemma 3.4.6 for $\lambda_\ell = \hat{\lambda}_\ell$, and

$$\hat{\delta} := \frac{1}{s-\hat{\iota}} \sum_{\ell=\hat{\iota}+1}^{k} \hat{\lambda}_\ell . \tag{3.29}$$

Note that $0 < \hat{\beta}_1 \le \hat{\beta}_2 \le \cdots \le \hat{\beta}_k$.

From Lemma 3.4.6, we have that $\hat{\iota} < s$. Then, as $s \le \hat{r}$,

$$\begin{aligned} -\sum_{\ell=1}^{s} \log\left(\hat{\beta}_\ell\right) &= -\sum_{\ell=1}^{\hat{\iota}} \log\left(\frac{1}{\hat{\lambda}_\ell}\right) - \sum_{\ell=\hat{\iota}+1}^{s} \log\left(\frac{1}{\hat{\delta}}\right) \\ &= \sum_{\ell=1}^{\hat{\iota}} \log\left(\hat{\lambda}_\ell\right) + (s-\hat{\iota}) \log(\hat{\delta}) = \Gamma_s(F(\hat{x})), \end{aligned} \tag{3.30}$$

where the last identity follows from the definition of Γ_s and (3.29).

Moreover, considering that $\hat{\lambda}_\ell = 0$, for $\ell > \hat{r}$, from (3.29), we also have

$$F(\hat{x}) \bullet \hat{\Theta} = \sum_{\ell=1}^{\hat{r}} \hat{\lambda}_\ell \hat{\beta}_\ell = \sum_{\ell=1}^{\hat{\iota}} \hat{\lambda}_\ell \frac{1}{\hat{\lambda}_\ell} + \sum_{\ell=\hat{\iota}+1}^{\hat{r}} \hat{\lambda}_\ell \frac{1}{\hat{\delta}} = \hat{\iota} + \frac{1}{\hat{\delta}} \sum_{\ell=\hat{\iota}+1}^{\hat{r}} \hat{\lambda}_\ell = \hat{\iota} + (s-\hat{\iota}) = s,$$

which, together with (3.30), proves of the identity in (3.28).

Therefore, it remains to show that $\hat{\Theta}$ is an optimal solution to (3.27), which we do in the following by showing that the subdifferential of the objective function of (3.27) contains 0 at $\hat{\Theta}$.

The second term in the objective function of (3.27) is differentiable with respect to Θ. In fact,

$$\frac{\mathrm{d}}{\mathrm{d}\Theta} \left(F(\hat{x}) \bullet \Theta\right) = F(\hat{x}).$$

So, we need to show that

$$-F(\hat{x}) \in \partial \Delta_s(\hat{\Theta}),$$

where

$$\Delta_s(\Theta) := -\sum_{\ell=k-s+1}^{k} \log\left(\lambda_\ell\left(\Theta\right)\right) .$$

We recall that $\Theta = \sum_{\ell=1}^{k} \beta_\ell(\Theta) v_\ell(\Theta) v_\ell(\Theta)^\mathsf{T}$, with $\beta_1(\Theta) \leq \beta_2(\Theta) \leq \cdots \leq \beta_k(\Theta)$. Then, if $s \in \mathcal{E} := \{\hat{\iota}+1, \ldots, \hat{r}\}$, where $\mathcal{E}$ is defined so that $\beta_{\hat{\iota}}(\Theta) < \beta_{\hat{\iota}+1}(\Theta) = \cdots = \beta_{\hat{r}}(\Theta) < \beta_{\hat{r}+1}(\Theta)$, we have from Proposition A.4.24 and Theorem A.4.25, that the subdifferential of Δ_s at Θ is given by

$$\partial\Delta_s(\Theta) = \left\{ V \operatorname{Diag}(\lambda) V^\mathsf{T} : \lambda \in \mathcal{U}(\beta(\Theta)),\ V^\mathsf{T} V = I,\ \Theta = V \operatorname{Diag}(\beta(\Theta)) V^\mathsf{T} \right\},$$

where $\mathcal{U}(\beta(\Theta))$ is the convex hull of the subgradients

$$\left(-\frac{1}{\beta_1(\Theta)}, -\frac{1}{\beta_2(\Theta)}, \ldots, -\frac{1}{\beta_{\hat{\iota}}(\Theta)}, g_{\hat{\iota}+1}, \ldots, g_{\hat{r}}, 0, \ldots, 0 \right)^\mathsf{T},$$

with

$$g_i := \begin{cases} -1/\beta_i(\Theta), & \text{for } i \in \mathcal{F}; \\ 0, & \text{for } i \in \mathcal{E} \setminus \mathcal{F}, \end{cases}$$

for all sets $\mathcal{F}$, such that $\mathcal{F} \subset \mathcal{E}$ and $|\mathcal{F}| = s - \hat{\iota}$.

Thus, we may write

$$\begin{aligned} \mathcal{U}(\beta(\Theta)) = \Big\{ & \lambda : \lambda_i = -1/\beta_i(\Theta), \quad 1 \leq i \leq \hat{\iota}, \\ & (\lambda_{\hat{\iota}+1}, \ldots, \lambda_{\hat{r}}) \in \tfrac{-1}{\beta_{\hat{r}}(\Theta)} \operatorname{conv} \left\{ x \in \mathbb{R}^{\hat{r}-\hat{\iota}} : \mathbf{e}^\mathsf{T} x = s - \hat{\iota},\ x \in \{0,1\}^{\hat{r}-\hat{\iota}} \right\}, \\ & \lambda_i = 0, \quad \hat{r}+1 \leq i \leq k \Big\}. \end{aligned}$$

Finally, considering that $-F(\hat{x}) = -\sum_{\ell=1}^{k} \hat{\lambda}_\ell \hat{u}_\ell \hat{u}_\ell^\mathsf{T}$, $\hat{\Theta} = \sum_{\ell=1}^{k} \hat{\beta}_\ell \hat{u}_\ell \hat{u}_\ell^\mathsf{T}$, $\hat{\delta} = \frac{1}{s-\hat{\iota}} \sum_{\ell=\hat{\iota}+1}^{\hat{r}} \hat{\lambda}_\ell$,

$$\begin{aligned} \hat{\lambda} &= \left(\hat{\lambda}_1, \ldots, \hat{\lambda}_{\hat{\iota}}, \hat{\lambda}_{\hat{\iota}+1}, \ldots, \hat{\lambda}_{\hat{r}}, 0, \ldots, 0\right)^\mathsf{T}, \\ \hat{\beta} &= \Big(\frac{1}{\hat{\lambda}_1}, \ldots, \frac{1}{\hat{\lambda}_{\hat{\iota}}}, \underbrace{\frac{1}{\hat{\delta}}, \ldots, \frac{1}{\hat{\delta}}}_{\mathcal{E}}, \frac{1+\epsilon}{\hat{\delta}}, \ldots, \frac{1+\epsilon}{\hat{\delta}} \Big)^\mathsf{T}, \end{aligned}$$

for a given $\epsilon > 0$, and noting that

$$\begin{aligned} & \hat{\delta} \times \operatorname{conv} \left\{ x \in \mathbb{R}^{\hat{r}-\hat{\iota}} : \mathbf{e}^\mathsf{T} x = s - \hat{\iota},\ x \in \{0,1\}^{\hat{r}-\hat{\iota}} \right\} \\ & \quad = \left\{ x \in \mathbb{R}^{\hat{r}-\hat{\iota}} : \mathbf{e}^\mathsf{T} x = (s-\hat{\iota})\hat{\delta},\ 0 \leq x_i \leq \hat{\delta},\ i = 1, \ldots, \hat{r}-\hat{\iota} \right\}, \end{aligned}$$

we can see that $-\hat{\lambda} \in \mathcal{U}(\hat{\beta})$, and, therefore, $-F(\hat{x}) \in \partial\Delta_s(\hat{\Theta})$, which completes the proof for our first case.

Now, we consider the second case, where we have $F(\hat{x}) \not\succeq 0$ or $\operatorname{rank}(F(\hat{x})) < s$. We use the spectral decomposition $F(\hat{x}) = \sum_{\ell=1}^{k} \hat{\lambda}_\ell \hat{u}_\ell \hat{u}_\ell^\mathsf{T}$, and define $\hat{\lambda}_0 := +\infty$. We assume that

$$\hat{\lambda}_0 \geq \hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_p > 0 = \hat{\lambda}_{p+1} = \cdots = \hat{\lambda}_q > \hat{\lambda}_{q+1} \geq \hat{\lambda}_{q+2} \geq \cdots \geq \hat{\lambda}_k,$$

for some $p, q \in [0, k]$.

In this case, it is straightforward to see that $\hat{\Theta} := \sum_{\ell=1}^{p} (1/\hat{\lambda}_\ell)\hat{u}_\ell \hat{u}_\ell^\mathsf{T} + \sum_{\ell=p+1}^{k} t\, \hat{u}_\ell \hat{u}_\ell$ is a feasible solution to (3.27) for any $t > 0$. When $t > 1/\hat{\lambda}_p$, the objective value of (3.27) at $\hat{\Theta}$ becomes

$$-\sum_{\ell=1}^{p} \log \frac{1}{\hat{\lambda}_\ell} - \sum_{\ell=p+1}^{s} \log t + p - \sum_{\ell=q}^{k} t|\hat{\lambda}_\ell|. \tag{3.31}$$

From the assumptions on $F(\hat{x})$, we see that either $q < k$ (so $\lambda_k < 0$) or $p < s$. Therefore, we can verify that, in both cases, as $t \to +\infty$, (3.31) goes to $-\infty$, which completes our proof. □

Theorem 3.4.8 For $(x, y, t) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^m$, we have

$$\inf_{\nu,\pi,\tau} \nu^\mathsf{T}(\mathbf{e} - x - y) + \pi^\mathsf{T}(b - Ax - t) + \tau(s - \mathbf{e}^\mathsf{T}x) = \begin{cases} 0, & \text{if } \mathbf{e} - x - y = 0,\ b - Ax - t = 0,\ s - \mathbf{e}^\mathsf{T}x = 0; \\ -\infty, & \text{otherwise.} \end{cases}$$

Proof The result follows from the fact that a linear function is bounded below only when it is identically zero. □

Considering Theorems 3.4.7 and 3.4.8, we see that the Lagrangian dual problem of DFact is equivalent to

(DDFact)
$$\begin{aligned} & z_{\text{DDFact}}(C, s, A, b; F) = \\ & \max\ \Gamma_s(F(x)) \\ & \quad \text{subject to:} \\ & \quad \mathbf{e}^\mathsf{T}x = s, \\ & \quad Ax \le b, \\ & \quad 0 \le x \le \mathbf{e}. \end{aligned}$$

Theorem 3.4.9 (see [Nik15, Theorem 14]) DDFact is a convex optimization problem.

Proof From Theorem 3.4.7, we can see that $\Gamma_s(F(\cdot))$ is the point-wise infimum of linear functions, and therefore it is a concave function. The result follows. □

Because DFact has a strictly feasible solution (e.g., given by $(\hat{\Theta} := I, \hat{\upsilon} := 0, \hat{\nu} := \mathrm{diag}(FF^{\mathsf{T}}) = \mathrm{diag}(C), \hat{\pi} := 0, \hat{\tau} := 0)$), if DDFact has a feasible solution with finite objective value, then we have strong duality between DDFact and DFact.

Finally, we demonstrate that every 0/1 solution $\hat{x}$ of DDFact has objective value equal to the ldet of the submatrix of C selected by $S(\hat{x})$.

Lemma 3.4.10 Let $\lambda \in \mathbb{R}^k_+$ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_t > \lambda_{t+1} = \cdots = \lambda_s > \lambda_{s+1} = \cdots = \lambda_k = 0$. Then, the ι satisfying (3.26) is precisely t.

Proof The result is immediate because

$$\frac{1}{s-t}\sum_{\ell=t+1}^{k}\lambda_\ell = \lambda_{t+1}\ .$$

□

Theorem 3.4.11 For $S \subset N$ with $|S| = s$ and $C[S,S]$ nonsingular, we have

$$\Gamma_s(F(x(S))) = \mathrm{ldet}\, C[S,S].$$

Proof Let $\lambda := \lambda(F(x(S)))$ and ι be defined as in Lemma 3.4.6. Then

$$\Gamma_s(F(x(S))) = \sum_{\ell=1}^{\iota}\log(\lambda_\ell) + (s-\iota)\log\left(\frac{1}{s-\iota}\sum_{\ell=\iota+1}^{k}\lambda_\ell\right) = \sum_{\ell=1}^{s}\log(\lambda_\ell) = \mathrm{ldet}\, C[S,S],$$

with the penultimate equation following from Lemma 3.4.10 and that $\mathrm{rank}(C[S,S]) = s$, and the final equation following from the proof of Theorem 3.4.1. □

3.4.3 Fixing variables in DDFact

Theorem 3.4.12 Let

- LB be the objective-function value of a feasible solution for CMESP,
- $(\hat{\Theta}, \hat{\upsilon}, \hat{\nu}, \hat{\pi}, \hat{\tau})$ be a feasible solution for DFact with objective-function value $\hat{\zeta}$.

Then, for every optimal solution x^* for CMESP, we have:

$$x^*_k = 0,\ \forall\, k \in N \text{ such that } \hat{\zeta} - \mathrm{LB} < \hat{\upsilon}_k\ ,$$
$$x^*_k = 1,\ \forall\, k \in N \text{ such that } \hat{\zeta} - \mathrm{LB} < \hat{\nu}_k\ .$$

Proof Analogous to the proof of Theorem 3.3.9. □

3.4.4 Computing DDFact and DFact solutions

3.4.4.1 A good feasible solution of DFact from DDFact

By strong duality, the same bound is produced from DDFact and DFact. A big advantage of solving DDFact compared to solving DFact is that it has many fewer variables. But, variable fixing (see Theorem 3.4.12) relies on a good feasible solution of DFact. Moreover, certifying the quality of a feasible solution of DDFact also requires a good feasible solution of DFact. Motivated by these points, we show how to construct a feasible solution of DFact from a feasible solution $\hat{x}$ of DDFact, with the goal of producing a small gap.

As in the proof of Theorem 3.4.7, we consider the spectral decomposition $F(\hat{x}) = \sum_{\ell=1}^{k} \hat{\lambda}_\ell \hat{u}_\ell \hat{u}_\ell^\mathsf{T}$, with $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_{\hat{r}} > \hat{\lambda}_{\hat{r}+1} = \cdots = \hat{\lambda}_k = 0$. We define $\hat{\Theta} := \sum_{\ell=1}^{k} \hat{\beta}_\ell \hat{u}_\ell \hat{u}_\ell^\mathsf{T}$, where

$$\hat{\beta}_\ell := \begin{cases} 1/\hat{\lambda}_\ell\,, & \text{for } 1 \leq \ell \leq \hat{\imath}; \\ 1/\hat{\delta}, & \text{for } \hat{\imath} < \ell \leq \hat{r}; \\ (1+\epsilon)/\hat{\delta}, & \text{for } \hat{r} < \ell \leq k, \end{cases} \tag{3.32}$$

where ϵ is a positive number to be determined, and $\hat{\imath}$ is the unique integer defined in Lemma 3.4.6 for $\lambda_\ell = \hat{\lambda}_\ell$, and $\hat{\delta} := \frac{1}{s-\hat{\imath}} \sum_{\ell=\hat{\imath}+1}^{k} \hat{\lambda}_\ell$. From Lemma 3.4.6, we have that $\hat{\imath} < s$. Then, as $s \leq \hat{r}$,

$$-\sum_{\ell=1}^{s} \log\left(\hat{\beta}_\ell\right) = \sum_{\ell=1}^{\hat{\imath}} \log\left(\hat{\lambda}_\ell\right) + (s-\hat{\imath}) \log(\hat{\delta}) = \Gamma_s(F(\hat{x})).$$

The minimum duality gap between $\hat{x}$ in DDFact and feasible solutions of DFact of the form $(\hat{\Theta}, \upsilon, \nu, \pi, \tau)$ is the optimal value of the linear program

$$(G(\hat{\Theta})) \qquad \begin{array}{l} \min\ \nu^\mathsf{T}\mathbf{e} + \pi^\mathsf{T} b + \tau s - s \\ \quad \text{subject to:} \\ \quad \upsilon - \nu - A^\mathsf{T}\pi - \tau\mathbf{e} = -\operatorname{diag}(F\hat{\Theta}F^\mathsf{T}), \\ \quad \upsilon \geq 0,\ \nu \geq 0,\ \pi \geq 0. \end{array}$$

Note that $G(\hat{\Theta})$ has a simple closed-form solution for MESP, that is when there are no $Ax \leq b$ constraints (see [LX20]). To construct this solution, we consider the following optimality conditions for $G(\hat{\Theta})$ in this particular case,

$$\begin{array}{l} \nu + \tau\mathbf{e} \geq \operatorname{diag}(F\hat{\Theta}F^\mathsf{T}),\ \nu \geq 0, \\ \mathbf{e}^\mathsf{T} x = s,\ 0 \leq x \leq \mathbf{e}, \\ \nu^\mathsf{T}\mathbf{e} + \tau s = \operatorname{diag}(F\hat{\Theta}F^\mathsf{T})^\mathsf{T} x. \end{array} \tag{3.33}$$

Further, we consider the permutation σ of the indices $\{1,\dots,n\}$, such that $\operatorname{diag}(F\hat{\Theta}F^{\mathsf{T}})_{\sigma(1)} \geq \cdots \geq \operatorname{diag}(F\hat{\Theta}F^{\mathsf{T}})_{\sigma(n)}$. Then, we can see that

$$\begin{aligned}
\tau^* &:= \operatorname{diag}(F\hat{\Theta}F^{\mathsf{T}})_{\sigma(s)}\ , \\
\nu^*_{\sigma(\ell)} &:= \begin{cases} \operatorname{diag}(F\hat{\Theta}F^{\mathsf{T}})_{\sigma(\ell)} - \operatorname{diag}(F\hat{\Theta}F^{\mathsf{T}})_{\sigma(s)}\ , & \text{for } 1 \leq \ell \leq s; \\ 0, & \text{for } s < \ell \leq n, \end{cases} \\
x^*_{\sigma(\ell)} &:= \begin{cases} 1, & \text{for } 1 \leq \ell \leq s; \\ 0, & \text{for } s < \ell \leq n \end{cases}
\end{aligned}$$

satisfies (3.33). Therefore, when there are no $Ax \leq b$ constraints, $(\upsilon^* := \nu^* + \tau^*\mathbf{e} - \operatorname{diag}(F\hat{\Theta}F^{\mathsf{T}}), \nu^*, \tau^*)$ is an optimal solution to $G(\hat{\Theta})$, with objective value $\mathbf{e}^{\mathsf{T}}\nu^* + s\tau^* - s$, corresponding to the minimum duality gap between $\hat{x}$ in DDFact and feasible solutions of DFact of the form $(\hat{\Theta}, \upsilon, \nu, \tau)$.

Next, for MESP, we consider the behavior of the optimal value of $G(\hat{\Theta})$ as a function of ϵ. Let $\hat{U}$ be the matrix having ℓ-th column equal to $\hat{u}_\ell$, for $1 \leq \ell \leq k$. We verify from the last equation in (3.33) that

$$\begin{aligned}
\mathbf{e}^{\mathsf{T}}\nu^* + s\tau^* &= \operatorname{diag}(F\hat{\Theta}F^{\mathsf{T}})^{\mathsf{T}}x^* \\
&= \operatorname{diag}(F\hat{U}\operatorname{Diag}(\hat{\beta})(F\hat{U})^{\mathsf{T}})^{\mathsf{T}}x^* \\
&= \operatorname{diag}((F\hat{U})^{\mathsf{T}}\operatorname{Diag}(x^*)(F\hat{U}))^{\mathsf{T}}\hat{\beta}.
\end{aligned}$$

It is easy to see that the diagonal elements of $(F\hat{U})^{\mathsf{T}}\operatorname{Diag}(x^*)(F\hat{U})$ are nonnegative. Therefore, with x^* fixed, $\operatorname{diag}((F\hat{U})^{\mathsf{T}}\operatorname{Diag}(x^*)(F\hat{U}))^{\mathsf{T}}\hat{\beta}$ is non-decreasing in ϵ. Now the optimal value of the dual of $G(\hat{\Theta})$, is the point-wise max, over the choices of $x^* \in \{0,1\}^n$ satisfying $\mathbf{e}^{\mathsf{T}}x^* = s$. So, the optimal value of the dual of $G(\hat{\Theta})$ is the point-wise max of linear functions, each of which is non-decreasing in ϵ. And so the optimal value of the dual is also non-decreasing in ϵ. Therefore, to obtain the minimum duality gap between $\hat{x}$ in DDFact and $(\hat{\Theta}, \upsilon^*, \nu^*, \tau^*)$ in DFact, we should choose ϵ very close to zero when defining $\hat{\Theta}$.

3.4.4.2 Solving DDFact

In the case of MESP, [LX20] advocate solving DDFact using a Frank-Wolfe style algorithm (see [FW56]). One alternative (see [CFL21b] for details) is to use (relatively-sophisticated) general-purpose nonlinear-programming software (e.g., Ipopt [WB06]). Such software generally needs function-value, function-gradient and optionally function-Hessian subroutines from the user. At points where the function is not differentiable, we have found that supplying a subgradient is satisfactory. For the DDFact objective function, we have the gradient calculation below (see Theorem 3.4.13). But the general nonsmoothness of the objective function of DDFact dissuades us from supplying Hessian information to a solver. Internally, the solver is updating an estimate of the Hessian, and such strategies have been found to be rather robust for nonsmooth functions (see [LO13]).

Theorem 3.4.13 Let $F(\hat{x}) = \sum_{\ell=1}^{k} \hat{\lambda}_\ell \hat{u}_\ell \hat{u}_\ell^\mathsf{T}$ be a spectral decomposition of $F(\hat{x})$. Let $\hat{\iota}$ be the value of ι in Lemma 3.4.6, where λ in Lemma 3.4.6 is $\hat{\lambda} := \lambda(F(\hat{x}))$. If $\frac{1}{s-\hat{\iota}} \sum_{\ell=\hat{\iota}+1}^{k} \hat{\lambda}_\ell > \hat{\lambda}_{\hat{\iota}+1}$, then, for $i = 1, 2, \ldots, n$,

$$\frac{\partial \Gamma_s(F(\hat{x}))}{\partial x_i} = \sum_{\ell=1}^{\hat{\iota}} \frac{1}{\hat{\lambda}_\ell} (F_i . \hat{u}_\ell)^2 + \sum_{\ell=\hat{\iota}+1}^{k} \frac{s-\hat{\iota}}{\sum_{j=\hat{\iota}+1}^{k} \hat{\lambda}_j} (F_i . \hat{u}_\ell)^2 \, .$$

Proof Under the hypothesis $\frac{1}{s-\hat{\iota}} \sum_{\ell=\hat{\iota}+1}^{k} \hat{\lambda}_\ell > \hat{\lambda}_{\hat{\iota}+1}$, in an open neighborhood of $\hat{\lambda}$, the value of $\hat{\iota}$ is constant. We can further check that $\hat{\lambda}_{\hat{\iota}} > \hat{\lambda}_{\hat{\iota}+1}$. Therefore, at the associated $\hat{x}$, we can employ Theorem A.4.16, and we calculate

$$\frac{\partial \Gamma_s(F(\hat{x}))}{\partial x_i} = \sum_{\ell=1}^{k} \frac{\partial \phi_s(\lambda(F(\hat{x})))}{\partial \lambda_\ell} h_i^\ell(\hat{x}),$$

where

$$h_i^\ell(\hat{x}) = \hat{u}_\ell^\mathsf{T} \frac{\partial F(\hat{x})}{\partial x_i} \hat{u}_\ell = \hat{u}_\ell^\mathsf{T} F_i^\mathsf{T} F_i . \hat{u}_\ell = (F_i . \hat{u}_\ell)^2 \, .$$

Calculating

$$\frac{\partial \phi_s(\hat{\lambda})}{\partial \lambda_\ell} = \begin{cases} 1/\hat{\lambda}_\ell \, , & \text{for } \ell \le \hat{\iota}; \\ (s-\hat{\iota}) \, / \, \sum_{j=\hat{\iota}+1}^{k} \hat{\lambda}_j \, , & \text{for } \ell > \hat{\iota}, \end{cases}$$

the result follows. □

Without the technical condition $\frac{1}{s-\hat{\iota}} \sum_{\ell=\hat{\iota}+1}^{k} \hat{\lambda}_\ell > \hat{\lambda}_{\hat{\iota}+1}$, the formulae above still gives a subgradient of Γ_s (see [LX20] for details).

3.4.5 Properties of the factorization bound

It turns out that the factorization bound has a close relationship with the spectral bound. Next, we establish that like the spectral bound for MESP, the factorization bound for CMESP is scaling invariant. Additionally, we will see that the factorization bound dominates the spectral bound. We will also see that the factorization bound and its ability to fix variables using the result of Theorem 3.4.12 are independent of the matrix factorization $C = FF^\mathsf{T}$.

Theorem 3.4.14 For all $\gamma > 0$ and factorizations $C = FF^\mathsf{T}$, we have

$$z_{\text{DFact}}(C, s, A, b; F) = z_{\text{DFact}}(\gamma C, s, A, b; \sqrt{\gamma} F) - s \log \gamma.$$

Proof We simply observe that for every feasible solution $(\hat{\Theta}, \hat{\upsilon}, \hat{\nu}, \hat{\pi}, \hat{\tau})$ of DFact, we have that $(\frac{1}{\gamma}\hat{\Theta}, \hat{\upsilon}, \hat{\nu}, \hat{\pi}, \hat{\tau})$ is a feasible solution of DFact with F replaced by $\sqrt{\gamma}F$. Then we observe that $\lambda_\ell\left(\frac{1}{\gamma}\hat{\Theta}\right) = \frac{1}{\gamma}\lambda_\ell(\hat{\Theta})$, for all ℓ. This mapping between feasible solutions is a bijection, so the result follows. □

Theorem 3.4.15 The factorization bound is independent of the particular factorization of C.

Proof As $z_{\text{DDFact}}(C, s, A, b; F) = z_{\text{DFact}}(C, s, A, b; F)$, it suffices to show that the objective value of DDFact at any feasible solution $\hat{x}$, does not dependent on the factorization $C = FF^{\mathsf{T}}$. We have that $F(\hat{x}) := F^{\mathsf{T}}\,\text{Diag}(\hat{x})F$ has the same non-zero eigenvalues as the matrix $(\text{Diag}\,(\hat{x}))^{\frac{1}{2}}\,FF^{\mathsf{T}}\,(\text{Diag}\,(\hat{x}))^{\frac{1}{2}}$, which is equal to $(\text{Diag}\,(\hat{x}))^{\frac{1}{2}}\,C\,(\text{Diag}\,(\hat{x}))^{\frac{1}{2}}$. The result follows. □

Not only is the factorization bound independent of the factorization, we can also establish a correspondence between dual solutions based on different factorizations.

Theorem 3.4.16 Let $C = F_j F_j^{\mathsf{T}}$, for $j = 1, 2$, be two different factorizations of C, and let $(\hat{\Theta}_1, \hat{\upsilon}, \hat{\nu}, \hat{\pi}, \hat{\tau})$ be a feasible solution to DFact, for $F := F_1$. Then, there is a feasible solution $(\hat{\Theta}_2, \hat{\upsilon}, \hat{\nu}, \hat{\pi}, \hat{\tau})$ to DFact, for $F := F_2$, such that $f(\hat{\Theta}_1, \hat{\nu}, \hat{\pi}, \hat{\tau}) = f(\hat{\Theta}_2, \hat{\nu}, \hat{\pi}, \hat{\tau})$.

Proof Let r be the rank of C, and let $C = \sum_{i=1}^{n} \lambda_i u_i u_i^{\mathsf{T}}$ be a spectral decomposition of C. Suppose that $C = FF^{\mathsf{T}}$, with $F \in \mathbb{R}^{n\times k}$, and $r \le k \le n$. Our preliminary goal is to build a special singular-value decomposition of F.

Let $\sigma_i := \sqrt{\lambda_i}$, for $1 \le i \le k$. Now define $v_i \in \mathbb{R}^k$, for $1 \le i \le r$ by $v_i := \frac{1}{\sigma_i}F^{\mathsf{T}}u_i$.

We can easily check that for $1 \le i \le j \le r$, we have

$$v_i^{\mathsf{T}}v_j = \frac{1}{\sigma_i\sigma_j}u_i^{\mathsf{T}}FF^{\mathsf{T}}u_j = \frac{1}{\sigma_i\sigma_j}u_i C u_j = \frac{\lambda_j}{\sigma_i\sigma_j}u_i^{\mathsf{T}}u_j = \begin{cases} 1, & \text{for } i = j; \\ 0, & \text{for } i < j. \end{cases}$$

That is, $\{v_i : 1 \le i \le r\}$ is a set of r orthonormal vectors in $\mathbb{R}^k$. So, for $r < i \le k$, we can now choose v_i so as to complete $\{v_i : 1 \le i \le r\}$ to an orthonormal basis of $\mathbb{R}^k$.

Next, we have

$$\sum_{i=1}^{k}\sigma_i u_i v_i^{\mathsf{T}} = \sum_{i=1}^{r}\sigma_i u_i v_i^{\mathsf{T}} = \sum_{i=1}^{r} u_i u_i^{\mathsf{T}}F = \sum_{i=1}^{n} u_i u_i^{\mathsf{T}}F = I_n F = F$$

(note that above we have use the fact that $\|u_i^{\mathsf{T}} F\|^2 = u_i^{\mathsf{T}} F F^{\mathsf{T}} u_i = \lambda_i = 0$ for $r < i \leq n$, which implies that $u_i^{\mathsf{T}} F = 0_k^{\mathsf{T}}$ for $r < i \leq n$), and so we can conclude that $F = \sum_{i=1}^{k} \sigma_i u_i v_i^{\mathsf{T}}$ is a singular-value decomposition for F.

The important takeaway is that the $u_i \in \mathbb{R}^n$ $(1 \leq i \leq n)$ and the nonzero σ_i $(1 \leq i \leq r)$ in the singular-value decomposition that we constructed for F only depend on C, not on the particular factorization $C = FF^{\mathsf{T}}$.

It is convenient now to establish that in a factorization matrix F, we can without loss of generality take $k = n$, by appending 0 columns to F if needed, and this will not affect the bound $z_{\text{DFact}}(C, s, A, b; F)$. Let $\bar{F} := [F \mid 0_{n\times(n-k)}]$, and consider

$$\bar{\Theta} = \begin{pmatrix} \hat{\Theta} & \times \\ \times & \times \end{pmatrix} \in \mathbb{S}_+^n .$$

It is trivial to check that $\bar{F}\bar{\Theta}\bar{F}^{\mathsf{T}} = F\hat{\Theta}F^{\mathsf{T}}$. And by Cauchy's eigenvalue interlacing inequalities (Theorem A.3.3), we have $\lambda_{\ell+n-k}(\bar{\Theta}) \leq \lambda_\ell(\hat{\Theta})$, for $1 \leq \ell \leq k$. Therefore, we have $z_{\text{DFact}}(C, s, A, b; \bar{F}) \geq z_{\text{DFact}}(C, s, A, b; F)$. In the other direction, suppose that $\hat{\Theta} \in \mathbb{S}_+^k$. Now define

$$\bar{\Theta} := \begin{pmatrix} \hat{\Theta} & 0^{\mathsf{T}} \\ 0 & \lambda_1(\hat{\Theta}) I_{n-k} \end{pmatrix} \in \mathbb{S}_+^n .$$

As above, we have $\bar{F}\bar{\Theta}\bar{F}^{\mathsf{T}} = F\hat{\Theta}F^{\mathsf{T}}$. And by construction, we have $\lambda_{\ell+n-k}(\bar{\Theta}) = \lambda_\ell(\hat{\Theta})$, for $1 \leq \ell \leq k$. And therefore, we have $z_{\text{DFact}}(C, s, A, b; \bar{F}) \leq z_{\text{DFact}}(C, s, A, b; F)$.

With this we can now conclude that if we have two different factorizations of C, say $C = F_j F_j^{\mathsf{T}}$ for $j = 1, 2$, we can without loss of generality assume for each that F_j has $k = n$ columns, and further that we can choose singular-value decompositions of the form $F_j = U\Sigma V_j$, where here we now take U, Σ, V_1 and V_2 to all be $n \times n$.

Now, right multiplying $U\Sigma V_1^{\mathsf{T}} = F_1$ by $V_1 V_2^{\mathsf{T}}$, we get $U\Sigma V_1^{\mathsf{T}} V_1 V_2^{\mathsf{T}} = F_1 V_1 V_2^{\mathsf{T}}$, and so we have $F_2 = F_1 V_1 V_2^{\mathsf{T}}$.

Finally, for $\Theta_1 \succ 0$, we have

$$F_1 \Theta_1 F_1^{\mathsf{T}} = F_2 V_2 V_1^{\mathsf{T}} \Theta_1 V_1 V_2^{\mathsf{T}} F_2^{\mathsf{T}} ,$$

and so by taking $\Theta_2 := V_2 V_1^{\mathsf{T}} \Theta_1 V_1 V_2^{\mathsf{T}}$, we get $F_1 \Theta_1 F_1^{\mathsf{T}} = F_2 \Theta_2 F_2^{\mathsf{T}}$, with Θ_2 being similar to Θ_1. Therefore, we have that $\lambda_\ell(\Theta_1) = \lambda_\ell(\Theta_2)$, for all ℓ, and so we can transform any feasible solution of DFact with respect to factor $F := F_1$ into a feasible solution of DFact having the same objective value with respect to factor $F := F_2$. The result follows. □

Remark 3.4.17 We note that the proof of Theorem 3.4.16 presents a methodology for constructing a feasible solution $(\hat{\Theta}_2, \hat{\upsilon}, \hat{\nu}, \hat{\pi}, \hat{\tau})$ to DFact for a given factorization of C, from a feasible solution $(\hat{\Theta}_1, \hat{\upsilon}, \hat{\nu}, \hat{\pi}, \hat{\tau})$ to DFact for any other factorization. Importantly, both of these solutions have the same objective value with respect to the corresponding factor. So, we can conclude that all factorizations $C = FF^{\mathsf{T}}$ have the same power to fix variables, based on the result of Theorem 3.4.12.

Next, we establish that the factorization bound for MESP dominates the spectral bound for MESP.

Theorem 3.4.18 Let $C \in \mathbb{S}^n_+$, with $r := \operatorname{rank}(C)$, and $s \leq r$. Then, for all factorizations $C = FF^\mathsf{T}$, we have

$$z_{\text{DFact}}(C, s, \cdot, \cdot\,; F) \leq \sum_{\ell=1}^{s} \log \lambda_\ell(C).$$

Proof Let $C = \sum_{\ell=1}^{r} \lambda_\ell(C) u_\ell u_\ell^\mathsf{T}$ be a spectral decomposition of C. By Theorem 3.4.15, it suffices to take F to be the symmetric matrix $\sum_{\ell=1}^{r} \sqrt{\lambda_\ell(C)} u_\ell u_\ell^\mathsf{T}$.

We consider the solution for DFact given by: $\hat{\Theta} := C^\dagger + \frac{1}{\lambda_r(C)} \left(I - CC^\dagger\right)$, where $C^\dagger := \sum_{\ell=1}^{r} \frac{1}{\lambda_\ell(C)} u_\ell u_\ell^\mathsf{T}$ is the Moore-Penrose pseudoinverse of C, $\hat{\upsilon} := \mathbf{e} - \operatorname{diag}(F\hat{\Theta}F^\mathsf{T})$, $\hat{\nu} := 0$, $\hat{\pi} := 0$, and $\hat{\tau} := 1$. We can verify that the r least eigenvalues of $\hat{\Theta}$ are $\frac{1}{\lambda_1(C)}, \frac{1}{\lambda_2(C)}, \ldots, \frac{1}{\lambda_r(C)}$ and the $n - r$ greatest eigenvalues are all equal to $\frac{1}{\lambda_r(C)}$. Therefore, $\hat{\Theta}$ is positive definite.

The equality constraint of DFact is clearly satisfied at this solution. Additionally, we can verify that $F\hat{\Theta}F^\mathsf{T} = \sum_{\ell=1}^{r} u_\ell u_\ell^\mathsf{T}$. As the positive semidefinite matrix $\sum_{\ell=r+1}^{n} u_\ell u_\ell^\mathsf{T}$ is equal to $I - \sum_{\ell=1}^{r} u_\ell u_\ell^\mathsf{T}$, we conclude that $\operatorname{diag}(F\hat{\Theta}F^\mathsf{T}) \leq \mathbf{e}$. Therefore, $\hat{\upsilon} \geq 0$, and the solution constructed is a feasible solution to DFact. Finally, we can see that the with objective value of this solution is equal to the spectral bound. The result then follows. □

Remark 3.4.19 We note that when C is nonsingular, then $F = C^{1/2}$, and using the symmetry of $C^{1/2}$, it is easy to directly check that with $\hat{\Theta} := C^{-1}$, $\hat{\tau} := 1$, $\hat{\upsilon} := \hat{\nu} := 0$, and $\hat{\pi} := 0$, we have a feasible solution of DFact with objective value equal to the spectral bound.

3.5 NLP bound

The "NLP bound", described in this section, was introduced in [AFLW99, AFLW96]. At the time, there was no other convex-optimization based bound for CMESP, and so the name "NLP bound" seemed reasonable. Like linx and DDFact, the optimization problem NLP is based only on the variable $x \in \mathbb{R}^n$. It depends on a set of parameters, for which we present a detailed analysis that establishes strategies for their selection, mainly guaranteeing the convexity of NLP, but additionally seeking the best possible bound. An important particularity of the NLP bound is that it is not amenable to solution using existing convex-solver languages (like CVX (see [BV04]) and YALMIP (see [Löf04])) and their underlying solvers (e.g., MOSEK (see [MOS19]) and SDPT3 (see [TTT99, TTT12])). These languages are attractive because they can handle ldet maximization terms, and as we have indicated, a perfectly viable

way to calculate the linx bound is with YALMIP/SDPT3. But neither YALMIP nor CVX can recognize the convexity of NLP. We could instead resort to general nonlinear-optimization codes, and this is a practical technique. Toward that end, we will provide formulae for gradient and Hessian of the NLP objective function (see §3.5.1).

Alternatively, [AFLW99] devised an interior-point algorithm for the solution of NLP. We will describe that algorithm, which also directly applies to linx. Our algorithm is a primal path-following algorithm, and we describe a methodology for constructing a feasible solution for the dual of NLP, from an approximate point on the "central path". As we did in §3.3.3 for linx and in §3.4.3 for DDFact, we discuss how a dual-feasible solution can be used to fix variables, during the execution of a branch-and-bound algorithm, based on the NLP. Finally, we address another particularity of NLP: because of the selection of different parameters, the monotonicity of its optimal objective value is not trivially obtained as we go deeper in the branch-and-bound tree. However, monotonicity can be established for some of the proposed parameter-selection strategies.

We assume that $d \in \mathbb{R}^n_{++}$ and $p \in \mathbb{R}^n_{++}$, and we let $D := \mathrm{Diag}(d)$. For $x \in [0,1]^n$, we let

$$f(x) := \begin{cases} \mathrm{ldet}\, R(x), & \text{for } R(x) \succ 0; \\ -\infty, & \text{otherwise,} \end{cases}$$

where

$$\begin{aligned} R(x) &:= \mathrm{Diag}(x^{p/2})(C - D)\,\mathrm{Diag}(x^{p/2}) + \mathrm{Diag}(d^x) \\ &= \mathrm{Diag}(x^{p/2})C\,\mathrm{Diag}(x^{p/2}) + \mathrm{Diag}(d_i^{x_i} - d_i x_i^{p_i} \ : \ i \in N). \end{aligned}$$

We define the **NLP bound** for CMESP:

$$\text{(NLP)} \qquad z_{\text{NLP}} := \max \left\{ f(x) \ : \ \mathbf{e}^{\mathsf{T}} x = s,\ Ax \le b,\ 0 \le x \le \mathbf{e} \right\}.$$

We will choose the parameters d and p so that $R(x) \succeq 0$ on the entire feasible region. Therefore f will be concave and continuous on the feasible region. Then we are justified in using maximum rather than supremum, by Proposition A.1.4 (using $\mathcal{K} := \{x \in \mathbb{R}^n \ : \ R(x) \in \mathbb{S}^n_{++}\}$ and $\mathcal{Q} := \left\{x \in \mathbb{R}^n \ : \ \mathbf{e}^{\mathsf{T}} x = s;\ Ax \le b;\ 0 \le x \le \mathbf{e}\right\}$).

The name "NLP bound" is justified by the following result.

Theorem 3.5.1 ([AFLW99])

$$z(C, s, A, b) \leq z_{\text{NLP}} .$$

Proof We note that for $x \in \{0,1\}^n$, we have $d_i^{x_i} - d_i x_i^{p_i} = 1 - x_i$, and then $R(x) = \text{Diag}(x) C \,\text{Diag}(x) + \text{Diag}(\mathbf{e} - x)$, and furthermore $f(x) = \text{ldet}\, C[S(x), S(x)]$. Therefore NLP is an exact relaxation of CMESP. □

A motivation for the formulation of NLP comes from the particular case where C is diagonal, the case of independent random variables. In this case, CMESP can be modeled as an integer linear optimization problem by setting $D = C$. This transforms the objective function of NLP into the linear function

$$\sum_{i \in N} \log(C[i,i]) \cdot x_i ,$$

so NLP becomes a linear optimization problem, and enforcing integrality on its variables gives us the natural integer linear optimization problem modeling CMESP. If, in addition to having C diagonal, we also have $m = 0$ (the case of MESP) or A being totally unimodular and $b \in \mathbb{Z}^n$, then any basic optimal solution of NLP (which is now a linear optimization problem) is an optimal solution to the corresponding CMESP as well.

Next, we discuss how to select the parameters $d \in \mathbb{R}^n_{++}$ and $p \in \mathbb{R}^n_{++}$ for NLP in order to guarantee that we have a tractable relaxation. That is, we want f to be concave on the feasible region of NLP, so that NLP is a convex relaxation.

3.5.1 Convexity of NLP

We will see that by selecting the parameters $d \in \mathbb{R}^n_{++}$ and $p \in \mathbb{R}^n_{++}$ for NLP, such that $D \geq C$, $p \geq \mathbf{e}$, and $0 < d_i \leq \exp(p_i - \sqrt{p_i})$, for $i \in N$, we will guarantee the convexity of NLP. We establish this result in Theorem 3.5.5, demonstrating that the Hessian of f is negative semidefinite on the feasible region of NLP.

In what follows, we assemble the tools used in the proof; we calculate the gradient and the Hessian of f, and we show that $R(x)$ is positive definite on the feasible region of NLP, when the parameters are selected appropriately.

Lemma 3.5.2 For $x \in \mathbb{R}^n_{++}$ such that $R(x) \succ 0$, we have

$$\frac{\partial f(x)}{\partial x_i} = \frac{p_i}{x_i} + \frac{d_i^{x_i} \left(\log(d_i^{x_i}) - p_i\right)}{x_i} R(x)^{-1}_{ii} . \tag{3.34}$$

Proof We define

$$\tilde{R}(x) := C - D + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}(d^x).$$

Then

$$R(x) = \mathrm{Diag}(x^{p/2})\tilde{R}(x)\,\mathrm{Diag}(x^{p/2})$$

and

$$\begin{aligned} f(x) &= \sum_{i\in N} p_i \log(x_i) + \mathrm{ldet}\,\tilde{R}(x) \\ &= \sum_{i\in N} p_i \log(x_i) + \mathrm{ldet}\,(C - D + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}(d^x))\,. \end{aligned}$$

Therefore,

$$\frac{\partial f(x)}{\partial x_i} = \frac{p_i}{x_i} + \frac{\partial}{\partial x_i}\,\mathrm{ldet}\,(C - D + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}(d^x))\,.$$

Finally, by the chain rule and Lemma A.4.13, we have

$$\begin{aligned} &\frac{\partial}{\partial x_i}\,\mathrm{ldet}\,(C - D + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}(d^x)) \\ &\quad= \left(C - D + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}(d^x)\right)^{-1}_{ii} \frac{\mathrm{d}}{\mathrm{d}x_i}\left(x_i^{-p_i} d_i^{x_i}\right) \\ &\quad= \left(-p_i x_i^{-p_i-1} d_i^{x_i} + x_i^{-p_i} d_i^{x_i}\log(d_i)\right)\left(C - D + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}(d^x)\right)^{-1}_{ii} \\ &\quad= d_i^{x_i} x_i^{-(p_i+1)}\left(\log(d_i^{x_i}) - p_i\right)\left(C - D + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}(d^x)\right)^{-1}_{ii} \\ &\quad= d_i^{x_i} x_i^{-1}\left(\log(d_i^{x_i}) - p_i\right) R(x)^{-1}_{ii}. \end{aligned}$$

□

Lemma 3.5.3 Let $l(x) \in \mathbb{R}^n$ be such that $l_i(x) := p_i - \log(d_i^{x_i})$. Then, for $x \in \mathbb{R}^n_{++}$ such that $R(x) \succ 0$, the Hessian of f is

$$\begin{aligned} \nabla^2 f(x) &= (\mathrm{Diag}(x))^{-1}\,\mathrm{Diag}(p)\left(\mathrm{Diag}(d^x)\,\mathrm{Diag}\left(R(x)^{-1}\right) - I\right)(\mathrm{Diag}(x))^{-1} \\ &+ (\mathrm{Diag}(x))^{-1}\mathrm{Diag}(l(x))\left(\mathrm{Diag}(d^x)\,\mathrm{Diag}\left(R(x)^{-1}\right)\right)\mathrm{Diag}(l(x))(\mathrm{Diag}(x))^{-1} \\ &- (\mathrm{Diag}(x))^{-1}\mathrm{Diag}(l(x))\left(\mathrm{Diag}(d^x)\left(R(x)^{-1}\right)^{(2)}\mathrm{Diag}(d^x)\right)\mathrm{Diag}(l(x))(\mathrm{Diag}(x))^{-1}. \end{aligned}$$

Proof Let $g_i(x) := \frac{\partial}{\partial x_i} f(x)$. Considering Lemma 3.5.2, Lemma A.2.5, and the chain rule, we have for $j \neq i$,

$$\begin{aligned}\frac{\partial g_i(x)}{\partial x_j} &= -\frac{d_i^{x_i}\left(\log(d_i^{x_i}) - p_i\right)}{x_i^{p_i+1}}\left((C - D + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}(d^x))^{-1}_{ij}\right)^2 \frac{\mathrm{d}}{\mathrm{d}x_j}\left(\frac{d_j^{x_j}}{x_j^{p_j}}\right)\\ &= -\frac{d_i^{x_i}\left(\log(d_i^{x_i}) - p_i\right)}{x_i^{p_i+1}}\frac{d_j^{x_j}\left(\log(d_j^{x_j}) - p_j\right)}{x_j^{p_j+1}}\left((C - D + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}(d^x))^{-1}_{ij}\right)^2\\ &= -\frac{d_i^{x_i} d_j^{x_j} l_i(x) l_j(x)}{x_i x_j}\left(R(x)^{-1}_{ij}\right)^2.\end{aligned}$$

Similarly, for $j = i$, we have

$$\begin{aligned}\frac{\partial g_i(x)}{\partial x_i} &= -\frac{p_i}{x_i^2} - \left(\frac{d_i^{x_i} l_i(x)}{x_i}\right)^2\left(R(x)^{-1}_{ii}\right)^2\\ &\quad + \left(C - D + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}(d^x)\right)^{-1}_{ii}\frac{\mathrm{d}}{\mathrm{d}x_i}\left(\frac{d_i^{x_i}\left(\log(d_i^{x_i}) - p_i\right)}{x_i^{p_i+1}}\right)\\ &= -\frac{p_i}{x_i^2} - \left(\frac{d_i^{x_i} l_i(x)}{x_i}\right)^2\left(R(x)^{-1}_{ii}\right)^2\\ &\quad + \left(C - D + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}(d^x)\right)^{-1}_{ii}\frac{d_i^{x_i}}{x_i^{p_i+2}}\left(p_i + \left(\log(d_i^{x_i}) - p_i\right)^2\right)\\ &= -\frac{p_i}{x_i^2} - \left(\frac{d_i^{x_i} l_i(x)}{x_i}\right)^2\left(R(x)^{-1}_{ii}\right)^2 + \frac{d_i^{x_i}}{x_i^2}\left(p_i + (l_i(x))^2\right)R(x)^{-1}_{ii}.\end{aligned}$$

□

Lemma 3.5.4 Suppose that $p \geq \mathbf{e}$, and $0 < d_i \leq \exp(p_i)$, for $i \in N$.

(i) For $C \geq 0$, we have $R(x) \succ 0$, for all $0 \leq x < \mathbf{e}$.
(ii) For $C \succ 0$, we have $R(x) \succ 0$, for all $0 \leq x \leq \mathbf{e}$.

Proof For $x = 0$, we have $R(0) = I \succ 0$. For $x \neq 0$, let $x_i > 0$, for all $i \in T \subset N$. Let $\bar{x} := x[T]$, $\bar{p} := p[T]$, $\bar{d} := d[T]$, $\bar{D} := \mathrm{Diag}(\bar{d})$, and $\bar{C} := C[T, T]$. Assume, without loss of generality, that $T := \{1, \ldots, \ell\}$, for $\ell \leq n$. We then have

$$R(x) := \left(\begin{array}{c|c} \mathrm{Diag}(\bar{x}^{\bar{p}/2})(\bar{C} - \bar{D})\,\mathrm{Diag}(\bar{x}^{\bar{p}/2}) + \mathrm{Diag}(\bar{d}^{\bar{x}}) & 0 \\ \hline 0 & I \end{array}\right),$$

and it suffices to show that

$$\mathrm{Diag}(\bar{x}^{\bar{p}/2})(\bar{C} - \bar{D})\,\mathrm{Diag}(\bar{x}^{\bar{p}/2}) + \mathrm{Diag}(\bar{d}^{\bar{x}}) \succ 0.$$

For $i \in T$, we have

$$\log(x_i) \leq x_i - 1, \tag{3.35}$$

and, from the hypotheses,

$$\log(d_i) \leq p_i \ .$$

Then,

$$\begin{array}{rcll} (x_i - 1)\log(d_i) & \geq & p_i \log(x_i) & \Rightarrow \\ d_i^{x_i - 1} & \geq & x_i^{p_i} & \Rightarrow \\ d_i^{x_i} & \geq & d_i x_i^{p_i} \ . & \end{array} \tag{3.36}$$

(i) For $C \geq 0$ and $0 \leq x < \mathbf{e}$,

$$\mathrm{Diag}(\bar{x}^{\bar{p}/2})\bar{C}\,\mathrm{Diag}(\bar{x}^{\bar{p}/2}) \geq 0,$$

and, as $x < \mathbf{e}$, the inequality (3.35) and all the inequalities in (3.36) become strict. Therefore,

$$\mathrm{Diag}(d_i^{x_i} - d_i x_i^{p_i} \ : \ i \in T) \succ 0.$$

(ii) For $C \succ 0$ and $0 \leq x \leq \mathbf{e}$,

$$\mathrm{Diag}(\bar{x}^{\bar{p}/2})\bar{C}\,\mathrm{Diag}(\bar{x}^{\bar{p}/2}) \succ 0$$

because $\bar{x} > 0$, and

$$\mathrm{Diag}(d_i^{x_i} - d_i x_i^{p_i} \ : \ i \in T) \geq 0.$$

The result follows. □

Theorem 3.5.5 Suppose that $D \geq C$, $p \geq \mathbf{e}$, and $0 < d_i \leq \exp(p_i - \sqrt{p_i})$, for $i \in N$. Then the function $\mathrm{ldet}(R(\cdot))$ is concave on $\{x \in [0,1]^n \ : \ R(x) \succ 0\}$.

Proof As $C \preceq D$, we have $R(x) \preceq \mathrm{Diag}(d^x)$. Then, considering Lemma 3.5.4, we have

$$R(x)^{-1} \succeq \mathrm{Diag}(d^x)^{-1} \quad \Rightarrow \tag{3.37}$$

$$R(x)^{-1} - \mathrm{Diag}(d^x)^{-1} \succeq 0. \tag{3.38}$$

From (3.37), we have

$$\mathrm{Diag}(d^x)R(x)^{-1}\,\mathrm{Diag}(d^x) \succeq \mathrm{Diag}(d^x), \tag{3.39}$$

and

$$\begin{array}{ll} & \mathrm{Diag}(d^x) \circ R(x)^{-1} \succeq \mathrm{Diag}(d^x) \circ \mathrm{Diag}(d^x)^{-1} = I \quad \Rightarrow \\ & \mathrm{Diag}(d^x) \circ R(x)^{-1} - I \succeq 0. \end{array} \tag{3.40}$$

From (3.38) and (3.39), we have

$$\begin{aligned}&\mathrm{Diag}(d^x)R(x)^{-1}\,\mathrm{Diag}(d^x)\circ\left(R(x)^{-1}-\mathrm{Diag}(d^x)^{-1}\right)\\&\quad\geq \mathrm{Diag}(d^x)\circ\left(R(x)^{-1}-\mathrm{Diag}(d^x)^{-1}\right),\end{aligned}$$

or, equivalently

$$\begin{aligned}&\mathrm{Diag}(d^x)\left(R(x)^{-1}\right)^{(2)}\mathrm{Diag}(d^x)-\mathrm{Diag}(d^x)\circ R(x)^{-1}\\&\quad\geq \mathrm{Diag}(d^x)\circ R(x)^{-1}-I.\end{aligned}\tag{3.41}$$

Let us consider $l_i(x) := p_i - \log(d_i^{x_i})$, for $i \in N$, as defined in Lemma 3.5.3. Considering also that $p_i \geq 1$, $0 < d_i \leq \exp(p_i - \sqrt{p_i})$, and $0 \leq x_i \leq 1$, we have

- If $\log(d_i) \leq 0$, then $l_i(x) = p_i - x_i \log(d_i) \geq p_i \geq \sqrt{p_i}$.
- If $\log(d_i) \geq 0$, then $x_i \log(d_i) \leq \log(d_i) \leq p_i - \sqrt{p_i}$. So, $l_i(x) = p_i - \log(d_i^{x_i}) \geq \sqrt{p_i}$.

Therefore,

$$l_i(x) \geq \sqrt{p_i}\,, \quad \text{for } i \in N.\tag{3.42}$$

From (3.40) and (3.42), we have

$$\begin{aligned}&\mathrm{Diag}(l(x))\left(\mathrm{Diag}(d^x)\circ R(x)^{-1}-I\right)\mathrm{Diag}(l(x))\\&\quad\geq \mathrm{Diag}(p)\left(\mathrm{Diag}(d^x)\circ R(x)^{-1}-I\right).\end{aligned}\tag{3.43}$$

From (3.41) and (3.43), we have

$$\begin{aligned}&\mathrm{Diag}(l(x))\Big(\mathrm{Diag}(d^x)\left(R(x)^{-1}\right)^{(2)}\mathrm{Diag}(d^x)-\mathrm{Diag}(d^x)\circ R(x)^{-1}\Big)\mathrm{Diag}(l(x))\\&\quad\geq \mathrm{Diag}(p)\left(\mathrm{Diag}(d^x)\circ R(x)^{-1}-I\right).\end{aligned}$$

Finally, from Lemma 3.5.3, we have that $\nabla^2 f(x) \preceq 0$, which completes the proof. □

Corollary 3.5.6 Suppose that $D \geq C$, $p \geq \mathbf{e}$, and $0 < d_i \leq \exp(p_i - \sqrt{p_i})$, for $i \in N$. Then, NLP is a convex optimization problem.

Proof As the constraints of NLP are all linear, it suffices to demonstrate that the objective function is concave on the feasible set. We proved in Theorem 3.5.5 that the objective function is concave on the convex set $\mathcal{F} := \{x \in [0,1]^n \,:\, R(x) \succ 0\}$, and because $f(x)$ is $-\infty$ on $\{x \in [0,1]^n \,:\, R(x) \succeq 0\} \setminus \mathcal{F}$. Therefore, the objective function is concave (in the extended sense) on all of the feasible set. □

3.5.2 Scaling for NLP

Considering the scaling introduced in §1.6, we define, for a positive constant γ,

$$f(\gamma;x) := \begin{cases} \operatorname{ldet} R(\gamma;x) - s\log(\gamma), & \text{for } R(\gamma;x) \succ 0; \\ -\infty, & \text{otherwise,} \end{cases} \tag{3.44}$$

$$\text{where } R(\gamma;x) := \gamma \operatorname{Diag}(x^{p/2})(C-D)\operatorname{Diag}(x^{p/2}) + \operatorname{Diag}((\gamma d)^x).$$

We note that $f(x) = f(\gamma;x)$ for all x such that $\mathbf{e}^{\mathsf{T}}x = s$, $x \in \{0,1\}^n$. We can therefore, replace $f(x)$ by $f(\gamma;x)$ in NLP, which is particularly relevant in our context, because the NLP bound depends on the scaling factor γ. We then define the **scaled NLP bound**

$$(\mathrm{NLP}_\gamma) \qquad z_{\mathrm{NLP}_\gamma} := \max\left\{ f(\gamma;x) \;:\; \mathbf{e}^{\mathsf{T}}x = s,\ Ax \le b,\ 0 \le x \le \mathbf{e} \right\}.$$

Scaling C, D and d in Theorem 3.5.5, we immediately have the following result

Corollary 3.5.7 Let $D \succeq C$, $\gamma > 0$, $p_i \ge 1$, and $0 < \gamma d_i \le \exp(p_i - \sqrt{p_i})$, for $i \in N$. Then the function $\operatorname{ldet}(R(\gamma;\cdot))$ is concave on $\{x \in [0,1]^n \;:\; R(x) \succ 0\}$.

3.5.3 Good parameters for NLP_γ

The conditions imposed in Corollary 3.5.7, on the parameters d, p, and γ, of NLP_γ, do not uniquely establish their values. Therefore, we present next, results that will guide us in the selection of the parameters, aiming to obtain the best possible bound for CMESP.

In the next lemma, we see that if the hypotheses of Corollary 3.5.7 hold, then for fixed d and γ, p_i should be chosen as small as possible, for all $i \in N$.

Lemma 3.5.8 Assume that p, d, and γ satisfy the hypothesis in Corollary 3.5.7, and that $\check{p} \geq p$. Let $\check{f}(\gamma, \cdot)$ be defined as in (3.44), but replacing p by $\check{p}$. Then, $\check{f}(\gamma; x) \geq f(\gamma; x)$, for all $0 < x \leq \mathbf{e}$ such that $R(x) \succ 0$.

Proof Let us consider $f(\gamma; x)$ as a function of p. In the following, we show that $\partial f(\gamma; x)/\partial p_i \geq 0$, for all $i \in N$, and $0 < x \leq \mathbf{e}$ such that $R(x) \succ 0$. We define

$$\tilde{R}(\gamma; x) := \gamma(C - D) + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}((\gamma d)^x).$$

Then

$$R(\gamma; x) = \mathrm{Diag}(x^{p/2})\tilde{R}(\gamma; x)\,\mathrm{Diag}(x^{p/2})$$

and

$$\begin{aligned} f(\gamma; x) &= \sum_{i \in N} p_i \log(x_i) + \mathrm{ldet}\,\tilde{R}(\gamma; x) - s\log(\gamma) \\ &= \sum_{i \in N} p_i \log(x_i) + \mathrm{ldet}\left(\gamma(C - D) + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}((\gamma d)^x)\right) - s\log(\gamma). \end{aligned}$$

Therefore,

$$\frac{\partial f(\gamma; x)}{\partial p_i} = \log(x_i) + \frac{\partial}{\partial p_i}\,\mathrm{ldet}\left(\gamma(C - D) + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}(\gamma d)^x)\right).$$

Moreover, from Lemma A.4.13, we have

$$\begin{aligned} &\frac{\partial}{\partial p_i}\,\mathrm{ldet}\left(\gamma(C - D) + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}((\gamma d)^x)\right) \\ &\quad = \left(\gamma(C - D) + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}((\gamma d)^x)\right)^{-1}_{ii} \frac{\mathrm{d}}{\mathrm{d}p_i}\left(x_i^{-p_i}(\gamma d_i)^{x_i}\right) \\ &\quad = -\log(x_i)x_i^{-p_i}(\gamma d_i)^{x_i}\left(\gamma(C - D) + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}((\gamma d)^x)\right)^{-1}_{ii} \\ &\quad = -\log(x_i)(\gamma d_i)^{x_i} R(\gamma; x)^{-1}_{ii}. \end{aligned}$$

So,

$$\frac{\partial f(\gamma; x)}{\partial p_i} = \log(x_i)\left(1 - (\gamma d_i)^{x_i} R(\gamma; x)^{-1}_{ii}\right).$$

Similarly to (3.40), we can verify that

$$R(\gamma; x)^{-1}_{ii}(\gamma d_i)^{x_i} \geq 1, \tag{3.45}$$

for all $i \in N$. Also, from $x_i \leq 1$, we have $\log(x_i) \leq 0$. Therefore, $\partial f(\gamma; x)/\partial p_i \geq 0$, and the result follows. □

Considering the result above, Corollary 3.5.9 establishes for given d and γ, and for all $i \in N$, the value of p_i as the smallest value such that $p_i \geq 1$ and $\gamma d_i \leq \exp(p_i - \sqrt{p_i})$.

Corollary 3.5.9 Let $D \succeq C$ and $\gamma > 0$. The minimum upper bound for CMESP, given by z_{NLP_γ}, is obtained by

$$p_i := \begin{cases} 1, & \text{for } \gamma d_i \leq 1; \\ \left(1 + \sqrt{1 + 4\log(\gamma d_i)}\right)^2 \Big/ 4, & \text{for } \gamma d_i > 1. \end{cases}$$

Remark 3.5.10 For any $D \succeq C$ and $\gamma > 0$, with the choice of p from Corollary 3.5.9, the hypothesis of Corollary 3.5.7 are satisfied.

Remark 3.5.11 From Corollary 3.5.9, we can see that in the particular case where $\gamma D \preceq I$, setting $p = \mathbf{e}$ leads to the best possible bound.

In Lemma 3.5.12, we see that for a given γ, and p set as in Corollary 3.5.9, the selection of the smallest d_i, for all $i \in N$, also leads to the best bound.

Lemma 3.5.12 Let $\gamma > 0$, $C \preceq D$, where $D = \mathrm{Diag}(d)$, and $d \in \mathbb{R}^n_{++}$. Let $p = p(d)$ be given as in Corollary 3.5.9. Assume that i is such that $\gamma d_i \neq 1$, and that $\check{d}_i \geq d_i$. Let $\check{f}(\gamma, \cdot)$ be defined as in (3.44), but replacing d_i by $\check{d}_i$. Then $\check{f}(\gamma; x) \geq f(\gamma; x)$ for all $0 < x \leq \mathbf{e}$ such that $R(x) > 0$.

Proof Let us consider $f(\gamma; x)$ as a function of d_i. In the following, we show that $\partial f(\gamma; x)/\partial d_i \geq 0$, for $0 < x \leq \mathbf{e}$ such that $R(x) > 0$. We define

$$\tilde{R}(\gamma; x) := \gamma(C - D) + \mathrm{Diag}(x^{-p})\,\mathrm{Diag}((\gamma d)^x).$$

Then

$$R(\gamma; x) = \mathrm{Diag}(x^{p/2})\tilde{R}(\gamma; x)\,\mathrm{Diag}(x^{p/2})$$

and

$$f(\gamma; x) = \sum_{i \in N} p_i \log(x_i) + \mathrm{ldet}\,\tilde{R}(\gamma; x) - s\log(\gamma).$$

Therefore,

$$\frac{\partial f(\gamma; x)}{\partial d_i} = \log(x_i)\frac{\mathrm{d}p_i(d_i)}{\mathrm{d}d_i} + \frac{\partial}{\partial d_i}\,\mathrm{ldet}\,\tilde{R}(\gamma; x).$$

- We first analyze the case where $\gamma d_i > 1$, and

$$p_i = p_i(d_i) = \left(1 + \sqrt{1 + 4\log(\gamma d_i)}\right)^2 \Big/ 4.$$

We have

$$\begin{aligned}\frac{\mathrm{d}p_i(d_i)}{\mathrm{d}d_i} &= \frac{2\left(1+\sqrt{1+4\log(\gamma d_i)}\right)}{4}\frac{1}{2\sqrt{1+\log(\gamma d_i)}}\frac{4}{\gamma d_i}\gamma \\ &= \frac{2\sqrt{p_i}}{d_i\left(2\sqrt{p_i}-1\right)},\end{aligned}$$

and we note that $\frac{\mathrm{d}}{\mathrm{d}d_i}p_i(d_i) \geq 0$.
By the chain rule and Lemma A.4.13, we have

$$\begin{aligned}\frac{\partial}{\partial d_i}\operatorname{ldet}\tilde{R}(\gamma;x) &= \tilde{R}(\gamma;x)_{ii}^{-1}\frac{\partial}{\partial d_i}\left(\gamma(C_{ii}-d_i)+x_i^{-p_i}(\gamma d_i)^{x_i}\right) \\ &= \tilde{R}(\gamma;x)_{ii}^{-1}\left(-\gamma+x_i^{-p_i}\gamma^{x_i}x_i d_i^{x_i-1}-(\gamma d_i)^{x_i}x_i^{-p_i}\log(x_i)\frac{\mathrm{d}p_i(d_i)}{\mathrm{d}d_i}\right) \\ &= \tilde{R}(\gamma;x)_{ii}^{-1}x_i^{-p_i}(\gamma d_i)^{x_i}\left(-\gamma\frac{x_i^{p_i}}{(\gamma d_i)^{x_i}}+\frac{x_i}{d_i}-\log(x_i)\frac{\mathrm{d}p_i(d_i)}{\mathrm{d}d_i}\right) \\ &= R(\gamma;x)_{ii}^{-1}(\gamma d_i)^{x_i}\left(-\gamma\frac{x_i^{p_i}}{(\gamma d_i)^{x_i}}+\frac{x_i}{d_i}-\log(x_i)\frac{\mathrm{d}p_i(d_i)}{\mathrm{d}d_i}\right).\end{aligned}$$

Then

$$\begin{aligned}\frac{\partial f(\gamma;x)}{\partial d_i} &= \log(x_i)\frac{\mathrm{d}p_i(d_i)}{\mathrm{d}d_i} \\ &\quad + R(\gamma;x)_{ii}^{-1}(\gamma d_i)^{x_i}\left(-\gamma\frac{x_i^{p_i}}{(\gamma d_i)^{x_i}}+\frac{x_i}{d_i}-\log(x_i)\frac{\mathrm{d}p_i(d_i)}{\mathrm{d}d_i}\right).\end{aligned} \tag{3.46}$$

We note that

$$\begin{aligned}\log(x_i d_i^{-1})-\log(\gamma x_i^{p_i}(\gamma d_i)^{-x_i}) &= \log(x_i)-\log(d_i) \\ &\quad -\left(\log(\gamma)+p_i\log(x_i)-x_i\log(\gamma d_i)\right) \\ &= \log(x_i)-\log(\gamma d_i)-p_i\log(x_i)+x_i\log(\gamma d_i) \\ &= \log(x_i)(1-p_i)+\log(\gamma d_i)(x_i-1) \\ &\geq (x_i-1)(1-p_i)+(p_i-1)(x_i-1) \\ &= 0,\end{aligned}$$

where we consider that $\log(x_i) \leq x_i - 1$, and $p_i \geq 1, x_i \leq 1, \log(\gamma d_i) \leq p_i - \sqrt{p_i} \leq p_i - 1$ (see Remark 3.5.10). So,

$$\log\left(x_i d_i^{-1}\right) \geq \log\left(\gamma x_i^{p_i}(\gamma d_i)^{-x_i}\right),$$

and

$$\frac{x_i}{d_i} \geq \gamma \frac{x_i^{p_i}}{(\gamma d_i)^{x_i}} \,. \tag{3.47}$$

Considering also that $\log(x_i) \leq 0$ and $\mathrm{d}p_i(d_i)/\mathrm{d}d_i \geq 0$, we have

$$-\gamma \frac{x_i^{p_i}}{(\gamma d_i)^{x_i}} + \frac{x_i}{d_i} - \log(x_i) \frac{\mathrm{d}p_i(d_i)}{\mathrm{d}d_i} \geq 0. \tag{3.48}$$

Therefore, from (3.40), (3.46), and (3.48), we have

$$\begin{aligned} \frac{\partial f(\gamma;x)}{\partial d_i} &\geq \log(x_i) \frac{\mathrm{d}p_i(d_i)}{\mathrm{d}d_i} + \left(-\gamma \frac{x_i^{p_i}}{(\gamma d_i)^{x_i}} + \frac{x_i}{d_i} - \log(x_i) \frac{\mathrm{d}p_i(d_i)}{\mathrm{d}d_i} \right) \\ &= -\gamma \frac{x_i^{p_i}}{(\gamma d_i)^{x_i}} + \frac{x_i}{d_i} \\ &\geq 0, \end{aligned} \tag{3.49}$$

where the last inequality comes from (3.47).

- Next, we analyze the case where $\gamma d_i < 1$ and $p_i = 1$.

Considering $\frac{\mathrm{d}}{\mathrm{d}d_i} p_i(d_i) = 0$, (3.49) still holds. Therefore, in this case we also have $\frac{\partial}{\partial d_i} f(\gamma;x) \geq 0$. □

In the following, we prove two more lemmas, from which we conclude that we should not select γ and D such that $\gamma D \prec I$ or $\gamma D \succ I$.

Lemma 3.5.13 Let $C \preceq D \preceq I$ and $p = \mathbf{e}$. Then $f(\gamma;x) \geq f(1,x)$ for all $0 \leq x \leq \mathbf{e}$ such that $R(x) \succ 0$, $\mathbf{e}^{\mathsf{T}} x = s$, and $0 < \gamma \leq 1$.

Proof We show that if $C \preceq D$ and p is constant, then $\partial f(\gamma;x)/\partial\gamma \leq 0$, for all $x \in [0,1]^n$ such that $R(x) \succ 0$. We define

$$\tilde{R}(\gamma;x) := \mathrm{Diag}(x^{p/2})(C - D)\,\mathrm{Diag}(x^{p/2}) + \mathrm{Diag}(\gamma\mathbf{e})^{x-\mathbf{e}}\,\mathrm{Diag}(d^x).$$

Then

$$R(\gamma;x) = \gamma \tilde{R}(\gamma;x)$$

and

$$f(\gamma;x) = n\log(\gamma) + \mathrm{ldet}\,\tilde{R}(\gamma;x) - s\log(\gamma).$$

Therefore,

$$\frac{\partial f(\gamma;x)}{\partial\gamma} = \frac{n-s}{\gamma} + \frac{\partial}{\partial\gamma}\,\mathrm{ldet}\,\tilde{R}(\gamma;x). \tag{3.50}$$

By the chain rule and Lemma A.4.13, we have

$$\begin{aligned}\frac{\partial}{\partial\gamma}\operatorname{ldet}\tilde{R}(\gamma;x) &= \sum_{i\in N}\frac{\partial\operatorname{ldet}\tilde{R}(\gamma;x)}{\partial\tilde{R}(\gamma;x)_{ii}}\frac{\partial\tilde{R}(\gamma;x)_{ii}}{\partial\gamma}\\ &= \sum_{i\in N}\tilde{R}(\gamma;x)_{ii}^{-1}(x_i-1)\gamma^{x_i-2}d_i^{x_i}\\ &= \sum_{i\in N}\tilde{R}(\gamma;x)_{ii}^{-1}(\gamma d_i)^{x_i}(x_i-1)\gamma^{-2}\\ &= \frac{1}{\gamma}\sum_{i\in N}R(\gamma;x)_{ii}^{-1}(\gamma d_i)^{x_i}(x_i-1)\\ &\le \frac{1}{\gamma}\sum_{i\in N}(x_i-1)\\ &= \frac{s-n}{\gamma}\,.\end{aligned}\tag{3.51}$$

where, considering that $x_i \le 1$, the inequality in (3.51) follows from (3.45). Then, from (3.50) and (3.51), we have $\partial f(\gamma;x)/\partial\gamma \le 0$. □

Lemma 3.5.14 Let $C \preceq D$ and $D \succeq I$. Then $f(\gamma;x) \ge f(1,x)$ for all $0 < x \le \mathbf{e}$ such that $R(x) \succ 0$, $\mathbf{e}^{\mathsf{T}}x = s$, and $\gamma \ge 1$, where $p = p(\gamma)$ is given as in Corollary 3.5.9 for all γ.

Proof We show that $\partial f(\gamma;x)/\partial\gamma \ge 0$, when the hypotheses of the theorem hold. We define

$$\tilde{R}(\gamma;x) := C - D + \operatorname{Diag}\left(x^{-p(\gamma)}\right)\operatorname{Diag}\left((\gamma\mathbf{e})^{x-\mathbf{e}}\right)\operatorname{Diag}(d^x).$$

Then

$$R(\gamma;x) = \gamma\operatorname{Diag}\left(x^{p(\gamma)/2}\right)\tilde{R}(\gamma;x)\operatorname{Diag}\left(x^{p(\gamma)/2}\right)$$

and

$$f(\gamma;x) = n\log(\gamma) + \sum_{i\in N}p_i(\gamma)\log(x_i) + \operatorname{ldet}\tilde{R}(\gamma;x) - s\log(\gamma).$$

Therefore,

$$\frac{\partial f(\gamma;x)}{\partial\gamma} = \frac{n-s}{\gamma} + \sum_{i\in N}\log(x_i)\frac{\mathrm{d}p_i(\gamma)}{\mathrm{d}\gamma} + \frac{\partial}{\partial\gamma}\operatorname{ldet}\tilde{R}(\gamma;x).$$

By the chain rule and Lemma A.4.13, we have

$$\frac{\partial}{\partial\gamma}\operatorname{ldet}\tilde{R}(\gamma;x) = \sum_{i\in N}\frac{\partial\operatorname{ldet}\tilde{R}(\gamma;x)}{\partial\tilde{R}(\gamma;x)_{ii}}\frac{\partial\tilde{R}(\gamma;x)_{ii}}{\partial\gamma}\,.$$

Moreover,

$$\tilde{R}(\gamma;x)_{ii} = C_{ii} - d_i + x_i^{-p_i(\gamma)}\gamma^{x_i-1}d_i^{x_i},$$

and

$$\begin{aligned}\frac{\mathrm{d}p_i(\gamma)}{\mathrm{d}\gamma} &= \frac{2\left(1+\sqrt{1+4\log(\gamma d_i)}\right)}{4}\frac{1}{2\sqrt{1+4\log(\gamma d_i)}}\frac{4}{\gamma d_i}d_i \\ &= \frac{2\sqrt{p_i(\gamma)}}{\gamma\left(2\sqrt{p_i(\gamma)}-1\right)}.\end{aligned}$$

Then,

$$\begin{aligned}\frac{\partial\tilde{R}(\gamma;x)_{ii}}{\partial\gamma} &= \gamma^{x_i-1}d_i^{x_i}x_i^{-p_i(\gamma)}\log(x_i)\left(-\frac{\mathrm{d}p_i(\gamma)}{\mathrm{d}\gamma}\right) \\ &\quad + x_i^{-p_i(\gamma)}(x_i-1)\gamma^{x_i-2}d_i^{x_i} \\ &= -x_i^{-p_i(\gamma)}\log(x_i)\frac{2\sqrt{p_i(\gamma)}}{2\sqrt{p_i(\gamma)}-1}\gamma^{x_i-2}d_i^{x_i} \\ &\quad + x_i^{-p_i(\gamma)}(x_i-1)\gamma^{x_i-2}d_i^{x_i} \\ &= x_i^{-p_i(\gamma)}\gamma^{x_i-2}d_i^{x_i}\left(-\log(x_i)\frac{2\sqrt{p_i(\gamma)}}{2\sqrt{p_i(\gamma)}-1}+x_i-1\right) \\ &= \gamma^{-1}x_i^{-p_i(\gamma)}(\gamma d_i)^{x_i}\left(\frac{2\sqrt{p_i(\gamma)}\,(x_i-1-\log(x_i))+1-x_i}{\gamma\left(2\sqrt{p_i(\gamma)}-1\right)}\right),\end{aligned}$$

and

$$\begin{aligned}&\frac{\partial}{\partial\gamma}\operatorname{ldet}\tilde{R}(\gamma;x) \\ &= \sum_{i\in N}\tilde{R}(\gamma;x)_{ii}^{-1}\gamma^{-1}x_i^{-p_i(\gamma)}(\gamma d_i)^{x_i}\left(\frac{2\sqrt{p_i(\gamma)}(x_i-1-\log(x_i))+1-x_i}{\gamma\left(2\sqrt{p_i(\gamma)}-1\right)}\right) \\ &= \sum_{i\in N}R(\gamma;x)_{ii}^{-1}(\gamma d_i)^{x_i}\left(\frac{2\sqrt{p_i(\gamma)}(x_i-1-\log(x_i))+1-x_i}{\gamma\left(2\sqrt{p_i(\gamma)}-1\right)}\right) \\ &\geq \sum_{i\in N}\left(\frac{2\sqrt{p_i(\gamma)}(x_i-1-\log(x_i))+1-x_i}{\gamma\left(2\sqrt{p_i(\gamma)}-1\right)}\right),\end{aligned}$$

where the above inequality follows from (3.45) and also from $x_i \leq 1$, $p_i(\gamma) \geq 1$, $\gamma > 0$, and $\log(x_i) \leq x_i - 1$, which together assures that the expression on the right-hand side of the inequality is nonnegative.

Then,

$$\begin{aligned}
\frac{\partial f(\gamma; x)}{\partial \gamma} &\geq \frac{n-s}{\gamma} + \sum_{i \in N} \log(x_i) \frac{2\sqrt{p_i(\gamma)}}{\gamma\left(2\sqrt{p_i(\gamma)} - 1\right)} \\
&\quad + \sum_{i \in N} \frac{2\sqrt{p_i(\gamma)}\left(x_i - 1 - \log(x_i)\right) + 1 - x_i}{\gamma\left(2\sqrt{p_i(\gamma)} - 1\right)} \\
&= \frac{n-s}{\gamma} + \sum_{i \in N} \frac{\left(2\sqrt{p_i(\gamma)} - 1\right)(x_i - 1)}{\gamma\left(2\sqrt{p_i(\gamma)} - 1\right)} \\
&= \frac{n-s}{\gamma} + \sum_{i \in N} \frac{x_i - 1}{\gamma} \\
&= 0.
\end{aligned}$$

□

3.5.4 Strategies to select parameters for NLP$_\gamma$

Next, we establish different ways of selecting the parameters in NLP$_\gamma$ aiming at obtaining the best possible bound from its solution. Considering the results presented in §3.5.3, we suggest three strategies to select d and γ. Once these parameters are determined, p is given as in Corollary 3.5.9.

The existence of different strategies, identified below as *Identity*, *Diagonal*, and *Trace*, shows that our theoretical analysis does not determine a single choice for d and γ, but gives conditions that should be satisfied by them, namely

$$D \geq C, \ \gamma D \not\prec I, \text{ and } \gamma D \not\succ I,$$

where $D = \text{Diag}(d)$.

The choice of d in the three strategies, first guarantees that $D \geq C$. The selection of γ then, assures that $\gamma D \not\prec I$, and $\gamma D \not\succ I$. In the first strategy, the selection of γ is unique, but in the other two strategies, we have an interval of values that satisfy the conditions.

(Identity)

$$D := \rho I,\ \rho := \lambda_{\max}(C), \qquad \gamma := 1/\rho$$

(Diagonal)

$$D := \rho\Lambda,\ \rho := \lambda_{\max}(\Lambda^{-1/2}C\Lambda^{-1/2}),\ \Lambda := \mathrm{Diag}(C), \quad \gamma \in \left[\frac{1}{d_{\max}}, \frac{1}{d_{\min}}\right]$$

(Trace)

$$D := \mathrm{argmin}\{\mathrm{tr}(D) : D - C \succeq 0,\ D \text{ diagonal}\}, \qquad \gamma \in \left[\frac{1}{d_{\max}}, \frac{1}{d_{\min}}\right]$$

In the first strategy γD is simply the identity matrix. In the second strategy, the diagonal matrix Λ contains the variances of the covariance matrix C, and ρ is the greatest eigenvalue of the correlation matrix. In case C is diagonal and $m = 0$, this strategy has the advantage of turning NLP_γ into a linear program that has a 0/1 optimal solution solving the original MESP. In the third strategy, aiming at selecting small values for each d_i , we minimize the trace of D, subject to $D - C$ being positive semidefinite. We can only apply this strategy when C is positive definite (otherwise D could have zero diagonal entries).

The time to compute D in the first two strategies is dominated by computing the greatest eigenvalues of an n symmetric matrix. The third strategy requires more computational effort. To apply it, we need to solve a semidefinite-programming problem.

3.5.5 Duality and the logarithmic-barrier problem for gNLP

Now, let us consider the more general convex nonlinear programming problem

$$\text{(gNLP)} \qquad z^* := \max\{f(x) \ :\ Ex = h,\ Ax \le b,\ 0 \le x \le \mathbf{e}\},$$

where f is concave and has continuous first and second derivatives on the feasible set, $f(x) = \mathrm{ldet}\, C[S(x), S(x)]$ for $x \in \{0,1\}^n$, E is a full row-rank $k \times n$ matrix, and A is an $m \times n$ matrix, where we assume that

$$\mathcal{F}^0 := \{x \in \mathbb{R}^n \ :\ Ex = h,\ Ax < b,\ 0 < x < \mathbf{e}\}$$

is nonempty and bounded. For example, $f(x) := \operatorname{ldet} R(x)$ (i.e., the NLP objective function) and $f(x) := \frac{1}{2} \operatorname{ldet} L(x)$ (i.e., the linx objective function) satisfy the assumptions for $f(x)$ in gNLP.

We will present an interior-point algorithm for gNLP, which requires the hypotheses concerning $\mathcal{F}^0$. To put NLP (or linx) into the format of gNLP, $Ex = h$ should contain the constraint $\mathbf{e}^\mathsf{T} x = s$. Furthermore, if some constraints of $Ax \leq b$ of NLP (or linx) hold as equations for all feasible x in NLP (or linx), we should identify those constraints and rewrite them as equality constraints, moving them to the set of equations $Ex = h$ in gNLP. Finally, if some constraints of $0 \leq x \leq \mathbf{e}$ in NLP (or linx) hold as equations for all feasible x in NLP (or linx), we would have to remove the associated variables from the formulation of gNLP by substituting in 0 or 1 for each such variable, as appropriate. In this way, we obtain a formulation satisfying the assumption of gNLP, that $\mathcal{F}^0$ is nonempty.

Next, we describe the interior-point algorithm applied to solve gNLP and show how to construct a feasible dual solution for the problem. We use the constructed dual solution to generate a valid upper bound and to fix primal variables in the execution of the branch-and-bound algorithm.

The Lagrangian function associated to gNLP is

$$\mathcal{L}(x, \upsilon, \nu, \pi, \tau) := f(x) + \upsilon^\mathsf{T} x + \nu^\mathsf{T}(\mathbf{e} - x) + \pi^\mathsf{T}(b - Ax) + \tau^\mathsf{T}(h - Ex),$$

with $\operatorname{dom}\mathcal{L} = \operatorname{dom} f \times \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^k$. The corresponding dual function is

$$\mathcal{L}^*(\upsilon, \nu, \pi, \tau) := \sup_{x \in \operatorname{dom} f} \mathcal{L}(x, \upsilon, \nu, \pi, \tau), \tag{3.52}$$

and the Lagrangian dual problem of gNLP is

$$\min\{\mathcal{L}^*(\upsilon, \nu, \pi, \tau) \ : \ \upsilon \geq 0, \nu \geq 0, \pi \geq 0\}.$$

Provided that f is concave and all constraints in gNLP are linear, the supremum in (3.52) occurs where the gradient of the Lagrangian function with respect to x is equal to zero. Then, the Lagrangian dual problem may be formulated as

(DgNLP)
$$\begin{aligned} \min\ & f(x) + \upsilon^\mathsf{T} x + \nu^\mathsf{T}(\mathbf{e} - x) + \pi^\mathsf{T}(b - Ax) + \tau^\mathsf{T}(h - Ex) \\ & \text{subject to:} \\ & \nabla f(x) + \upsilon - \nu - A^\mathsf{T}\pi - E^\mathsf{T}\tau = 0, \\ & \upsilon \geq 0, \nu \geq 0, \pi \geq 0. \end{aligned}$$

The necessary and sufficient Karush-Kuhn-Tucker (KKT) optimality conditions for gNLP are given by

$$Ex = h,\ Ax \le b,\ 0 \le x \le \mathbf{e}, \tag{3.53}$$
$$\upsilon - \nu - A^{\mathsf{T}}\pi - E^{\mathsf{T}}\tau = -\nabla f(x), \tag{3.54}$$
$$\upsilon \ge 0, \nu \ge 0, \pi \ge 0, \tag{3.55}$$
$$\upsilon^{\mathsf{T}}x = 0,\ \nu^{\mathsf{T}}(\mathbf{e} - x) = 0,\ \pi^{\mathsf{T}}(b - Ax) = 0, \tag{3.56}$$

where (3.53) represents primal feasibility, (3.54)-(3.55) represents dual feasibility, and (3.56) represents the complementarity conditions.

Proposition 3.5.15 (Weak Duality) If $(x, \upsilon, \nu, \pi, \tau)$ satisfies (3.53)-(3.55), then

$$f(x) \le z^* \le f(x) + \upsilon^{\mathsf{T}}x + \nu^{\mathsf{T}}(\mathbf{e} - x) + \pi^{\mathsf{T}}(b - Ax). \tag{3.57}$$

Now, we consider, for $\mu > 0$, the logarithmic-barrier function

$$F_\mu(x) := -f(x) - \mu \sum_{i \in N} \log(x_i) - \mu \sum_{i \in N} \log(1 - x_i) - \mu \sum_{i \in M} \log\left(b_i - A[i, \cdot]x\right), \tag{3.58}$$

and the logarithmic-barrier problem

$$\min\left\{F_\mu(x) \ :\ Ex = h,\ Ax < b,\ 0 < x < \mathbf{e}\right\}. \tag{P_μ}$$

For fixed μ, the KKT optimality conditions for P_μ are given by

$$E^{\mathsf{T}}\tau = \nabla f(x) + \upsilon - \nu - A^{\mathsf{T}}\pi, \tag{3.59}$$
$$Ex = h, \tag{3.60}$$
$$\upsilon \circ x = \mu\mathbf{e},\ \nu \circ (\mathbf{e} - x) = \mu\mathbf{e},\ \pi \circ (b - Ax) = \mu\mathbf{e}. \tag{3.61}$$

Therefore, if $x(\mu)$ solves P_μ , there is a $\tau(\mu)$, such that

$$\left(x(\mu), \upsilon(\mu) := \mu x(\mu)^{-1}, \nu(\mu) := \mu(\mathbf{e} - x(\mu))^{-1}, \pi(\mu) := \mu(b - Ax(\mu))^{-1}, \tau(\mu)\right)$$

satisfies (3.59)-(3.61), where for any vector $w > 0$, w^{-1} denotes the vector whose components are the reciprocals of the components of w. Clearly, it also satisfies (3.53)-(3.55), that is, $x(\mu)$ is a primal feasible solution and $(\upsilon(\mu), \nu(\mu), \pi(\mu), \tau(\mu))$ is a dual feasible solution of gNLP. Thus, replacing these solutions in (3.57), we obtain the following bounds for z^*.

$$f(x(\mu)) \leq z^* \leq f(x(\mu)) + (2n+m)\mu. \tag{3.62}$$

The logarithmic-barrier function is strictly convex in the interior of the feasible region of gNLP. Then, let $(x(\mu), \upsilon(\mu), \nu(\mu), \pi(\mu), \tau(\mu))$, be the (unique) solution of (3.59)-(3.61). These values form the primal-dual central path of gNLP, defined as

$$C := \left\{ \left(x(\mu), \upsilon(\mu), \nu(\mu), \pi(\mu), \tau(\mu) \right) \ : \ 0 < \mu < \infty \right\}. \tag{3.63}$$

Furthermore, $(x(\mu), \upsilon(\mu), \nu(\mu), \pi(\mu), \tau(\mu))$ converges to a point satisfying the KKT conditions (3.53)-(3.56) as $\mu \to 0$. The central path for linear programming was analyzed in [Meg89, BL89a, BL89b, Son86], and for the more general convex case in [FM68, McL80].

We conclude that, from a point on the primal-dual central path, for a given $\mu > 0$, we can compute an upper bound for gNLP that converges to z^* as $\mu \to 0$.

Nevertheless, in practice, the logarithmic-barrier problem is only solved approximately. So, in the following, we show how to compute a dual feasible solution and an upper bound for gNLP, from an approximate solution of P_μ .

Let $\hat{x}$ be a feasible solution of P_μ that is only an approximate minimizer. We set $\hat{\pi} := \mu(b - A\hat{x})^{-1}$, and consider the problem

$$\begin{aligned} \min \ & \|\upsilon \circ \hat{x} - \mu \mathbf{e}\|^2 + \|\nu \circ (\mathbf{e} - \hat{x}) - \mu \mathbf{e}\|^2 \\ & \text{subject to:} \\ & \upsilon - \nu - E^{\mathsf{T}}\tau = -\nabla f(\hat{x}) + A^{\mathsf{T}}\hat{\pi}. \end{aligned} \tag{3.64}$$

If $\hat{x}$ is an optimal solution for P_μ , the optimal solution value of (3.64) is zero. Otherwise, we can interpret problem (3.64) as a search for $\hat{\upsilon}$, $\hat{\nu}$, and $\hat{\tau}$, that together with the given $\hat{x}$ and $\hat{\pi}$, satisfy (3.53)-(3.55) and minimize the distance of the primal-dual solution to the central path.

The necessary and sufficient KKT optimality conditions for (3.64) are given by

$$(\mathrm{Diag}(\hat{x}))^2 \upsilon - \mu\hat{x} - y = 0, \tag{3.65}$$

$$(\mathrm{Diag}(\mathbf{e} - \hat{x}))^2 \nu - \mu(\mathbf{e} - \hat{x}) + y = 0, \tag{3.66}$$

$$\upsilon - \nu - E^{\mathsf{T}}\tau = -\nabla f(\hat{x}) + A^{\mathsf{T}}\hat{\pi}, \tag{3.67}$$

$$Ey = 0, \tag{3.68}$$

where y is the dual variable corresponding to the equation in (3.64). We see that the convex optimization problem (3.64) has only linear constraints, is feasible and its optimal value is bounded below (by zero). Therefore optimal solutions are characterized by its KKT conditions.

Using (3.65) and (3.66), we obtain

$$\upsilon - \nu = \left((\mathrm{Diag}(\hat{x}))^{-2} + (\mathrm{Diag}(\mathbf{e} - \hat{x}))^{-2} \right) y + \mu\hat{x}^{-1} - \mu(\mathbf{e} - \hat{x})^{-1}. \tag{3.69}$$

Replacing (3.69) in (3.67), and considering that $\hat{\pi} := \mu(b - A\hat{x})^{-1}$ and

$$\nabla F_\mu(\hat{x}) := -\nabla f(\hat{x}) - \mu\hat{x}^{-1} - \mu(\mathbf{e}-\hat{x})^{-1} - \mu A^{\mathsf{T}}(b - A\hat{x})^{-1}, \tag{3.70}$$

we rewrite (3.67)-(3.68) as

$$\left((\mathrm{Diag}(\hat{x}))^{-2} + (\mathrm{Diag}(\mathbf{e}-\hat{x}))^{-2}\right) y - E^{\mathsf{T}}\tau = \nabla F_\mu(\hat{x}), \tag{3.71}$$

$$Ey = 0. \tag{3.72}$$

Finally, letting $T := \left((\mathrm{Diag}(\hat{x}))^{-2} + (\mathrm{Diag}(\mathbf{e}-\hat{x}))^{-2}\right)^{-2}$, we rewrite (3.71)-(3.72) as

$$T^{-1}y - (ET)^{\mathsf{T}}\tau = T\nabla F_\mu(\hat{x}), \tag{3.73}$$

$$(ET)(T^{-1}y) = 0. \tag{3.74}$$

From (3.74) we see that $T^{-1}y$ is in the null space of ET, and therefore, from (3.73), we see that $T^{-1}y$ is the orthogonal projection of $T\nabla F_\mu(\hat{x})$ onto the null space of ET. Computing $t := T^{-1}y$, $\phi := E^{\mathsf{T}}\tau$ can then be computed from (3.71). Finally, considering (3.67), or equivalently, (3.69), we set

$$\hat{v} := \left[-\nabla f(\hat{x}) + \mu A^{\mathsf{T}}\hat{\pi} + \phi\right]^+ = \left[\mu\hat{x}^{-1} - \mu(\mathbf{e}-\hat{x})^{-1} + T^{-1}t\right]^+,$$
$$\hat{v} := \left[-\nabla f(\hat{x}) + \mu A^{\mathsf{T}}\hat{\pi} + \phi\right]^- = \left[\mu\hat{x}^{-1} - \mu(\mathbf{e}-\hat{x})^{-1} + T^{-1}t\right]^-.$$

It is easy to verify that $\hat{x}, \hat{v}, \hat{v}, \hat{\pi}$, and $\phi(= E^{\mathsf{T}}\tau)$ satisfy (3.67)-(3.68). They also satisfy (3.53)-(3.55), and can be used in (3.57) to generate an upper bound to gNLP.

In Algorithm 4, we present the procedure to obtain, from a feasible solution $\hat{x}$ of P_μ , a feasible solution $\hat{v}, \hat{v}, \hat{\pi}, \hat{\tau}$ of DgNLP. Using this solution of DgNLP, the procedure also computes an upper bound for z^* (the optimal value of gNLP).

Algorithm 4: Dual solution and upper bound for gNLP

Input: Feasible solution $\hat{x}$ of P_μ , $\mu > 0$;

Output: Feasible solution $\hat{v}, \hat{v}, \hat{\pi}, \hat{\tau}$ of DgNLP and upper bound $\hat{\zeta}$ for z^* ;

$\hat{\pi} := \mu(b - A\hat{x})^{-1}$;

$T := \left((\mathrm{Diag}(\hat{x}))^{-2} + (\mathrm{Diag}(\mathbf{e}-\hat{x}))^{-2}\right)^{-2}$;

$t :=$Proj$_{\text{n.s.}(ET)}\, T\nabla F_\mu(\hat{x})$;

$\phi := \nabla F_\mu(\hat{x}) - T^{-1}t$;

Compute a solution $\hat{\tau}$ of $\phi = E^{\mathsf{T}}\tau$;

$\hat{v} := \left[\mu\hat{x}^{-1} - \mu(\mathbf{e}-\hat{x})^{-1} + T^{-1}t\right]^+$;

$\hat{v} := \left[\mu\hat{x}^{-1} - \mu(\mathbf{e}-\hat{x})^{-1} + T^{-1}t\right]^-$;

$\hat{\zeta} := f(\hat{x}) + \hat{v}^{\mathsf{T}}\hat{x} + \hat{v}^{\mathsf{T}}(\mathbf{e}-\hat{x}) + m\mu$;

3.5.6 Fixing variables in gNLP

Theorem 3.5.16 Let:

- LB be the objective-function value of a feasible solution of CMESP,
- $\hat{x}$ be a feasible solution of the logarithmic-barrier problem P_μ , for given μ,
- $\hat{\upsilon}, \hat{\nu}, \hat{\pi}$ and $\hat{\zeta}$ be obtained by Algorithm 4, for $\hat{x}$ and μ.

Then, for every optimal solution x^* for CMESP, we have:

$$x_k^* = 0, \ \forall\, k \in N \text{ such that } \hat{\zeta} - \text{LB} < \hat{\upsilon}_k \ ,$$
$$x_k^* = 1, \ \forall\, k \in N \text{ such that } \hat{\zeta} - \text{LB} < \hat{\nu}_k \ .$$

Proof Consider gNLP with the additional constraint $x_k = 1$. The dual becomes then,

$$\begin{aligned} \min\ & f(x) + \upsilon^\mathsf{T} x + \nu^\mathsf{T}(\mathbf{e} - x) + \pi^\mathsf{T}(b - Ax) + \tau^\mathsf{T}(h - Ex) - \omega \\ & \text{subject to:} \\ & \nabla f(x) + \upsilon - \nu - A^\mathsf{T}\pi - E^\mathsf{T}\tau + \omega \mathbf{e}_k = 0, \\ & \upsilon \geq 0, \nu \geq 0, \pi \geq 0, \end{aligned}$$

where ω is the new dual variable. Notice that if $(\hat{x}, \hat{\upsilon}, \hat{\nu}, \hat{\pi}, \hat{\tau})$ is a feasible solution for DgNLP, then $(\hat{x}, \hat{\upsilon} - \omega\mathbf{e}_k, \hat{\nu}, \hat{\pi}, \hat{\tau}, \omega)$ is a feasible solution of this modified dual, *as long as* $\hat{\upsilon}_k - \omega \geq 0$.The objective value of this modified dual is $\hat{\zeta} - \omega$. So, to minimize the objective value of our feasible solution of this modified dual, we set ω equal to $\hat{\upsilon}_k$. We conclude that $\hat{\zeta} - \hat{\upsilon}_k$ is an upper bound on objective value of all solutions of CMESP that satisfy $x_k = 1$. So if $\hat{\zeta} - \hat{\upsilon}_k < LB$, then no optimal solution of CMESP can have $x_k = 1$.

Similarly, consider gNLP with the additional constraint $x_k = 0$. The dual becomes instead,

$$\begin{aligned} \min\ & f(x) + \upsilon^\mathsf{T} x + \nu^\mathsf{T}(\mathbf{e} - x) + \pi^\mathsf{T}(b - Ax) + \tau^\mathsf{T}(h - Ex) \\ & \text{subject to:} \\ & \nabla f(x) + \upsilon - \nu - A^\mathsf{T}\pi - E^\mathsf{T}\tau + \omega \mathbf{e}_k = 0, \\ & \upsilon \geq 0, \nu \geq 0, \pi \geq 0, \end{aligned}$$

where ω is again the new dual variable. Notice that if $(\hat{x}, \hat{\upsilon}, \hat{\nu}, \hat{\pi}, \hat{\tau})$ is a feasible solution for DgNLP, then $(\hat{x}, \hat{\upsilon}, \hat{\nu}+\omega\mathbf{e}_k, \hat{\pi}, \hat{\tau}, \omega)$ is a feasible solution of this modified dual, *as long as* $\hat{\nu}_k + \omega \geq 0$.The objective value of this modified dual is $\hat{\zeta} + \omega$. So, to minimize the objective value of our feasible solution of this modified dual, we set ω equal to $-\hat{\nu}_i$. We conclude that $\hat{\zeta} - \hat{\nu}_k$ is an upper bound on objective value of all solutions of CMESP that satisfy $x_k = 0$. So if $\hat{\zeta} - \hat{\nu}_k < LB$, then no optimal solution of CMESP can have $x_k = 0$. □

3.5.7 The logarithmic-barrier algorithm for gNLP

Next, we present the logarithmic-barrier algorithm, which obtains a feasible solution of gNLP, such that the difference between the objective function value at this solution and the optimal value of the problem, is no more than a given $\epsilon > 0$.

The logarithmic-barrier algorithm deals with the inequality constraints of gNLP, by adding a barrier term to the objective function, and solving the logarithm-barrier problem defined in P_μ . The barrier term keeps the solution in the interior of the feasible set, and therefore, prevents it from becoming infeasible. In order to produce a sequence of iterates that converges to the optimum, the influence of the barrier, measured by the barrier parameter μ, successively decreases as the optimization process progresses.

The logarithmic-barrier function was introduced by Frisch [Fri56, Fri61] and the technique used in the algorithm, known as Sequential Unconstrained Minimization Technique (SUMT), was proposed by Fiacco and McCormick [FM68]. Generally speaking, we can say that this algorithm follows the central path defined in (3.63), and forms the core of interior-point methods.

Den Hertog [dH92] analyzed the logarithmic-barrier algorithm for three important classes of problems: Linear Programming, Convex Quadratic Programming and Convex Programming where the objective function satisfies the self-concordance condition introduced by Nesterov and Nemirovsky [NN94].

The algorithm consists of outer and inner iterations. Starting with the barrier parameter $\mu := \mu_0$, at each outer iteration, we decrease its value by setting $\mu := (1 - \theta)\mu$, for a given $\theta \in (0, 1)$. Depending on how θ is selected, the algorithm is classified as a short, medium or long-step path following algorithm. We consider here a long-step path following algorithm. In this case, θ is a constant that depends neither on the number of variables nor on the tolerance of the solution. Although stronger theoretical results can be proved for short and medium-step path-following algorithms, long steps are, in general, more efficient in practice.

We stop the outer iterations when $\mu \leq \epsilon/(4n+2m)$. Denoting by $x(\mu)$, the optimal solution of the barrier problem for the final μ in the algorithm, we have from (3.62), that

$$f(x(\mu)) \leq z^* \leq f(x(\mu)) + \epsilon/2. \tag{3.75}$$

At each outer iteration of the algorithm, we execute inner iterations. At each inner iteration, we perform a line search on the Newton direction given by

$$\xi := -\left(\nabla^2 F_\mu(x)\right)^{-1} \nabla F_\mu(x),$$

to approximately minimize the barrier function for the current value of μ. We note that not even in the theoretical analysis of the algorithm, is it necessary to minimize the barrier function exactly.

The number of inner iterations is determined by the same stopping criterion in all outer iterations, except the last one. The stopping criterion is $\|\xi\|_{H_B} \leq \tau$,

for a given $\tau > 0$, where $H_B := \nabla^2 F_\mu(x)$ and $\|\xi\|_{H_B} := \sqrt{\xi^\top H_B \xi}$. The stopping criterion measures the proximity of the approximate solution x to $x(\mu)$. We note that $\|\xi\|_{H_B} = 0$ if and only if $x = x(\mu)$.

On the last outer iteration, we continue to perform damped Newton steps until the precision on the approximate solution x of the barrier problem is enough to lead to an upper bound, constructed by Algorithm 4, that is smaller than $f(x) + \epsilon$. We note that (3.75) assures the termination of the algorithm.

The logarithmic-barrier algorithm is detailed in Algorithm 5.

Algorithm 5: Logarithmic-barrier algorithm

Input: $\epsilon > 0$, $\tau > 0$, $\theta \in (0, 1)$, $\mu_0 > 0$, x^0;
Output: $\hat{x}$, μ;
$\hat{x} := x^0$;
$\xi := -\left(\nabla^2 F_\mu(\hat{x})\right)^{-1} \nabla F_\mu(\hat{x})$;
$\mu := \mu_0$;
while $\mu > \epsilon/(4n + 2m)$ **do**
  $\mu := (1 - \theta)\mu$;
  while $\|\xi\|_{H_B} \geq \tau$ **do**
    $\tilde{\alpha} := \operatorname{argmin} \left\{F_\mu(\hat{x} + \alpha\xi) \ : \ \hat{x} + \alpha\xi \in \mathcal{F}^0\right\}$;
    $\hat{x} := \hat{x} + \tilde{\alpha}\xi$;
    $\xi := -\left(\nabla^2 F_\mu(\hat{x})\right)^{-1} \nabla F_\mu(\hat{x})$;
  end while
end while
Compute $\hat{v}$ and $\hat{\nu}$ with Algorithm 4;
while $\hat{x}^\top \hat{v} + (\mathbf{e} - \hat{x})^\top \hat{\nu} + m\mu > \epsilon$ **do**
  $\tilde{\alpha} := \operatorname{argmin} \left\{F_\mu(\hat{x} + \alpha\xi) \ : \ \hat{x} + \alpha\xi \in \mathcal{F}^0\right\}$;
  $\hat{x} := \hat{x} + \tilde{\alpha}\xi$;
  $\xi := -\left(\nabla^2 F_\mu(\hat{x})\right)^{-1} \nabla F_\mu(\hat{x})$;
  Compute $\hat{v}$ and $\hat{\nu}$ with Algorithm 4;
end while

3.5.8 NLP$_\gamma$ in the branch-and-bound algorithm

In a branch-and-bound algorithm, the division of a problem into subproblems is accomplished by fixing indices in and out of S. In this section, we analyze cases where it is possible to guarantee that the upper bound given by NLP$_\gamma$ for a parent in the branch-and-bound search is greater than or equal to the upper bound given for its child. We note that this result is not trivial, as the parameters of the relaxation

may change after the indices are fixed. In the following, we denote the objective functions of the parent and child subproblems, respectively by $f^{\mathfrak{p}}$ and $f^{\mathfrak{c}}$, and the optimal solutions to the parent and child subproblems, respectively by y^* and x^*. Note that assuming that these optimal exist, we have that $R(y^*) \succ 0$ and $R(x^*) \succ 0$.

3.5.8.1 The "Identity" case

- Fixing the index i out of S.
  In this case, we have

$$f^{\mathfrak{p}}(\gamma_{\mathfrak{p}}; y) = \operatorname{ldet}(\operatorname{Diag}(y^{\mathbf{e}/2})(\gamma_{\mathfrak{p}} C^{\mathfrak{p}} - I)\operatorname{Diag}(y^{\mathbf{e}/2}) + I) - s\log(\gamma_{\mathfrak{p}}), \tag{3.76}$$

$$f^{\mathfrak{c}}(\gamma_{\mathfrak{c}}; x) = \operatorname{ldet}(\operatorname{Diag}(x^{\mathbf{e}/2})(\gamma_{\mathfrak{c}} C^{\mathfrak{c}} - I)\operatorname{Diag}(x^{\mathbf{e}/2}) + I) - s\log(\gamma_{\mathfrak{c}}), \tag{3.77}$$

  where $C^{\mathfrak{c}}$ is obtained from $C^{\mathfrak{p}}$ by removing its row and column i, $\gamma_{\mathfrak{p}} = 1/\lambda_{\max}(C^{\mathfrak{p}})$, and $\gamma_{\mathfrak{c}} = 1/\lambda_{\max}(C^{\mathfrak{c}})$.

Theorem 3.5.17 Let $f^{\mathfrak{p}}$ and $f^{\mathfrak{c}}$ be defined as in (3.76) and (3.77). Then

$$f^{\mathfrak{p}}(\gamma_{\mathfrak{p}}; y^*) \geq f^{\mathfrak{c}}(\gamma_{\mathfrak{c}}; x^*).$$

Proof Let

$$\hat{y}_j := \begin{cases} x^*_j, & \text{for } j < i; \\ 0, & \text{for } j = i; \\ x^*_{j-1}, & \text{for } j > i. \end{cases}$$

As $\hat{y}$ is a feasible solution to the parent problem, it suffices to show that $f^{\mathfrak{p}}(\gamma_{\mathfrak{p}}; \hat{y}) \geq f^{\mathfrak{c}}(\gamma_{\mathfrak{c}}; x^*)$. We note that

$$\begin{aligned} f^{\mathfrak{p}}(\gamma_{\mathfrak{p}}; \hat{y}) &= \operatorname{ldet}\left(\operatorname{Diag}\left(x^{*\mathbf{e}/2}\right)(\gamma_{\mathfrak{p}} C^{\mathfrak{c}} - I)\operatorname{Diag}\left(x^{*\mathbf{e}/2}\right) + I\right) - s\log(\gamma_{\mathfrak{p}}) \\ &= f^{\mathfrak{c}}(\gamma_{\mathfrak{p}}; x^*). \end{aligned}$$

As $0 < \gamma_{\mathfrak{p}} \leq \gamma_{\mathfrak{c}}$, it suffices to show that $\partial f^{\mathfrak{c}}(\gamma; x)/\partial\gamma \leq 0$, for $\gamma > 0$ and $0 \leq x \leq \mathbf{e}$, such that $\mathbf{e}^{\mathsf{T}} x = s$, which we prove in the following.
We define

$$\tilde{R}(\gamma; x) := \operatorname{Diag}\left(x^{\mathbf{e}/2}\right)\left(C^{\mathfrak{c}} - \gamma^{-1} I\right)\operatorname{Diag}\left(x^{\mathbf{e}/2}\right) + \gamma^{-1} I.$$

Then,

$$R(\gamma; x) = \gamma \tilde{R}(\gamma; x)$$

and

$$f^{\mathfrak{c}}(\gamma; x) = n\log(\gamma) + \operatorname{ldet}\tilde{R}(\gamma; x) - s\log(\gamma).$$

Therefore,

$$\frac{\partial}{\partial\gamma} f^{\mathfrak{c}}(\gamma;x) = \frac{n-s}{\gamma} + \frac{\partial}{\partial\gamma} \operatorname{ldet} \tilde{R}(\gamma;x). \tag{3.78}$$

By the chain rule and Lemma A.4.13, we have

$$\begin{aligned}
\frac{\partial}{\partial\gamma} \operatorname{ldet} \tilde{R}(\gamma;x) &= \sum_{i\in N} \frac{\partial \operatorname{ldet} \tilde{R}(\gamma;x)}{\partial \tilde{R}(\gamma;x)_{ii}} \frac{\partial \tilde{R}(\gamma;x)_{ii}}{\partial\gamma} \\
&= \sum_{i\in N} \tilde{R}(\gamma;x)_{ii}^{-1}(x_i-1)\gamma^{-2} \\
&= \tfrac{1}{\gamma} \sum_{i\in N} R(\gamma;x)_{ii}^{-1}(x_i-1) \qquad (3.79) \\
&\le \tfrac{1}{\gamma} \sum_{i\in N} (x_i-1) \\
&= \tfrac{s-n}{\gamma},
\end{aligned}$$

where, the inequality in (3.79) follows from $x_i \le 1$ and $R(\gamma;x)_{ii}^{-1} \ge 1$ (consider (3.45), with $\gamma d_i = 1$).
Then, from (3.78) and (3.79), we have $\partial f^{\mathfrak{c}}(\gamma;x)/\partial\gamma \le 0$. □

- Fixing the index i in S.
  In this case, we have

$$f^{\mathfrak{p}}(\gamma_{\mathfrak{p}};y) = \operatorname{ldet}\left(\operatorname{Diag}\left(y^{\mathbf{e}/2}\right)(\gamma_{\mathfrak{p}}C^{\mathfrak{p}} - I)\operatorname{Diag}\left(y^{\mathbf{e}/2}\right) + I\right) - s\log(\gamma_{\mathfrak{p}}), \tag{3.80}$$

$$\begin{aligned}
f^{\mathfrak{c}}(\gamma_{\mathfrak{c}};x) = \operatorname{ldet}&\left(\operatorname{Diag}\left(x^{\mathbf{e}/2}\right)(\gamma_{\mathfrak{c}}C^{\mathfrak{c}} - I)\operatorname{Diag}\left(x^{\mathbf{e}/2}\right) + I\right) \\
&- (s-1)\log(\gamma_{\mathfrak{c}}) + \log\left(C^{\mathfrak{p}}_{ii}\right),
\end{aligned} \tag{3.81}$$

where $C^{\mathfrak{c}}$ is the Schur complement of $C^{\mathfrak{p}}_{ii}$ in $C^{\mathfrak{p}}$, $\gamma_{\mathfrak{p}} = 1/\lambda_{\max}(C^{\mathfrak{p}})$, and $\gamma_{\mathfrak{c}} = 1/\lambda_{\max}(C^{\mathfrak{c}})$.

Theorem 3.5.18 Let $f^{\mathfrak{p}}$ and $f^{\mathfrak{c}}$ be defined as in (3.80) and (3.81). Then

$$f^{\mathfrak{p}}(\gamma_{\mathfrak{p}};y^*) \ge f^{\mathfrak{c}}(\gamma_{\mathfrak{c}};x^*).$$

Proof Let

$$\hat{y}_j := \begin{cases} x^*_j, & \text{for } j < i; \\ 1, & \text{for } j = i; \\ x^*_{j-1}, & \text{for } j > i. \end{cases}$$

As $\hat{y}$ is a feasible solution to the parent problem, it suffices to show that $f^{\mathfrak{p}}(\gamma_{\mathfrak{p}}; \hat{y}) \geq f^{\mathfrak{c}}(\gamma_{\mathfrak{c}}; x^*)$.
We note that

$$\begin{aligned} f^{\mathfrak{p}}(\gamma_{\mathfrak{p}}; \hat{y}) &= \log\left(\gamma_{\mathfrak{p}} C^{\mathfrak{p}}_{ii}\right) + \operatorname{ldet}\left(\operatorname{Diag}\left(x^{*\mathbf{e}/2}\right)(\gamma_{\mathfrak{p}} C^{\mathfrak{c}} - I)\operatorname{Diag}\left(x^{*\mathbf{e}/2}\right) + I\right) \\ &\quad - s\log(\gamma_{\mathfrak{p}}), \\ &= \log\left(C^{\mathfrak{p}}_{ii}\right) + \operatorname{ldet}\left(\operatorname{Diag}\left(x^{*\mathbf{e}/2}\right)(\gamma_{\mathfrak{p}} C^{\mathfrak{c}} - I)\operatorname{Diag}\left(x^{*\mathbf{e}/2}\right) + I\right) \\ &\quad - (s-1)\log(\gamma_{\mathfrak{p}}) \\ &= f^{\mathfrak{c}}(\gamma_{\mathfrak{p}}; x^*). \end{aligned}$$

As $C^{\mathfrak{p}}$ is positive semidefinite, $\lambda_{\max}(C^{\mathfrak{p}}) \geq \lambda_{\max}(C^{\mathfrak{c}})$ (see Theorem A.4.7). Therefore, $0 < \gamma_{\mathfrak{p}} \leq \gamma_{\mathfrak{c}}$ and it suffices to show that $\partial f^{\mathfrak{c}}(\gamma; x)/\partial\gamma \leq 0$, for $\gamma > 0$ and $0 \leq x \leq \mathbf{e}$, such that $\mathbf{e}^{\mathsf{T}} x = s - 1$. The remaining of the proof is analogous to the proof of $\partial f^{\mathfrak{c}}(\gamma; x)/\partial\gamma \leq 0$ in Theorem 3.5.17. □

3.5.8.2 The "Diagonal" case

In the "Diagonal" case we have $D = \operatorname{Diag}(C)$, $\gamma \in [1/d_{\max}, 1/d_{\min}]$, and p given as in Corollary 3.5.9. Here, we are only able to analyze the selection $\gamma = 1/d_{\max}$.

- Fixing the index i out of S.
  In this case, we have

$$f^{\mathfrak{p}}(\gamma_{\mathfrak{p}}, \rho_{\mathfrak{p}}; y) = \operatorname{ldet}\left(\gamma_{\mathfrak{p}} \operatorname{Diag}\left(y^{\mathbf{e}/2}\right)(C^{\mathfrak{p}} - D_{\mathfrak{p}})\operatorname{Diag}\left(y^{\mathbf{e}/2}\right) + \operatorname{Diag}\left((\gamma_{\mathfrak{p}} d_{\mathfrak{p}})^{y}\right)\right) - s\log(\gamma_{\mathfrak{p}}), \tag{3.82}$$

$$f^{\mathfrak{c}}(\gamma_{\mathfrak{c}}, \rho_{\mathfrak{c}}; x) = \operatorname{ldet}\left(\gamma_{\mathfrak{c}} \operatorname{Diag}\left(x^{\mathbf{e}/2}\right)(C^{\mathfrak{c}} - D_{\mathfrak{c}})\operatorname{Diag}\left(x^{\mathbf{e}/2}\right) + \operatorname{Diag}\left((\gamma_{\mathfrak{c}} d_{\mathfrak{c}})^{x}\right)\right) - s\log(\gamma_{\mathfrak{c}}), \tag{3.83}$$

  where

  - $C^{\mathfrak{c}}$ is obtained from $C^{\mathfrak{p}}$ by removing its row and column i,
  - $D_{\mathfrak{p}} = \rho_{\mathfrak{p}} \operatorname{Diag}(C^{\mathfrak{p}})$, $D_{\mathfrak{c}} = \rho_{\mathfrak{c}} \operatorname{Diag}(C^{\mathfrak{c}})$, with

$$\rho_{\mathfrak{p}} = \lambda_{\max}\left(\operatorname{Diag}(C^{\mathfrak{p}})^{-\frac{1}{2}} C^{\mathfrak{p}} \operatorname{Diag}(C^{\mathfrak{p}})\right)^{-\frac{1}{2}},$$

$$\rho_{\mathfrak{c}} = \lambda_{\max}\left(\operatorname{Diag}(C^{\mathfrak{c}})^{-\frac{1}{2}} C^{\mathfrak{c}} \operatorname{Diag}(C^{\mathfrak{c}})\right)^{-\frac{1}{2}},$$

  - $d_{\mathfrak{p}} = \operatorname{diag}(D_{\mathfrak{p}})$, $d_{\mathfrak{c}} = \operatorname{diag}(D_{\mathfrak{c}})$,
  - $\gamma_{\mathfrak{p}} = 1/d_{\mathfrak{p}_{\max}}$, $\gamma_{\mathfrak{c}} = 1/d_{\mathfrak{c}_{\max}}$.

Theorem 3.5.19 Let $f^{\mathfrak{p}}$ and $f^{\mathfrak{c}}$ be defined as in (3.82) and (3.83). Then

$$f^{\mathfrak{p}}(\gamma_{\mathfrak{p}}, \rho_{\mathfrak{p}}; y^*) \geq f^{\mathfrak{c}}(\gamma_{\mathfrak{c}}, \rho_{\mathfrak{c}}; x^*).$$

Proof Let

$$\hat{y}_j := \begin{cases} x^*_j\,, & \text{for } j < i; \\ 0, & \text{for } j = i; \\ x^*_{j-1}\,, & \text{for } j > i. \end{cases}$$

As $\hat{y}$ is a feasible solution to the parent problem, it suffices to show that $f^{\mathfrak{p}}(\gamma_{\mathfrak{p}}, \rho_{\mathfrak{p}}; \hat{y}) \geq f^{\mathfrak{c}}(\gamma_{\mathfrak{c}}, \rho_{\mathfrak{c}}; x^*)$. We note that

$$f^{\mathfrak{p}}(\gamma_{\mathfrak{p}}, \rho_{\mathfrak{p}}; \hat{y}) = \operatorname{ldet}\left(\operatorname{Diag}\left(x^{*\mathbf{e}/2}\right)\left(\gamma_{\mathfrak{p}} C^{\mathfrak{c}} - \gamma_{\mathfrak{p}} \frac{\rho_{\mathfrak{p}}}{\rho_{\mathfrak{c}}} D_{\mathfrak{c}}\right) \operatorname{Diag}\left(x^{*\mathbf{e}/2}\right) + \operatorname{Diag}\left(\left(\gamma_{\mathfrak{p}} \frac{\rho_{\mathfrak{p}}}{\rho_{\mathfrak{c}}} d_{\mathfrak{c}}\right)^{x^*}\right)\right) - s \log(\gamma_{\mathfrak{p}}) = f^{\mathfrak{c}}(\gamma_{\mathfrak{p}}, \rho_{\mathfrak{p}}; x^*).$$

Matrix $\operatorname{Diag}(C^{\mathfrak{c}})^{-\frac{1}{2}} C^{\mathfrak{c}} \operatorname{Diag}(C^{\mathfrak{c}})^{-\frac{1}{2}}$ is obtained from $\operatorname{Diag}(C^{\mathfrak{p}})^{-\frac{1}{2}} C^{\mathfrak{p}} \operatorname{Diag}(C^{\mathfrak{p}})^{-\frac{1}{2}}$ by removing its row and column i, and therefore $\rho_{\mathfrak{p}} \geq \rho_{\mathfrak{c}} > 0$. It is also straightforward to see that $0 < \gamma_{\mathfrak{p}} \leq \gamma_{\mathfrak{c}}$. So, now it suffices to show that $\partial f^{\mathfrak{c}}(\gamma, \rho; x)/\partial \gamma \leq 0$ for $\gamma > 0$ and $\partial f^{\mathfrak{c}}(\gamma, \rho; x)/\partial \rho \geq 0$ for $\rho > 0$, both for all $0 \leq x \leq \mathbf{e}$, such that $\mathbf{e}^{\mathsf{T}} x = s$. The former was verified in the proof of Lemma 3.5.13. The latter was verified in the proof of Lemma 3.5.12, as increasing ρ in $f^{\mathfrak{c}}$ is equivalent to increasing the components of $d_{\mathfrak{c}}$. □

- Fixing the index i in S.
  In this case, we have

$$f^{\mathfrak{p}}(\gamma_{\mathfrak{p}}, \rho_{\mathfrak{p}}; y) = \operatorname{ldet}\left(\gamma_{\mathfrak{p}} \operatorname{Diag}\left(y^{\mathbf{e}/2}\right)(C^{\mathfrak{p}} - D_{\mathfrak{p}}) \operatorname{Diag}\left(y^{\mathbf{e}/2}\right) + \operatorname{Diag}\left((\gamma_{\mathfrak{p}} d_{\mathfrak{p}})^{y}\right)\right) - s \log(\gamma_{\mathfrak{p}}), \tag{3.84}$$

$$f^{\mathfrak{c}}(\gamma_{\mathfrak{c}}, \rho_{\mathfrak{c}}; x) = \operatorname{ldet}\left(\gamma_{\mathfrak{c}} \operatorname{Diag}\left(x^{\mathbf{e}/2}\right)(C^{\mathfrak{c}} - D_{\mathfrak{c}}) \operatorname{Diag}\left(x^{\mathbf{e}/2}\right) + \operatorname{Diag}\left((\gamma_{\mathfrak{c}} d_{\mathfrak{c}})^{x}\right)\right) - (s-1) \log(\gamma_{\mathfrak{c}}) + \log\left(C^{\mathfrak{p}}_{ii}\right), \tag{3.85}$$

  where

  - $C^{\mathfrak{c}}$ is the Schur complement of $C^{\mathfrak{p}}[i,i]$ in $C^{\mathfrak{p}}$, i.e.

$$C^{\mathfrak{c}} = C^{\mathfrak{p}}[T,T] - \frac{C^{\mathfrak{p}}[T,i] C^{\mathfrak{p}}[i,T]}{C^{\mathfrak{p}}[i,i]}\,, \tag{3.86}$$

- $D_{\mathfrak{p}} = \rho_{\mathfrak{p}} \operatorname{Diag}(C^{\mathfrak{p}})$, $D_{\mathfrak{c}} = \rho_{\mathfrak{c}} \operatorname{Diag}(C^{\mathfrak{c}})$, with

$$\rho_{\mathfrak{p}} = \lambda_{\max}\left(\operatorname{Diag}(C^{\mathfrak{p}})^{-\frac{1}{2}} C^{\mathfrak{p}} \operatorname{Diag}(C^{\mathfrak{p}})^{-\frac{1}{2}}\right), \tag{3.87}$$

$$\rho_{\mathfrak{c}} = \lambda_{\max}\left(\operatorname{Diag}(C^{\mathfrak{c}})^{-\frac{1}{2}} C^{\mathfrak{c}} \operatorname{Diag}(C^{\mathfrak{c}})^{-\frac{1}{2}}\right), \tag{3.88}$$

- $d_{\mathfrak{p}} = \operatorname{diag}(D_{\mathfrak{p}})$, $d_{\mathfrak{c}} = \operatorname{diag}(D_{\mathfrak{c}})$,
- $\gamma_{\mathfrak{p}} = 1/d_{\mathfrak{p}_{\max}}$, $\gamma_{\mathfrak{c}} = 1/d_{\mathfrak{c}_{\max}}$.

Theorem 3.5.20 Let $f^{\mathfrak{p}}$ and $f^{\mathfrak{c}}$ be defined as in (3.84) and (3.85). Let $\rho_{\mathfrak{p}}$ and $\rho_{\mathfrak{c}}$ be defined as in (3.87) and (3.88). Then, if $\rho_{\mathfrak{p}} \geq \rho_{\mathfrak{c}}$,

$$f^{\mathfrak{p}}(\gamma_{\mathfrak{p}}, \rho_{\mathfrak{p}}; y^*) \geq f^{\mathfrak{c}}(\gamma_{\mathfrak{c}}, \rho_{\mathfrak{c}}; x^*).$$

Proof Let

$$\hat{y}_j := \begin{cases} x^*_j, & \text{for } j < i; \\ 1, & \text{for } j = i; \\ x^*_{j-1}, & \text{for } j > i. \end{cases}$$

As $\hat{y}$ is a feasible solution to the parent problem, it suffices to show that $f^{\mathfrak{p}}(\gamma_{\mathfrak{p}}, \rho_{\mathfrak{p}}; \hat{y}) \geq f^{\mathfrak{c}}(\gamma_{\mathfrak{c}}, \rho_{\mathfrak{c}}; x^*)$. We note that

$$\begin{aligned} f^{\mathfrak{p}}(\gamma_{\mathfrak{p}}, \rho_{\mathfrak{p}}; \hat{y}) = \operatorname{ldet}\Big(&\gamma_{\mathfrak{p}} \operatorname{Diag}\left(x^{*\mathbf{e}/2}\right)(C^{\mathfrak{c}} - D_{\mathfrak{p}}[T,T]) \operatorname{Diag}\left(x^{*\mathbf{e}/2}\right) \\ &+ \operatorname{Diag}\left((\gamma_{\mathfrak{p}} d_{\mathfrak{p}}[T])^{x^*}\right)\Big) - (s-1)\log(\gamma_{\mathfrak{p}}) + \log\left(C^{\mathfrak{p}}_{ii}\right). \end{aligned}$$

We also have

$$\begin{aligned} D_{\mathfrak{p}}[T,T] &= \rho_{\mathfrak{p}} \operatorname{Diag}(C^{\mathfrak{p}}[T,T]), \\ \frac{\rho_{\mathfrak{p}}}{\rho_{\mathfrak{c}}} D_{\mathfrak{c}} &= \rho_{\mathfrak{p}} \operatorname{Diag}(C^{\mathfrak{c}}) = \rho_{\mathfrak{p}} \operatorname{Diag}\left(C^{\mathfrak{p}}[T,T] - \frac{C^{\mathfrak{p}}[T,i]C^{\mathfrak{p}}[i,T]}{C^{\mathfrak{p}}[i,i]}\right) \\ &= \rho_{\mathfrak{p}} \operatorname{Diag}(C^{\mathfrak{p}}[T,T]) - \rho_{\mathfrak{p}} \operatorname{Diag}\left(\frac{C^{\mathfrak{p}}[T,i]C^{\mathfrak{p}}[i,T]}{C^{\mathfrak{p}}[i,i]}\right). \end{aligned}$$

Then, defining $D := D_{\mathfrak{p}}[T,T] - \frac{\rho_{\mathfrak{p}}}{\rho_{\mathfrak{c}}} D_{\mathfrak{c}}$, we have $D = \rho_{\mathfrak{p}} \operatorname{Diag}\left(\frac{C^{\mathfrak{p}}[T,i]C^{\mathfrak{p}}[i,T]}{C^{\mathfrak{p}}[i,i]}\right)$, and

$$\begin{aligned} D[j,j] = \rho_{\mathfrak{p}} \frac{C^{\mathfrak{p}}[j,i]C^{\mathfrak{p}}[i,j]}{C^{\mathfrak{p}}[i,i]} = \rho_{\mathfrak{p}} \frac{C^{\mathfrak{p}}[i,j]^2}{C^{\mathfrak{p}}[i,i]} \qquad &\Rightarrow \\ \gamma_{\mathfrak{p}} D[j,j] = \rho_{\mathfrak{p}} \frac{C^{\mathfrak{p}}[i,j]^2}{d_{\mathfrak{p}_{\max}} C^{\mathfrak{p}}[i,i]} \leq \frac{C^{\mathfrak{p}}[i,j]^2}{C^{\mathfrak{p}}[j,j]C^{\mathfrak{p}}[i,i]} &< 1, \end{aligned}$$

where the last inequality follows from the fact that $C^{\mathfrak{p}}$ is positive semidefinite. Then, as $0 \le x_j^* \le 1$ and $\gamma_{\mathfrak{p}} d_j < 1$, we have

$$
\begin{aligned}
&\left(\gamma_{\mathfrak{p}} d_j\right)^{x_j^*} \ge x_j^* \gamma_{\mathfrak{p}} d_j && \Rightarrow \\
&\operatorname{Diag}\left((\gamma_{\mathfrak{p}} d)^{x^*}\right) \ge \operatorname{Diag}\left(x^{*\mathbf{e}/2}\right) \gamma_{\mathfrak{p}} D \operatorname{Diag}\left(x^{*\mathbf{e}/2}\right) && \Rightarrow \\
&\operatorname{Diag}\left(\left(\gamma_{\mathfrak{p}} d_{\mathfrak{p}}[T]\right)^{x^*}\right) - \operatorname{Diag}\left(x^{*\mathbf{e}/2}\right) \gamma_{\mathfrak{p}} D_{\mathfrak{p}}[T,T] \operatorname{Diag}\left(x^{*\mathbf{e}/2}\right) \\
&\quad \ge \operatorname{Diag}\left(\left(\gamma_{\mathfrak{p}} \frac{\rho_{\mathfrak{p}}}{\rho_{\mathfrak{c}}} d_{\mathfrak{c}}\right)^{x^*}\right) - \operatorname{Diag}\left(x^{*\mathbf{e}/2}\right) \gamma_{\mathfrak{p}} \frac{\rho_{\mathfrak{p}}}{\rho_{\mathfrak{c}}} D_{\mathfrak{c}} \operatorname{Diag}\left(x^{*\mathbf{e}/2}\right) && \Rightarrow \\
&f^{\mathfrak{p}}(\gamma_{\mathfrak{p}}, \rho_{\mathfrak{p}}; \hat{y}) \ge \operatorname{ldet}\left(\gamma_{\mathfrak{p}} \operatorname{Diag}\left(x^{*\mathbf{e}/2}\right)\left(C^{\mathfrak{c}} - \frac{\rho_{\mathfrak{p}}}{\rho_{\mathfrak{c}}} D_{\mathfrak{c}}\right) \operatorname{Diag}\left(x^{*\mathbf{e}/2}\right)\right. \\
&\qquad \left. + \operatorname{Diag}\left(\left(\gamma_{\mathfrak{p}} \frac{\rho_{\mathfrak{p}}}{\rho_{\mathfrak{c}}} d_{\mathfrak{c}}\right)^{x^*}\right)\right) - (s-1)\log(\gamma_{\mathfrak{p}}) + \log\left(C^{\mathfrak{p}}_{ii}\right) \\
&\quad = f^{\mathfrak{c}}(\gamma_{\mathfrak{p}}, \rho_{\mathfrak{p}}; x^*).
\end{aligned}
$$

Now, defining $k := \operatorname{argmax}_j \left\{C^{\mathfrak{p}}[j,j] - \frac{C^{\mathfrak{p}}[j,i]C^{\mathfrak{p}}[i,j]}{C^{\mathfrak{p}}[i,i]}\right\}$, then $d_{\mathfrak{p}_{\max}} \ge \rho_{\mathfrak{p}} C^{\mathfrak{p}}[k,k] \ge \rho_{\mathfrak{c}} C^{\mathfrak{p}}[k,k] \ge d_{\mathfrak{c}_{\max}}$. Therefore $0 < \gamma_{\mathfrak{p}} \le \gamma_{\mathfrak{c}}$. So, it suffices to show that $\partial f^{\mathfrak{c}}(\gamma, \rho; x)/\partial\gamma \le 0$ for $\gamma > 0$ and $\partial f^{\mathfrak{c}}(\gamma, \rho; x)/\partial\rho \ge 0$ for $\rho > 0$, both for all $0 \le x \le \mathbf{e}$, such that $\mathbf{e}^{\mathsf{T}} x = s - 1$. The proof of the former is similar to the proof of Lemma 3.5.13, after replacing $\mathbf{e}^{\mathsf{T}} x = s$ by $\mathbf{e}^{\mathsf{T}} x = s - 1$. The proof of the latter is equivalent to the proof of Lemma 3.5.12, as increasing ρ in $f^{\mathfrak{c}}$ is equivalent to increasing the components of $d_{\mathfrak{c}}$. □

Remark 3.5.21 Let $\rho_{\mathfrak{p}}$ and $\rho_{\mathfrak{c}}$ be defined respectively by (3.87) and (3.88). In general, we do not have $\rho_{\mathfrak{p}} \ge \rho_{\mathfrak{c}}$, as assumed in Theorem 3.5.20. For example, for fixing in $i = 1$ with

$$
C = \begin{pmatrix} 2 & 1.5 & 0 & 0 \\ 1.5 & 6 & -0.5 & 3 \\ 0 & -0.5 & 1 & -0.5 \\ 0 & 3 & -0.5 & 2 \end{pmatrix},
$$

we have $\rho_{\mathfrak{p}} = 2.0949$ and $\rho_{\mathfrak{c}} = 2.1125$. However, we note that although the sufficient condition stated in Theorem 3.5.20 does not hold for this matrix C, we still have $f^{\mathfrak{p}}(\gamma_{\mathfrak{p}}, \rho_{\mathfrak{p}}; y^*) \ge f^{\mathfrak{c}}(\gamma_{\mathfrak{c}}, \rho_{\mathfrak{c}}; x^*)$, for both $s = 1, 2$.

3.6 BQP bound

We can lift $x \in \mathbb{R}^n$ to $(x, X) \in \mathbb{R}^n \times \mathbb{S}^n$ via the map $X := xx^{\mathsf{T}}$. Now consider the convex sets

$$P(n) := \left\{(x, X) \in \mathbb{R}^n \times \mathbb{S}^n \ : \ X - xx^\mathsf{T} \succeq 0, \ \operatorname{diag}(X) = x\right\},$$
$$P(n, s) := \left\{(x, X) \in P(n) \ : \ \mathbf{e}^\mathsf{T} x = s, \ X\mathbf{e} = sx\right\}.$$

The SDP inequality $X - xx^\mathsf{T} \succeq 0$ is the well-known (convex) **Shor relaxation** of $X - xx^\mathsf{T} = 0$ (see Corollary A.4.5 and [Sho87, Sho90]). The equation $\operatorname{diag}(X) = x$ lifts the equations $x_j^2 = x_j$ (for $j \in N$), which is implied by $x \in \{0, 1\}^n$. The equation $X\mathbf{e} = sx$ arises by left-multiplying the equation $x^\mathsf{T}\mathbf{e} = s$ by x, and then substituting X for xx^T; while $X\mathbf{e} = sx$ is implied by $\mathbf{e}^\mathsf{T} x = s$ and $X = xx^\mathsf{T}$, it is not implied by $X = X^\mathsf{T}$, $X - xx^\mathsf{T} \succeq 0$, $\operatorname{diag}(X) = x$ and $\mathbf{e}^\mathsf{T} x = s$.

Proposition 3.6.1 For $(x, X) \in P(n)$, we have $0 \leq x \leq \mathbf{e}$.

Proof Because $X \succeq xx^\mathsf{T} \succeq 0$, we have $x = \operatorname{diag}(X) \geq 0$. Furthermore, $X - xx^\mathsf{T} \succeq 0$ implies that $\operatorname{diag}(X - xx^\mathsf{T}) \geq 0$; that is, $x_i(1 - x_i) \geq 0$, for $i = 1, 2, \ldots, n$. This directly implies now that $x \leq \mathbf{e}$. □

For $x \in [0, 1]^n$ and $X \in \mathbb{S}^n_+$, we define

$$f(x, X) := \operatorname{ldet} M(x, X),$$
$$\text{where } M(x, X) := C \circ X + \operatorname{Diag}(\mathbf{e} - x).$$

We define the **BQP bound** for CMESP:

$$\text{(BQP)} \qquad z_{\text{BQP}} := \max \{f(x, X) \ : \ Ax \leq b, \ (x, X) \in P(n, S)\},$$

We are justified in using maximum rather than supremum, using the fact that $M(x, X) \in \mathbb{S}^n_+$ for all $(x, X) \in P(n)$ and Proposition A.1.4 (using $\mathcal{K} := \left\{(x, X) \in \mathbb{R}^n \times \mathbb{S}^n \ : \ M(x, X) \in \mathbb{S}^n_{++}\right\}$ and $\mathcal{Q} := \{(x, X) \in \mathbb{R}^n \times \mathbb{S}^n \ : \ Ax \leq b; \ (x, X) \in P(n, s)\}$).

The name "BQP bound" is justified by the following result.

Theorem 3.6.2

$$z(C, s, A, b) \leq z_{\text{BQP}} .$$

Proof For $x \in \{0,1\}^n$ and $X = xx^{\mathsf{T}}$, let $S := S(x)$, and let $T := N \setminus S$. After a symmetric permutation of indices, we can write

$$C \circ X = \left(\begin{array}{c|c} C[S,S] & C[S,T] \\ \hline C[S,T]^{\mathsf{T}} & C[T,T] \end{array}\right) \circ \left(\begin{array}{c|c} J & 0 \\ \hline 0 & 0 \end{array}\right) = \left(\begin{array}{c|c} C[S,S] & 0 \\ \hline 0 & 0 \end{array}\right).$$

So,

$$C \circ X + \text{Diag}(\mathbf{e} - x) = \left(\begin{array}{c|c} C[S,S] & 0 \\ \hline 0 & I \end{array}\right)$$

and we have $f(x) = \text{ldet}\, C[S,S]$. Therefore, BQP is an exact relaxation of CMESP. □

The name "BQP bound" relates to the fact that the **boolean quadric polytope** is defined as the convex hull of the $(x, X) \in \{0,1\}^n \times \mathbb{S}^n$ satisfying $X = xx^{\mathsf{T}}$ (see [Pad89]). We note that the BQP is also known as the **correlation polytope**.

Helmberg (private communication) suggested (essentially) the BQP bound in 1995 (see [Lee12, FL00]) to Anstreicher and Lee, but no one developed it at that time. [CHB08, CHB13] developed an outer-approximation scheme connected with (essentially) the BQP bound. [Ans18] developed the BQP bound extensively, drawing in and significantly extending some techniques from [AFLW99]. Part of the reason that the idea was not developed and experimented with when first conceived is that there was no convenient and efficient software for maximizing ldet objective functions subject to semidefiniteness constraints. Later, with the inclusion of ldet maximization in SDPT3 Version 4.0 (released February 2009) and convenient access via YALMIP, the time was ripe for returning to Helmberg's nice idea.

The boolean quadric polytope and equivalently the "cut polytope" (see [DS90, DS96]) have an extensive literature (see for example [DL97, SLA95, LS14, Nik16, LS20]), and so quite a lot is known about the facial structure of this polytope. Of practical significance for us is some classes of valid inequalities, linear in (x, X), that can be used to strengthen $P(n, s)$ as a tractable convex approximation of the boolean quadric polytope.

Considering, without loss of generality, the upper-triangle entries X_{jk}, for $1 \leq j < k \leq n$, we can derive the well-known **McCormick inequalities**:

$$\begin{aligned} X_{jk} &\leq x_j \ ; \\ X_{jk} &\leq x_k \ ; \\ x_j + x_k &\leq 1 + X_{jk} \ , \end{aligned}$$

which are all necessary for describing the boolean quadric polytope via linear inequalities (for $n \geq 2$). These constraints can all be seen to arise by multiplying linear nonnegativity constraints in x together, and then linearizing via the equations $X_{jk} = x_j x_k$ (which are part of $X = xx^{\mathsf{T}}$). For example, multiplying $x_j \geq 0$ and $1 - x_k \geq 0$ together, we get $x_j - x_j x_k \geq 0$, which linearizes to $X_{jk} \leq x_j$. Multiplying $1 - x_j \geq 0$ and $1 - x_k \geq 0$ together, we get $1 - x_j - x_j + x_j x_k \geq 0$, which linearizes to the McCormick inequality $x_j + x_k \leq 1 + X_{jk}$.

Going a little further, considering $1 \leq i < j < k \leq n$, we have the **triangle inequalities**:

$$
\begin{aligned}
& x_i + x_j + x_k \leq 1 + X_{ij} + X_{ik} + X_{jk} \; ; \\
& X_{ij} + X_{ik} \leq X_{jk} + x_i \; ; \\
& X_{ij} + X_{jk} \leq X_{ik} + x_j \; ; \\
& X_{ij} + X_{jk} \leq X_{ij} + x_k \; ,
\end{aligned}
$$

which are all necessary for describing the boolean quadric polytope via linear inequalities (for $n \geq 3$). In fact, for $n \leq 3$, these inequalities above are all that we need to describe the boolean quadric polytope. As we did for the McCormick inequalities, the triangle inequalities have simple derivations using the McCormick inequalities. For example $(1 - x_i)(x_j - X_{jk}) + (x_i)(x_k - X_{jk}) \geq 0$ because each of the four parenthesized terms is nonnegative. Then simplifying, noticing that the terms $x_i X_{jk}$ cancel out, and then linearizing $x_i x_j$ and $x_i x_k$, we arrive at the triangle inequality $X_{ij} + X_{jk} \leq X_{ik} + x_j$.

For $n \geq 4$, the complexity of the boolean quadric polytope rapidly increases (see [Pad89]). But for our use, we would mainly consider judiciously using only the McCormick inequalities and the triangle inequalities to improve the BQP bound for CMESP. We have to be judicious because the difficulty of calculating such an improved BQP bound increased quickly with the number of linear inequalities utilized. We could get further inequalities by linearizing inequalities from $x_j(b - Ax) \geq 0$ or $(1 - x_j)(b - Ax) \geq 0$ or even $(b - Ax)(b - Ax)^{\mathsf{T}} \geq 0$ (see [SA99] for much more on this so called "RLT" (reformulation-linearization technique), but again, this may be computationally prohibitive (see [Ans18]). One way to handle such a large number of potentially useful constraints is via the conic-bundle method (see [FGRS06], [BELW17], and [CFLL21] for its use in our context using the Conic Bundle Library [Hel21] and SDPT3 [TTT99, TTT12]).

3.6.1 Convexity of BQP

Lemma 3.6.3

(i) For $C \geq 0$, we have $M(x, X) \succ 0$, for all $(x, X) \in P(n)$, such that $x < \mathbf{e}$.
(ii) For $C \succ 0$, we have $M(x, X) \succ 0$, for all $(x, X) \in P(n)$.

Proof From Proposition 3.6.1, we have that $(x, X) \in P(n)$ implies that $0 \leq x \leq \mathbf{e}$. Moreover, in both cases, if $x < \mathbf{e}$, then $\mathrm{Diag}(\mathbf{e} - x) \succ 0$. Therefore $M(x, X) \succ 0$ because we have $C \circ X \succeq 0$. In particular, we have established case (i).

For case (ii), consider first when $x = \mathbf{e}$. Because $C \succ 0$ and $X - xx^{\mathsf{T}} \succeq 0$, we have $C \circ (X - xx^{\mathsf{T}}) \succeq 0$. So, $C \circ X \succeq C \circ xx^{\mathsf{T}} = C \circ J = C \succ 0$. Therefore, we have $M(x, X) \succ 0$ because $\mathrm{Diag}(\mathbf{e} - x) = 0$.

So, what remains is case (ii) with $x \neq \mathbf{e}$ and $x \not< \mathbf{e}$. Now suppose that U and V form a nontrivial partition of N, and that $x_i = 1$, for $i \in U$, and $x_i < 1$, for $i \in V$. After a symmetric permutation of indices we can write

$$\begin{aligned} C \circ X &= \left(\begin{array}{c|c} C[U,U] & C[U,V] \\ \hline C[V,U] & C[V,V] \end{array}\right) \circ \left(\begin{array}{c|c} J & X[U,V] \\ \hline X[V,U] & X[V,V]) \end{array}\right) \\ &= \left(\begin{array}{c|c} C[U,U] & C[U,V] \circ X[U,V] \\ \hline C[V,U] \circ X[V,U] & C[V,V] \circ X[V,V] \end{array}\right). \end{aligned}$$

As $C \circ X \succeq 0$ and $C[U,U] \succ 0$, from Theorem A.4.4 we have that the Schur complement of $C[U,U]$ in $C \circ X$ is positive semidefinite, i.e.,

$$C[V,V] \circ X[V,V] - C[V,U] \circ X[V,U] C[U,U]^{-2} C[U,V] \circ X[U,V]) \succeq 0. \tag{3.89}$$

We also have

$$C \circ X + \mathrm{Diag}(\mathbf{e} - x) = \left(\begin{array}{c|c} C[U,U] & C[U,V] \circ X[U,V] \\ \hline C[V,U] \circ X[V,U] & C[V,V] \circ X[V,V] + \mathrm{Diag}(\mathbf{e} - x[V]) \end{array}\right).$$

As $\mathrm{Diag}(\mathbf{e} - x[V]) \succ 0$, from (3.89) we see that

$$\begin{aligned} C[V,V] \circ X[V,V] + \mathrm{Diag}(\mathbf{e} - x[V]) \\ - C[V,U] \circ X[V,U] C[U,U]^{-2} C[U,V] \circ X[U,V] \succ 0, \end{aligned} \tag{3.90}$$

which is the Schur complement of $C[U,U]$ in $C \circ X + \mathrm{Diag}(\mathbf{e} - x)$. Finally, considering (3.90) and $C[U,U] \succ 0$, we apply Theorem A.4.4 and so case (ii) holds. □

Theorem 3.6.4
The function $\mathrm{ldet}\, M(\cdot, \cdot)$ is concave on $\{(x, X) \in P(n) \ : \ M(x, X) \succ 0\}$.

Proof For each $(x, X) \in P(n)$ such that $M(x, X) \succ 0$, we can write

$$\mathrm{ldet}\, M(x, X) = \max_{Z \succ 0} \{\mathrm{ldet}\, Z \ : \ Z \preceq M(x, X)\}. \tag{3.91}$$

Considering Lemma 3.6.3 and the monotonicity of ldet, we have that the maximum $Z \succ 0$ satisfying the equality in the constraint in (3.91) gives the value of $\operatorname{ldet} M(x, X)$, for each $(x, X) \in P(n)$ such that $M(x, X) \succ 0$. Considering that ldet is strictly concave on $\mathbb{S}^n_{++}$ (see Theorem A.4.12) and Proposition A.1.4, we have that the optimal solution of (3.91), denoted in the following by $Z^*(x, X)$, exists and is unique for all $(x, X) \in P(n)$ such that $M(x, X) \succ 0$.

Now, considering the linearity of the function M, the optimality of $Z^*(\cdot, \cdot)$, and the concavity of ldet, we have

$$\begin{aligned} &\operatorname{ldet} M(\tau(x, X) + (1-\tau)(y, Y)) \\ &\quad = \max_{Z \succ 0} \{\operatorname{ldet} Z \ : \ Z \preceq M(\tau(x, X) + (1-\tau)(y, Y))\} \\ &\quad = \max_{Z \succ 0} \{\operatorname{ldet} Z \ : \ Z \preceq \tau M(x, X) + (1-\tau) M(y, Y)\} \\ &\quad \geq \operatorname{ldet} (\tau Z^*(x, X) + (1-\tau) Z^*(y, Y)) \\ &\quad \geq \tau \operatorname{ldet} Z^*(x, X) + (1-\tau) \operatorname{ldet} Z^*(y, Y), \end{aligned} \tag{3.92}$$

for all $\tau \in [0, 1]$ and $(x, X), (y, Y) \in P(n)$ such that $M(x, X), M(y, Y) \succ 0$. The result follows. □

Corollary 3.6.5 BQP is a convex optimization problem.

Proof As the feasible region of BQP is a convex set, it suffices to demonstrate that the objective function is concave on the feasible set. We proved in Theorem 3.6.4 that the objective function is concave on the convex set $\mathcal{F} := \{(x, X) \in P(n) \ : \ M(x, X) \succ 0\}$, and because $f(x, X)$ is $-\infty$ on $\{(x, X) \in P(n) \ : \ M(x, X) \succeq 0\} \setminus \mathcal{F}$. Therefore, the objective function is concave (in the extended sense) on all of the feasible set. □

Remark 3.6.6 As is the case for linx, the formulation BQP can be conveniently modeled using CVX or YALMIP. In particular, YALMIP in conjunction with the solver SDPT3 provides a very efficient algorithmic treatment of the ldet function.

In Corollary 3.6.5, we established that BQP is a convex relaxation of CMESP. In Proposition 3.6.9, we will see that under certain assumptions, BQP has a unique optimal solution as well. In Lemma 3.6.7, we introduce these assumptions.

Lemma 3.6.7 If $C_{ii} \neq 1$, for all i and $C_{ij} \neq 0$ for all $i \neq j$, then $M(\cdot, \cdot)$ is a one-to-one function for $x \in \mathbb{R}^n$, $X \in \mathbb{S}^n$, $\operatorname{diag}(X) = x$.

Proof For $x, y \in \mathbb{R}^n$, $X, Y \in \mathbb{S}^n$, such that $\operatorname{diag}(X) = x$ and $\operatorname{diag}(Y) = y$, we have

$$\begin{aligned} &C \circ X + \operatorname{Diag}(\mathbf{e} - x) = C \circ Y + \operatorname{Diag}(\mathbf{e} - y) \iff \\ &C \circ (X - Y) - \operatorname{Diag}(x - y) = 0. \end{aligned} \tag{3.93}$$

□

Remark 3.6.8 Note that Lemma 3.6.7 and its proof do not need $C \geq 0$.

Proposition 3.6.9 If $C_{ii} \neq 1$ for all i, and $C_{ij} \neq 0$ for all $i \neq j$, and $z_{\text{BQP}} > -\infty$, then BQP has a unique optimal solution.

Proof With $z_{\text{BQP}} > -\infty$, any optimal solution $(\hat{x}, \hat{X})$ of BQP has $M(\hat{x}, \hat{X}) \succ 0$. Therefore, it suffices to prove that BQP has a unique optimal solution on the convex set $\mathcal{F} := \{(x, X) \in P(n, s) : Ax \leq b;\ M(x, X) \succ 0\}$. Considering that ldet is strictly concave on $\mathbb{S}^n_{++}$ (see Theorem A.4.12), the last inequality in (3.92) is strict for all $\tau \in (0, 1)$ and $(x, X), (y, Y)$ with $M(x, X) \neq M(y, Y)$. Therefore, if $C_{ii} \neq 1$, for all i, and $C_{ij} \neq 0$ for all $i \neq j$, then M is a one-to-one function on $\mathcal{F}$, and then we conclude that ldet $M(\cdot, \cdot)$ is strictly concave on $\mathcal{F}$. The result thus follows from Proposition A.1.4. □

3.6.2 Duality for BQP

We have established that BQP is a convex optimization problem. In order to formulate the dual problem for BQP, we initially rewrite it using the following matrices in $\mathbb{S}^{n+1}$:

$$\tilde{X} := \left(\begin{array}{c|c} X & x \\ \hline x^{\mathsf{T}} & 1 \end{array}\right), \quad \tilde{C} := \left(\begin{array}{c|c} C - I_n & 0 \\ \hline 0 & 0 \end{array}\right).$$

Then, BQP can be rewritten as

$$\begin{array}{l} \max \text{ ldet } W \\ \quad \text{subject to:} \\ \quad \tilde{C} \circ \tilde{X} + I = W, \\ \quad H_i^{\leq} \bullet \tilde{X} \leq h_i^{\leq},\ i \in \mathcal{M}_{\leq}, \\ \quad H_i^{=} \bullet \tilde{X} = h_i^{=},\ i \in \mathcal{M}_{=}, \\ \quad \tilde{X} \succeq 0, \end{array} \tag{3.94}$$

where $H_i^{\leq} \in \mathbb{S}^{n+1}$, $h_i^{\leq} \in \mathbb{R}$, for $i \in \mathcal{M}_{\leq}$, and $H_i^{=} \in \mathbb{S}^{n+1}$, $h_i^{=} \in \mathbb{R}$, for $i \in \mathcal{M}_{=}$. Note that $H_i^{\leq} \bullet \tilde{X} \leq h_i^{\leq}$ ($i \in \mathcal{M}_{\leq}$) includes the $Ax \leq b$ constraints of CMESP; and $H_i^{=} \bullet \tilde{X} = h_i^{=}$ ($i \in \mathcal{M}_{=}$) includes: diag$(X) = x$, $\mathbf{e}^{\mathsf{T}} x = s$, and $X\mathbf{e} = sx$ (from $P(n, s)$), all written of course in the $\tilde{X}$ variable, and also $\tilde{X}_{n+1,n+1} = 1$. Corollary A.4.5 gives us the equivalence of $\tilde{X} \succeq 0$, $\tilde{X}_{n+1,n+1} = 1$ with $X - xx^{\mathsf{T}} \succeq 0$.

We consider the Lagrangian function

$$\begin{aligned} \mathcal{L}(W, \tilde{X}, \Theta, \Psi, \pi, \tau) := \text{ldet } W &+ \Theta \bullet \left(\tilde{C} \circ \tilde{X} + I - W\right) + \Psi \bullet \tilde{X} \\ &+ \sum_{i \in \mathcal{M}_{\leq}} \pi_i \left(h_i^{\leq} - H_i^{\leq} \bullet \tilde{X}\right) + \sum_{i \in \mathcal{M}_{=}} \tau_i \left(h_i^{=} - H_i^{=} \bullet \tilde{X}\right), \end{aligned}$$

with dom $\mathcal{L} = \mathbb{S}^{n+1}_{++} \times \mathbb{S}^{n+1} \times \mathbb{S}^{n+1} \times \mathbb{S}^{n+1} \times \mathbb{R}^{|\mathcal{M}_\leq|} \times \mathbb{R}^{|\mathcal{M}_=|}$. The corresponding dual function is

$$\mathcal{L}^*(\Theta, \Psi, \pi, \tau) := \sup_{W \in \mathbb{S}^{n+1}_{++}, \tilde{X} \in \mathbb{S}^{n+1}} \mathcal{L}(W, \tilde{X}, \Theta, \Psi, \pi, \tau),$$

and the Lagrangian dual problem of (3.94) is

$$\min\{\mathcal{L}^*(\Theta, \Psi, \pi, \tau) \ : \ \Psi \geq 0, \ \pi \geq 0\}.$$

We note that the Lagrangian function is concave in $W \succ 0$ and $\tilde{X}$. Then the supremum in (3.7) occurs where the gradient of the Lagrangian function with respect to W and $\tilde{X}$ is equal to zero. So, considering Lemma A.4.13, the Lagrangian dual problem may be written as

$$\begin{aligned} \min \quad & \operatorname{ldet} W + \Theta \bullet \left(\tilde{C} \circ \tilde{X} + I - W\right) + \Psi \bullet \tilde{X} \\ & \quad + \sum_{i \in \mathcal{M}_\leq} \pi_i \left(h_i^\leq - H_i^\leq \bullet \tilde{X}\right) + \sum_{i \in \mathcal{M}_=} \tau_i \left(h_i^= - H_i^= \bullet \tilde{X}\right) \\ & \text{subject to:} \\ & W^{-1} - \Theta = 0, \\ & \Theta \circ \tilde{C} + \Psi - \sum_{i \mathcal{M}_\leq} \pi_i H_i^\leq - \sum_{i \in \mathcal{M}_=} \tau_i H_i^= = 0, \\ & W \succ 0, \ \pi \geq 0, \ \Psi \geq 0. \end{aligned} \tag{3.95}$$

Theorem 3.6.10 The following equation is valid for all solutions of (3.95):

$$\begin{aligned} & \Theta \bullet \left(\tilde{C} \circ \tilde{X} + I - W\right) + \Psi \bullet \tilde{X} + \sum_{i \in \mathcal{M}_\leq} \pi_i \left(h_i^\leq - H_i^\leq \bullet \tilde{X}\right) + \sum_{i \in \mathcal{M}_=} \tau_i \left(h_i^= - H_i^= \bullet \tilde{X}\right) \\ & \quad = \pi^{\mathsf{T}} h^\leq + \tau^{\mathsf{T}} h^= + \operatorname{tr}(\Theta) - (n+1). \end{aligned}$$

Proof From the first constraint in (3.95), we have $\Theta = W^{-1}$, and from the second constraint we have

$$\begin{aligned} & \left(\Theta \circ \tilde{C} + \Psi - \sum_{i \in \mathcal{M}_\leq} \pi_i H_i^\leq - \sum_{i \in \mathcal{M}_=} \tau_i H_i^=\right) \bullet \tilde{X} = 0 \quad \Rightarrow \\ & \Theta \bullet \left(\tilde{C} \circ \tilde{X}\right) + \Psi \bullet \tilde{X} - \sum_{i \in \mathcal{M}_\leq} \pi_i H_i^\leq \bullet \tilde{X} - \sum_{i \in \mathcal{M}_=} \tau_i H_i^= \bullet \tilde{X} = 0 \quad \Rightarrow \\ & \Theta \bullet \left(\tilde{C} \circ \tilde{X} + I - W\right) + \Psi \bullet \tilde{X} + \sum_{i \in \mathcal{M}_\leq} \pi_i \left(h_i^\leq - H_i^\leq \bullet \tilde{X}\right) + \sum_{i \in \mathcal{M}_=} \tau_i \left(h_i^= - H_i^= \bullet \tilde{X}\right) \end{aligned}$$

$$= \Theta \bullet (I - W) + \sum_{i \in \mathcal{M}_\leq} \pi_i h_i^\leq + \sum_{i \in \mathcal{M}_=} \tau_i h_i^= \\ = \mathrm{tr}(\Theta) - (n+1) + \pi^\mathsf{T} h^\leq + \tau^\mathsf{T} h^= .$$

□

Considering Theorem 3.6.10, we finally re-cast the Lagrangian dual problem (3.95) of (3.94), as

(DBQP)

$$\begin{array}{l} \min -\mathrm{ldet}(\Theta) + \mathrm{tr}(\Theta) + \pi^\mathsf{T} h^\leq + \tau^\mathsf{T} h^= - (n+1) \\ \quad \text{subject to:} \\ \Theta \circ \tilde{C} - \sum_{i \in \mathcal{M}_\leq} \pi_i H_i^\leq - \sum_{i \in \mathcal{M}_=} \tau_i H_i^= \preceq 0, \\ \Theta \succ 0, \ \pi \geq 0. \end{array}$$

Proposition 3.6.11 If BQP is feasible, then we have strong duality for BQP and DBQP.

Proof From general principles concerning convex duality, if DBQP is strictly feasible, then the maximum value in BQP is attained, and we have strong duality, assuming that BQP is feasible. Therefore, it suffices to construct a strictly feasible solution for DBQP.

Referring to the general linear equations $H_i^= \bullet \tilde{X} = h_i^=$ of (3.94), we formulate the constraints $\mathrm{diag}(X) = x$ and $\mathbf{e}^\mathsf{T} x = s$ of BQP as follows.

We formulate the constraints $X_{ii} = x_i$, $i \in N$, with

$$H_i^= := \left(\begin{array}{c|c} E_{ii} & -\frac{1}{2}\mathbf{e}_i \\ \hline -\frac{1}{2}\mathbf{e}_i^\mathsf{T} & 0 \end{array} \right), \quad h_i^= := 0.$$

We formulate the constraint $\mathbf{e}^\mathsf{T} x = s$ with

$$H_{n+1}^= := \left(\begin{array}{c|c} 0 & \frac{1}{2}\mathbf{e} \\ \hline \frac{1}{2}\mathbf{e}^\mathsf{T} & 1 \end{array} \right), \quad h_{n+1}^= := s + 1.$$

Let $\hat{\tau}_i := 1$, for $i = 1, \ldots, n+1$. Notice that $\sum_{i=1}^{n+1} \hat{\tau}_i H_i^= = I_{n+1}$. Let $\hat{\tau}_i := 0$, for all $i \in \mathcal{M}_= \setminus \{1, \ldots, n+1\}$.

We define a diagonal matrix $\hat{\Theta} \in \mathbb{S}_{++}^{n+1}$ by

$$\hat{\Theta}_{ii} := \begin{cases} 1/(2\tilde{C}_{ii}), & \text{if } \tilde{C}_{ii} \neq 0; \\ 1/2, & \text{if } \tilde{C}_{ii} = 0, \end{cases}$$

for all $i \in \{1, \ldots, n+1\}$. Let $\hat{\pi}_i := 0$, $i \in \mathcal{M}_{\leq}$. Then

$$\hat{\Theta} \circ \tilde{C} - \sum_{i \in \mathcal{M}_{\leq}} \hat{\pi}_i H_i^{\leq} - \sum_{i \in \mathcal{M}_{=}} \hat{\tau}_i H_i^{=} =: \Psi,$$

where $\Psi < 0$ is a diagonal matrix with

$$\hat{\Psi}_{ii} := \begin{cases} -1/2, & \text{if } \tilde{C}_{ii} \neq 0; \\ -1, & \text{if } \tilde{C}_{ii} = 0, \end{cases}$$

for all $i \in \{1, \ldots, n+1\}$. Therefore $(\hat{\Theta}, \hat{\pi}, \hat{\tau})$ is strictly feasible in DBQP. □

As we mentioned in Remark 3.6.6, a convenient and efficient means for solving BQP is with the software YALMIP/SDPT3. Conveniently for us, YALMIP also returns to us an optimal solution of the dual, DBQP. We will see in the next section how we can use any dual feasible solution in a variable-fixing methodology for CMESP.

3.6.3 Fixing variables in BQP

For the linx and NLP relaxations, it is fairly straightforward to develop a methodology for fixing x_i variables at zero or one, based on a dual-feasible solution; see §3.3.3 and §3.5.6. In both of those relaxations, we explicitly have the bound constraints $0 \leq x \leq \mathbf{e}$, and associated dual variables. For the BQP relaxation, the constraints $0 \leq x \leq \mathbf{e}$ are implied by $X - xx^{\mathsf{T}} \geq 0$ and $\operatorname{diag}(X) = x$ (see Proposition 3.6.1), and so we do not have them explicitly, and so we have no dual variables associated with them. Still, we can carry out fixing, but it is more complicated to derive suitable rules.

Theorem 3.6.12 (see [Ans18, "Methodology 1"]) Let

- LB be the objective-function value of a feasible solution of CMESP,
- $(\hat{\Theta}, \hat{\pi}, \hat{\tau})$ be a feasible solution of DBQP with objective-function value $\hat{\zeta}$.

Define

- $\tilde{N} := \{k \in N \,:\, \tilde{C}_{kk} \neq 0\}$,
- $\upsilon_k := \frac{\tilde{C}_{kk}+1}{(\hat{\Theta}^{-1})_{kk}} - \log\left(\frac{\tilde{C}_{kk}+1}{(\hat{\Theta}^{-1})_{kk}}\right) - 1, \ \forall\, k \in \tilde{N}$,
- $\nu_k := \frac{1}{(\hat{\Theta}^{-1})_{kk}} - \log\left(\frac{1}{(\hat{\Theta}^{-1})_{kk}}\right) - 1, \ \forall\, k \in \tilde{N}$.

Then, for every optimal solution x^* of MESP, we have:

$$x_k^* = 0, \forall\, k \in \tilde{N} \ \text{ such that } \ \hat{\zeta} - \text{LB} < \upsilon_k \ ,$$
$$x_k^* = 1, \forall\, k \in \tilde{N} \ \text{ such that } \ \hat{\zeta} - \text{LB} < \nu_k \ .$$

Proof We first consider the addition of the constraint $x_k = 1$ to BQP, which is equivalently modeled as $\mathbf{e}_k\mathbf{e}_k^\mathsf{T} \bullet \tilde{X} = 1$ in (3.94), where here $\mathbf{e}_k$ is a standard unit vector in $\mathbb{R}^{n+1}$. The dual problem of (3.94) with this constraint added can be written as

$$\begin{aligned} &\min -\operatorname{ldet}(\Theta) + \operatorname{tr}(\Theta) + \pi^\mathsf{T} h^{\leq} + \tau^\mathsf{T} h^{=} - (n+1) - \omega \\ &\quad \text{subject to:} \\ &\quad \Theta \circ \tilde{C} - \sum_{i \in \mathcal{M}_{\leq}} \pi_i H_i^{\leq} - \sum_{i \in \mathcal{M}_{=}} \tau_i H_i^{=} + \omega \mathbf{e}_k \mathbf{e}_k^\mathsf{T} \preceq 0, \\ &\quad \Theta \succ 0,\ \pi \geq 0. \end{aligned} \tag{3.96}$$

Next, we verify that a feasible solution to (3.96) can be easily obtained from the given feasible solution $(\hat{\Theta}, \hat{\pi}, \hat{\tau})$ of DBQP. Define

$$\Theta(\omega) := \hat{\Theta} - \frac{\omega}{\tilde{C}_{kk}} \mathbf{e}_k \mathbf{e}_k^\mathsf{T} = \hat{\Theta}^{\frac{1}{2}} \left(I - \frac{\omega}{\tilde{C}_{kk}} \hat{\Theta}^{-\frac{1}{2}} \mathbf{e}_k \mathbf{e}_k^\mathsf{T} \hat{\Theta}^{-\frac{1}{2}} \right) \hat{\Theta}^{\frac{1}{2}}.$$

Then, for any ω, we have that $(\Theta(\omega), \hat{\pi}, \hat{\tau}, \omega)$ is a feasible solution to (3.96).

Now, we let $\eta := \operatorname{diag}(\hat{\Theta}^{-1})$ and denote by $\zeta(\omega)$, the objective-function value of (3.96) at $(\Theta(\omega), \hat{\pi}, \hat{\tau}, \omega)$. So,

$$\begin{aligned} \zeta(\omega) &= \hat{\zeta} - \omega - (\operatorname{ldet}(\Theta(\omega)) - \operatorname{ldet}(\hat{\Theta})) + \operatorname{tr}(\Theta(\omega) - \hat{\Theta}) \\ &= \hat{\zeta} - \omega - \log\left(1 - \tfrac{\eta_k \omega}{\tilde{C}_{kk}}\right) - \tfrac{\omega}{\tilde{C}_{kk}} \\ &= \hat{\zeta} - \omega \left(1 + \tfrac{1}{\tilde{C}_{kk}}\right) - \log\left(1 - \tfrac{\eta_k \omega}{\tilde{C}_{kk}}\right). \end{aligned} \tag{3.97}$$

To obtain the value of ω that minimizes $\zeta(\omega)$, we solve

$$\frac{d\zeta}{d\omega} = -\left(1 + \frac{1}{\tilde{C}_{kk}}\right) + \frac{\eta_k}{\tilde{C}_{kk} - \eta_k \omega} = 0$$

and obtain the solution

$$\omega^* := \tilde{C}_{kk} \left(\frac{1}{\eta_k} - \frac{1}{\tilde{C}_{kk} + 1} \right).$$

Substituting ω^* into (3.97), we obtain, after simplifying,

$$\zeta(\omega^*) = \hat{\zeta} - \upsilon_k\ .$$

Finally, considering that LB is a lower bound for the optimal solution of CMESP and $\hat{\zeta} - \upsilon_k$ is an upper bound for its optimal solution when the constraint $x_k = 1$ is included, it is clear that if $\hat{\zeta} - \upsilon_k < \text{LB}$, then $x_k^* = 0$.

Similarly, we consider instead the addition of the constraint $x_k = 0$ to BQP. In this case, the dual problem of (3.94) with the added constraint $\mathbf{e}_k\mathbf{e}_k^\mathsf{T} \bullet \tilde{X} = 0$, differs

from (3.96) only in that the variable ω does not appear in the objective function. Therefore, we can write $\zeta(\omega)$ as in (3.97), but removing ω from its right-hand side. So, we have now

$$\zeta(\omega) = \hat{\zeta} - \frac{\omega}{\tilde{C}_{kk}} - \log\left(1 - \frac{\eta_k \omega}{\tilde{C}_{kk}}\right)$$

and

$$\frac{d\zeta}{d\omega} = -\frac{1}{\tilde{C}_{kk}} + \frac{\eta_k}{\tilde{C}_{kk} - \eta_k \omega} = 0.$$

Then, the minimizer of $\hat{\zeta}$ is

$$\omega^* := \tilde{C}_{kk}\left(\frac{1-\eta_k}{\eta_k}\right).$$

and

$$\zeta(\omega^*) = \hat{\zeta} - \nu_k \ .$$

Similarly to what we discussed for the first case, we can see that if $\hat{\zeta} - \nu_k < \text{LB}$, then $x_k^* = 1$. □

Remark 3.6.13 If BQP is solved to optimality and has optimal solution $(\hat{x}, \hat{X})$, then the diagonal of $\hat{\Theta}$ in Theorem 3.6.12 (which is all we need from the dual solution) can be directly computed from $\hat{x}$. In this case, using complementarity between the positive definite $\hat{\Theta}$ and the positive semidefinite $\tilde{C} \circ \hat{\tilde{X}} + I - \hat{\Theta}^{-1}$, we have that

$$(\hat{\Theta}^{-1})_{kk} = \tilde{C}_{kk}\hat{x}_k + 1, \text{ for } k \in N.$$

Therefore, the strategy for fixing variables described in Theorem 3.6.12 can be applied with the knowledge of the primal optimal solution of BQP alone.

Remark 3.6.14 The cutoffs υ_k and ν_k in Theorem 3.6.12 are nonnegative (because $t - \log(t) - 1 \geq 0$, for all $t > 0$). However, they depend on the dual feasible solution that we construct in the proof. It is possible that with a better choice, we could get higher cutoffs.

3.6.4 A good feasible solution of DBQP from BQP

Variable fixing (see Theorem 3.6.12) relies on a good feasible solution of DBQP. Moreover, certifying the quality of a feasible solution of BQP also requires a good feasible solution of DBQP. Motivated by these points, we show how to construct a feasible solution of DBQP from a feasible solution $(\hat{x}, \hat{X})$ of BQP such that $M(\hat{x}, \hat{X}) \succ 0$, with the goal of producing a small gap. We define

$$\hat{\tilde{X}} := \left(\begin{array}{c|c} \hat{X} & \hat{x} \\ \hline \hat{x}^\mathsf{T} & 1 \end{array}\right).$$

We note that $\tilde{C} \circ \hat{\hat{X}} + I \succ 0$ because $M(\hat{x}, \hat{X}) \succ 0$, and we define $\hat{\Theta} := (\tilde{C} \circ \hat{\hat{X}} + I)^{-1}$. The minimum duality gap between $(\hat{x}, \hat{X})$ in BQP, or equivalently, between $(\hat{\hat{X}}, \hat{W} := \tilde{C} \circ \hat{\hat{X}} + I)$ in (3.94), and feasible solutions of DBQP of the form $(\hat{\Theta}, \pi, \tau)$ is the optimal value of the semidefinite program

$$
(G(\hat{\Theta})) \qquad \begin{array}{l} \min \ \operatorname{tr}(\hat{\Theta}) + \pi^{\mathsf{T}} h^{\leq} + \tau^{\mathsf{T}} h^{=} - (n+1) \\ \quad \text{subject to:} \\ \quad \displaystyle\sum_{i \in \mathcal{M}_{\leq}} \pi_i H_i^{\leq} + \sum_{i \in \mathcal{M}_{=}} \tau_i H_i^{=} \succeq \hat{\Theta} \circ \tilde{C}, \\ \quad \pi \geq 0. \end{array}
$$

We note that $G(\hat{\Theta})$ is feasible: take $\hat{\pi} := 0$, $\hat{\tau}$ as in the proof of Proposition 3.6.11, so that $\sum_{i \in \mathcal{M}_{=}} \hat{\tau}_i H_i^{=} = I$. Then, we scale $\hat{\tau}$ by the maximum eigenvalue of $\hat{\Theta} \circ \tilde{C}$ to obtain a feasible solution of $G(\hat{\Theta})$.

From the Lagrangian dual of BQP, formulated as (3.95), we see that if $(\hat{x}, \hat{X})$ is an optimal solution to BQP, then we have $\Theta = (\tilde{C} \circ \hat{\hat{X}} + I)^{-1}$ in an optimal solution of DBQP. Therefore, an optimal solution of $G(\hat{\Theta})$ gives the optimal values of the remaining variables (π, τ) of DBQP. In this case, due to strong duality for BQP and DBQP, the optimal objective value of $G(\hat{\Theta})$ is equal to zero.

3.6.5 Scaling for BQP

Considering the scaling introduced in §1.6, we define, for a positive constant γ,

$$
\begin{array}{l} f(\gamma; x, X) := \operatorname{ldet} M(\gamma; x, X) - s \log(\gamma), \\ \qquad \text{where } M(\gamma; x, X) := \gamma C \circ X + \operatorname{Diag}(\mathbf{e} - x). \end{array}
$$

We note that $f(x, X) = f(\gamma; x, X)$ for all (x, X) such that $\mathbf{e}^{\mathsf{T}} x = s$, $x \in \{0, 1\}^n$, $X = xx^{\mathsf{T}}$. So we can replace $f(x, X)$ by $f(\gamma; x, X)$ in BQP, defining the scaled BQP relaxation

$$(\mathrm{BQP}_\gamma) \qquad z_{\mathrm{BQP}_\gamma} := \max\{f(\gamma; x, X) \ : \ Ax \le b,\ (x, X) \in P(n, s)\}\,.$$

In the following, we discuss how we select the scaling parameter γ in BQP_γ, aiming at the best possible bound.

We define $\psi := \log(\gamma)$ and formulate the problem

$$\min_{\psi} \{h(\psi)\}, \tag{3.98}$$

where

$$h(\psi) := f(\exp(\psi); x^*, X^*),$$

and where $(x^*, X^*) := (x^*(\psi), X^*(\psi))$ is a maximizer of BQP_γ, with $\gamma(= \exp(\psi))$ fixed.

In Theorem 3.6.15, we show that the objective function of (3.98) is convex in the logarithm of the scaling parameter, which is particularly relevant for the computation of the optimal scaled BQP bound.

Theorem 3.6.15 The function h is convex on $\mathbb{R}$.

Proof $h(\psi)$ is the point-wise maximum of $f(\exp(\psi); x, X)$ over feasible (x, X) for BQP_γ. That is, $h(\psi) = \max\{f(\exp(\psi); x, X) \ : \ Ax \le b,\ (x, X) \in P(n, s)\}$. So, to show convexity of h, it suffices to show that $f(\exp(\psi); \hat{x}, \hat{X})$ is convex in ψ, for all fixed $(\hat{x}, \hat{X})$ in the feasible set of BQP_γ such that $M(\hat{x}, \hat{X}) \succ 0$ (which holds if and only if $M(\exp(\psi); \hat{x}, \hat{X}) \succ 0$ for all $\psi \in \mathbb{R}$).

Using the chain rule and Lemma A.4.13, we have

$$\begin{aligned}
\frac{\partial}{\partial\gamma} f(\gamma; \hat{x}, \hat{X}) &= \frac{\partial}{\partial\gamma}\Big(\operatorname{ldet} M(\gamma; \hat{x}, \hat{X}) - s\log\gamma\Big) \\
&= \frac{\partial}{\partial\gamma}\Big(\operatorname{ldet}\big(\gamma C \circ \hat{X} + \operatorname{Diag}(\mathbf{e} - \hat{x})\big) - s\log\gamma\Big) \\
&= M(\gamma; \hat{x}, \hat{X})^{-1} \bullet (C \circ \hat{X}) - \frac{s}{\gamma} \\
&= \frac{1}{\gamma}\Big(M(\gamma; \hat{x}, \hat{X})^{-1} \bullet (\gamma C \circ \hat{X}) - s\Big) \\
&= \frac{1}{\gamma}\Big(M(\gamma; \hat{x}, \hat{X})^{-1} \bullet M(\gamma; \hat{x}, \hat{X}) - M(\gamma; \hat{x}, \hat{X})^{-1} \bullet \operatorname{Diag}(\mathbf{e} - \hat{x}) - s\Big)
\end{aligned}$$

$$= \frac{1}{\gamma}\left(n - s - M(\gamma;\hat{x},\hat{X})^{-1} \bullet \mathrm{Diag}(\mathbf{e}-\hat{x})\right).$$

Letting $\psi := \log\gamma$, by the chain rule we have

$$\frac{\partial f}{\partial \gamma} = \frac{\partial f}{\partial \psi}\frac{d\psi}{d\gamma} = \frac{\partial f}{\partial \psi}\frac{1}{\gamma}.$$

Then,

$$\begin{aligned}\frac{\partial}{\partial \psi} f(\gamma;\hat{x},\hat{X}) &= \gamma\frac{\partial}{\partial \gamma} f(\gamma;\hat{x},\hat{X}) \\ &= n - s - M(\exp(\psi);\hat{x},\hat{X})^{-1} \bullet \mathrm{Diag}(\mathbf{e}-\hat{x}) =: g(\gamma).\end{aligned}$$

Next, we calculate

$$\begin{aligned}\frac{\partial^2}{\partial \gamma^2} f(\gamma;\hat{x},\hat{X}) &= \frac{\partial}{\partial \gamma}\left(\frac{1}{\gamma}\left(n - s - M(\gamma;\hat{x},\hat{X})^{-1} \bullet \mathrm{Diag}(\mathbf{e}-\hat{x})\right)\right) \\ &= -\frac{1}{\gamma^2}\left(n - s - M(\gamma;\hat{x},\hat{X})^{-1} \bullet \mathrm{Diag}(\mathbf{e}-\hat{x})\right) \\ &\quad + \frac{1}{\gamma}(\mathbf{e}-\hat{x})^{\mathsf{T}}\,\mathrm{diag}\left(M(\gamma;\hat{x},\hat{X})^{-1}(C\circ\hat{X})M(\gamma;\hat{x},\hat{X})^{-1}\right).\end{aligned}$$

Then,

$$\begin{aligned}\gamma^2\frac{\partial^2}{\partial \gamma^2} f(\gamma;\hat{x},\hat{X}) &= -n + s + M(\gamma;\hat{x},\hat{X})^{-1} \bullet \mathrm{Diag}(\mathbf{e}-\hat{x}) \\ &\quad + \gamma(\mathbf{e}-\hat{x})^{\mathsf{T}}\,\mathrm{diag}\left(M(\gamma;\hat{x},\hat{X})^{-1}(C\circ\hat{X})M(\gamma;\hat{x},\hat{X})^{-1}\right).\end{aligned}$$

Again taking $\psi := \log\gamma$ and using the chain rule, we have

$$\frac{\partial^2 f}{\partial \psi^2} = \frac{\partial g}{\partial \psi} = \gamma\frac{\partial g}{\partial \gamma} = \gamma\left(\frac{\partial f}{\partial \gamma} + \gamma\frac{\partial^2 f}{\partial \gamma^2}\right) = \gamma\frac{\partial f}{\partial \gamma} + \gamma^2\frac{\partial^2 f}{\partial \gamma^2}.$$

Finally, we have

$$\begin{aligned}&\frac{\partial^2}{\partial \psi^2} f(\gamma;\hat{x},\hat{X}) \\ &\quad = n - s - M(\exp(\psi);\hat{x},\hat{X})^{-1} \bullet \mathrm{Diag}(\mathbf{e}-\hat{x}) \\ &\qquad - n + s + M(\exp(\psi);\hat{x},\hat{X})^{-1} \bullet \mathrm{Diag}(\mathbf{e}-\hat{x}) \\ &\qquad + \exp(\psi)(\mathbf{e}-\hat{x})^{\mathsf{T}}\,\mathrm{diag}\left(F(\exp(\psi);\hat{x},\hat{X})^{-1}(C\circ\hat{X})M(\exp(\psi);\hat{x},\hat{X})^{-1}\right) \\ &\quad = \exp(\psi)(\mathbf{e}-\hat{x})^{\mathsf{T}}\,\mathrm{diag}\left(M(\exp(\psi);\hat{x},\hat{X})^{-1}(C\circ\hat{X})M(\exp(\psi);\hat{x},\hat{X})^{-1}\right).\end{aligned}$$

It remains to demonstrate that this last expression is nonnegative. We have $C \succeq 0$ and $\hat{X} \succeq 0$, and therefore $C \circ \hat{X} \succeq 0$ by Theorem A.4.8. Therefore, from Lemma A.4.2,

$$M(\exp(\psi); \hat{x}, \hat{X})^{-1}(C \circ \hat{X})M(\exp(\psi); \hat{x}, \hat{X})^{-1} \succeq 0.$$

So we have

$$\frac{\partial^2}{\partial \psi^2} f(\exp(\psi); \hat{x}, \hat{X}) \geq 0,$$

for all $\psi \in \mathbb{R}$, and we can conclude that $f(\exp(\psi); \hat{x}, \hat{X})$ is convex in ψ. □

Considering Theorem 3.6.15, we apply a quasi-Newton method to solve (3.98). We note that the objective function h is not everywhere differentiable, but, as we have observed in §3.3.5, BFGS has nice convergence properties even in nonsmooth situations (see [LO13]), and Proposition 3.6.16 shows how to obtain a subgradient for h.

Proposition 3.6.16 For $\psi \in \mathbb{R}$, we have that $g_\psi(\psi) := \partial f(\exp(\psi); x^*, X^*)/\partial \psi$ is a subgradient of h at ψ.

Proof From the proof of Theorem 3.6.15, we have that f is convex in ψ for the optimal (x, X) in BQP_γ . Then, for arbitrary $\tilde{\psi}$,

$$\begin{aligned} h(\psi) + (\tilde{\psi} - \psi)g_\psi(\psi) &= f(\exp(\psi); x^*, X^*) + (\tilde{\psi} - \psi)g_\psi(\psi) \\ &\leq f(\exp(\tilde{\psi}); x^*, X^*) \leq h(\tilde{\psi}), \end{aligned}$$

where the second inequality holds because (x^*, X^*) need not be an optimal solution of BQP_γ when $\gamma = \exp(\tilde{\psi})$. □

We can use the subgradient presented in Proposition 3.6.16 to replace the gradient in a quasi-Newton method that we could apply to select the scaling parameter γ. However, we should note that as f is continuously differentiable in $\psi (= \log(\gamma))$, then h is differentiable at every ψ for which BQP_γ has a *unique* optimum (see [Fia83, Corollary 3.4.2] or [FI90, Theorem 4.1]). In this case, the subgradient of h is the gradient. In Proposition 3.6.9, we have presented conditions over which we can guarantee that BQP has unique optimal solution. From the result, it is straightforward to see that for each given order-n covariance matrix C, such that $C_{ij} \neq 0$, for $i \neq j$, there are at most n values of ψ where h is not differentiable. This result is stated in the next corollary.

Corollary 3.6.17 If $C_{ij} \neq 0$, for $i \neq j$, and $z_{\text{BQP}} > -\infty$, we have that h is differentiable at every ψ such that $\psi \neq -\log(C_{ii})$, for all $i \in N$.

3.7 Mixing bounds

> "I might sing a gospel song in Arabic or do something in Hebrew. I want to mix it up and do it differently than one might imagine."
>
> - Stevie Wonder, 2013

In this section, we describe a general framework for "mixing" convex-optimization bounds for CMESP, recently described in [CFLL21]. The technique combines bounds to achieve better ones. In fact, the technique is very general and is not confined to CMESP. But CMESP motivated and is particularly suited to the approach, because of its many different convex-optimization bounds (e.g., linx, DDFact, complementary DDFact, NLP, complementary NLP, BQP, complementary BQP, and masked versions of these).

3.7.1 The mixing framework

We start with a general combinatorial maximization problem

$$z := \max\{f(S) \ : \ S \in \mathcal{F}\},$$

where $\mathcal{F}$ is an arbitrary subset of the power set of $\{1, 2, \ldots, n\}$. We consider upper bounds for z based on convex relaxation in a possibly lifted space of variables. The compact and convex set $\mathcal{P}_i$ uses variables $(x, \mathcal{X}^i)$ (ignore the i for now; we will use it later). The vector $x \in [0,1]^n$ relaxes $x \in \{0,1\}^n$ and is used to model sets in $\mathcal{F}$. Specifically, we assume that if we project $\mathcal{P}_i$ onto the x coordinates (i.e., project out the $\mathcal{X}^i$ variables), we get a subset of $[0,1]^n$, and then if we intersect with $\mathbb{Z}^n$, we get precisely the characteristic vectors of sets in $\mathcal{F}$.

Next, we have a concave function f_i, possibly depending on a parameter (vector) ψ_i, taking $(x, \mathcal{X}^i) \in \mathcal{P}_i$ to $\mathbb{R}$. We assume that for $(x, \mathcal{X}^i) \in \mathcal{P}_i$ such that $x \in \{0,1\}^n$, we have $f_i(\psi_i; x, \mathcal{X}^i) = f(\underline{x})$. In this sense, f_i on $\mathcal{P}_i$ is an *exact relaxation* (possibly in an extended space) of f on $\mathcal{F}$.

Now, we consider having $m \geq 2$ relaxations, indexed by $i = 1, \ldots, m$. So, for $i = 1, 2, \ldots, m$, we have the convex programs

$$v_i(\psi_i) := \max\left\{f_i(\psi_i; x, \mathcal{X}^i) \ : \ (x, \mathcal{X}^i) \in \mathcal{P}_i\right\},$$

yielding m upper bounds on z. Associated with each of these bounds $v_i(\psi_i)$ is the convex relaxation $\max\left\{f_i(\psi_i; x, \mathcal{X}^i) \ : \ (x, \mathcal{X}^i) \in \mathcal{P}_i\right\}$.

Next, for $\alpha \in \mathbb{R}^m_+$ such that $\mathbf{e}^\mathsf{T}\alpha = 1$ and $\psi := (\psi_1^\mathsf{T}, \ldots, \psi_m^\mathsf{T})^\mathsf{T}$, we define the **mixing bound**

$$v(\alpha, \psi) := \max\left\{\sum_{i=1}^m \alpha_i f_i(\psi_i; x, \mathcal{X}^i) \ : \ (x, \mathcal{X}^i) \in \mathcal{P}_i \ , \ 1 = 1, 2, \ldots, m\right\}.$$

It is a natural goal to optimize the mixing bound over both of the parameters α and ψ, but generally this is not tractable. We will soon see that generally we can optimize on α, and in some situations we can optimize on ψ.

The following is very simple to establish.

Proposition 3.7.1 For every fixed ψ, the function $v(\alpha, \psi)$ is convex on $\{\alpha \in \mathbb{R}^m \ : \ \alpha \geq 0\}$, and for all $\alpha \in \mathbb{R}^m$ such that $\mathbf{e}^\mathsf{T}\alpha = 1$, we have $v(\alpha, \psi) \geq z$.

Owing to this, a natural goal is to seek the best mixing bound (for fixed ψ) by solving the convex problem

$$\min_\alpha \left\{ v(\alpha, \psi) \ : \ \mathbf{e}^\mathsf{T}\alpha = 1, \ \alpha \in \mathbb{R}^m_+ \right\}. \tag{3.99}$$

The power of the mixing bound is that the same variable x is appearing in each of the $\mathcal{P}_i$. If it were not for this, then the minimum value in (3.99) would trivially be

In some cases, the individual $\mathcal{P}_i$ can be strengthened to improve the mixing bound. But additionally, the mixing bound can be strengthened by introducing valid equations and inequalities across the entire variable space: $x, \mathcal{X}_1, \ldots, \mathcal{X}_m$. We will show how to exploit these observations later, in the context of MESP.

3.7.2 Optimizing the mixing parameters

Next, we discuss some strategies for solving (3.99). Let

$$(x^*, \mathcal{X}^*) := (x^*, \mathcal{X}^{1*}, \ldots, \mathcal{X}^{m*}) := (x^*(\alpha, \psi), \mathcal{X}^{1*}(\alpha, \psi), \ldots, \mathcal{X}^{m*}(\alpha, \psi))$$

be an optimal solution of (3.7.1), for given α and ψ, and let $f_i^*(\alpha, \psi) := f_i(\psi_i; x^*, \mathcal{X}^{i*})$, $i = 1, \ldots, m$. Then,

$$v(\alpha, \psi) = \sum_{i=1}^m \alpha_i f_i^*(\alpha, \psi).$$

We have the following simple but useful result.

Proposition 3.7.2 For every fixed ψ,

$$g_\alpha(\alpha, \psi) := (f_1^*(\alpha, \psi), f_2^*(\alpha, \psi), \ldots, f_m^*(\alpha, \psi))^\mathsf{T}$$

is a subgradient of v (with respect to α) at $\alpha \in \mathbb{R}^m_+$ such that $\mathbf{e}^\mathsf{T}\alpha = 1$.

Proof For $\tilde{\alpha} \in \mathbb{R}^m$ such that $\mathbf{e}^\mathsf{T}\tilde{\alpha} = 1$,

$$v(\alpha,\psi) + (\tilde{\alpha}-\alpha)^\mathsf{T} g_\alpha(\alpha,\psi) = \sum_{i=1}^m \alpha_i f_i^*(\alpha,\psi) + \sum_{i=1}^m \tilde{\alpha}_i f_i^*(\alpha,\psi) - \sum_{i=1}^m \alpha_i f_i^*(\alpha,\psi)$$
$$= \sum_{i=1}^m \tilde{\alpha}_i f_i^*(\alpha,\psi) \le \sum_{i=1}^m \tilde{\alpha}_i f_i^*(\tilde{\alpha},\psi) = v(\tilde{\alpha},\psi),$$

where the inequality holds because $(x^*, \mathcal{X}^*)$ need not be optimal for $\tilde{\alpha}$. □

Via the standard projected subgradient algorithm (optimizing $v(\alpha,\psi)$ on the simplex $\{\alpha \in \mathbb{R}^m_+ \ : \ \mathbf{e}^\mathsf{T}\alpha = 1\}$), this gives us a convergent algorithm for (3.99). Additionally, when $m = 2$, we can rewrite v as a function of α_1 alone ($\alpha_2 = 1 - \alpha_1$), and then the subgradient for the now univariate v becomes the scalar $f_1^*(\alpha_1,\psi) - f_2^*(\alpha_1,\psi)$. With this, we can use a simple bisection search for $\alpha_1 \in [0,1]$, considering the sign of $f_1^*(\alpha_1,\psi) - f_2^*(\alpha_1,\psi)$ at each iteration, to get an improved algorithm when $m = 2$.

We note that v is not everywhere differentiable, but it is at α for which (3.7.1) has a unique optimum (see [Fia83, Corollary 3.4.2] or [FI90, Theorem 4.1]), in which case the subgradient identified is the gradient. With this observation, we can devise algorithms for (3.99) that are typically faster; see [CFLL21] for details.

Next we consider situations in which $f_i(\psi_i; x, \mathcal{X}^i)$ is convex in ψ_i , for each fixed $(x, \mathcal{X}^i)$, $i = 1, \ldots, m$, and we wish to optimize the bound $v(\alpha,\psi)$ over the parameter ψ, for fixed $\alpha \in \mathbb{R}^m_+$ such that $\mathbf{e}^\mathsf{T}\alpha = 1$:

$$\min_\psi \{v(\alpha,\psi)\}. \tag{3.100}$$

This is the case when $\psi_i = \exp(\gamma)$, for the scaling parameter γ for the linx_γ bound, the BQP_γ bound, and the complementary BQP_γ bound.

Proposition 3.7.3 For every fixed $\alpha \in \mathbb{R}^m_+$ such that $\mathbf{e}^\mathsf{T}\alpha = 1$, let g_{ψ_i} be a subgradient of $f_i(\psi_i; x^*, \mathcal{X}^{i*})$ with respect to ψ_i . Then

$$g_\psi(\alpha,\psi) := \left(\alpha_1 g_{\psi_1}^\mathsf{T}, \ldots, \alpha_m g_{\psi_m}^\mathsf{T}\right)^\mathsf{T}$$

is a subgradient of v (with respect to ψ) at ψ.

Proof For arbitrary $\tilde{\psi}$,

$$v(\alpha,\psi) + (\tilde{\psi}-\psi)^\mathsf{T} g_\psi(\alpha,\psi) = \sum_{i=1}^m \alpha_i f_i^*(\alpha,\psi) + \sum_{i=1}^m (\tilde{\psi}_i - \psi_i)^\mathsf{T} \alpha_i g_{\psi_i}$$

$$\leq \sum_{i=1}^{m} \alpha_i f_i(\tilde{\psi}_i; x^*, \mathcal{X}^{i*}) \leq \sum_{i=1}^{m} \alpha_i f_i^*(\alpha, \tilde{\psi}) = v(\alpha, \tilde{\psi}),$$

where the first inequality holds because g_{ψ_i} is a subgradient of $f_i(\psi_i; x^*, \mathcal{X}^{i*})$ (for $i = 1, \ldots, m$), and the second holds because $(x^*, \mathcal{X}^*)$ need not be optimal for $\tilde{\psi}$. □

We note that when the f_i are continuously differentiable, v is differentiable at ψ for which (3.100) has a unique solution; in which case the subgradient referred to in Proposition 3.7.3 is the gradient. With this observation, we can devise practical algorithms for (3.99); see [CFLL21] for details.

3.8 Duality for mixing

In this section, we develop convex duality for mixing, with an eye toward carrying out variable fixing in the context of mixing bounds.

We consider $m \geq 1$ convex relaxations for CMESP, indexed by $i = 1, \ldots, m$:

$$v_i := \max \left\{ f_i(M_i(x)) \ : \ \mathbf{e}^\mathsf{T} x = s, \ Ax \leq b, \ 0 \leq x \leq \mathbf{e} \right\},$$

where, for $i = 1, \ldots, m$, $k_i \leq n$, $M_i : \mathbb{R}^n \to \mathbb{S}_+^{k_i}$ are affine functions, and $f_i : \mathbb{S}_+^{k_i} \to \mathbb{R}$ are concave functions. We write $M_i(x) := M_{i0} + M_{i1}x_1 + \cdots + M_{in}x_n$ and $M_{ij} \in \mathbb{S}^{k_i}$, for $i = 1, \ldots, m$ and $j = 0, \ldots, n$, and we note that the objective functions of DDFact, complementary DDFact, and linx can be written as $f_i(M_i(x))$ (see §3.8.2.1).

For $\alpha \in \mathbb{R}_+^m$ such that $\mathbf{e}^\mathsf{T}\alpha = 1$, we define the following mixing bound:

$$\text{(mix)} \qquad v(\alpha) := \max \left\{ \sum_{i=1}^{m} \alpha_i f_i(M_i(x)) \ : \ \mathbf{e}^\mathsf{T} x = s, \ Ax \leq b, \ 0 \leq x \leq \mathbf{e} \right\}.$$

Next, we construct the Lagrangian dual of mix for a broad class of cases that covers some of our applications of mixing. We recast mix as

$$\max \left\{ \sum_{i=1}^{m} \alpha_i f_i(W_i) \ : \ \mathbf{e}^\mathsf{T} x = s, \ Ax \leq b, \ 0 \leq x \leq \mathbf{e}, \ \alpha_i M_i(x) = \alpha_i W_i \ , \ i = 1, \ldots, m \right\},$$

and we consider the Lagrangian function

$$\mathcal{L}(W_1, \ldots, W_m, x, \Theta_1, \ldots, \Theta_m, \upsilon, \nu, \pi, \tau) :=$$
$$\sum_{i=1}^{m} \alpha_i \Big(f_i(W_i) + \Theta_i \bullet (M_i(x) - W_i) \Big) + \upsilon^\mathsf{T} x + \nu^\mathsf{T}(\mathbf{e} - x) + \pi^\mathsf{T}(b - Ax) + \tau(s - \mathbf{e}^\mathsf{T} x),$$

with dom $\mathcal{L} = \left(\mathbb{S}_+^{k_i}\right)^m \times \mathbb{R}^n \times \left(\mathbb{S}^{k_i}\right)^m \times \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}$. The dual function is

$$\mathcal{L}^*(\Theta_1, \ldots, \Theta_m, \upsilon, \nu, \pi, \tau) := \sup_{W_i \succeq 0, \ x} \mathcal{L}(W_1, \ldots, W_m, x, \Theta_1, \ldots, \Theta_m, \upsilon, \nu, \pi, \tau),$$

and the Lagrangian dual problem is

$$\min\{\mathcal{L}^*(\Theta_1,\ldots,\Theta_m,\upsilon,\nu,\pi,\tau)\ :\ \upsilon\geq 0,\ \nu\geq 0,\ \pi\geq 0\}.$$

We note that

$$\sup_{W_i\succeq 0,\ x}\ \mathcal{L}(W_1,\ldots,W_m,x,\Theta_1,\ldots,\Theta_m,\upsilon,\nu,\pi,\tau)=$$
$$\sum_{i=1}^{m}\alpha_i\sup_{W_i\succeq 0}\Big(f_i(W_i)-\Theta_i\bullet W_i\Big)$$
$$+\sup_{x}\left\{\sum_{i=1}^{m}\alpha_i\Big(\Theta_i\bullet M_i(x)\Big)+\upsilon^{\mathsf{T}}x+\nu^{\mathsf{T}}(\mathbf{e}-x)+\pi^{\mathsf{T}}(b-Ax)+\tau(s-\mathbf{e}^{\mathsf{T}}x)\right\}.$$

For $i=1,\ldots,m$, we assume that for any given $\hat{\Theta}_i\in\mathbb{S}^{k_i}_{++}$, there is a closed-form solution $\hat{W}_i$ to $\sup\{f_i(W_i)-\hat{\Theta}_i\bullet W_i\ :\ W_i\succeq 0\}$, such that $\hat{\Theta}_i\bullet\hat{W}_i=:\rho_i\in\mathbb{R}$ and $\Omega_i:\mathbb{S}^{k_i}_{++}\to\mathbb{R}$, is defined by $\Omega_i(\hat{\Theta}_i):=f_i(\hat{W}_i)$. Furthermore, we assume that the supremum is $+\infty$ if $\hat{\Theta}_i\not\succ 0$. Then we have

$$\alpha_i\sup_{W_i\succeq 0}\Big(f_i(W_i)-\Theta_i\bullet W_i\Big)=\begin{cases}\alpha_i\Big(\Omega_i(\Theta_i)-\rho_i\Big), & \text{if } \Theta_i\succ 0;\\ +\infty, & \text{otherwise,}\end{cases}$$

and we also have

$$\sup_{x}\left\{\sum_{i=1}^{m}\alpha_i\Big(\Theta_i\bullet M_i(x)\Big)+\upsilon^{\mathsf{T}}x+\nu^{\mathsf{T}}(\mathbf{e}-x)+\pi^{\mathsf{T}}(b-Ax)+\tau(s-\mathbf{e}^{\mathsf{T}}x)\right\}$$
$$=\begin{cases}\displaystyle\sum_{i=1}^{m}\alpha_i\Big(\Theta_i\bullet M_{i0}\Big)+\nu^{\mathsf{T}}\mathbf{e}+\pi^{\mathsf{T}}b+\tau s,\\ \qquad\text{if } \displaystyle\sum_{i=1}^{m}\alpha_i\Big(\Theta_i\bullet M_{ij}\Big)+\upsilon_j-\nu_j-\pi^{\mathsf{T}}A_{\cdot j}-\tau=0,\ \text{ for } j\in N;\\ +\infty,\quad\text{otherwise.}\end{cases}$$

Then, we see that the Lagrangian dual problem of mix is equivalent to

(Dmix)
$$\begin{aligned}&z_{\text{Dmix}}(C,s,A,b):=\\ &\min\ \sum_{i=1}^{m}\alpha_i\Big(\Omega_i(\Theta_i)-\rho_i+\Theta_i\bullet M_{i0}\Big)+\nu^{\mathsf{T}}\mathbf{e}+\pi^{\mathsf{T}}b+\tau s\\ &\text{subject to:}\\ &\sum_{i=1}^{m}\alpha_i\Big(\Theta_i\bullet M_{ij}\Big)+\upsilon_j-\nu_j-\pi^{\mathsf{T}}A_{\cdot j}-\tau=0,\ \text{ for } j\in N,\\ &\Theta_1\succ 0,\ldots,\Theta_m\succ 0,\ \upsilon\geq 0,\ \nu\geq 0,\ \pi\geq 0.\end{aligned}$$

3.8.1 Fixing variables in mix

Theorem 3.8.1 Let

- LB be the objective-function value of a feasible solution for CMESP,
- $(\hat{\Theta}_1, \ldots, \hat{\Theta}_m, \hat{\upsilon}, \hat{\nu}, \hat{\pi}, \hat{\tau})$ be a feasible solution for Dmix with objective-function value $\hat{\zeta}$.

Then, for every optimal solution x^* for CMESP, we have:

$$x^*_k = 0, \ \forall\, k \in N \text{ such that } \hat{\zeta} - \text{LB} < \hat{\upsilon}_k \ ,$$
$$x^*_k = 1, \ \forall\, k \in N \text{ such that } \hat{\zeta} - \text{LB} < \hat{\nu}_k \ .$$

Proof Analogous to the proof of Theorem 3.3.9. □

3.8.2 A good feasible solution of Dmix from mix

Variable fixing (see Theorem 3.8.1) relies on a good feasible solution of Dmix. Moreover, certifying the quality of a feasible solution of mix also requires a good feasible solution of Dmix. Motivated by these points, we show how to construct a feasible solution of Dmix from a feasible solution $\hat{x}$ of mix, with the goal of producing a small gap.

We let $\hat{x}$ be a feasible solution of mix in the domain of f_i and define $\hat{W}_i := M_i(\hat{x})$, for $i = 1, \ldots, m$. First, we assume that it is possible to compute $\hat{\Theta}_i$, such that $\Omega_i(\hat{\Theta}_i) = f_i(\hat{W}_i)$, for $i = 1, \ldots, m$ (see §3.8.2.1, for example). Then, the minimum duality gap between $\hat{x}$ in mix and feasible solutions of Dmix of the form $(\hat{\Theta}_1, \ldots, \hat{\Theta}_m, \upsilon, \nu, \pi, \tau)$ is the optimal value of the linear program

$$(G(\hat{\Theta}_1, \ldots, \hat{\Theta}_m)) \qquad \begin{aligned} \min \ & \nu^{\mathsf{T}}\mathbf{e} + \pi^{\mathsf{T}} b + \tau s - \sum_{i=1}^{m} \alpha_i \left(\rho_i - \hat{\Theta}_i \bullet M_{i0}\right) \\ & \text{subject to:} \\ & \upsilon_j - \nu_j - \pi^{\mathsf{T}} A_{\cdot j} - \tau = -\sum_{i=1}^{m} \alpha_i \left(\hat{\Theta}_i \bullet M_{ij}\right), \quad \text{for } j \in N, \\ & \upsilon \geq 0, \ \nu \geq 0, \ \pi \geq 0. \end{aligned}$$

Analogously to $G(\hat{\Theta})$ (see §3.3.4), we can verify that $G(\hat{\Theta}_1, \ldots, \hat{\Theta}_m)$ is always feasible and has a simple closed-form solution for MESP. In fact, the only differences between $G(\hat{\Theta})$ and $G(\hat{\Theta}_1, \ldots, \hat{\Theta}_m)$ are the constant in the objective function and the right-hand side of the constraints.

3.8.2.1 Considering DDFact, complementary DDFact, and linx in mix

Considering $f_i(M_i(x))$ as the objective function of DDFact, we have $f_i(\cdot) := \Gamma_s(\cdot)$ and $M_i(x) := F^T \operatorname{Diag}(x)F$, so $k_i := k$, $M_{i0} := 0$ and $M_{ij} := F_{j\cdot}^T F_{j\cdot}$, for $j = 1, \ldots, n$. We also have $\rho_i := s$ and $\Omega_i(\Theta_i) := -\sum_{\ell=k-s+1}^{k} \log(\lambda_\ell(\Theta_i))$. For a given feasible solution $\hat{x}$ of mix in the domain of f_i and $\hat{W}_i := M_i(\hat{x})$, we construct $\hat{\Theta}_i$ as discussed in §3.4.2, and we see in (3.30), that $\Omega_i(\hat{\Theta}_i) = f_i(\hat{W}_i)$.

Considering $f_i(M_i(x))$ as the objective function of complementary DDFact, we have $f_i(\cdot) := \Gamma_{n-s}(\cdot) + \operatorname{ldet} C$ and $M_i(x) := F^{-1} \operatorname{Diag}(\mathbf{e} - x)F^{-T}$, so $k_i := k(= n)$, $M_{i0} := F^{-1}F^{-T}$ and $M_{ij} := -F_{j\cdot}^{-1}F_{j\cdot}^{-T}$, for $j = 1, \ldots, n$. We also have $\rho_i := n - s$ and $\Omega_i(\Theta_i) := -\sum_{\ell=k-n+s+1}^{k} \log(\lambda_\ell(\Theta_i)) + \operatorname{ldet} C$. For a given feasible solution $\hat{x}$ of mix in the domain of f_i and $\hat{W}_i := M_i(\hat{x})$, we construct $\hat{\Theta}_i$ as discussed in §3.4.2, and we see in (3.30), that $\Omega_i(\hat{\Theta}_i) = f_i(\hat{W}_i)$.

Considering $f_i(M_i(x))$ as the objective function of linx, we have $f_i(\cdot) := \frac{1}{2}(\operatorname{ldet}(\cdot) - s\log\gamma)$ and $M_i(x) := \gamma C \operatorname{Diag}(x) C + \operatorname{Diag}(\mathbf{e} - x)$, so $k_i := n$, $M_{i0} := I$ and $M_{ij} := \gamma C_{j\cdot}^T C_{j\cdot} - \mathbf{e}_j \mathbf{e}_j^T$, for $j = 1, \ldots, n$. We also have $\rho_i := n/2$ and $\Omega_i(\Theta_i) := -\frac{1}{2}(\operatorname{ldet}(2\Theta_i) + s\log\gamma)$. For a given feasible solution $\hat{x}$ of mix in the domain of f_i and $\hat{W}_i := M_i(\hat{x})$, we set $\hat{\Theta}_i := \frac{1}{2}\hat{W}_i^{-1}$, and we see that $\Omega_i(\hat{\Theta}_i) = f_i(\hat{W}_i)$.

3.8.3 Mixing the BQP$_\gamma$ bound with the complementary BQP$_\gamma$ bound

In this section, we apply the mixing idea to the combination of the BQP$_\gamma$ bound and the complementary BQP$_\gamma$ bound. Because the BQP$_\gamma$ bound uses an additional (matrix) variable X (beyond the x variable used in the other convex-optimization bounds), the mixing idea elaborates more in such a context.

We assume that $C \succ 0$, as we need to when working with the complementary problem. Recall

$$P(n,s) := \{(x,X) \in \mathbb{R}^n \times \mathbb{S}^n \ : \\ X - xx^T \succeq 0,\ \operatorname{diag}(X) = x,\ \mathbf{e}^T x = s,\ X\mathbf{e} = sx\},$$

and we now also define

$$Q(n,n-s) := \{(y,Y) \in \mathbb{R}^n \times \mathbb{S}^n \ : \\ Y - yy^T \succeq 0,\ \operatorname{diag}(Y) = y,\ \mathbf{e}^T y = n - s,\ Y\mathbf{e} = (n-s)y\}.$$

These are the feasible regions for the BQP$_\gamma$ relaxation and the complementary BQP$_\gamma$ relaxation of CMESP.

We define the **mixed BQP bound**:

$$
\begin{aligned}
& v(C,s;\alpha,\gamma_1,\gamma_2) := \\
& \quad \max\ (1-\alpha)\left(\operatorname{ldet}\left(\gamma_1 C \circ X + \operatorname{Diag}(\mathbf{e}-x)\right) - s\log\gamma_1\right) \\
\text{(mBQP)} \qquad & \quad + \alpha\left(\operatorname{ldet}\left(\gamma_2 C^{-1} \circ Y + \operatorname{Diag}(\mathbf{e}-y)\right) - (n-s)\log\gamma_2 + \operatorname{ldet} C\right) \\
& \quad \text{subject to:} \\
& \quad Ax \le b,\ (x,X) \in P(n,s),\ (y,Y) \in Q(n,n-s),\ x+y=\mathbf{e},
\end{aligned}
$$

where $0 \le \alpha \le 1$ is a weighting parameter, and $\gamma_1, \gamma_2 > 0$ are the BQP_γ scaling parameters.

We will see that this mBQP bound is a manifestation of mixing the BQP_γ bound with its complement. It is almost immediate that the mBQP bound is a mixing in the precise sense that was defined in general, but because of the way that we have formulated it with different variables for the complementary part, there is a little checking to do.

We define an invertible linear map Υ by $\Upsilon(x,X) := (\mathbf{e}-x, X + \mathbf{e}\mathbf{e}^{\mathsf{T}} - \mathbf{e}x^{\mathsf{T}} - x\mathbf{e}^{\mathsf{T}})$. Notice that if $(\hat{y},\hat{Y}) := \Upsilon(\hat{x},\hat{X})$, then $\hat{Y}_{ij} = \hat{X}_{ij} + 1 - \hat{x}_j - \hat{x}_i$. We then have the following useful result.

Lemma 3.8.2 $(\hat{x},\hat{X}) \in P(n,s)$ if and only if $\Upsilon(\hat{x},\hat{X}) \in Q(n,n-s)$.

Proof We check the constraints:

$$
\begin{aligned}
& \hat{Y} - \hat{y}\hat{y}^{\mathsf{T}} = \hat{X} + \mathbf{e}\mathbf{e}^{\mathsf{T}} - \mathbf{e}\hat{x}^{\mathsf{T}} - \hat{x}\mathbf{e}^{\mathsf{T}} - (\mathbf{e}-\hat{x})(\mathbf{e}-\hat{x})^{\mathsf{T}} = \hat{X} - \hat{x}\hat{x}^{\mathsf{T}} \succeq 0. \\
& \operatorname{diag}(\hat{Y}) = \operatorname{diag}(\hat{X}) + \operatorname{diag}(\mathbf{e}\mathbf{e}^{\mathsf{T}}) - \operatorname{diag}(\mathbf{e}\hat{x}^{\mathsf{T}}) - \operatorname{diag}(\hat{x}\mathbf{e}^{\mathsf{T}}) = \hat{x} + \mathbf{e} - \hat{x} - \hat{x} = \hat{y}. \\
& \mathbf{e}^{\mathsf{T}}\hat{y} = \mathbf{e}^{\mathsf{T}}(\mathbf{e}-\hat{x}) = n - s. \\
& \hat{Y}\mathbf{e} = \left(\hat{X} + \mathbf{e}\mathbf{e}^{\mathsf{T}} - \mathbf{e}\hat{x}^{\mathsf{T}} - \hat{x}\mathbf{e}^{\mathsf{T}}\right)\mathbf{e} = s\hat{x} + n\mathbf{e} - s\mathbf{e} - n\hat{x} = (n-s)(\mathbf{e}-\hat{x}) = (n-s)\hat{y}.
\end{aligned}
$$

The other direction is similar. □

Proposition 3.8.3 $v(C, s; \alpha = 0, \gamma_1, \gamma_2)$ is equal to the scaled BQP bound

$$\begin{aligned} -s\log\gamma_1 \;+\; \max\;\; & \operatorname{ldet}\left(\gamma_1 C \circ X + \operatorname{Diag}(\mathbf{e} - x)\right) \\ & \text{subject to:} \\ & Ax \leq b,\;\; (x, X) \in P(n, s), \end{aligned}$$

and $v(C, s; \alpha = 1, \gamma_1, \gamma_2)$ is equal to the scaled complementary BQP bound

$$\begin{aligned} \operatorname{ldet} C - (n-s)\log\gamma_2 \;+\; \max\;\; & \operatorname{ldet}\left(\gamma_2 C^{-1} \circ Y + \operatorname{Diag}(\mathbf{e} - y)\right) \\ & \text{subject to:} \\ & A(\mathbf{e} - y) \leq b,\;\; (y, Y) \in Q(n, n-s). \end{aligned}$$

Proof When $\alpha = 0$, for any $(\hat{x}, \hat{X}) \in P(n, s)$, Lemma 3.8.2 allows us to always be able to choose a $(\hat{y}, \hat{Y})$, which together with $(\hat{x}, \hat{X})$ is feasible for mBQP when $A\hat{x} \leq b$. And because $\alpha = 0$, the choice of $(\hat{y}, \hat{Y})$ has no impact on the mBQP objective function. Similarly, when $\alpha = 1$, for any $(\hat{y}, \hat{Y}) \in Q(n, n-s)$, Lemma 3.8.2 allows us to always be able to choose a $(\hat{x}, \hat{X})$ which together with $(\hat{y}, \hat{Y})$ is feasible for mBQP when $A(\mathbf{e} - \hat{y}) \leq b$. And because $\alpha = 1$, the choice of $(\hat{x}, \hat{X})$ has no impact on the mBQP objective function. □

So we can conclude now that the mBQP bound is equivalent to mixing the BQP_γ bound (using $\gamma = \gamma_1$) with the complementary BQP_γ bound, (using $\gamma = \gamma_2$), and furthermore we can optimize the scaling parameters for mBQP using the techniques of §3.7.2.

Next, we will see that we can strengthen the mBQP bound, using equations that link the extended variables from the two bounds that we mix, and then even eliminate the variables (y, Y). We define the **strengthened mixed BQP bound**:

(smBQP)

$$\begin{aligned} \check{v}(C, s; \alpha, \gamma_1, \gamma_2) := & \\ \max\;\; & (1-\alpha)\left(\operatorname{ldet}\left(\gamma_1 C \circ X + \operatorname{Diag}(\mathbf{e} - x)\right) - s\log\gamma_1\right) \\ & +\alpha\Big(\operatorname{ldet}\left(\gamma_2 C^{-1} \circ (X + \mathbf{e}\mathbf{e}^\mathsf{T} - \mathbf{e}x^\mathsf{T} - x\mathbf{e}^\mathsf{T}) + \operatorname{Diag}(x)\right) \\ & \qquad -(n-s)\log\gamma_2 + \operatorname{ldet} C\Big) \\ & \text{subject to:} \\ & Ax \leq b,\;\; (x, X) \in P(n, s). \end{aligned}$$

Proposition 3.8.4

$$v(C, s; \alpha, \gamma_1, \gamma_2) \;\geq\; \check{v}(C, s; \alpha, \gamma_1, \gamma_2).$$

Proposition 3.8.4 follows from Lemma 3.8.2 and the following simple lemma.

Lemma 3.8.5 For the solutions of $x + y = \mathbf{e}$, $X = xx^{\mathsf{T}}$, $Y = yy^{\mathsf{T}}$, the equation $Y = X + \mathbf{e}\mathbf{e}^{\mathsf{T}} - \mathbf{e}x^{\mathsf{T}} - x\mathbf{e}^{\mathsf{T}}$ is valid.

Proof Under $x + y = \mathbf{e}$, we have that

$$0 = Y - yy^{\mathsf{T}} = Y - (\mathbf{e} - x)(\mathbf{e} - x)^{\mathsf{T}} = Y - \mathbf{e}\mathbf{e}^{\mathsf{T}} + \mathbf{e}x^{\mathsf{T}} + x\mathbf{e}^{\mathsf{T}} - xx^{\mathsf{T}}.$$

Subtracting $0 = X - xx^{\mathsf{T}}$, we obtain the desired equation. □

Remark 3.8.6 As is the case for BQP, the formulation smBQP can be conveniently modeled using CVX or YALMIP. In particular, YALMIP in conjunction with the solver SDPT3 provides a very efficient algorithmic treatment of the ldet terms. Furthermore, It is easy to see that we can optimize the scaling parameters for smBQP by adapting the techniques of §3.7.2.

Remark 3.8.7 [CFLL21] contains computational results on several mixings for CMESP, including the mixing that we presented in this section. All of those computational results are on mixing a pair of bounds. An interesting three-way mixing would combine the linx_γ bound, the NLP_γ bound, and the complementary NLP_γ bound. With scaling parameters $\gamma_0, \gamma_1, \gamma_2$, respectively, this bound is the same as as the same three-way mixing for the complementary problem, but now with scaling parameters $1/\gamma_0, \gamma_2, \gamma_1$, respectively. So, such a bound is "self-complementary" in a sense similar to that of the linx_γ bound (cf. Proposition 3.3.16). But an advantage of this three-way mixing, as compared to the linx_γ bound is that it inherits the property from NLP_γ bound that it is exact (i.e., equal to the optimal value of MESP) when C is a diagonal matrix. Of course, this three-way mixing is expensive to compute — in fact, we do not even have a convenient software framework for mixing the linx_γ bound with either the NLP_γ bound or the complementary NLP_γ bound (see [CFLL21] for details).

3.8.4 Duality for smBQP

In §3.6.2, we developed the dual for BQP, and because of the simple relationship between BQP_γ and BQP, it is then very easy to write down the dual for BQP_γ . Similarly, for the complementary versions. In what follows, we develop the dual of smBQP, assuming $0 < \alpha < 1$. In this situation, we carry along the scaling parameters

γ_1, γ_2 because the treatment of the two problems that are mixed is not symmetric (as it would be for mBQP).

In order to formulate the dual problem for smBQP, we define the following matrices in $\mathbb{S}^{n+1}$:

$$\tilde{X} = \left(\begin{array}{c|c} X & x \\ \hline x^\mathsf{T} & 1 \end{array}\right), \ \tilde{C} = \left(\begin{array}{c|c} \gamma_1 C - I_n & 0 \\ \hline 0 & 0 \end{array}\right),$$
$$\bar{Y} = \left(\begin{array}{c|c} Y & y \\ \hline y^\mathsf{T} & 1 \end{array}\right), \ \bar{C} = \left(\begin{array}{c|c} \gamma_2 C^{-1} - I_n & 0 \\ \hline 0 & 0 \end{array}\right).$$

Then, smBQP can be rewritten as

$$\begin{array}{ll} \max & (1-\alpha)(\operatorname{ldet}\tilde{W} - s\log\gamma_1) \\ & \quad + \alpha(\operatorname{ldet}\bar{W} - (n-s)\log\gamma_2 + \operatorname{ldet} C) \\ & \text{subject to:} \\ & (1-\alpha)(\tilde{C}\circ\tilde{X} + I - \tilde{W}) = 0, \\ & \alpha(\bar{C}\circ\bar{Y} + I - \bar{W}) = 0, \\ (3.101) \quad & H_i^{\leq} \bullet \tilde{X} \leq h_i^{\leq}, \ \text{for } i \in \mathcal{M}_{\leq}, \\ & H_i^{=} \bullet \tilde{X} = h_i^{=}, \ \text{for } i \in \mathcal{M}_{=}, \\ & \tilde{X} \succeq 0, \\ & \bar{Y} = \left(\begin{array}{c|c} \tilde{X}[N,N] + J - J(I\circ\tilde{X}[N,N]) - (I\circ\tilde{X}[N,N])J & \mathbf{e} - \tilde{X}[N,n+1] \\ \hline \mathbf{e}^\mathsf{T} - \tilde{X}[n+1,N] & 1 \end{array}\right), \end{array}$$

where each $H_i^{\leq}$ and each $H_i^{=}$ is a symmetric matrix. As in §3.6.2, $H_i^{\leq} \bullet \tilde{X} \leq h_i^{\leq}$ ($i \in \mathcal{M}_{\leq}$) includes the $Ax \leq b$ constraints of CMESP; and $H_i^{=} \bullet \tilde{X} = h_i^{=}$ ($i \in \mathcal{M}_{=}$) includes: $\operatorname{diag}(X) = x$, $\mathbf{e}^\mathsf{T} x = s$, and $X\mathbf{e} = sx$ (from $P(n,s)$), all written in the $\tilde{X}$ variable, and also $\tilde{X}_{n+1,n+1} = 1$. The last equation models the constraint $\bar{Y}_{n+1,n+1} = 1$ and the correspondences $y = \mathbf{e} - x$ and $Y = X + \mathbf{e}\mathbf{e}^\mathsf{T} - \mathbf{e}x^\mathsf{T} - x\mathbf{e}^\mathsf{T}$ seen in Lemma 3.8.5.

We define $E_{ij} := \left(\mathbf{e}_i\mathbf{e}_j^\mathsf{T} + \mathbf{e}_j\mathbf{e}_i^\mathsf{T}\right)/2 \in \mathbb{S}^{n+1}$, for all $i, j = 1, \ldots, n+1$, and we consider the Lagrangian function

$$\begin{aligned} \mathcal{L}(\tilde{W}, \bar{W}, \tilde{X}, \bar{Y}, \Theta, \Phi, \Psi, \Omega, \pi, \tau, \rho, \sigma) := \\ (1-\alpha)(\operatorname{ldet}\tilde{W} - s\log\gamma_1) + \alpha(\operatorname{ldet}\bar{W} - (n-s)\log\gamma_2 + \operatorname{ldet} C) \\ + \ (1-\alpha)\Theta \bullet (\tilde{C}\circ\tilde{X} + I - \tilde{W}) + \alpha\,\Phi \bullet (\bar{C}\circ\bar{Y} + I - \bar{W}) + \Psi \bullet \tilde{X} \\ + \sum_{i\in\mathcal{M}_{\leq}} \pi_i\left(h_i^{\leq} - H_i^{\leq} \bullet \tilde{X}\right) + \sum_{i\in\mathcal{M}_{=}} \tau_i\left(h_i^{=} - H_i^{=} \bullet \tilde{X}\right) \\ + \sum_{i,j\in N} \Omega_{ij}(1 - E_{ij} \bullet \bar{Y} + E_{ij} \bullet \tilde{X} - E_{ii} \bullet \tilde{X} - E_{jj} \bullet \tilde{X}) \\ + \sum_{i\in N} \rho_i(1 - E_{i,n+1} \bullet \bar{Y} - E_{i,n+1} \bullet \tilde{X}) + \sigma(1 - E_{n+1,n+1} \bullet \bar{Y}), \end{aligned}$$

with $\mathrm{dom}\,\mathcal{L} = \mathbb{S}^{n+1}_{++} \times \mathbb{S}^{n+1}_{++} \times \mathbb{S}^{n+1} \times \mathbb{S}^{n+1} \times \mathbb{S}^{n+1} \times \mathbb{S}^{n+1} \times \mathbb{S}^{n+1} \times \mathbb{S}^{n} \times \mathbb{R}^{|\mathcal{M}_{\leq}|} \times \mathbb{R}^{|\mathcal{M}_{=}|} \times \mathbb{R}^{n} \times \mathbb{R}$.

As we did in §3.6.2, we write the Lagrangian dual problem of (3.101) as

$$
\begin{aligned}
&\min\,(1-\alpha)(\mathrm{ldet}\,\tilde{W} - s\log\gamma_1) + \alpha(\mathrm{ldet}\,\bar{W} - (n-s)\log\gamma_2 + \mathrm{ldet}\,C) \\
&\quad + \ (1-\alpha)\Theta \bullet (\tilde{C} \circ \tilde{X} + I - \tilde{W}) + \alpha\,\Phi \bullet (\bar{C} \circ \bar{Y} + I - \bar{W}) + \Psi \bullet \tilde{X} \\
&\quad + \sum_{i\in\mathcal{M}_{\leq}} \pi_i \left(h_i^{\leq} - H_i^{\leq} \bullet \tilde{X}\right) + \sum_{i\in\mathcal{M}_{=}} \tau_i \left(h_i^{=} - H_i^{=} \bullet \tilde{X}\right) \\
&\quad + \sum_{i,j\in N} \Omega_{ij}(1 - E_{ij} \bullet \bar{Y} + E_{ij} \bullet \tilde{X} - E_{ii} \bullet \tilde{X} - E_{jj} \bullet \tilde{X}) \\
&\quad + \sum_{i\in N} \rho_i(1 - E_{i,n+1} \bullet \bar{Y} - E_{i,n+1} \bullet \tilde{X}) + \sigma(1 - E_{n+1,n+1} \bullet \bar{Y}) \\
&\text{subject to:} \\
&(1-\alpha)\tilde{W}^{-1} - (1-\alpha)\Theta = 0, \\
&\alpha\bar{W}^{-1} - \alpha\Phi = 0, \\
&(1-\alpha)\Theta \circ \tilde{C} + \Psi - \sum_{i\mathcal{M}_{\leq}} \pi_i H_i^{\leq} - \sum_{i\in\mathcal{M}_{=}} \tau_i H_i^{=} \\
&\qquad + \sum_{i,j\in N} \Omega_{ij}(E_{ij} - E_{ii} - E_{jj}) - \sum_{i\in N} \rho_i E_{i,n+1} = 0, \\
&\alpha\Phi \circ \bar{C} - \sum_{i,j\in N} \Omega_{ij} E_{ij} - \sum_{i\in N} \rho_i E_{i,n+1} - \sigma E_{n+1,n+1} = 0, \\
&\tilde{W} \succ 0,\ \bar{W} \succ 0,\ \Psi \geq 0,\ \pi \geq 0.
\end{aligned}
\tag{3.102}
$$

Theorem 3.8.8 The following equation is valid for all feasible solutions of (3.102):

$$
\begin{aligned}
&(1-\alpha)\Theta \bullet (\tilde{C} \circ \tilde{X} + I - \tilde{W}) + \alpha\,\Phi \bullet (\bar{C} \circ \bar{Y} + I - \bar{W}) + \Psi \bullet \tilde{X} \\
&\quad + \sum_{i\in\mathcal{M}_{\leq}} \pi_i \left(h_i^{\leq} - H_i^{\leq} \bullet \tilde{X}\right) + \sum_{i\in\mathcal{M}_{=}} \tau_i \left(h_i^{=} - H_i^{=} \bullet \tilde{X}\right) \\
&\quad + \sum_{i,j\in N} \Omega_{ij}(1 - E_{ij} \bullet \bar{Y} + E_{ij} \bullet \tilde{X} - E_{ii} \bullet \tilde{X} - E_{jj} \bullet \tilde{X}) \\
&\quad + \sum_{i\in N} \rho_i(1 - E_{i,n+1} \bullet \bar{Y} - E_{i,n+1} \bullet \tilde{X}) + \sigma(1 - E_{n+1,n+1} \bullet \bar{Y}) \\
&= (1-\alpha)\,(\mathrm{tr}(\Theta) - (n+1)) + \pi^{\mathsf{T}} h^{\leq} + \tau^{\mathsf{T}} h^{=} + \alpha\left(\mathrm{tr}(\Phi) - (n+1) + \Phi \bullet \bar{C}\right).
\end{aligned}
$$

Proof From the first constraint in (3.102), we have $\Theta = \tilde{W}^{-1}$. So, from the third constraint, we can obtain

$$
\begin{aligned}
&(1-\alpha)\Theta \bullet (\tilde{C} \circ \tilde{X} + I - \tilde{W}) + \Psi \bullet \tilde{X} \\
&\quad + \sum_{i \in \mathcal{M}_{\leq}} \pi_i \left(h_i^{\leq} - H_i^{\leq} \bullet \tilde{X}\right) + \sum_{i \in \mathcal{M}_{=}} \tau_i \left(h_i^{=} - H_i^{=} \bullet \tilde{X}\right) \\
&\quad + \sum_{i,j \in N} \Omega_{ij} (E_{ij} \bullet \tilde{X} - E_{ii} \bullet \tilde{X} - E_{jj} \bullet \tilde{X}) - \sum_{i \in N} \rho_i (E_{i,n+1} \bullet \tilde{X}) \\
&= (1-\alpha)\Theta \bullet (I - \tilde{W}) + \sum_{i \in \mathcal{M}_{\leq}} \pi_i h_i^{\leq} + \sum_{i \in \mathcal{M}_{=}} \tau_i h_i^{=} \\
&= (1-\alpha)\left(\mathrm{tr}(\Theta) - (n+1)\right) + \pi^{\mathsf{T}} h^{\leq} + \tau^{\mathsf{T}} h^{=} .
\end{aligned}
$$

From the second constraint in (3.102), we obtain $\Phi = \bar{W}^{-1}$. Moreover, from the fourth constraint, we can see that $\rho = 0$ and $\sigma = 0$, leading to $\alpha\ \Phi \circ \bar{C} = \sum_{i,j \in N} \Omega_{ij} E_{ij}$. Finally, from the fourth constraint, we can then obtain

$$
\begin{aligned}
&\alpha\ \Phi \bullet (\bar{C} \circ \bar{Y} + I - \bar{W}) \\
&\quad + \sum_{i,j \in N} \Omega_{ij} (1 - E_{ij} \bullet \bar{Y}) + \sum_{i \in N} \rho_i (1 - E_{i,n+1} \bullet \bar{Y}) + \sigma (1 - E_{n+1,n+1} \bullet \bar{Y}) \\
&= \alpha\ \Phi \bullet (I - \bar{W}) + \sum_{i,j \in N} \Omega_{ij} \\
&= \alpha \left(\mathrm{tr}(\Phi) - (n+1) + \Phi \bullet \bar{C}\right) .
\end{aligned}
$$

□

Considering Theorem 3.8.8, we finally re-cast the Lagrangian dual problem (3.102) of (3.101), as

$$
\begin{aligned}
&\min\ \pi^{\mathsf{T}} h^{\leq} + \tau^{\mathsf{T}} h^{=} + (1-\alpha)\left(-\mathrm{ldet}\,\Theta + \mathrm{tr}(\Theta) - (n+1) - s \log \gamma_1\right) \\
&\qquad + \alpha \Big(- \mathrm{ldet}\,\Phi + \mathrm{tr}(\Phi) + \Phi \bullet \bar{C} \\
&\qquad\quad - (n+1) - (n-s)\log \gamma_2 + \mathrm{ldet}\, C \Big) \\
&\qquad \text{subject to:} \\
&(1-\alpha)\Theta \circ \tilde{C} + \alpha\ \Phi \circ \bar{C} - 2\alpha \sum_{i,j \in N} \Phi[i,j] \bar{C}[i,j] E_{ii} \\
&\qquad - \sum_{i \mathcal{M}_{\leq}} \pi_i H_i^{\leq} - \sum_{i \in \mathcal{M}_{=}} \tau_i H_i^{=} \preceq 0, \\
&\qquad \Theta \succ 0,\ \Phi \succ 0,\ \pi \geq 0.
\end{aligned}
\tag{DsmBQP}
$$

Proposition 3.8.9 If smBQP is feasible, then we have strong duality for smBQP and DsmBQP.

Proof From general principles concerning convex duality, if DsmBQP is strictly feasible, then the maximum value in smBQP is attained, and we have strong duality, assuming that smBQP is feasible. Therefore, it suffices to construct a strictly feasible solution for DsmBQP.

Referring to the general linear equations $H_i^= \bullet \tilde{X} = h_i^=$ of (3.101), we formulate the constraints $\mathrm{Diag}(x) = x$ and $\mathbf{e}^{\mathsf{T}} x = s$ of smBQP as follows.

We formulate the constraints $X_{ii} = x_i$, $i \in N$, with

$$H_i^= := \left(\begin{array}{c|c} E_{ii} & -\frac{1}{2}\mathbf{e}_i \\ \hline -\frac{1}{2}\mathbf{e}_i^{\mathsf{T}} & 0 \end{array}\right), \quad h_i^= := 0.$$

We formulate the constraint $\mathbf{e}^{\mathsf{T}} x = s$ with

$$H_{n+1}^= := \left(\begin{array}{c|c} 0 & \frac{1}{2}\mathbf{e} \\ \hline \frac{1}{2}\mathbf{e}^{\mathsf{T}} & 1 \end{array}\right), \quad h_{n+1}^= := s + 1.$$

Let $\hat{\tau}_i := 1$, for $i = 1, \ldots, n+1$. Notice that $\sum_{i=1}^{n+1} \hat{\tau}_i H_i^= = I_{n+1}$. Let $\hat{\tau}_i := 0$, for all $i \in \mathcal{M}_= \setminus \{1, \ldots, n+1\}$.

We define a diagonal matrix $\hat{\Theta} \in \mathbb{S}_{++}^{n+1}$ by

$$\hat{\Theta}_{ii} := \begin{cases} 1/\big(2\tilde{C}_{ii}(1-\alpha)\big), & \text{if } \tilde{C}_{ii} \neq 0; \\ 1/2, & \text{if } \tilde{C}_{ii} = 0, \end{cases}$$

for all $i \in \{1, \ldots, n+1\}$.

We define a diagonal matrix $\hat{\Phi} \in \mathbb{S}_{++}^{n+1}$ by

$$\hat{\Phi}_{ii} := \begin{cases} 1/\big(\tilde{C}_{ii}\alpha\big), & \text{if } \tilde{C}_{ii} \neq 0; \\ 1, & \text{if } \tilde{C}_{ii} = 0, \end{cases}$$

for all $i \in \{1, \ldots, n+1\}$.

Let $\hat{\pi}_i := 0$, $i \in \mathcal{M}_\leq$. Then

$$\begin{aligned} &(1-\alpha)\hat{\Theta} \circ \tilde{C} + \alpha\, \hat{\Phi} \circ \bar{C} - 2\alpha \sum_{i,j \in N} \hat{\Phi}[i,j]\bar{C}[i,j]E_{ii} - \sum_{i \in \mathcal{M}_\leq} \hat{\pi}_i H_i^\leq - \sum_{i \in \mathcal{M}_=} \hat{\tau}_i H_i^= \\ &= (1-\alpha)\hat{\Theta} \circ \tilde{C} - \alpha\, \hat{\Phi} \circ \bar{C} - \sum_{i \in \mathcal{M}_=} \hat{\tau}_i H_i^= =: \Psi, \end{aligned}$$

where $\Psi \prec 0$ is a diagonal matrix with

$$\hat{\Psi}_{ii} := \begin{cases} -1, & \text{if } \tilde{C}_{ii} = 0 \text{ and } \bar{C}_{ii} = 0; \\ -1/2, & \text{if } \tilde{C}_{ii} \neq 0 \text{ and } \bar{C}_{ii} = 0; \\ -2, & \text{if } \tilde{C}_{ii} = 0 \text{ and } \bar{C}_{ii} \neq 0; \\ -3/2, & \text{if } \tilde{C}_{ii} \neq 0 \text{ and } \bar{C}_{ii} \neq 0, \end{cases}$$

for all $i \in \{1, \ldots, n+1\}$. Therefore, we have that $(\hat{\Theta}, \hat{\Phi}, \hat{\pi}, \hat{\tau})$ is strictly feasible for DsmBQP. □

As we mentioned in Remark 3.8.6, a convenient and efficient means for solving smBQP is with YALMIP/SDPT3. Conveniently for us, YALMIP also returns to us an optimal solution of DsmBQP. We will see in the next section what we can do with any dual feasible solution.

3.8.5 Fixing variables in smBQP

Next, we develop a methodology for fixing x_i variables at zero or one, based on a feasible solution of DsmBQP and a feasible solution of CMESP. Already this was complicated in the context of DBQP (see §3.6.3). Here, it gets even a bit more complicated, but the general idea is the same.

Theorem 3.8.10 Let

- LB be the objective-function value of a feasible solution of CMESP,
- $(\hat{\Theta}, \hat{\Phi}, \hat{\pi}, \hat{\tau})$ be a feasible solution of DsmBQP with objective-function value $\hat{\zeta}$.

Define

- $\tilde{N} := \{k \in N \ : \ \tilde{C}_{kk} \neq 0, \ \bar{C}_{kk} \neq 0\}$,
- $\tilde{\eta} := \operatorname{diag} \hat{\Theta}^{-1}$, $\bar{\eta} := \operatorname{diag} \hat{\Phi}^{-1}$,
- $\upsilon_k := (1-\alpha)\left(\frac{\tilde{C}_{kk}+1}{\tilde{\eta}_k} - \log\left(\frac{\tilde{C}_{kk}+1}{\tilde{\eta}_k}\right) - 1\right) + \alpha\left(\frac{1}{\hat{\eta}_k} - \log\left(\frac{1}{\hat{\eta}_k}\right) - 1\right), \ \forall\, k \in \tilde{N}$,
- $\nu_k := (1-\alpha)\left(\frac{1}{\tilde{\eta}_k} - \log\left(\frac{1}{\tilde{\eta}_k}\right) - 1\right) + \alpha\left(\frac{\bar{C}_{kk}+1}{\hat{\eta}_k} - \log\left(\frac{\bar{C}_{kk}+1}{\hat{\eta}_k}\right) - 1\right), \ \forall\, k \in \tilde{N}$.

Then, for every optimal solution x^* of CMESP, we have:

$$x^*_k = 0, \forall\, k \in \tilde{N} \ \text{ such that } \ \hat{\zeta} - \text{LB} < \upsilon_k \ ,$$
$$x^*_k = 1, \forall\, k \in \tilde{N} \ \text{ such that } \ \hat{\zeta} - \text{LB} < \nu_k \ .$$

Proof We first consider the addition of the constraint $x_k = 1$ to smBQP, for $k \in \tilde{N}$, which is equivalently modeled as $\mathbf{e}_k \mathbf{e}_k^{\mathsf{T}} \bullet \tilde{X} = 1$ in (3.101), where here $\mathbf{e}_k$ is a standard unit vector in $\mathbb{R}^{n+1}$. The dual problem of (3.101) with this constraint added can be written as

$$
\begin{aligned}
&\min \pi^{\mathsf{T}} h^{\leq} + \tau^{\mathsf{T}} h^{=} + (1-\alpha)\Big(- \operatorname{ldet}(\Theta) + \operatorname{tr}(\Theta) - (n+1) - s\log\gamma_1\Big) \\
&\quad +\alpha\Big(- \operatorname{ldet}\Phi + \operatorname{tr}(\Phi) + \Phi\bullet\bar{C} \\
&\qquad -(n+1) - (n-s)\log\gamma_2 + \operatorname{ldet} C\Big) - \omega \\
&\text{subject to:} \\
&(1-\alpha)\Theta\circ\tilde{C} + \alpha\Phi\circ\bar{C} - 2\alpha \sum_{i,j\in N} \Phi[i,j]\bar{C}[i,j]E_{ii} \\
&\quad - \sum_{i\mathcal{M}_{\leq}} \pi_i H_i^{\leq} - \sum_{i\in\mathcal{M}_{=}} \tau_i H_i^{=} + \omega \mathbf{e}_k \mathbf{e}_k^{\mathsf{T}} \preceq 0, \\
&\Theta \succ 0,\ \Phi \succ 0,\ \pi \geq 0.
\end{aligned}
\tag{3.103}
$$

Next, we verify that feasible solutions to (3.103) can be obtained from the given feasible solution $(\hat{\Theta}, \hat{\Phi}, \hat{\pi}, \hat{\tau})$ to DsmBQP. We define

$$
\begin{aligned}
\Theta(\omega) &:= \hat{\Theta} - \frac{1}{(1-\alpha)\tilde{C}_{kk}}\omega \mathbf{e}_k \mathbf{e}_k^{\mathsf{T}} = \hat{\Theta}^{\frac{1}{2}} \left(I - \frac{1}{(1-\alpha)\tilde{C}_{kk}}\omega \hat{\Theta}^{-\frac{1}{2}} \mathbf{e}_k \mathbf{e}_k^{\mathsf{T}} \hat{\Theta}^{-\frac{1}{2}} \right) \hat{\Theta}^{\frac{1}{2}}, \\
\Phi(\omega) &:= \hat{\Phi} + \frac{1}{\alpha\bar{C}_{kk}}\omega \mathbf{e}_k \mathbf{e}_k^{\mathsf{T}} = \hat{\Phi}^{\frac{1}{2}} \left(I + \frac{1}{\alpha\bar{C}_{kk}}\omega \hat{\Phi}^{-\frac{1}{2}} \mathbf{e}_k \mathbf{e}_k^{\mathsf{T}} \hat{\Phi}^{-\frac{1}{2}} \right) \hat{\Phi}^{\frac{1}{2}},
\end{aligned}
$$

for all ω. Then, we can easily see that, for any ω, both $(\Theta(\omega), \hat{\Phi}, \hat{\pi}, \hat{\tau}, \omega)$ and $(\hat{\Theta}, \Phi(\omega), \hat{\pi}, \hat{\tau}, \omega)$ are feasible solutions to (3.103).

Now, denote by $\zeta_1(\omega)$ and $\zeta_2(\omega)$, the objective-function value of (3.103) at $(\Theta(\omega), \hat{\Phi}, \hat{\pi}, \hat{\tau}, \omega)$ and $(\hat{\Theta}, \Phi(\omega), \hat{\pi}, \hat{\tau}, \omega)$, respectively. So,

$$
\begin{aligned}
\zeta_1(\omega) &= \hat{\zeta} - \omega + (1-\alpha)\left(- \left(\operatorname{ldet}\Theta(\omega) - \operatorname{ldet}\hat{\Theta} \right) + \operatorname{tr}(\Theta(\omega) - \hat{\Theta} \right) \\
&= \hat{\zeta} - \omega + (1-\alpha)\left(- \log\left(1 - \frac{\tilde{\eta}_k \omega}{(1-\alpha)\tilde{C}_{kk}} \right) - \frac{\omega}{(1-\alpha)\tilde{C}_{kk}} \right), \\
\zeta_2(\omega) &= \hat{\zeta} - \omega + \alpha\left(- \left(\operatorname{ldet}\Phi(\omega) - \operatorname{ldet}\hat{\Phi} \right) + \operatorname{tr}\left(\Phi(\omega) - \hat{\Phi} \right) + \left(\Phi(\omega)\bullet\bar{C} - \hat{\Phi}\bullet\bar{C} \right) \right) \\
&= \hat{\zeta} - \omega + \alpha\left(- \log\left(1 + \frac{\bar{\eta}_k \omega}{\alpha\bar{C}_{kk}} \right) + \frac{\omega}{\alpha\bar{C}_{kk}} + \frac{\omega}{\alpha} \right).
\end{aligned}
$$

Next, we obtain the value of ω that minimizes ζ_1, by solving

$$
\frac{d\zeta_1}{d\omega} = -1 + (1-\alpha)\left(\frac{\frac{\tilde{\eta}_k}{(1-\alpha)\tilde{C}_{kk}}}{1 - \frac{\tilde{\eta}_k \omega}{(1-\alpha)\tilde{C}_{kk}}} - \frac{1}{(1-\alpha)\tilde{C}_{kk}} \right) = 0,
$$

and obtaining the solution

$$
\omega_1^* = (1-\alpha)\tilde{C}_{kk}\left(\frac{1}{\tilde{\eta}_k} - \frac{1}{\tilde{C}_{kk}+1} \right).
$$

Similarly, we obtain the value of ω that minimizes ζ_2, by solving

$$\frac{d\zeta_2}{d\omega} = -1 + \alpha \left(- \frac{\frac{\bar{\eta}_k}{\alpha \bar{C}_{kk}}}{1 + \frac{\bar{\eta}_k \omega}{\alpha \bar{C}_{kk}}} + \frac{1}{\alpha \bar{C}_{kk}} (1 + \bar{C}_{kk}) \right) = 0,$$

and obtaining the solution

$$\omega_2^* = \alpha \bar{C}_{kk} \left(1 - \frac{1}{\bar{\eta}_k} \right).$$

We can now construct a feasible solution $(\Theta(\omega_1^*), \Phi(\omega_2^*), \hat{\pi}, \hat{\tau}, \omega_1^* + \omega_2^*)$ to (3.103). The objective-function value of (3.103) at this solution, is given by

$$\begin{aligned}
\zeta(\omega_1^*, \omega_2^*) &= \hat{\zeta} - \omega_1^* - \omega_2^* + (1 - \alpha) \left(- \log \left(1 - \frac{\tilde{\eta}_k \omega_1^*}{(1 - \alpha) \tilde{C}_{kk}} \right) - \frac{\omega_1^*}{(1 - \alpha) \tilde{C}_{kk}} \right) \\
&\quad + \alpha \left(- \log \left(1 + \frac{\bar{\eta}_k \omega_2^*}{\alpha \bar{C}_{kk}} \right) + \frac{\omega_2^*}{\alpha \bar{C}_{kk}} \left(1 + \bar{C}_{kk} \right) \right) \\
&= \hat{\zeta} + (1 - \alpha) \left(1 - \frac{\tilde{C}_{kk} + 1}{\tilde{\eta}_k} + \log \left(\frac{\tilde{C}_{kk} + 1}{\tilde{\eta}_k} \right) \right) + \alpha \left(1 - \frac{1}{\bar{\eta}_k} + \log \left(\frac{1}{\bar{\eta}_k} \right) \right) \\
&= \hat{\zeta} - \upsilon_k \, .
\end{aligned}$$

As LB is a lower bound for the optimal solution of CMESP and $\hat{\zeta} - \upsilon_k^1$ is an upper bound for its optimal solution when the constraint $x_k = 1$ is included, it is clear that if $\hat{\zeta} - \upsilon_k <$ LB, then $x_k^* = 0$.

Now, we consider the addition of the constraint $x_k = 0$ to smBQP, which is equivalently modeled as $\mathbf{e}_k \mathbf{e}_k^\mathsf{T} \bullet \tilde{X} = 0$ in (3.101). In this case, the dual problem of (3.101) with this constraint added is similar to (3.103), but without the variable ω added to the objective function. We then have the functions ζ_1 and ζ_2 rewritten as

$$\zeta_1(\omega) = \hat{\zeta} + (1 - \alpha) \left(- \log \left(1 - \frac{\tilde{\eta}_k \omega}{(1 - \alpha) \tilde{C}_{kk}} \right) - \frac{\omega}{(1 - \alpha) \tilde{C}_{kk}} \right),$$

$$\zeta_2(\omega) = \hat{\zeta} + \alpha \left(- \log \left(1 + \frac{\bar{\eta}_k \omega}{\alpha \bar{C}_{kk}} \right) + \frac{\omega}{\alpha \bar{C}_{kk}} \left(1 + \bar{C}_{kk} \right) \right).$$

Then,

$$\frac{d\zeta_1}{d\omega} = (1 - \alpha) \left(\frac{\frac{\tilde{\eta}_k}{(1 - \alpha) \tilde{C}_{kk}}}{1 - \frac{\tilde{\eta}_k \omega}{(1 - \alpha) \tilde{C}_{kk}}} - \frac{1}{(1 - \alpha) \tilde{C}_{kk}} \right) = 0$$

has solution

$$\omega_1^* = (1 - \alpha) \tilde{C}_{kk} \left(-1 + \frac{1}{\tilde{\eta}_k} \right).$$

Similarly,

$$\frac{d\zeta_2}{d\omega} = \alpha\left(-\frac{\frac{\bar{\eta}_k}{\alpha\bar{C}_{kk}}}{1+\frac{\bar{\eta}_k\omega}{\alpha\bar{C}_{kk}}} + \frac{1}{\alpha\bar{C}_{kk}}(1+\bar{C}_{kk})\right) = 0$$

has solution

$$\omega_2^* = \alpha\bar{C}_{kk}\left(\frac{1}{\bar{C}_{kk}+1} - \frac{1}{\bar{\eta}_k}\right).$$

The objective value of the dual at the solution $\left(\Theta(\omega_1^*), \Phi(\omega_2^*), \hat{\pi}, \hat{\tau}, \omega_1^* + \omega_2^*\right)$ is

$$\begin{aligned}\zeta(\omega_1^*, \omega_2^*) &= \hat{\zeta} + (1-\alpha)\left(-\log\left(1 - \frac{\tilde{\eta}_k\omega_1^*}{(1-\alpha)\tilde{C}_{kk}}\right) - \frac{\omega_1^*}{(1-\alpha)\tilde{C}_{kk}}\right) \\ &\quad + \alpha\left(-\log\left(1+\frac{\bar{\eta}_k\omega_2^*}{\alpha\bar{C}_{kk}}\right) + \frac{\omega_2^*}{\alpha\bar{C}_{kk}}\left(1+\bar{C}_{kk}\right)\right) \\ &= \hat{\zeta} - \nu_k\ .\end{aligned}$$

Using the same argument as above, we see that if $\hat{\zeta} - \nu_k < \text{LB}$, then $x_k^* = 1$. □

Remark 3.8.11 The cutoffs υ_k and ν_k in Theorem 3.8.10 are nonnegative (because $t - \log(t) - 1 \geq 0$, for all $t > 0$). However, they depend on the dual feasible solution that we construct in the proof. It is possible that with a better choice, we could get higher cutoffs.

3.8.6 A good feasible solution of DsmBQP from smBQP

Variable fixing (see Theorem 3.8.10) relies on a good feasible solution of DsmBQP. Moreover, certifying the quality of a feasible solution of smBQP also requires a good feasible solution of DsmBQP. Motivated by these points, we show how to construct a feasible solution of DsmBQP from a feasible solution $(\hat{x}, \hat{X})$ of smBQP such that $\gamma_1 C \circ \hat{X} + \text{Diag}(\mathbf{e} - \hat{x}) \succ 0$, with the goal of producing a small gap. We define $(\hat{y}, \hat{Y}) := \Upsilon(\hat{x}, \hat{X})$ and

$$\hat{\hat{X}} := \left(\begin{array}{c|c} \hat{X} & \hat{x} \\ \hline \hat{x}^\mathsf{T} & 1 \end{array}\right), \qquad \hat{\hat{Y}} := \left(\begin{array}{c|c} \hat{Y} & \hat{y} \\ \hline \hat{y}^\mathsf{T} & 1 \end{array}\right).$$

We note that $\tilde{C} \circ \hat{\hat{X}} + I \succ 0$ and $\bar{C} \circ \hat{\hat{Y}} + I \succ 0$ because $\gamma_1 C \circ \hat{X} + \text{Diag}(\mathbf{e} - \hat{x}) \succ 0$ and therefore also $\gamma_2 C^{-1} \circ (\hat{X} + \mathbf{e}\mathbf{e}^\mathsf{T} - \mathbf{e}\hat{x}^\mathsf{T} - \hat{x}\mathbf{e}^\mathsf{T}) + \text{Diag}(\hat{x}) \succ 0$. We define $\hat{\Theta} := (\tilde{C} \circ \hat{\hat{X}} + I)^{-1}$ and $\hat{\Phi} := (\bar{C} \circ \hat{\hat{Y}} + I)^{-1}$. The minimum duality gap between $(\hat{x}, \hat{X})$ in smBQP and feasible solutions of DsmBQP of the form $(\hat{\Theta}, \hat{\Phi}, \pi, \tau)$ is the optimal value of the semidefinite program

$$
\begin{aligned}
& \min\ \pi^{\mathsf{T}} h^{\leqslant} + \tau^{\mathsf{T}} h^{=} + (1-\alpha)\,\mathrm{tr}(\hat{\Theta}) + \alpha\left(\mathrm{tr}(\hat{\Phi}) + \Phi \bullet \bar{C}\right) - (n+1) \\
& \text{subject to:} \\
(G(\hat{\Theta},\hat{\Phi})) \qquad & \sum_{i \mathcal{M}_{\leqslant}} \pi_i H_i^{\leqslant} + \sum_{i \in \mathcal{M}_{=}} \tau_i H_i^{=} \\
& \qquad \geq (1-\alpha)\hat{\Theta} \circ \tilde{C} + \alpha\, \hat{\Phi} \circ \bar{C} - 2\alpha \sum_{i,j \in N} \hat{\Phi}[i,j]\bar{C}[i,j]E_{ii}\,, \\
& \pi \geq 0.
\end{aligned}
$$

We note that $G(\hat{\Theta},\hat{\Phi})$ is feasible: take $\hat{\pi} := 0$, $\hat{\tau}$ as in the proof of Proposition 3.8.9, so that $\sum_{i \in \mathcal{M}_{=}} \hat{\tau}_i H_i^{=} = I$. Then, we scale $\hat{\tau}$ by the maximum eigenvalue of $(1-\alpha)\hat{\Theta} \circ \tilde{C} + \alpha\, \hat{\Phi} \circ \bar{C} - 2\alpha \sum_{i,j \in N} \hat{\Phi}[i,j]\bar{C}[i,j]E_{ii}$ to obtain a feasible solution of $G(\hat{\Theta},\hat{\Phi})$.

From the Lagrangian dual of smBQP, formulated as (3.102), we see that if $(\hat{x},\hat{X})$ is an optimal solution to smBQP, then we have $(\Theta = (\tilde{C} \circ \hat{\tilde{X}} + I)^{-1}, \Phi = (\bar{C} \circ \hat{\tilde{Y}} + I)^{-1})$ in an optimal solution of DsmBQP. Therefore, an optimal solution of $G(\hat{\Theta},\hat{\Phi})$ gives the optimal values of the remaining variables (π,τ) of DsmBQP. In this case, due to strong duality for smBQP and DsmBQP, the optimal objective value of $G(\hat{\Theta},\hat{\Phi})$ is equal to zero.

3.9 Comparison of bounds

We have described several upper bounds for MESP and CMESP. Taking some risk at oversimplifying, we will make some summary remarks and comparisons.

- Spectral bounds:
  - For MESP, the spectral bound is very fast to compute.
  - The spectral bound is invariant under scaling.
  - The spectral bound is self complementary.
  - The spectral bound can be adapted for CMESP, but at a significant computational cost.
- Convex-optimization bounds:
  - General properties:
    - As compared to spectral bounding methods, bounding methods based on convex optimization immediately adapt to CMESP (i.e., including linear side constraints in the convex relaxation).

  - Bounding methods based on convex optimization admit variable fixing methodology based on convex duality.
  - Bounding methods based on convex optimization are amenable to the mixing methodology.
- The linx bound:
  - Its formulation linx is convex, smooth, and efficiently solvable with even general-purpose nonlinear-programming software.
  - The linx bound is sensitive to a scale factor, but finding the optimal scale factor can be cast as a univariate convex minimization problem.
  - The linx bound is self complementary.
- The factorization bound:
  - Its formulation DDFact is convex and efficiently solvable with even general-purpose nonlinear-programming software.
  - The factorization bound is independent of the chosen factorization.
  - The factorization bound provably dominates the spectral bound.
  - The factorization bound is invariant under scaling.
  - Complementation is useful for the factorization bound.
- The NLP bound:
  - With appropriate choices of parameters, its formulation NLP is convex, smooth, and efficiently solvable with nonlinear-programming software.
  - The NLP bound is sensitive to a scale factor.
  - Complementation is useful for the NLP bound.
- The BQP bound:
  - Its formulation BQP is convex, smooth, and efficiently solvable with nonlinear-programming software.
  - The BQP bound is sensitive to a scale factor, but finding the optimal scale factor can be cast as a univariate convex minimization problem.
  - Complementation is useful for the BQP bound.
  - An apparent shortcoming of the BQP bound is its large number of variables. But for instances of CMESP that happen to include quadratic constraints on the binary variables x_i, the BQP might be advantageous to work with.

A general remark is that masks can be quite useful for improving possibly all upper-bounding methods, but finding a good (non-optimal) mask (for any particular upper-bounding method) can be quite expensive. More work on this is needed.

Finally, we do not have definitive conclusions as to which bounding methods can be the most effective in a branch-and-bound context. But based on all of our experiences, we should all try to get along.

Chapter 4
Environmental monitoring

> "Seldom do more than a few of nature's secrets give way at one time. It will be all too easy for our somewhat artificial prosperity to collapse overnight when it is realized that the use of a few exciting words like *information, entropy, redundancy,* do not solve all our problems... I personally believe that many of the concepts of information theory will prove useful in these other fields — and, indeed, some results are already quite promising — but the establishing of such applications is not a trivial matter of translating words to a new domain, but rather the slow tedious process of hypothesis and experimental verification."
>
> - Claude Shannon, 1956 [Sha56]

4.1 The setting

4.1.1 MESP within statistics and optimal experimental design

Mainly, our starting point for MESP has been the short seminal paper by Shewry and Wynn, [SW87]. But it is useful to put that paper into some context. There is the very broad and well-developed field of *optimal experimental design*. At the center of this theory is the "Fisher information matrix". It is a device for measuring the information that some observable random vector X carries regarding some parameter vector θ of the density $f(x, \theta)$ for X. As such, the concept is tightly coupled with a distributional form. In fact, under suitable conditions, the **Fisher information matrix** is the negative of the Hessian of the log likelihood at θ (while the density $f(x, \theta)$ is normally viewed as a function of x for fixed parameter vector θ, the **likelihood** is just viewing it as a function of θ for fixed observations x). The usual criteria for minimization in optimal experimental design are functions of the eigenvalues of the Fisher information matrix.

At around the same time of [SW87], we have an interesting work by Ylvisaker, [Ylv87] (even referenced in [SW87] as "to appear"). This work is concerned with optimum design when a stochastic process is finitely sampled, over say time or space.

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It ties optimal design into the field of spatial statistics (see [GDFG10]), which aims at estimation/prediction of a spatial process at locations that do not have observations. A prominent tool for this is "kriging", also known as "Gaussian process regression". [Ylv87] looks at a Gaussian process X_T (where T is a finite set of "sites") and $E[X_T] = 0$. Conditioned on a finite set of observations X_S, $S \subset T$, he defines the "conditional process" and the difference as the "error process". Then he frames the idea of a design problem in this context:

> "For a sample of limited side, it is natural to consider the design questions which accompany prediction. Thus, corresponding to [the processes conditioned on a finite set of observations], determine $S \subset T$, with $|S| = n$, to minimize the error process. Evidently [this] requires some notion of a small process."

And then [SW87], drawing on [Lin56] and others, advanced the idea of using entropy as the measure. Note that in a closed system, the entropy of the conditional process plus the entropy of the observed part of the process is the entropy of the entire system. As [SW87] puts it,

> "This criterion is different in philosophy from those which arise in classical optimum design. Here we wish to absorb into the sample the maximum amount of variability so that conditionally on the sample the unsampled population has minimum variability."

So [SW87] follows the same idea as Ylvisaker, which naturally leads to adopting the information-theoretic concept of entropy, when the system is closed.

Remark 4.1.1 Notably, at the time of [SW87], the only methodology available for solving MESP problems was local search:

> "... amenable to a simple exchange algorithm in which a single point in S is exchanged for a point in $\bar{S}$. Extension to multiple exchange or 'excursion' algorithms following the ideas of [Mit74] and others is under way. This includes close collaboration with J. Sacks at the University of Illinois, Urbana-Champaign, and co-workers, who have carried out extensive work with the simulated annealing algorithm. It is typical of these computational problems in spatial sampling that local optima are prevalent and experience is needed with combinations of exchange and stochastic relaxation methods."

This all changed, beginning with [KLQ95], when mathematical optimizers started getting involved.

4.1.2 MESP and environmental statistics

For a broad reference to the field of environmental statistics, an enticing reference would appear to be the six-volume (3510 page) [ESP12] — but at the jaw-dropping price of $2,314, we have never had the opportunity to see more than [Lee12]. Much more modestly priced is the nice book [LZ06] (and the free companion software `EnviroStat` [LZWC19]); in particular, Chapter 11: Environmental Network Design, includes material on the applications of MESP and CMESP to environmental monitoring.

We should mention that in the context of monitoring networks, [Hus79, HC80, CZ84] are entropy-based precursors to the precise [SW87] Gaussian formulation of MESP. Later, there was extensive application of MESP for monitoring networks; see [CKZ92, WZ92, GLSZ93, BLZ94, WLZ20, ZSL00].

In the simplest setting, we have univariate time-series $\mathcal{Y}_j(t)$ at spatially dispersed locations ("monitoring sites") $j \in N$, for discrete time points $t = 1, 2, \ldots, T$. The **network-contraction** problem is to choose a subset $S \subset N$, with $|S| = s$ (given), with the idea that future observations will only be collected at S. From the data, we can calculate a sample covariance matrix C, and then we can formulate an instance of MESP to solve our network-contraction problem. Of course there are substantial issues to consider and confront. For this approach to be sound, we have to believe that there is an underlying Gaussian random vector $\mathcal{Y} = (\mathcal{Y}_1, \ldots \mathcal{Y}_n)$, and that the samples across time at a given location are i.i.d. With large enough T, we might start to convince ourselves that the Gaussian assumption is realistic, but the situation is really not as convenient as that; in particular, even for the univariate time series at each monitoring site, there can be trends and seasonality over time that have to be modeled. We will confront this issue later, in §4.2.

We can also well imagine a similar **network-expansion** problem. Here, we would have data for previously monitored sites, as well as specific candidate locations for potential monitoring sites. We would need to set up a joint model for the random variables at monitoring sites where we have data and potential locations where we do not have data. This would require some kind of spatial modeling, which is a big and challenging area within the field of forecasting. Of course we could combine contraction and expansion to realize a more general **network-redesign** problem.

Next, we will describe a concrete setting that has been of intense interest in this field. The NADP (National Acidic Deposition Program) maintains the NTN (National Trends Network); see [NAD22]). The NTN provides a long-term record of precipitation chemistry across the United States. The NTN measures the chemistry of precipitation at 379 monitoring sites across the United States, with some data available as far back as 1978; at present, 255 sites are active; see Figure 4.1[3], where active sites are indicated with blue markers and inactive ones with white markers. Monitoring sites are mostly located away from urban areas and specific sources of pollution. such as coal-fired power plants and other industrial operations. Automatic collection occurs during precipitation events. Samples accumulate for one week, and then are collected from each device — on Tuesday. Each site also has a weighing-bucket rain gauge, providing a continuous (independent) record of rainfall quantity. The equipment can be seen in Figure 4.2. Extensive details on the collection process can be found in [DB99].

Power plants burning fossil fuels produces a substantial fraction of the sulfur dioxide (SO_2) emissions in the United States. Emissions of nitrogen oxides (NO_x) come from transportation (e.g., cars, buses, trucks), power plants, and other industrial sources. Through a series of reactions, sulphur dioxide ultimately leads to sulfuric acid (H_2SO_4). Similarly, nitrogen oxides lead to nitric acid (HNO_3). Presence of

[3] reproduced by courtesy of the National Atmospheric Deposition Program (NRSP-3), 2022. NADP Program Office, Wisconsin State Laboratory of Hygiene, 465 Henry Mall, Madison, WI 53706.

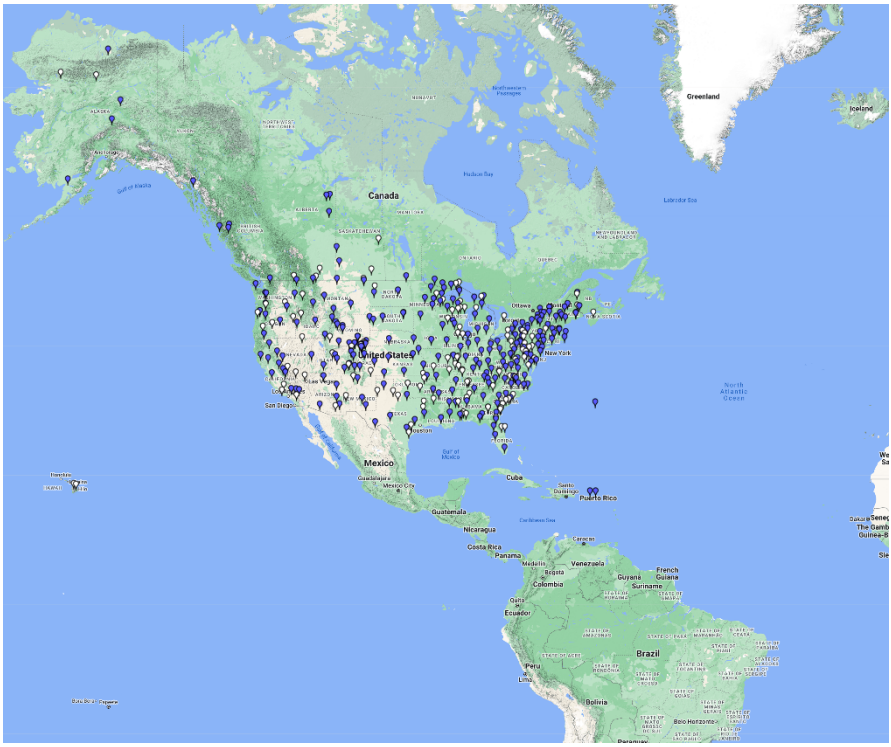


Fig. 4.1 National Trends Network

these acids in significant amounts decreases the pH [4] of precipitation to 4.0 and less, and this is considered "acid rain" (normal clean precipitation is slightly acidic, with a pH of around 5.6–5.7 (due to dissolved carbonic acid, from the reaction of CO_2 with water); note that a pH of 4.0 is 10 times the acidity compared to a pH of 5.0!). See Figure 4.3[3]. Acid rain has adverse impacts on terrestrial and aquatic creatures and also on infrastructure.

Toward monitoring for acid rain, the NTN analyzes samples for: hydrogen ion (H^+ measured as pH), sulfate (SO_4^{2-}), and nitrate (NO_3^-). Additional analysis is carried out for ammonium (NH_4^+), chloride (Cl^-), calcium (Ca^{2+}), magnesium (Mg^{2+}), potassium (K^+), ortho-phosphate (PO_4^{3-}). and sodium (Na^+). A full description of the weekly data can be found at `https://nadp.slh.wisc.edu/wp-content/uploads/2021/10/ntn-weekly-Meta.pdf`.

Figure 4.4[3] depicts the acid-rain situation in 1987, the year of the publication of [SW87]. The more recent situation is much better; see Figure 4.5[3]. The changes can be attributed to awareness, monitoring and legislation (in particular, the 1990

[4] pH (potential of hydrogen) is more complicated than you think; but roughly speaking: the practical range is 0 to 14, with values below 7 being acidic, 7 being neutral, and values above 7 being basic; it is essentially a logarithmic scale.



Fig. 4.2 Aerochem Metrics Precipitation Collector

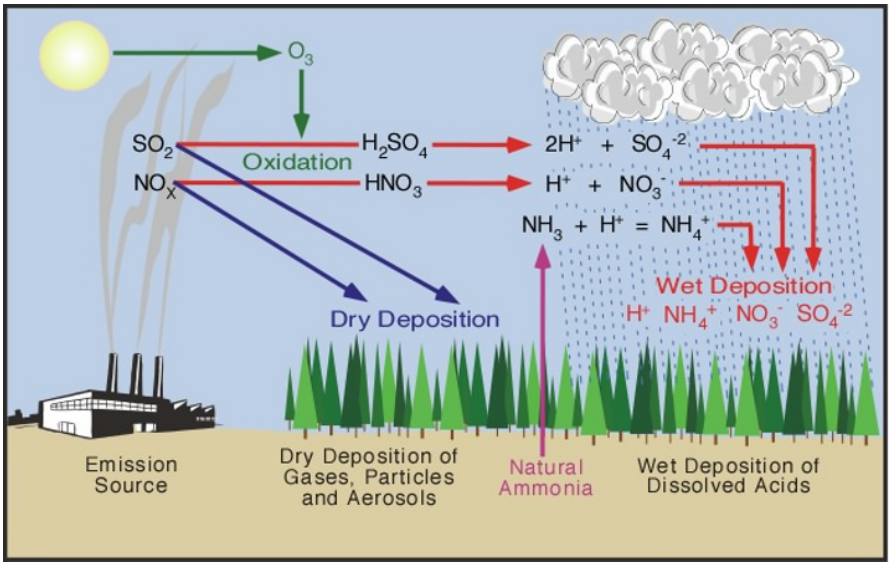


Fig. 4.3 The "acid rain" process

amendments of the U.S. Clean Air Act increased federal oversight, and new stringent regulatory programs were authorized for controlling acid rain).

Remark 4.1.2 There is a certain irony in the fact that "heat maps" like Figures 4.4 and 4.5 are displays of spatial models. After all, we only have data at the monitoring sites, indicated by black dots, so we have to estimate values at other points to make

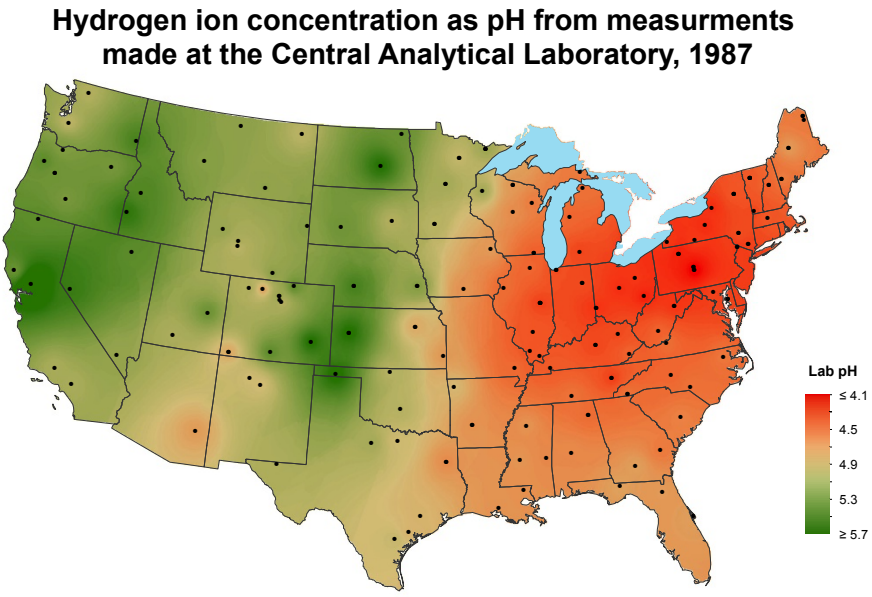


Fig. 4.4 NTN "acid rain" data: 1987

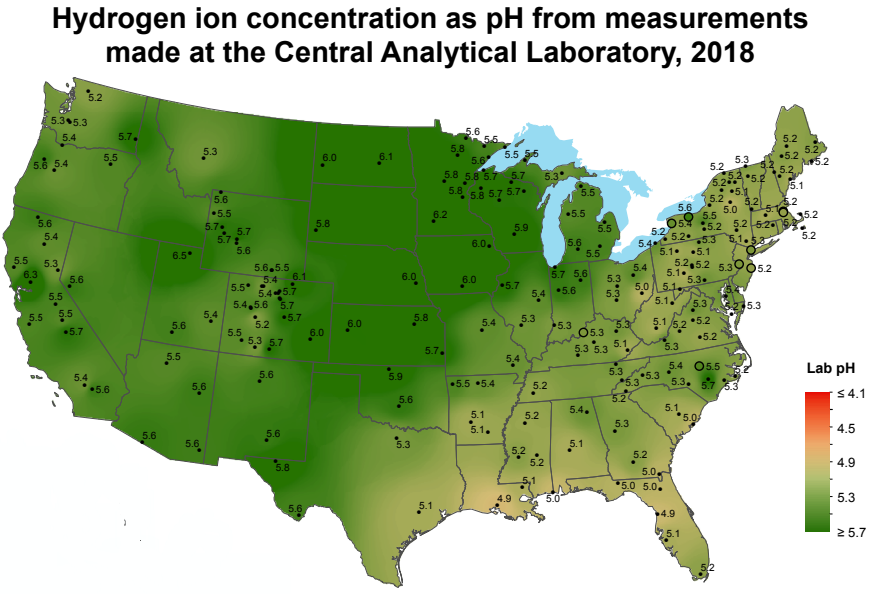


Fig. 4.5 NTN "acid rain" data: 2018

a heat map. The irony is that the entropy approach to environmental-monitoring network contraction (our main motivation) shuns spatial models! For the curious,

the usual spatial interpolation employed to form a heat map is **Inverse Distance Weighting (IDW) interpolation**:

$$\mathcal{Y}_j := \sum_{i \in \mathcal{N}(j)} \alpha_i \mathcal{Y}_i \,,$$

$$\text{where } \alpha_i := \frac{\frac{1}{d(i,j)}}{\sum_{k \in \mathcal{N}(j)} \frac{1}{d(k,j)}},$$

and $\mathcal{N}(j)$ indicates some "neighbors" i (with observations $\mathcal{Y}_i$) of a location j with no data. The value $d(i, j)$ is the distance from i to j, or possibly a small power (> 1) of that — which will accentuate peaks and troughs.

4.2 From raw data to covariance matrices

There are several issues to consider and confront in starting from raw data — even high-quality and voluminous data like that available from the NADP/NTN — and eventually arriving at a meaningful covariance matrix $C[N, N]$. We will summarize and expand on the methodology of [ATL20b], as implemented in the R package `MESgenCov` (see [ATL20a]), which is directly aimed at the network contraction problem, and is tightly coupled with the NADP/NTN data.

Multivariate data. Precipitation chemistry data is multivariate. Practically speaking, if we choose to monitor at a location, we will get data for all of the chemicals that we analyze for. Typically, in formulating and solving an appropriate MESP, we would confine our analysis to data from one chemical. Alternatively, we could have a separate element of N for each chemical-location pair, and then enforce the all-or-nothing nature of sampling at a location through constraints, leading to a (large) CMESP.

Missing data. In any large data set, we can assume that there will be missing data points. And we must have a coherent procedure for dealing with this. For the NADP/NTN data, we carry this out by appropriate averaging, passing from weeks to months. In the NADP/NTN data, missing weekly concentration values are coded as "-9.00". We calculate the level of a chemical's concentration by summing weekly quantities (mg) of the chemical, over a month, and dividing the monthly total by total precipitation (L), over dates in that month, to get monthly values of chemical concentration (mg/L). We use monthly concentrations instead of the available weekly concentrations because there is a large proportion of missing data for individual weeks compared to full months. Furthermore the univariate models are significantly better at predicting average monthly concentrations than they were are at predicting weekly concentrations.

For a given monitoring site $j \in N$, chemical, and month $t = 1, 2, \ldots, T$, let

$$W(t) := \text{set of weeks in month } t,$$

$D(w)$:= set of days in week w,
c_w := recorded chemical concentration (mg/L) for week w
($c_w = *$ denotes an unrecorded value),
p_d := recorded precipitation quantity (L) for day d,
p_w := precipitation quantity (L) for week w; $p_w = \sum_{d \in D(w)} p_d$.

Then the chemical concentration (mg/L) for month t at site j is calculated as

$$y_j(t) := \frac{\sum_{w \in W(t) \,:\, c_w \neq *} p_w c_w}{\sum_{w \in W(t) \,:\, c_w \neq *} \sum_{d \in w} p_d} .$$

When there is no weekly value available for the chemical quantity, we do not use the precipitation values for any of the days in such a week (so as to not artificially dilute the chemical-concentration level for the month). Finally, we note that that in the NADP/NTN data, H^+ concentration is recorded as pH, so that is converted to H^+ concentration and then aggregated over each month.

Spatial non-stationarity. We could consider a spatial model, and in fact this is necessary for network expansion and redesign. But for the simpler case of network contraction, we do not consider a spatial model, but rather we let the entropy handle that. This approach is sound, as long as the errors left, after fitting appropriate per-site models, do follow a multivariate Gaussian distribution — as required for MESP to be meaningful.

Data transformations. There are many possible transformations that can be applied to the times-series data $y_j(t)$, $t = 1, 2, \ldots, T$. It is important that we only consider univariate transformations; that is, we apply functions for each site $j \in N$ separately. This is due to the practical nature of sampling. The cost and effort of sampling is *per site*, so it makes sense in the optimal-design problem to restrict our attention to choices that sample or don't sample on a per-site basis. Applying a more complicated transformation and change-of-variables would not respect the combinatorial nature of our problem and model. We do our best to transform the time series on a per-site basis, and then we simply pray that the resulting transformed data follows a multivariate Gaussian model — as required for MESP to be meaningful.

Heavy tails and skewness: I. Like a lot of time-series data, there can be variability that is not easily captured by parametric distributions that are symmetric and have rapidly decaying tails. So a common technique is to make a logarithmic transformation. Conveniently, our data has no zero concentrations, so there is no issue of $\log(0)$. We note that pH is already essentially on a logarithmic scale, so for pH the logarithm is implicit.

Trends over time and seasonality. Next, we can quickly notice that there are long-term trends present in the data. We could already see a glimpse of this in comparing Figures 4.4 and 4.5. See Figure 4.6 for a finer-grain illustration of his phenomenon.

Looking further at some data, but over a shorter time widow, we can easily identify yearly periodicity; see Figure 4.7[3].

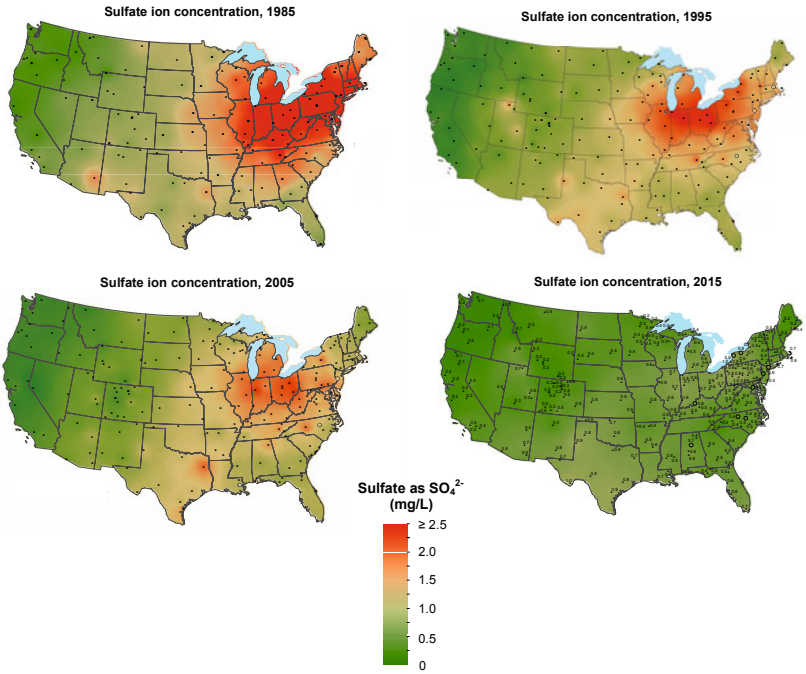


Fig. 4.6 Sulfate concentration over time

Seeking a good model to capture these phenomena, we arrive at:

$$\widehat{\log(y_j(t))} = \sum_{i=0}^{r} \beta_i t^i + \sum_{\ell=1}^{k} \left[a_\ell \cos\left(\frac{2\pi\ell t}{12}\right) + b_\ell \sin\left(\frac{2\pi\ell t}{12}\right) \right],$$

with the parameters β_i, a_ℓ, and b_ℓ fit by ordinary linear regression. A user can specify the degree r for the polynomial part of the model, which we think of as a truncated Taylor series, aimed at capturing long-term aperiodic trends. Periodic trends are captured via a truncated Fourier series, truncated at level k. Typically, parameter choices such as $r = 1$ and $k = 3$ provide a good fit. We note that morally it seems reasonable to have the same parameter choices for each site $j \in N$, but there is no practical impediment to varying these parameters across sites.

Heavy tails and skewness: II. The log transformation does not seem to be enough to fully tame heavy tails and skewness. Fortunately there are techniques (see [Goe11]) and associated software (see [Goe16]) available for taking care of these phenomena. The R function `lambertWtransform()` from the package "LambertW: Probabilistic Models to Analyze and Gaussianize Heavy-Tailed, Skewed Data" is directly aimed at this situation. And its use is integrated into the R package `MESgenCov` (see [ATL20a]).

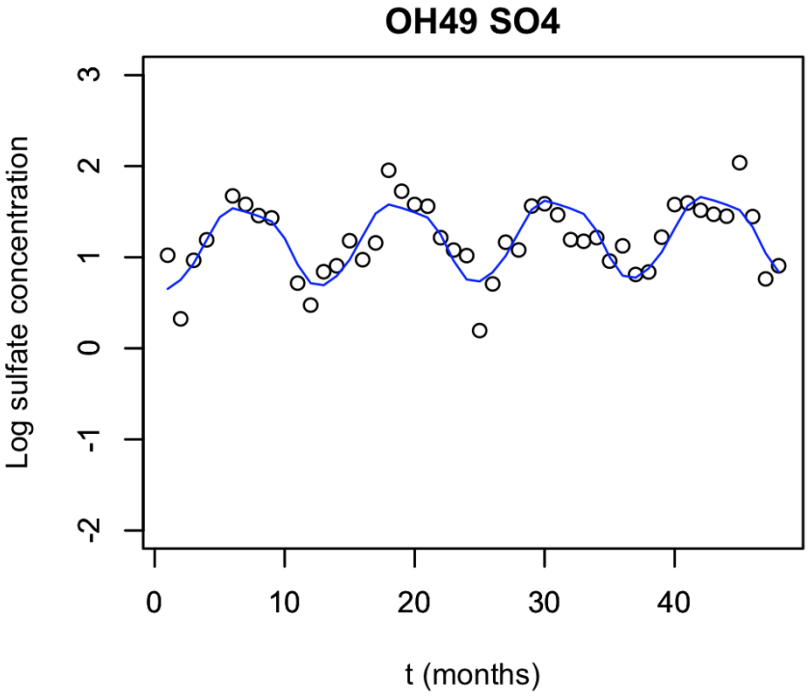


Fig. 4.7 Log sulfate concentration over a four-year period at a site

Are we there yet? Finally, we advocate applying univariate normality tests (e.g., the Shapiro-Wilk test) to the per-site residuals and multivariate tests (as available in the R package `MVN`; see [KGZ14]) to the residuals across sites. This functionality is all integrated into `MESgenCov`; for more details, see [ATL20b].

4.3 An example

Using the methodology described in §4.2, we prepared a covariance matrix C from hydrogen ion (H^+ measured as pH) data collected at a set of $n = 60$ sites, for the time span 2003–2014[5]. For the per-site univariate time series, we used $r = 1$, $k = 3$. All n fitted models reasonably passed the univariate (Shapiro-Wilk) test for normality (the minimum p-value is 0.0644, and 35 sites have a p-value of more than 0.4). We did observe that an appropriate multivariate test for normality was somewhat ambiguous.

[5] special thanks to Hessa Al-Thani for preparing the covariance matrix.

In Figure 4.8[6], we indicate the optimal MESP solution, calculated by branch-and-bound (using the code of [Ans20]), for $s = 30$. The optimal sites are indicated with yellow markers. It might seem surprising that, for example: (i) the rather isolated site in western Tennessee (NTN Site TN14, Hatchie National Wildlife Refuge, near Memphis) is not selected, (ii) all four sites in (northern) Colorado are selected, and (iii) in a large area centered on Wisconsin, all 16 sites are selected. But this is simply indicative of the fact that selection based on spatial considerations alone is inadequate. The data matters!



Fig. 4.8 An optimal MESP solution

[6] special thanks to Gabriel Ponte for preparing the figure.

Chapter 5
Opportunities

5.1 Developing algorithmic advances for MESP/CMESP

In this section, we give a miscellany of current and future projects related to MESP/CMESP.

5.1.1 Variable fixing and branch-and-bound: state of the art

Variable fixing turns out to be a very powerful methodology for branch-and-bound. Furthermore, even without branching, the variable-fixing methodology can be used to solve difficult instances to optimality: an "alternating fixing scheme" (switching between bounding methods), developed in [CFL21b], has been used to solve a very-large benchmark MESP instance with $n = 2,000$ to optimality for $s = 20, 40, 60$.

Using a linx-based branch-and-bound with variable fixing, [Ans20] reported: (i) optimal solutions of an $n = 124$ benchmark MESP instance for $s = 20, 30, \ldots, 110$; (ii) optimal solutions of an $n = 90$ benchmark MESP instance for $s = 10, 20, 70, 80$, and only near-optimal solutions for $s = 30, 40, 50, 60$. [CFL21b] found for this $n = 90$ instance: (i) superiority of the factorization bound for $s \lesssim 50$; (ii) applying the mixing methodology in the range $40 \lesssim s \lesssim 60$ gives improved bounds; (iii) variable fixing based on the factorization bound is quite effective in the range $1 \lesssim s \lesssim 18$. These observations, together with the success of the alternating fixing scheme (mentioned above), suggests that the $n = 90$ benchmark MESP instance could soon be solved to optimality for all s.

Putting all of the known ideas together, perhaps most practical CMESP instances with say $n \lesssim 100$ could soon be routinely solved using the branch-and-bound paradigm. A reasonable "challenge goal" is to push this to $n \lesssim 200$.

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5.1.2 Optimizing γ for NLP$_\gamma$

Although we have developed a strategy to select the scaling-parameter for NLP$_\gamma$, we wish to highlight that we do not know how to exactly optimize it. For linx$_\gamma$ and BQP$_\gamma$ (and also mBQP and smBQP), we have convexity of the bound in the logarithms of the scaling parameters; but we do not have such a property for NLP$_\gamma$.

5.1.3 Solvable cases of MESP and mask optimization

There is some very recent and promising work on polynomially-solvable cases of MESP. [ATL21a] found an efficient polynomial-time dynamic-programming algorithm for MESP when C is a tridiagonal matrix (see §1.4) and exploited that heuristically for mask optimization. [ATL21b] extended the tridiagonal solvability further, when the support graph of C is a "spider with a bounded number of legs". [CFL21a]) is developing mask optimization more generally, and [CFL22] demonstrated, analytically, the improvements that can be achieved by masking the linx$_\gamma$ bound.

5.1.4 OA for CMESP

We have been pursuing an outer-approximation (OA) algorithm for CMESP. Our goal is to take the best upper-bounding methods based on convex relaxation and instead of using them within the branching-and-bound scheme, we will marry them with the successful OA scheme for general convex mixed-integer nonlinear optimization (see [DG86]). We do note that [CHB08, CHB13] developed an OA scheme connected with the BQP bound.

Regarding OA, our needs are very specific. For simplicity of exposition, we restrict our attention to convex-optimization bounds that use only the variables $x \in \mathbb{R}^n$, and we assume that we are working with a smooth exact relaxation; that is, $f : \mathbb{R}^n \to \mathbb{R}$ is a differentiable concave function, and $f(x) = \text{ldet}\, C[\underline{x}, \underline{x})]$, for $x \in \{0,1\}^n$ such that $\mathbf{e}^\mathsf{T} x = s$; For example, the linx and NLP bounds satisfy these requirements. So CMESP is equivalent to

$$(P) \qquad z := \max \left\{ f(x) \ : \ Ax \le b,\ \mathbf{e}^\mathsf{T} x = s,\ x \in \{0,1\}^n \right\},$$

Notably, we have only 0/1 variables, and our only nonlinearity is the objective function. A convenient reference that covers such a situation for OA is [FL94] (note that they work with minimization, their y variables are the integer ones, and they have continuous variables additionally, which they call x). While the framework of [FL94] does cover our situation, there appears to be no successful experience in the literature of successfully applying OA in the context of such a problem as P.

At a high level, at each iteration we solve a 0/1 linear-optimization "main problem". The main problem at iteration ℓ has the form,

$$
(M^\ell)\qquad
\begin{aligned}
z(\mathcal{T}^\ell) := \quad & \max\ \eta \\
& \text{subject to:} \\
& \eta \le f(\tilde{x}) + \nabla f(\tilde{x})^\mathsf{T}(x - \tilde{x}),\ \text{for } \tilde{x} \in \mathcal{T}^\ell, \\
& A^\ell x \le b^\ell, \\
& \mathbf{e}^\mathsf{T} x = s, \\
& x \in \{0,1\}^n.
\end{aligned}
$$

The set of linearization points $\mathcal{T}^\ell$ grows at each iteration, and also the $A^\ell x \le b^\ell$ constraints accumulate additional inequalities, starting from $A^0 := A$ and $b^0 := b$. Therefore $z(\mathcal{T}^\ell)$ is nonincreasing (in ℓ) as the algorithm progresses and *in spirit* these values are upper bounds on z. But to be precise, we will maintain a lower bound LB, which is the incumbent objective value — the best solution that we have seen so far for P. And actually $UB := \max\{z(\mathcal{T}^\ell), LB\}$ is an upper bound on z, because even though we will accumulate inequalities in $A^\ell x \le b^\ell$ to exclude solutions that we have already seen, the best of those is always captured in LB.

The OA algorithm for P. At iteration ℓ, we solve M^ℓ. If M^ℓ is infeasible, then we stop the algorithm with the determination that $z = LB$. If not, we let the optimal solution of M^ℓ be x^ℓ. We update $LB \leftarrow \max\{LB, f(x^\ell)\}$. We stop if $UB - LB < \tau$, an input optimality tolerance. If not, we update our set of linearization points $\mathcal{T}^{\ell+1} \leftarrow \mathcal{T}^\ell \cup \{x^\ell\}$. Next, we update $A^\ell x \le b^\ell$ to form $A^{\ell+1} x \le b^{\ell+1}$ by appending the linear inequality:

$$\sum_{j \in S^\ell} x_j \le s - 1,$$

where S^ℓ denotes the support of x^ℓ. The inequality excludes (only) x^ℓ from being visited again — even if it turns out that x^ℓ is optimal, its value was already captured when we updated LB.

An enhancement for MESP. First, we can potentially improve LB at iteration ℓ by running a best-improvement swapping/interchange heuristic, starting from S^ℓ, using the true objective function $f(x(S))$, and using the neighbors $S \leftarrow S - i + j$, for $i \in S$, $j \in N \setminus S$ (see §2.3.2). The local search visits a sequence of sets $S^\ell =: S^\ell_0, S^\ell_1, \ldots, S^\ell_p =: S^\ell_*$, starting at S^ℓ, finally reaching some locally-optimal S^ℓ_*. For each set S^ℓ_k visited along the way, $k \le p$, we can do the following:

- update the linearization set: $\mathcal{T}^{\ell+1} \leftarrow \mathcal{T}^{\ell+1} \cup \{x(S^\ell_k)\}$, and
- update $A^{\ell+1} x \le b^{\ell+1}$, for $2 \le s \le n-2$, by appending the linear inequality

$$\sum_{j \in S^\ell_k} x_j \le s - 2, \tag{5.1}$$

which rules out precisely S_k^ℓ and its immediate neighbors (which we know cannot be improving on S_k^ℓ, because we are performing a best-improvement local search). Note that if $s = n - 1$, then (5.1) is not valid because, in this case, there is only one element in $N \setminus S_k^\ell$. Thus, it is not possible to construct a feasible solution, by selecting only $n - 3$ elements from S_k^ℓ.

Finally, we update the lower bound: $LB \leftarrow \max\{LB, f(x(S_*^\ell))\}$; note that $f(x(S_*^\ell))$ is the best function value observed (over all solutions visited and their neighbors) in the current application of local search.

With more work, we can improve on the cuts (5.1), for $3 \le s \le n - 3$. We can choose a set $S^+ \subset N$, with $|S^+| = s + 1$. Next, we search over all $S = S^+ - e$, for all $e \in S^+$, and all neighbors of such S. Taking the best of all of these solutions, we update LB, and then we have the inequality

$$\sum_{j \in S^+} x_j \le s - 2, \tag{5.2}$$

which is valid for any solution with objective value greater than LB. This inequality is stronger than each $\sum_{j \in N \setminus S} x_j \ge 2$, for $S = S^+ - e$ with $e \in S^+$. We can see the validity directly, or by

$$\sum_{\substack{S = S^+ - e: \\ e \in S^+}} \frac{1}{s} \left(\sum_{j \in S} x_j \le s - 2 \right) = \sum_{j \in S^+} x_j \le (s - 1) - \frac{2}{s},$$

and the right-hand side rounds down to $s - 2$ for $s \ge 3$.

Under arbitrary inequalities of the form (5.1), it is very easy to produce a relative-interior-point solution in all non-trivial cases. This can be important for interior-point algorithms in solving the relaxations efficiently (see [AFLW99], for example).

Proposition 5.1.1 For integers $6 < s < n - 2$, the point $\hat{x} := \frac{s}{n}\mathbf{e}$ strictly satisfies all inequalities of the form (5.1).

Proof Plugging in $\hat{x}$ to (5.1) written strictly, we can manipulate it to the form $s + \frac{4}{s-2} < n - 2$, which is equivalent to $s < n - 2$ for $s > 6$. □

Similarly, we have the following result.

Proposition 5.1.2 For integers $8 < s < n - 3$, the point $\hat{x} := \frac{s}{n}\mathbf{e}$ strictly satisfies all inequalities of the form (5.2).

We are currently developing a disjunctive-cut framework to enhance our OA approach for CMESP; see [FL22].

5.2 MESP/CMESP variations and cousins

There are variations on and close cousins of MESP/CMESP where the surface has only been scratched with regard to exact discrete optimization. We highlight a few to provoke more research:

- [KLW98] considered the **D-optimality problem**: given $m \leq s < n$ and full column-rank $A \in \mathbb{R}^{m \times n}$, find $S \subset \{1, 2, \ldots, m\}$, with $|S| = s$, so as to maximize $\det\left(A[S,N]^{\mathsf{T}} A[S,N]\right)$. This corresponds to minimizing the "generalized variance" of the least-squares parameter estimates x of the linear model $A[S,N]\, x \approx b$. There is a huge literature on many variations on this "optimal experimental design" problem; see [Puk06]. Recent work on this problem includes [PFL22] and [LFL+22].
- [LL20] looked at the **generalized maximum-entropy sampling problem**: given $0 < t \leq s < n$ and $C \in \mathbb{S}^n_+$, find $S \subset N$ with $|S| = s$, so as to maximize $\prod_{\ell=1}^{t} \lambda_\ell(C[S,S])$. For $t = s$ this is equivalent to MESP, while for $t = m$ we can model the D-optimality problem by taking $C := AA^{\mathsf{T}}$.
- [AFLW01] looked at the **maximum-entropy remote-sampling problem** (also see [BAA98] and [Lee98b]): given n, $C \in \mathbb{S}^n_+$, $\emptyset \neq T \subsetneq N$, and $0 < s < n - |T|$, find $S \subset N \setminus T$, with $|S| = s$, so as to maximize $\operatorname{ldet} C_S[T,T] = \operatorname{ldet} C[S \cup T, S \cup T] - \operatorname{ldet} C[S,S]$. For a random Gaussian vector $\mathcal{Y}_N \in \mathbb{R}^n$ having covariance matrix C, this corresponds to choosing S so as to minimize the uncertainty on $\mathcal{Y}_T$ (T is fixed and is disjoint from allowable S), after observing $\mathcal{Y}_S$.
- Given $0 < s < n$ and $C \in \mathbb{S}^n_+$, the associated **k-determinental point process** (k-DPP) has

$$\mathbf{Pr}(S) := \frac{\det C[S,S]}{\sum\limits_{\substack{T \subset N \\ |T|=s}} \det C[T,T]},$$

  for all $S \subset N$ such that $|S| = s$. A "most likely event" is called a **mode**, and so mode determination (also known as **maximum a posteriori (MAP) inference**; see [CZZ18], for example) of a k-DPP is precisely MESP. There is a big literature on these DPPs (see [KT11], in particular), mainly concerned with different issues (e.g., sampling from such a process), and [CRLZW18] explores connections with MESP in the context of the environmental-monitoring application.

5.3 Applications

[ATL20a, ATL20b] is a first step in providing a tool for producing a meaningful covariance matrix for MESP/CMESP from raw environmental-monitoring data. Critically, the success of this — and the very applicability of MESP/CMESP — depends on being able to fit a model that is plausibly multivariate Gaussian.[7] It is a

[7] Of course one can raise a similar concern about many parametric statistical methods.

challenge to make the process more seamless and reliable, and to build an end-to-end system, from raw data to MESP/CMESP solutions, that scales to large instances.

Everything in [ATL20a, ATL20b] addresses network contraction — where we have data for all sites. To handle network redesign, what we did for MESP/CMESP has to be applied after an effective spatial model is created and validated. This is an even greater challenge for seamless integration. Additionally, the multivariate setting (i.e., several chemicals monitored at each location) has not been treated well (i.e., in a truly scalable way; not mapping it into a large CMESP).

A further challenge is to see if some of the more recent algorithmic ideas for MESP/CMESP can have some impact on other application areas. Additionally, perhaps data-processing frameworks, like what we developed for environmental monitoring ([ATL20a, ATL20b]) could be developed for other application areas. Areas where MESP has been applied and with high further potential include: the design of (costly) computer experiments (which was an early application area for MESP; see [SWMW89, CMMY88, MM92, YW19]), sensor-network location (see [PBA00]), and the "phasor measurement unit (PMU) placement problem" in electric power systems (see [LCW+12]). Other application areas that seem ripe for application of MESP/CMESP technology include: seismic monitoring, and infrastructure (e.g., water network) protection.

Appendix A
Basic formulae and inequalities

In this appendix, we have gathered many formulae and inequalities that we use. Rather than simply list them, we have made an effort to organize and develop them in a logical way, exposing relationships between many of them, while endeavouring to be reasonably concise. Only a few results are stated without proof (e.g., Theorems A.3.1, A.3.2 and A.4.16).

A.1 Preliminary miscellany

Our starting point is a version of the well-known Jensen's inequality.

Theorem A.1.1 (Jensen's inequality) For a random $\mathcal{Y} = (\mathcal{Y}_1, \ldots, \mathcal{Y}_d)$ and a concave function $\phi : \mathbb{R}^d \to \mathbb{R}$, we have

$$\phi(E[\mathcal{Y}]) \geq E[\phi(\mathcal{Y})] \, .$$

Proof Define $g : \mathbb{R}^d \to \mathbb{R}$ such that $g(y)$ is a subgradient of the concave ϕ at y in the domain of ϕ. If ϕ is assumed to be differentiable, then $g(y)$ is simply the gradient of ϕ at y. By concavity of ϕ, for all $y, \tilde{y}$ in the domain of ϕ, we have

$$\phi(y) + g(y)^{\mathsf{T}}(\tilde{y} - y) \geq \phi(\tilde{y}).$$

Therefore,

$$\phi(E[\mathcal{Y}]) + g(E[\mathcal{Y}])^{\mathsf{T}}(\mathcal{Y} - E[\mathcal{Y}]) \geq \phi(\mathcal{Y}).$$

Taking the expected value on both sides, we obtain

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$$\phi(E[\mathcal{Y}]) + g(E[\mathcal{Y}])^{\mathsf{T}} \underbrace{(E[\mathcal{Y}] - E[\mathcal{Y}])}_{=0} \geq E[\phi(\mathcal{Y})],$$

and the result follows. □

From Jensen's inequality, we get the following result.

Theorem A.1.2 (arithmetic-geometric mean inequality)

$$\frac{\sum_{i=1}^{n} x_i}{n} \geq \left(\prod_{i=1}^{n} x_i\right)^{1/n}.$$

Proof By the concavity of $\log(\cdot)$ and (a very special case of) Jensen's inequality ($d = 1$ and a discrete probability space), we have

$$\log\left(\frac{\sum_{i=1}^{n} x_i}{n}\right) \geq \frac{\sum_{i=1}^{n} \log(x_i)}{n} = \log\left(\left(\prod_{i=1}^{n} x_i\right)^{\frac{1}{n}}\right).$$

The result follows by exponentiating the left and right. □

A convenient set of inequalities, with an elementary proof, describes how to minimize or maximize the dot product of a pair of vectors $x, y \in \mathbb{R}^n$, by sorting the components independently.

Theorem A.1.3 (Hardy-Littlewood-Pólya rearrangement inequalities) If $x, y \in \mathbb{R}^n$, with $n \geq 2$, then

$$\sum_{i=1}^{n} x_{[i]} y_{[n-i]} \leq \sum_{i=1}^{n} x_i y_i \leq \sum_{i=1}^{n} x_{[i]} y_{[i]}.$$

Proof It is easy to verify the result for $n = 2$. The general result can then be proved by first sorting the components of x in nonincreasing order, and then using repeated pair-wise interchanges of components of y to either increase or decrease the dot product. □

Finally, several times we will be working with special concave supremum problems where we employ a useful sufficient condition for the supremum to be attained.

Proposition A.1.4 Let $Q \subset \mathbb{R}^n$ be a compact convex set, let $\mathcal{K} \subset \mathbb{R}^n$ be a convex set, and assume that $Q \cap \mathcal{K} \neq \emptyset$. Let $f : Q \cap \mathcal{K} \to \mathbb{R}$ be a continuous and concave function. Let $\bar{\mathcal{K}}$ denote the closure of $\mathcal{K}$. Suppose that for all $\hat{x} \in Q \cap \mathcal{K}$ and all $z \in Q \cap (\bar{\mathcal{K}} \setminus \mathcal{K})$, we have $\lim_{t \to 0^+} f(t\hat{x} + (1-t)z) = -\infty$. Then $f(x^*) = \sup\{f(x) \ : \ x \in Q \cap \bar{\mathcal{K}}\}$, for some $x^* \in Q \cap \mathcal{K}$, and the maximizer x^* is unique when f is strictly concave on $Q \cap \mathcal{K}$.

Proof The hypotheses imply that

$$\sup\{f(x) \ : \ x \in Q \cap \bar{\mathcal{K}}\} = \sup\{f(x) \ : \ x \in Q \cap \mathcal{K}, \ f(x) \geq f(\hat{x})\},$$

where $\hat{x}$ is some point in $Q \cap \mathcal{K}$. Note that the function f is continuous and proper on $Q \cap \mathcal{K}$, so the (convex) upper level set $\{x \in Q \cap \mathcal{K} \ : \ f(x) \geq f(\hat{x})\}$ is closed and hence compact. Therefore, the supremum is attained at some $x^* \in Q \cap \mathcal{K}$ (which is unique if f is strictly concave). □

A.2 Square matrices

The **Schur complement** of invertible $C[S,S]$ in

$$C[N,N] := \left(\begin{array}{c|c} C[S,S] & C[S, N \setminus S] \\ \hline C[N \setminus S, S] & C[N \setminus S, N \setminus S] \end{array} \right)$$

is

$$C_S[N \setminus S, N \setminus S] := C[N \setminus S, N \setminus S] - C[N \setminus S, S](C[S,S])^{-1} C[S, N \setminus S].$$

A wealth of information about the Schur complement can be found in [Zha05]. Importantly for us, when C is the covariance matrix of a random vector $\mathcal{Y} \in \mathbb{R}^n$, $C_S[N \setminus S, N \setminus S]$ is the covariance matrix for $\mathcal{Y}_{N \setminus S}$ *conditioned on* $\mathcal{Y}_S$.
It is easy to check the following result.

Proposition A.2.1 For $C \in \mathbb{R}^{n\times n}$ and $S \subset N$ with $C[S,S]$ invertible,

$$\left(\begin{array}{c|c} C[S,S] & 0 \\ \hline 0 & C_S[N\setminus S, N\setminus S] \end{array}\right)$$
$$= \left(\begin{array}{c|c} I & 0 \\ \hline X & I \end{array}\right)\left(\begin{array}{c|c} C[S,S] & C[S,N\setminus S] \\ \hline C[N\setminus S,S] & C[N\setminus S, N\setminus S] \end{array}\right)\left(\begin{array}{c|c} I & Z \\ \hline 0 & I \end{array}\right),$$

where $X := -C[N\setminus S,S]\,(C[S,S])^{-1}$ and $Z := -\,(C[S,S])^{-1}\,C[S,N\setminus S]$.

From Proposition A.2.1, we get a very useful formula:

Theorem A.2.2 (Schur complement determinant formula) For $C \in \mathbb{R}^{n\times n}$ and $S \subset N$ with $C[S,S]$ invertible,

$$\det C = \det C[S,S] \times \det C_S[N\setminus S, N\setminus S].$$

We can also see a useful formula for the inverse of a partitioned C.

Theorem A.2.3 (Banachiewicz-Schur formula) For $C \in \mathbb{R}^{n\times n}$ and $S \subset N$,

$$\left(\begin{array}{c|c} C[S,S] & C[S,N\setminus S] \\ \hline C[N\setminus S,S] & C[N\setminus S, N\setminus S] \end{array}\right)^{-1}$$
$$= \left(\begin{array}{c|c} (C_{N\setminus S}[S,S])^{-1} & -(C[S,S])^{-1}C[S,T](C_S[T,T])^{-1} \\ \hline -(C_S[T,T])^{-1}C[T,S](C[S,S])^{-1} & (C_S[N\setminus S, N\setminus S])^{-1} \end{array}\right),$$

when all the inverses on the right-hand side of the equation exist.

And we easily obtain the following result from these last two.

Corollary A.2.4 (complementary principal minors) For invertible $C \in \mathbb{R}^{n\times n}$ and $S \subset N$, we have

$$\det C[S,S] = \det C \times \det(C^{-1})[N \setminus S, N \setminus S].$$

Finally, it is useful to understand how the inverse of a nonsingular matrix varies when we perturb the entries of the matrix.

Lemma A.2.5 Let V be a nonsingular $n \times n$ matrix. Then

$$\frac{\partial V^{-1}}{\partial V_{ij}} = -V^{-1}\mathbf{e}_i\mathbf{e}_j^{\mathsf{T}}V^{-1}.$$

Proof We have that $VV^{-1} = I$ and $\frac{\partial V}{\partial V_{ij}} = \mathbf{e}_i\mathbf{e}_j^{\mathsf{T}}$. Then

$$\frac{\partial V}{\partial V_{ij}}V^{-1} + V\frac{\partial V^{-1}}{\partial V_{ij}} = 0 \quad \Rightarrow$$
$$\frac{\partial V^{-1}}{\partial V_{ij}} = -V^{-1}\frac{\partial V}{\partial V_{ij}}V^{-1} = -V^{-1}\mathbf{e}_i\mathbf{e}_j^{\mathsf{T}}V^{-1}.$$

□

A.3 Symmetric matrices

Let A be a symmetric $n \times n$ real matrix. That is $A \in \mathbb{S}^n$. If $v \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$ satisfy $Av = \lambda v$, then λ is known as an **eigenvalue** of A, and the associated v is known as an **eigenvector** of A belonging to λ. For symmetric A, the **characteristic polynomial** $\det(A - \lambda I)$ has all real roots, and these roots are precisely the eigenvalues of A.

Symmetric A have nice representations via their eigenvalues and eigenvectors.

Theorem A.3.1 (spectral decomposition) For $A \in \mathbb{S}^n$, we can write $A = \sum_{\ell=1}^n \lambda_\ell(A) v_\ell v_\ell^\mathsf{T}$, with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$, and $\{v_\ell \ : \ \ell = 1, 2, \ldots, n\}$ is an orthonormal basis for $\mathbb{R}^n$.

This decomposition is not unique, except for the eigenvalues and the "eigenspaces" belonging to an eigenvalue. That is, if $\lambda_{\ell-1} > \lambda_\ell = \cdots = \lambda_k > \lambda_{k+1}$ for some $\ell \leq k$, then $\text{span}(\{v_\ell, \cdots, v_k\})$ (a $(k - \ell + 1)$-dimensional subspace of $\mathbb{R}^n$) is unique. In this way, the spectral decomposition is "essentially unique".

In the spectral decomposition, sometimes it is convenient to define $\tilde{v}_\ell := \sqrt{\lambda_\ell(A)}\, v_\ell$, so that we have $A = \sum_{\ell=1}^n \tilde{v}_\ell \tilde{v}_\ell^\mathsf{T}$.

Using elementary facts and calculations, it is easy to check:

- $\det(A) = \prod_{\ell=1}^n \lambda_\ell(A)$;
- $\text{tr}(A) = \sum_{\ell=1}^n \lambda_\ell(A)$;
- $\lambda_\ell(I + A) = 1 + \lambda_\ell(A)$, for $\ell = 1, 2, \ldots, n$;
- if B is invertible, the the eigenvalues of $B^{-1}AB$ and A are the same — we say that the matrix $B^{-1}AB$ is **similar** to A.

Next, we wish to get a relationship between the eigenvalues of a symmetric matrix and the eigenvalues of a principal submatrix. The ordered sequence of real numbers $\mu_1 \leq \cdots \leq \mu_{n-1}$ **interlaces** the ordered sequence of real numbers $\lambda_1 \leq \cdots \leq \lambda_n$ if

$$\lambda_1 \leq \mu_1 \leq \lambda_2 \leq \mu_2 \cdots \leq \lambda_{n-1} \leq \mu_{n-1} \leq \lambda_n \ .$$

We make use of a special case of a beautiful and powerful theorem from algebra.

Theorem A.3.2 (special case of the Hermite-Biehler theorem see [RS02, p. 197]) Suppose that polynomials f and g have all real roots. The ordered roots of g interlace the ordered roots of f if and only if $f + \alpha g$ has all real roots for all $\alpha \in \mathbb{R}$.

From this elegant and powerful theorem, we can the following very useful result.

Theorem A.3.3 (Cauchy's eigenvalue interlacing inequalities / inclusion principle) Let $C[S,S]$ be an order s principal submatrix of an order t symmetric matrix $C[T,T]$; so $S \subset T$. Then, for $1 \le \ell \le s$, we have

$$(i) \quad \lambda_\ell(C[T,T]) \ge \lambda_\ell(C[S,S]) \qquad (\ell\text{-th greatest eigenvalue});$$
$$(ii) \quad \lambda_{t+1-\ell}(C[T,T]) \le \lambda_{s+1-\ell}(C[S,S]) \qquad (\ell\text{-th least eigenvalue}).$$

Proof It suffices to prove this for the case of $S = N$ and $T = N-1$. It is easy to check that

$$\det\left(\begin{array}{c|c} c_{1,1}+\alpha-\lambda & C[1,N-1] \\ \hline C[N-1,1] & C[N-1,N-1]-\lambda I \end{array}\right)$$
$$= \det\left(\begin{array}{c|c} c_{1,1}-\lambda & C[1,N-1] \\ \hline C[N-1,1] & C[N-1,N-1]-\lambda I \end{array}\right)$$
$$+ \det\left(\begin{array}{c|c} \alpha & 0 \\ \hline C[N-1,1] & C[N-1,N-1]-\lambda I \end{array}\right).$$

On the left-hand side of the equation, we have the characteristic polynomial of a symmetric matrix, and so it has all real roots, for all $\alpha \in \mathbb{R}$. Therefore, $\det(C-\lambda I) + \alpha \det(C[N-1,N-1]-\lambda I)$. has all real roots, for all for all $\alpha \in \mathbb{R}$. Therefore, by Theorem A.3.2, the eigenvalues of $C[N-1,N-1]$ interlace the eigenvalues of C. □

A vector $\alpha \in \mathbb{R}^n$ **majorizes** a vector $\beta \in \mathbb{R}^n$ if

$$\sum_{i=1}^{k} \alpha_{[i]} \ge \sum_{i=1}^{k} \beta_{[i]} ,$$

for $k = 1, 2, \ldots, n$, and with equality for $k = n$.

Theorem A.3.4 (Schur's Theorem: majorization by eigenvalues; see [MOA11]) For $A \in \mathbb{S}^n$, the vector $\lambda(A)$ majorizes the vector $\mathrm{diag}(A)$.

Proof First, we note that $\mathrm{tr}(A) = \sum_{i=1}^n A[i,i] = \sum_{i=1}^n \lambda_i(A)$, so we can concentrate on the inequalities. The proof is by induction on n. The base case of $n = 1$ is trivial. Now assume that the result is true for symmetric matrices of order $k < n$. For $A \in \mathbb{S}^n$, let $\hat{A}$ be the principal submatrix of A obtained by deleting the row and column of

a least diagonal entry of A. Then, for $k = 1, 2, \ldots, n-1$, the k greatest diagonal elements of A and $\hat{A}$ are identical.

By Theorem A.3.3, we have

$$\lambda_1(A) \geq \lambda_1(\hat{A}) \geq \lambda_2(A) \geq \lambda_2(\hat{A}) \geq \cdots \geq \lambda_{n-1}(\hat{A}) \geq \lambda_n(A).$$

Therefore,

$$\sum_{i=1}^{k} \lambda_i(A) \geq \sum_{i=1}^{k} \lambda_i(\hat{A}),$$

for $k = 1, 2, \ldots, n-1$.

By the inductive hypothesis, we have that the vector $\lambda(\hat{A})$ majorizes the vector $\mathrm{diag}(\hat{A})$. So we can conclude that $\lambda(A)$ majorizes $\mathrm{diag}(A)$. □

A.4 Positive definite and semidefinite matrices

Symmetric C is **positive definite** if $\lambda_n(C) > 0$, and we write $C \succ 0$ (or $C \in \mathbb{S}^n_{++}$). Symmetric C is **positive semidefinite** if $\lambda_n(C) \geq 0$, and we write $C \succeq 0$ (or $C \in \mathbb{S}^n_{+}$).

It is easy to check that $C \succ 0$ if and only if $x^\mathsf{T} C x > 0$ for all $x \in \mathbb{R}^n - 0$, and $C \succeq 0$ if and only if $x^\mathsf{T} C x \geq 0$ for all $x \in \mathbb{R}^n$. From this, we can easily verify that the set of positive semidefinite matrices $\mathbb{S}^n_+$ is a full-dimensional closed convex cone in $\mathbb{S}^n$, with the set of positive definite matrices $\mathbb{S}^n_{++}$ being its interior. As the set of full-rank positive-semidefinite matrices is a convex set, it is natural to look at the set of positive-semidefinite matrices with an arbitrary rank lower bound. For integer $0 \leq s \leq n$, let $\mathbb{S}^{n,s}_+$ be the set of positive-semidefinite matrices having rank at least s. Of course, we have

$$\mathbb{S}^n_+ = \mathbb{S}^{n,0}_+ \supset \mathbb{S}^{n,1}_+ \supset \cdots \mathbb{S}^{n,n-1} \supset \cdots \supset \mathbb{S}^{n,n} = \mathbb{S}^n_{++} .$$

The set $\mathbb{S}^n_+$ is the closure of each $\mathbb{S}^{n,s}_+$, and $\mathbb{S}^n_{++}$ is the interior of each $\mathbb{S}^{n,s}_+$.

Proposition A.4.1 For $0 \leq s \leq n$, the set $\mathbb{S}^{n,s}_+$ is a convex set.

Proof Suppose that $A, B \in \mathbb{S}^n_+$, with $\mathrm{rank}(A) \geq s$ and $\mathrm{rank}(B) \geq s$. Clearly, $\frac{1}{2}A + \frac{1}{2}B \in \mathbb{S}^n_+$. Now suppose that $x \in \mathrm{n.s.}(A+B)$. Then $0 = x^\mathsf{T}(A+B)x = x^\mathsf{T}Ax + x^\mathsf{T}Bx$, with each summand on the right-hand side being nonnegative, because $A, B \in \mathbb{S}^n_+$. Therefore $x^\mathsf{T}Ax = 0$ and $x^\mathsf{T}Bx = 0$. Now let $\sum_\ell \lambda_\ell(A) v_\ell v_\ell^\mathsf{T}$ be a spectral decomposition of A. Then $0 = x^\mathsf{T}Ax = \sum_\ell \lambda_\ell(A)(x^T v_\ell)^2$, which implies that $x^T v_\ell = 0$, for all ℓ. Therefore $x \in \mathrm{n.s.}(A)$, and hence $\mathrm{n.s.}(A+B) \subset \mathrm{n.s.}(A)$. So we conclude that $\mathrm{rank}(\frac{1}{2}A + \frac{1}{2}B) = \mathrm{rank}(A+B) \geq \mathrm{rank}(A) \geq s$. □

The following result is very simple but useful.

Lemma A.4.2 Let $A \in \mathbb{S}^n_+$ and $B \in \mathbb{S}^n$. Then $BAB \succeq 0$. Furthermore, if $A, B \in \mathbb{S}^n_{++}$, then $BAB \succ 0$.

Proof For any $x \in \mathbb{R}^n$, we have $x^\mathsf{T} BABx = y^\mathsf{T} Ay \geq 0$, where $y := Bx$ and the inequality holds because $A \succeq 0$. Thus, $BAB \succeq 0$.

Considering now, $B \succ 0$ in the above, we see that $y = 0$ if and only if $x = 0$. Therefore, if $A \succ 0$, then for any $x \neq 0$, we have $x^\mathsf{T} BABx = y^\mathsf{T} Ay > 0$. Thus, $BAB \succ 0$. □

It is also very easy to see the following.

Proposition A.4.3 (principal minors of positive definite/semidefinite matrices) If $C \in \mathbb{R}^{n\times n}$ is positive definite/semidefinite and $S \subset N$, then $C[S, S]$ is positive definite/semidefinite.

From the Schur complement determinant formula (Theorem A.2.2), we easily obtain the following useful results:

Theorem A.4.4 (positive definiteness/semidefiniteness of the Schur complement) For $C \in \mathbb{S}_n$ and $S \subset N$ such that $C[S, S]$ is invertible, C is positive definite/semidefinite if and only if $C[S, S]$ is positive definite and $C_S[N \setminus S, N \setminus S]$ is positive/definite semidefinite.

Corollary A.4.5 For the vector variable $x \in \mathbb{R}^n$ and the matrix variable $X \in \mathbb{S}^n$, the constraint $X - xx^\mathsf{T} \succeq 0$ is equivalent to the constraint

$$\left(\begin{array}{c|c} X & x \\ \hline x^\mathsf{T} & 1 \end{array}\right) \succeq 0.$$

From the Banachiewicz-Schur formula (Theorem A.2.3), we obtain the following useful result:

Corollary A.4.6 For $C \in \mathbb{S}^n_+$ and $S \subset N$, we have

$$(C^{-1})[S,S] \geq (C[S,S])^{-1} .$$

Proof From the Banachiewicz-Schur formula, (Theorem A.2.3), we see that

$$(C^{-1})[S,S] = (C_{N\setminus S}[S,S])^{-1}.$$

So, it suffices to prove that

$$\begin{aligned} &C[S,S] \geq C_{N\setminus S}[S,S] \\ &\qquad (\ = C[S,S] - C[S,N\setminus S](C[N\setminus S, N\setminus S])^{-1}C[N\setminus S,S]\ ). \end{aligned}$$

But this is implied by

$$C[S,N\setminus S](C[N\setminus S,N\setminus S])^{-1}C[N\setminus S,S] \geq 0,$$

which is clear from

$$C \geq 0 \Longrightarrow C[N\setminus S,N\setminus S] \geq 0 \Longrightarrow C[N\setminus S,N\setminus S])^{-1} \geq 0.$$

□

When C is positive semidefinite, there is an interlacing of the eigenvalues of Schur complements with the eigenvalues of C.

Theorem A.4.7 (eigenvalue interlacing for the Schur complement) For $C \in \mathbb{S}^n_+$ and $S \subset N$ with $C[S,S]$ invertible, we have

$$\begin{aligned} &(i) \quad \lambda_\ell(C) \geq \lambda_\ell(C_S[N\setminus S, N\setminus S]) && (\ell\text{-th greatest eigenvalue}); \\ &(ii) \quad \lambda_{n+1-\ell}(C) \leq \lambda_{|S|+1-\ell}(C_S[N\setminus S, N\setminus S]) && (\ell\text{-th least eigenvalue}), \end{aligned}$$

for $1 \leq \ell \leq |S|$.

Proof When C is symmetric, the equation of Proposition A.2.1 is then a similarity (because then $Z = X^{-1}$). So the eigenvalues of C and

$$\left(\begin{array}{c|c} C[S,S] & 0 \\ \hline 0 & C_S[N\setminus S, N\setminus S] \end{array}\right)$$

are the same. Because $C_S[N \setminus S, N \setminus S]$ is a principal submatrix of this latter matrix, the result follows from Theorem A.3.3. □

A straightforward consequence of the spectral decomposition (Theorem A.3.1) is the following nice property that positive semidefiniteness is preserved under Hadamard (i.e., element-wise) products.

Theorem A.4.8 (Schur product theorem) If $A, B \in \mathbb{S}^n_+$, then $A \circ B \in \mathbb{S}^n_+$.

Proof Via the spectral decomposition (Theorem A.3.1), we write $A = \sum_i u_i u_i^\mathsf{T}$ and $B = \sum_j v_j v_j^\mathsf{T}$. Then $A \circ B = \sum_i \sum_j (u_i u_i^\mathsf{T}) \circ (v_j v_j^\mathsf{T})$. We claim that $(u_i u_i^\mathsf{T}) \circ (v_j v_j^\mathsf{T}) = (u_i \circ v_j)(u_i \circ v_j)^\mathsf{T}$. To see this, just check that the entry in row ℓ and column k of $(u_i u_i^\mathsf{T}) \circ (v_j v_j^\mathsf{T})$ is simply $u_{i\ell} u_{ik} v_{j\ell} v_{jk}$, and the entry in row ℓ and column k of $(u_i \circ v_j)(u_i \circ v_j)^\mathsf{T}$ is simply $u_{i\ell} v_{j\ell} u_{ik} v_{jk}$, which is the same. Therefore, $A \circ B = \sum_i \sum_j (u_i \circ v_j)(u_i \circ v_j)^\mathsf{T}$, which is clearly positive semidefinite, because it is the sum of positive-semidefinite matrices. □

The spectral decomposition of positive semidefinite matrices also allows us to define functions of such matrices. For example, with the spectral decomposition $A = \sum_{\ell=1}^n \lambda_\ell(A) v_\ell v_\ell^\mathsf{T}$, we can define functions of A from univariate functions via: $f(A) := \sum_{\ell=1}^n f(\lambda_\ell(A)) v_\ell v_\ell^\mathsf{T}$. The essential uniqueness of the spectral decomposition ensures for us that the functions are well defined. It is easy to check consistency with common usage. For example, for $f : \mathbb{R} \to \mathbb{R}$ defined by $f(w) := w^p$, for $p > 1$, we do get that $f(A) = A^{-1}$, when $p = -1$. That is, A and A^{-1} have the same eigenvectors, and the eigenvalues of A and A^{-1} associated with a given eigenvector are reciprocals of one another. Importantly, we have the **matrix square root** when $p = 1/2$ (This idea is useful in the proof of the next result). When A is symmetric positive definite, $A^{1/2}$ is the unique symmetric positive definite matrix B satisfying $A = BB$. Also, the familiar rules for exponents apply: $A^p A^q = A^{p+q}$, $(A^p)^q = A^{pq}$, etc.

Theorem A.4.9 (Minkowski's determinant inequality) If $A, B \in \mathbb{S}^n_+$, then

$$(\det(A + B))^{1/n} \geq (\det(A))^{1/n} + (\det(B))^{1/n}.$$

Proof The result is trivial if both A and B are singular, so without loss of generality assume that A is nonsingular, hence positive definite. Then its matrix square-root $A^{1/2}$ exists, and likewise the inverse $A^{-1/2}$ of $A^{1/2}$ exists. It is easy to check that multiplying on left and right by $\left((\det(A^{-1/2}))\right)^{1/n}$ and simplifying, the inequality that we want to prove is equivalent to

$$(\det(I + A^{-1/2}BA^{-1/2}))^{1/n} \geq 1 + (\det(A^{-1/2}BA^{-1/2}))^{1/n}.$$

But now we can redefine $A^{-1/2}BA^{-1/2} \to B$ and without loss of generality prove

$$(\det(I + B))^{1/n} \geq 1 + (\det(B))^{1/n}.$$

We have $\lambda_i(I + B) = 1 + \lambda_i(B)$, for $i = 1, 2, \ldots, n$. Let $\lambda_i := \lambda_i(B)$, $i = 1, 2, \ldots, n$. What we want to prove is that

$$\prod_{i=1}^{n}(1 + \lambda_i)^{1/n} \geq 1 + \prod \lambda_i^{1/n}.$$

We have

$$\begin{aligned}\frac{1 + \prod \lambda_i^{1/n}}{\prod_{i=1}^{n}(1 + \lambda_i)^{1/n}} &= \prod \left(\frac{1}{1 + \lambda_i}\right)^{1/n} + \prod \left(\frac{\lambda_i}{1 + \lambda_i}\right)^{1/n} \\ &\leq \frac{1}{n}\sum_{i=1}^{n}\frac{1}{1 + \lambda_i} + \frac{1}{n}\sum_{i=1}^{n}\frac{\lambda_i}{1 + \lambda_i} \;=\; 1,\end{aligned}$$

the inequality coming from two applications of the arithmetic-geometric mean inequality (Theorem A.1.2). □

A convenient weakening of Minkowski's determinant inequality (Theorem A.4.9) is:

Corollary A.4.10 If $A, B \in \mathbb{S}_+^n$, then

$$\det(A + B) \geq \det(A) + \det(B).$$

Proof

$$\begin{aligned}\det(A + B) &\geq \left((\det(A))^{1/n} + (\det(B))^{1/n}\right)^n \\ &= \sum_{i=0}^{n}\binom{n}{i}(\det(A))^{i/n}(\det(B))^{(n-i)/n} \\ &\geq \det(A) + \det(B).\end{aligned}$$

□

We will use this weakening of Minkowski's determinant inequality in the proof of the next result.

Theorem A.4.11 (Oppenheim's inequality ([HJ85, Theorem 7.8.6] and [Opp30])). For $A, B \in \mathbb{S}_+^n$,

$$(\det A) \prod_{i=\ell}^{n} b_{\ell\ell} \leq \det(A \circ B).$$

Proof Our proof is by induction on n, using Schur complements to decrease n. Clearly the result is true for $n = 1$, so we assume that $n > 1$ and that the results hold for matrices of order $n - 1$.

Let $C := A \circ B$. Using the Schur complement determinant formula (Theorem A.2.2), we have

$$\begin{aligned}\det(C) &= c_{1,1} \times \det C_{\{1\}}[N-1, N-1] \\ &= a_{1,1} b_{1,1} \times \det \underbrace{\Big(C[N-1, N-1] - C[N-1, 1](c_{1,1})^{-1} C[1, N-1] \Big)}_{(*)}.\end{aligned}$$

Next, a bit magically, we will rewrite $(*)$ as a sum of a pair of Hadamard products, one with a (Hadamard) factor of $A_{\{1\}}[N-1, N-1]$ and the other with a (Hadamard) factor of $B_{\{1\}}[N-1, N-1]$.

$$\begin{aligned}&C[N-1, N-1] - C[N-1, 1](c_{1,1})^{-1} C[1, N-1] \\ &= B[N-1, N-1] \circ A_{\{1\}}[N-1, N-1] \\ &\quad + \left(\frac{1}{a_{1,1}} A[N-1, 1] A[1, N-1] \right) \circ B_{\{1\}}[N-1, N-1].\end{aligned}$$

By the Schur product theorem (Theorem A.4.8), both summands are positive semidefinite. By the corollary of Minkowski's determinant formula (Corollary A.4.10), we then obtain

$$\det(*) \geq \det \Big(B[N-1, N-1] \circ A_{\{1\}}[N-1, N-1] \Big),$$

and so we have

$$\det(C) \geq a_{1,1} b_{1,1} \det \Big(B[N-1, N-1] \circ A_{\{1\}}[N-1, N-1] \Big).$$

Applying the inductive hypothesis, and then the Schur complement determinant formula (Theorem A.2.2), we have now

$$\begin{aligned}\det(C) &\geq a_{1,1}b_{1,1}\left(\prod_{\ell=1}^{n-1} b_{\ell\ell}\right) \times \det\left(A_{\{1\}}[N-1, N-1]\right) \\ &= (\det A)\prod_{\ell=1}^{n} b_{\ell\ell} \ .\end{aligned}$$

□

When $B = \mathrm{Diag}(B_1, \ldots, B_p)$, with each B_ℓ a square block of ones, $p \geq 1$, Oppenheim's inequality becomes **Fischer's inequality**: For $A = \mathrm{Diag}(A_1, \ldots, A_p)$, with each A_ℓ a square block, $p \geq 1$,

$$\det A \leq \prod_{\ell=1}^{p} \det(A_\ell) \ .$$

The further special case of $B = I_n$ is known as **Hadamard's inequality**:

$$\det A \leq \prod_{\ell=1}^{n} a_{\ell\ell} \ .$$

Finally, we move on to a couple of useful results concerning the ldet function.

Theorem A.4.12 The function ldet is strictly concave on $\mathbb{S}^n_{++}$.

Proof It suffices to demonstrate that ldet is strictly concave on every line in $\mathbb{S}^n_{++}$. For $X \in \mathbb{S}^n_{++}$, $Z \in \mathbb{S}^n_{+}$ and $t \in \mathbb{R}$, we have

$$\begin{aligned}f(t) &:= \mathrm{ldet}(X + tZ) \\ &= \mathrm{ldet}(\left(X^{\frac{1}{2}}\left(I + tX^{-\frac{1}{2}}ZX^{-\frac{1}{2}}\right)X^{\frac{1}{2}}\right) \\ &= \mathrm{ldet}\,(X) + \mathrm{ldet}\left(tX^{-\frac{1}{2}}ZX^{-\frac{1}{2}}\right) \\ &= \mathrm{ldet}\,(X) + \sum_{i=1}^{n} \log\left(\lambda_i\left(I + tX^{-\frac{1}{2}}ZX^{-\frac{1}{2}}\right)\right) \\ &= \mathrm{ldet}\,(X) + \sum_{i=1}^{n} \log\left(1 + t\ \lambda_i\left(X^{-\frac{1}{2}}ZX^{-\frac{1}{2}}\right)\right) .\end{aligned}$$

The result now follows from the strict concavity of the log function. □

Lemma A.4.13 Let V be an $n \times n$ matrix with positive determinant. Then

$$\frac{\partial}{\partial V_{ij}} \operatorname{ldet} V = (V^{-\mathsf{T}})_{ij} \, .$$

Proof For $i \in N$, we have the Laplace expansion (on row i)

$$\det V = \sum_{j=1}^{n} V_{ij} \mathcal{V}_{ij} \, ,$$

where $\mathcal{V}_{ij}$ is the cofactor[8] of the element V_{ij} in V. Then,

$$\frac{\partial \operatorname{ldet} V}{\partial V_{ij}} = \frac{1}{\det V} \frac{\partial \det V}{\partial V_{ij}} = \frac{1}{\det V} \mathcal{V}_{ij} = (V^{-\mathsf{T}})_{ij} \, ,$$

where in the last step we use the well-known fact that the transpose of the matrix of cofactors of V (known as the **adjugate** of V or the **classical adjoint** of V) is equal to $\det V$ times the inverse of V. □

Finally, we have a bit more calculus for functions of the eigenvalues of symmetric matrices. First, we should introduce the following definitions.

Let $f : \mathbb{R}^n \to \mathbb{R}$ and let Π be a partition of $N := \{1, 2, \ldots, n\}$. We say that f is a **symmetric function with respect to the partition** Π if f is invariant under the permutation of the variables induced by permutations of the indices within the blocks of Π.

For a vector $\hat{z} \in \mathbb{R}^n$, we define $\Pi_{\hat{z}}$, **the partition of** N **induced by** $\hat{z}$, as the partition of N having i and j in the same block if and only if $\hat{z}_i = \hat{z}_j$.

In the next examples, we clarify these definitions.

Example A.4.14 For some $0 < s < n$, let $f : \mathbb{R}^n \to \mathbb{R}$ be defined by $f(z) := \prod_{\ell=1}^{s} z_\ell$. Then,

- f is clearly symmetric with respect to $\Pi := \{\{1, 2, \ldots, s\}, \{s+1, s+2, \ldots, n\}\}$;
- given $\hat{z} \in \mathbb{R}^n$, such that $\hat{z}_1 > \hat{z}_2 > \cdots > \hat{z}_n$, then $\Pi_{\hat{z}} = \{\{1\}, \{2\}, \ldots, \{n\}\}$. In this case, $f(\hat{z})$ is the product of the s greatest components of $\hat{z}$, and f is trivially symmetric with respect to $\Pi_{\hat{z}}$ — because each block of $\Pi_{\hat{z}}$ has only one element;

[8] $\mathcal{V}_{ij} := (-1)^{i+j} \det M_{ij}$, where M_{ij} is the matrix obtained from V by striking out row i and column j.

- given $\hat{z} \in \mathbb{R}^n$, such that $\min\{\hat{z}_1, \hat{z}_2, \ldots, \hat{z}_s\} > \max\{\hat{z}_{s+1}, \hat{z}_{s+2}, \ldots, \hat{z}_n\}$, then $f(\hat{z})$ is the product of the s greatest components of $\hat{z}$, and f is symmetric with respect to $\Pi_{\hat{z}}$, which must be a refinement of the partition $\Pi := \{\{1, 2, \ldots, s\}, \{s+1, s+2, \ldots, n\}\}$;
- we note that f is *not* symmetric with respect to any partition of N that puts any elements of $\{1, 2, \ldots, s\}$ in the same block as an element from $\{s+1, s+2, \ldots, n\}$. For example, if for a given $\hat{z}$, we have $\min\{\hat{z}_1, \hat{z}_2, \ldots, \hat{z}_s\} = \max\{\hat{z}_{s+1}, \hat{z}_{s+2}, \ldots, \hat{z}_n\}$, then f is *not* symmetric with respect to $\Pi_{\hat{z}}$. ♠

Example A.4.15 For $z \in \mathbb{R}^n$, let $z_{[\ell]}$ be the ℓ-the greatest component of z. For some $0 < s < n$, let $f : \mathbb{R}^n \to \mathbb{R}$ be defined by $f(z) := \prod_{\ell=1}^{s} z_{[\ell]}$. In contrast to Example A.4.14, built into the present function f is a sorting of the components of z. This function f is continuous and symmetric with respect to $\Pi := \{\{1, 2, \ldots, n\}\}$, in contrast to Example A.4.14, where symmetry is with respect to a partition if and only if the partition is a refinement of $\{\{1, 2, \ldots, s\}, \{s + 1, s + 2, \ldots, n\}\}$. Now consider the subdivision of $\mathbb{R}^n$ determined by all permutations of $\{1, 2, \ldots, n\}$. That is, we associate with each permutation π, the cone

$$S(\pi) := \left\{ z \in \mathbb{R}^n \ : \ z_{\pi(1)} \geq z_{\pi(2)} \geq \cdots \geq z_{\pi(n)} \right\}.$$

These $n!$ cones partition $\mathbb{R}^n$ in such a way that the intersection of any pair is a face of each. On a given cone $S(\pi)$, we have that $f(z) = \prod_{\ell=1}^{s} z_{\pi(\ell)}$. And then it is clear that f is differentiable on the interior of each $S(\pi)$. So the only place where f could be non-differentiable is on the boundaries of the $S(\pi)$; that is, at z for which some components are equal. A closer look at f reveals that it is only at places where $z_{[s]} = z_{[s+1]}$ where f is non-differentiable. ♠

Now we are prepared to state a key result. We consider $M : \mathbb{R}^m \to \mathbb{S}^n$, and we assume that the elements of $M(x)$ are analytic functions of x, for all $x \in \mathbb{R}^m$. Let $\lambda_1(x) \geq \cdots \geq \lambda_n(x)$ be the nonincreasing sequence of eigenvalues of $M(x)$, and we write $\lambda(x) := (\lambda_1(x), \lambda_2(x), \ldots, \lambda_n(x))$. Continuity of $M(x)$ implies continuity of its eigenvalues, but differentiability is not so clear, when there are eigenvalues with multiplicity (see the Proposition A.4.22 and its proof). But we have the following nice and useful result.

Theorem A.4.16 (see [TFV94, part of Theorem 3.1]) Suppose that $M : \mathbb{R}^m \to \mathbb{S}^n$, and assume that the elements of $M(x)$ are analytic functions of x, for all $x \in \mathbb{R}^m$. Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is analytic and symmetric with respect to the partition $\Pi_{\lambda(\hat{x})}$, for a given $\hat{x} \in \mathbb{R}^m$. Writing $\lambda(x) := \lambda(M(x))$, the composite function $g(x) := f(\lambda(x))$ is analytic at $\hat{x}$. Furthermore, we have

$$\frac{\partial g(\hat{x})}{\partial x_i} = \sum_{k=1}^{n} \frac{\partial f(\lambda(\hat{x}))}{\partial \lambda_k} h_i^k(\hat{x}),$$

where

$$h_i^k(\hat{x}) := u_k^{\mathsf{T}}(\hat{x}) \frac{\partial M(\hat{x})}{\partial x_i} u_k(\hat{x}),$$

and where the $u_k(\hat{x})$ are any set of orthonormal eigenvectors associated to $\lambda_k(\hat{x})$, for $k = 1, 2, \ldots, n$.

Example A.4.17 Let $f(z) := \prod_{\ell=1}^{s} z_\ell$. Then f is analytic and is symmetric for every refinement of $\{\{1, 2, \ldots, s\}, \{s+1, s+2, \ldots, n\}\}$. Therefore, if M satisfies the hypotheses of Theorem A.4.16, then $f(\lambda(x))$ is analytic at $\hat{x}$ whenever $\lambda_{s+1}(M(\hat{x})) > \lambda_s(M(\hat{x}))$ (because then $\Pi_{\lambda(\hat{x})}$ is a refinement of $\{\{1, 2, \ldots, s\}, \{s+1, s+2, \ldots, n\}\}$). ♠

Remark A.4.18 [TFV94, Theorem 3.1] also provides (complicated) formulae for the second derivatives of $g(x) := f(\lambda(x))$, under the same technical conditions of Theorem A.4.16.

Remark A.4.19 The complete proof of Theorem A.4.16 is complicated, and we refer readers to [TFV94]. But in the special case that $\lambda(\hat{x})$ has all distinct components, then each $\lambda_k(\hat{x})$ is differentiable. In that case, it is well known (but also see the proof below) that

$$\frac{\partial \lambda_k(\hat{x})}{\partial x_i} = h_i^k(\hat{x}),$$

and we can then see

$$\frac{\partial g(\hat{x})}{\partial x_i} = \sum_{k=1}^{n} \frac{\partial f(\lambda(\hat{x}))}{\partial \lambda_k} \frac{\partial \lambda_k(\hat{x})}{\partial x_i}$$

(the conclusion of Theorem A.4.16 in this special case) as simply an application of the chain rule.

To see how in this case we get the simple formula above for the derivative of an eigenvalue, we carry out some elementary calculations. Below, we write $M := M(x)$, $\lambda := \lambda_k(x)$, and $u := u_k(x)$, and we do not explicitly repeat the needed assumption in these derivations that $\lambda_k(\hat{x})$ has multiplicity one in $\lambda(\hat{x})$, and justification that all of the needed derivatives exist is via the implicit function theorem.

Lemma A.4.20 In an open neighborhood of $\hat{x}$, we have $u^\mathsf{T} \frac{\partial u}{\partial x_i} = 0$.

Proof $u^\mathsf{T} u = 1$ implies that $\frac{\partial u^\mathsf{T}}{\partial x_i} u + u^\mathsf{T} \frac{\partial u}{\partial x_i} = 0$. Because this is a sum of equal dot products, the result follows. □

Lemma A.4.21 In an open neighborhood of $\hat{x}$, we have $u^\mathsf{T} M \frac{\partial u}{\partial x_i} = 0$.

Proof Transposing $Mu = \lambda u$, we have $u^\mathsf{T} M = \lambda u^\mathsf{T}$, which implies $u^\mathsf{T} M \frac{\partial u}{\partial x_i} = \lambda u^\mathsf{T} \frac{\partial u}{\partial x_i}$, which is 0 by Lemma A.4.20. □

Proposition A.4.22

$$\frac{\partial \lambda_k(\hat{x})}{\partial x_i} = u_k^\mathsf{T}(\hat{x}) \frac{\partial M(\hat{x})}{\partial x_i} u_k(\hat{x}),$$

for $k = 1, \ldots, m$ and $i = 1, \ldots, n$.

Proof In an open neighborhood of $\hat{x}$, from $\lambda u = Mu$, we obtain

$$\frac{\partial \lambda}{\partial x_i} u + \lambda \frac{\partial u}{\partial x_i} = \frac{\partial M}{\partial x_i} u + M \frac{\partial u}{\partial x_i}.$$

Left multiplying by u^T, we obtain

$$\frac{\partial \lambda}{\partial x_i} \underbrace{u^\mathsf{T} u}_{=1} + \lambda \underbrace{u^\mathsf{T} \frac{\partial u}{\partial x_i}}_{=0} = u^\mathsf{T} \frac{\partial M}{\partial x_i} u + \underbrace{u^\mathsf{T} M \frac{\partial u}{\partial x_i}}_{=0},$$

which, using the lemmas and then evaluating at $\hat{x}$, is our desired result. □

For a convex function $f : \mathbb{R}^n \to \mathbb{R}$, recall that g is a **subgradient** of f at a point $\hat{x}$ in the (convex) domain of f, if

$$f(x) \geq f(\hat{x}) + g^\mathsf{T}(x - \hat{x}), \text{ for all } x \text{ in the domain of } f.$$

The set of subgradients of f at $\hat{x}$ is the **subdifferential** of f at $\hat{x}$, which we denote by $\partial f(\hat{x})$. If f is differentiable at $\hat{x}$, then there is a unique subgradient of f at $\hat{x}$, which is precisely the gradient $\nabla f(\hat{x})$. The following result is easy to show.

Lemma A.4.23 If $f_i : \mathbb{R}^n \to \mathbb{R}$ are differentiable convex functions, for $i = 1, 2, \ldots, p$, then the pointwise-maximum function

$$f(x) := \max_{i=1,\ldots,p} f_i(x)$$

is convex on the intersection of the domains of the f_i , and

$$\partial f(\hat{x}) = \text{conv}\left\{\nabla f_i(\hat{x}) \ : \ i \in \{1, 2, \ldots, p\} \text{ and } f_i(\hat{x}) = f(\hat{x})\right\}.$$

Proposition A.4.24 Let $f : \mathbb{R}^k \to \mathbb{R}$ be defined by $f(z) := \sum_{i=k-s+1}^{k} g(z_{[i]})$, where g is a univariate nonincreasing convex function on $\mathbb{R}_{++}$. Then f is convex. Moreover, if $z_1 \le z_2 \le \cdots \le z_k$, with $s \in \mathcal{E} := \{\ell+1, \ldots, t\}$ defined so that $z_\ell < z_{\ell+1} = \cdots = z_t < z_{t+1}$, and $\mathcal{F} \subset \mathcal{E}$ with $|\mathcal{F}| = s - \ell$, we define

$$\nabla^{\mathcal{F}}(z) := (g'(z_1), g'(z_2), \ldots, g'(z_\ell), \underbrace{\overset{\ell+1}{*}, *, \ldots, \overset{t}{*}}_{\mathcal{E}}, 0, 0, \ldots, 0)^{\mathsf{T}},$$

where we set the "$*$" entries, depending on the choice of $\mathcal{F}$, as follows

$$\nabla^{\mathcal{F}}_i(z) = \begin{cases} g'(z_i), & \text{for } i \in \mathcal{F}; \\ 0, & \text{for } i \in \mathcal{E} \setminus \mathcal{F}. \end{cases}$$

$\nabla^{\mathcal{F}}(z)$ is a subgradient of f at z, and the subdifferential of f at z is the convex hull of all such subgradients.

Proof The key point to observe is that, because g is nonincreasing on $\mathbb{R}_{++}$,

$$f(z) = \max_{\substack{S \subset \{1,\ldots,k\} \\ : \, |S| = s}} \left\{ \sum_{i \in S} g(z_i) \right\}.$$

That is, f is the point-wise maximum of convex functions. Calculating the gradient for each of the ones that are active at a given point z, we obtain the result by applying Lemma A.4.23. □

An **absolutely symmetric function** $f : \mathbb{R}^n \to \mathbb{R}$ is one satisfying $f(\pi(z)) = f(z)$ for every permutation π and $f(z) = f(|z|)$, where $|z|$ denotes component-wise absolute value. When f is a norm that is absolutely symmetric, it is called a **symmetric gauge function**. An interesting example is the **Ky Fan k norm**

$$f(z) := \text{the sum of the } k \text{ greatest elements of } \{|z_1|, \ldots, |z_n|\}.$$

Another example is the function f of Proposition A.4.24. This is useful for us (see proof of Theorem 3.4.7) for the concrete case where the univariate g (in the definition of this f) is the negative of the log function.

We are interested in the subdifferential of the composition of an absolutely symmetric function f with the function λ that gives the eigenvalues of a symmetric matrix.

Theorem A.4.25 (see [Lew95, special case of Corollary 2.5]) Let $f : \mathbb{R}^n \to \mathbb{R}$ be an absolutely symmetric convex and continuous function. Let $\lambda : \mathbb{S}^n_+ \to \mathbb{R}^n$ be the function giving the eigenvalues of a positive-semidefinite symmetric matrix, in nonincreasing order. Then $f \circ \lambda$ is convex and continuous, and its subdifferential is given by

$$\partial(f \circ \lambda)(\Theta) = \left\{ V \operatorname{Diag}(\mu) V^{\mathsf{T}} \ : \ \mu \in \partial(f(\lambda(\Theta)),\ V^{\mathsf{T}} V = I,\ \Theta = V \operatorname{Diag}(\lambda(\Theta)) V^{\mathsf{T}} \right\}.$$

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